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Day Month Year

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = ready to send to Harry

- Introduction Chapter (1): Not started. Will write last/almost last
- IFO Chapter (2): Made first set of Ken's edits.
- PSL Chapter (3): Done?
- BHD Chapter (5): Editing resulted in this chapter being almost completely rewritten.
- BHD Chapter (6): Edited, need more comments.
- SRM beam parameter Chapter (4): Needs basic grammar check, all structure and style changes have been made
- MZ chapter Chapter (??): Not written, vaguely planned
- Photodiode Chapter (8): Need to finish adding references.

- Conclusion Chapter (9): Not started. Will write last/almost last

Chapter 1

Introduction

Chapter 2

The Fundamentals of Gravitational Wave Interferometers

2.1 Introduction

On the 14th of September 2015, the detection of a gravitational wave emanating from two coalescing black holes signalled the beginning of a new era for astronomy [1]. The information carried by a gravitational wave is undisturbed by the matter between the observer and the astrophysical body that is emitting them, thus it gives astronomers valuable information which cannot be obtained from the electromagnetic radiation emitted by that body. Gravitational wave astronomy also allows for direct observations of objects that do not emit electromagnetic radiation, e.g. black

holes. Therefore, it is important that large numbers of gravitational wave signals are measured to develop our understanding of the universe.

However, gravitational waves are incredibly weak and the detection of them is a science in itself. This chapter will outline how a gravitational wave interferometer, such as LIGO [2], can be used to detect these signals and the fundamental noise sources these detectors face. While there are other important sources of noise, the suppression of quantum noise is crucial to achieving the sensitivity required to detect gravitational waves; therefore, quantum noise is the focus of this chapter.

2.2 The Astrophysical Origins of Gravitational Waves

Gravitational waves produced by inspiraling pairs of massive objects, such as binary black holes (BBH) (e.g. [1,3–5]), binary neutron stars (BNS) [6,7], and a source which is either a BBH or a binary consisting of a black hole and a neutron star (NSBH) [8], have been detected. Every confirmed gravitational wave signal to date (November 2020) is summarised in [9]. The waveform of the gravitational radiation emitted from these systems is known as a chirp because the frequency of the gravitational wave signal increases as the two objects get closer to merging [10]. The cut-off frequency is determined by the radius of the objects that will merge; for the signals which have been measured, the cut-off frequencies were between 100 Hz and 10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away

from Earth, is $h \sim 10^{-21} - 10^{-23}$.

Einstein's general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo interferometers [11]. Gravitational waves are emitted from mass quadrupole systems (see e.g. [12] or [13]). As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses moving at relativistic velocities.

In addition to binary systems, other astrophysical phenomena will emit gravitational waves. Asymmetries in pulsars [14, 15] and magnetars [16] will cause them to emit a gravitational wave with a slowly evolving frequency. When the core of a star collapses as it goes supernova, a burst of gravitational radiation may be emitted [17, 18]. Burst signals may also be generated immediately after a BNS merges [19]. Weaker sources that future gravitational wave interferometers may detect include: a stochastic background level of gravitational waves due to the merging of compact binaries from all over the universe [20], dark matter [21, 22], and sources that existed during the early universe [23, 24].

To improve statistical models of the systems that emit gravitational waves, the number of detections needs to be increased. To detect continuous wave signals, the signal to noise ratio (SNR) improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades. The previous, current

Obs. Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 30/04/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	2025	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 2.1: Summary of observation runs for LIGO, Virgo and KAGRA [2, 26, 27]. This table includes detections that were made up to the time of writing (November 2020), and analysis of the O3 data is ongoing.

and future observation runs are summarised in Table 2.1; a more comprehensive overview can be found in [25].

2.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (e.g. [12] or [13]). The effect these waves have, in the local Lorentz frame, on free test masses on Earth is to move them by a small amount ¹.

¹In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses due to the strong equivalence principle: All non-gravitational laws of physics which can be expressed in a special relativistic form have the same form in a local Lorentz frame (e.g. [12]). In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{4\pi L^2}{\lambda_{\text{EM}}} h$. The

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These mirrors are referred to as *end test masses* (ETMs). A suspended mass will behave in the same way as a free mass when it is driven above its resonant frequency, so the ETMs need to be suspended [28]. A sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 2.1. The amount test masses move, ΔL , when they are at a distance of L away from the origin of the local Lorentz frame due to an incoming plus polarised gravitational wave of strain h is

$$\Delta L = \frac{1}{2}Lh. \quad (2.1)$$

Interferometers are sensitive to arm length changes that are on the order of one wavelength, λ_{EM} , of the light they are operated with. As a plus polarised gravitational wave will move the test masses in anti-phase, the difference between the phase of light when it recombines at the beam splitter after reflecting from the test masses, $\Delta\phi$, is

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}}Lh. \quad (2.2)$$

$\Delta\phi$ is known as the *differential phase*, and differential effects result in the light acquiring differential phase. The wavelength current interferometers operate with, $\lambda_{\text{EM}} = 1064 \text{ nm}$, was selected because test masses with low noise and mirror coatings with low loss and low thermal noise can be fabricated for this wavelength [29].

Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = hL$. Maxwell's equations are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{4\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{4\pi L}{\lambda_{\text{EM}}}$ is small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses.

The gravitational wave strain is converted into intensity fluctuations of light at the *antisymmetric* (AS) port due to the two light fields from the arms interfering at the beam splitter (see Figure 2.1). When on the *dark fringe*, there is a small amount of light at the AS port for the signal light to beat with and so the variation in the intensity of the light depends linearly on the gravitational wave strain. The interferometer is set to operate on the dark fringe by introducing a small static differential offset, ϕ_0 , in the arm lengths. This technique is known as DC readout [30, 31]. At the beam splitter, the electric field, Ψ_{AS} , emerging from the AS port due to the interference of the beams returning from the ETMs, Ψ_y and Ψ_x , due to a differential phase, $\Delta\phi = \phi_0 + \frac{4\pi}{\lambda_{\text{EM}}}Lh$, is

$$\Psi_{\text{AS}} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (2.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i\Delta\phi/2}}{\sqrt{2}} - \frac{e^{-i\Delta\phi/2}}{\sqrt{2}} \right), \quad (2.4)$$

where I_0 is the power of the beam injected into the beam splitter by the laser. The intensity of the beam at the AS port, I_{AS} , is

$$I_{\text{AS}} = \Psi_{\text{AS}}^* \Psi_{\text{AS}}, \quad (2.5)$$

$$\simeq \frac{I_0 \phi_0^2}{4} + \frac{I_0 \phi_0}{2} \frac{4\pi}{\lambda_{\text{EM}}} Lh. \quad (2.6)$$

In the absence of a gravitational wave signal, the DC light level at the AS port is related to the static offset by $\phi_0 = 2\sqrt{\frac{I_{\text{AS}}}{I_0}}$, where $\overline{I_{\text{AS}}}$ is the average of I_{AS} . As we are interested in a fluctuating signal, the DC offset from Equation (2.6) can be

removed, allowing for the signal at the AS port to be written as

$$I_{\text{AS}} = \sqrt{I_0 I_{\text{AS}}} \frac{4\pi}{\lambda_{\text{EM}}} Lh. \quad (2.7)$$

Therefore, it is possible to measure the differential motion of the ETMs due to a gravitational wave's strain by placing a photodiode in the beam emerging from the AS port of a Michelson interferometer.

2.3.1 Using Fabry-Perot Cavities in the Arms of a Michelson Interferometer to Enhance Its Sensitivity

The optimal arm length of the interferometer is determined by $1/4$ of the wavelength of a gravitational wave as this gives the maximal $\Delta\phi$. Gravitational wave signals from BBH and BNS have frequencies of ~ 100 Hz; therefore, the wavelength of the gravitational wave will be of the order ~ 3000 km. It is not possible to construct a Michelson interferometer with ~ 1000 km arms on Earth.

The sites that were available to construct the LIGO detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off of the ETMs, B , needs to be increased. Numerically, B should be ~ 250 . This means that Equation (2.2) becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} BLh. \quad (2.8)$$

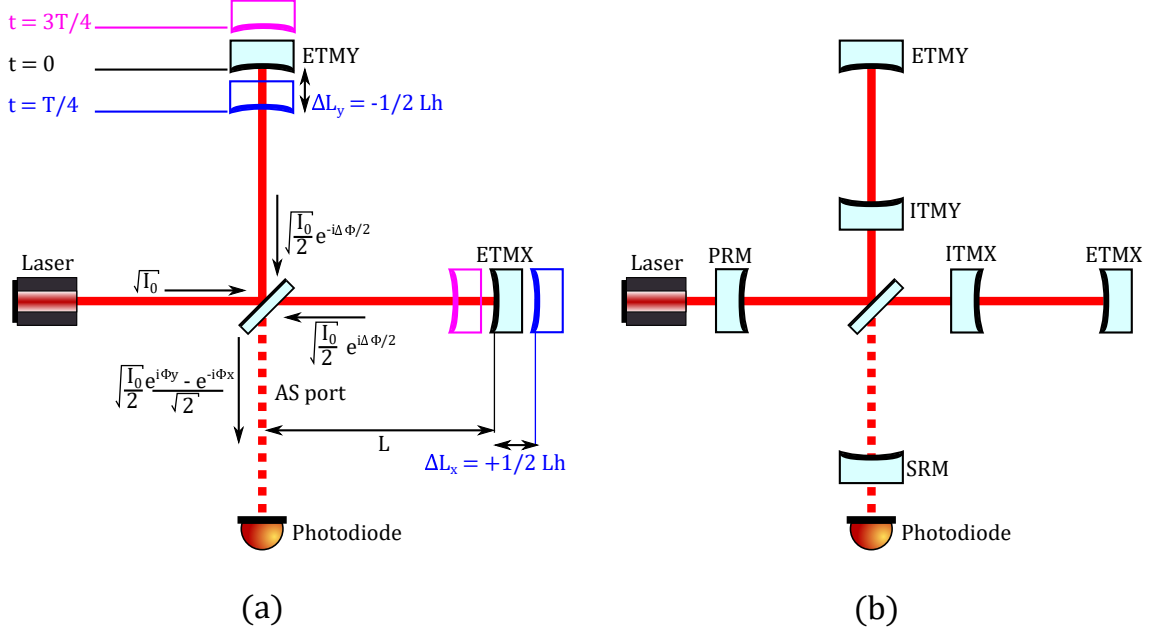


Figure 2.1: (a) A sketch of the effect a gravitational wave has on a Michelson interferometer. The positions of the test masses, ETMX and ETMY, at different times during the gravitational wave's period is shown by the blue and pink mirrors. The electric fields within the interferometer are indicated: $\sqrt{I_0}$ is the laser's amplitude and $\pm\Delta\phi/2$ is the phase acquired by the light as it travels up and down the arms. (b) To reach the sensitivity needed to detect gravitational waves, several mirrors need to be added to the basic Michelson interferometer: ITMX and ITMY boost the signal generated in the arms, the PRM increases the power at the beam splitter, and the SRM can be used to shape the quantum noise of the interferometer.

Fabry-Perot cavities are used to increase the number of times light bounces off of the ETMs². Cavities are formed by placing an *input test mass* (ITM) in each arm near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. Interferometers measure the difference in phase of the electromagnetic fields in the arms, $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry-Perot cavity is near or on resonance, the Fabry-Perot cavity enhances the detectors sensitivity as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much greater for a Fabry-Perot cavity than it is for a mirror. This is illustrated in Figure 2.2. The exact way this happens is dependent on the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity acting as a low pass filter (e.g. see [32, Chapter 3], [33, Appendix D], [34] or [35]).

2.4 Sources of Noise in Gravitational Wave Interferometers.

As the strain induced by a gravitational wave causes a minuscule change in the phase accumulated by light in the arm cavities, suppressing the noise below this signal is the main obstacle faced when trying to measure a gravitational wave. From Equation (2.8), for a gravitational wave strain of $h_{\text{rms}} = 10^{-23}$ interacting with LIGO, the beams interfering at the beam splitter will have a difference in phase of $\sim 10^{-11}$ rad.

²Herriott delay lines were considered as a means of achieving more bounces from the ETMs, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry-Perot cavities would perform equally well due to the Mizuno limit [32].

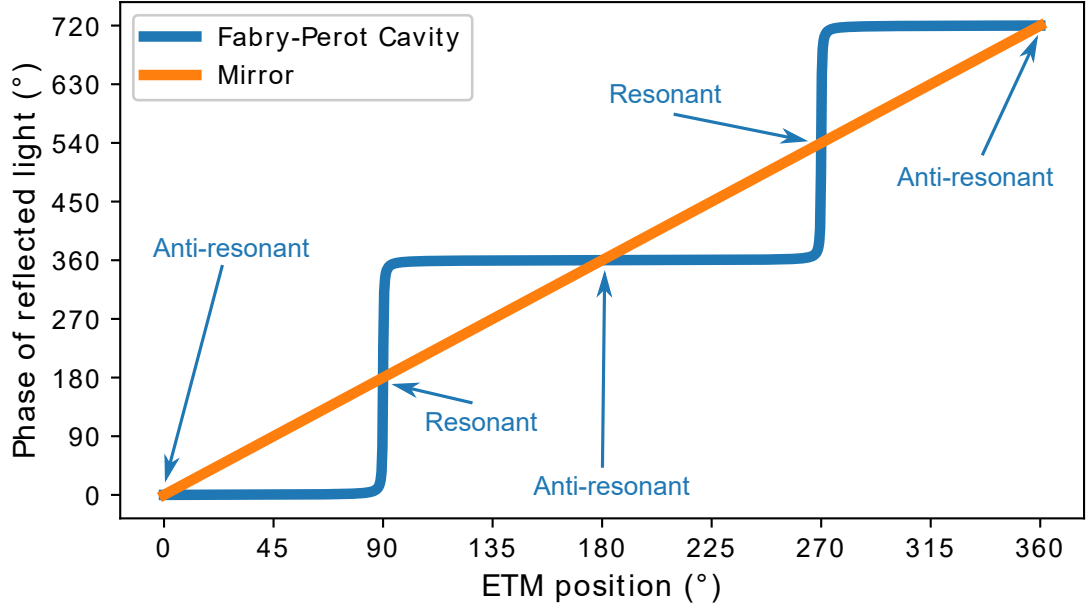


Figure 2.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity as the position of the mirror/end mirror is changed. The orange line shows the extra round trip phase acquired by light in the arm if the ETM is moved one wavelength (360°) and the blue line shows the same thing except for an Fabry-Perot cavity. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The interferometers sensitivity is proportional to the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$, and so a Fabry-Perot cavity can increase the sensitivity of the interferometer.

2.4.1 Laser Power and Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the interferometer. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The uncertainty in energy, ΔE , that associated with a measurement of the time between two events, ΔT , is determined by the time-energy uncertainty principle, $\Delta E \Delta T \geq \hbar/2$. The uncertainty in the phase, $\Delta \phi$, of a light beam with angular frequency ω_{EM} is related to the measurement time by $\Delta \phi = \omega_{EM} \Delta T$ [36]. The phase is averaged over a time corresponding to half the gravitational wave's period, so $\Delta T = \frac{1}{2f_{GW}}$. The energy used in the measurement is $E = I_0 \frac{1}{2f_{GW}} = \hbar \omega_{EM} N$, where I_0 is the power of the laser and N is the number of photons detected in the averaging time. As the laser is in a coherent state (e.g. [37–39]), the uncertainty in detecting N photons is $\Delta N = \sqrt{N}$; therefore, the uncertainty in the energy used in the measurement is $\Delta E = \hbar \omega_{EM} \sqrt{N}$. Using the expressions for ΔE and $\Delta \phi$, the uncertainty principle may be used to obtain

$$N \geq \frac{1}{4(\Delta \phi)^2}. \quad (2.9)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the AS port will depend on the amount of light at the dark port,

$$I_{ASrms} = \sqrt{2\bar{I}_{AS}\hbar\omega_{EM}\Delta f}. \quad (2.10)$$

The conversion from light power at the AS port to gravitational wave strain can be

inferred from Equation (2.7),

$$\frac{h}{I_{\text{AS}}} = \frac{1}{\sqrt{I_0 I_{\text{AS}}}} \frac{\lambda_{\text{EM}}}{4\pi L B}, \quad (2.11)$$

and so the amplitude spectrum of the strain corresponding to the shot noise of the light at the AS port is

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar c \lambda_{\text{EM}} \Delta f}{4\pi I_0 L^2 B^2}}. \quad (2.12)$$

Equation 2.12 assumed the frequency response of the interferometer to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity. This can be seen in the spectral density of the LIGO detector's noise (Figure 2.3).

To beat the photon shot noise by a factor of 10, Equation (2.8) and Equation (2.9) show that, for a LIGO-like interferometer, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave signal with a frequency of 100 Hz, the laser power incident on the beam splitter would need to be $\sim 1\text{MW}$. No such laser exists.

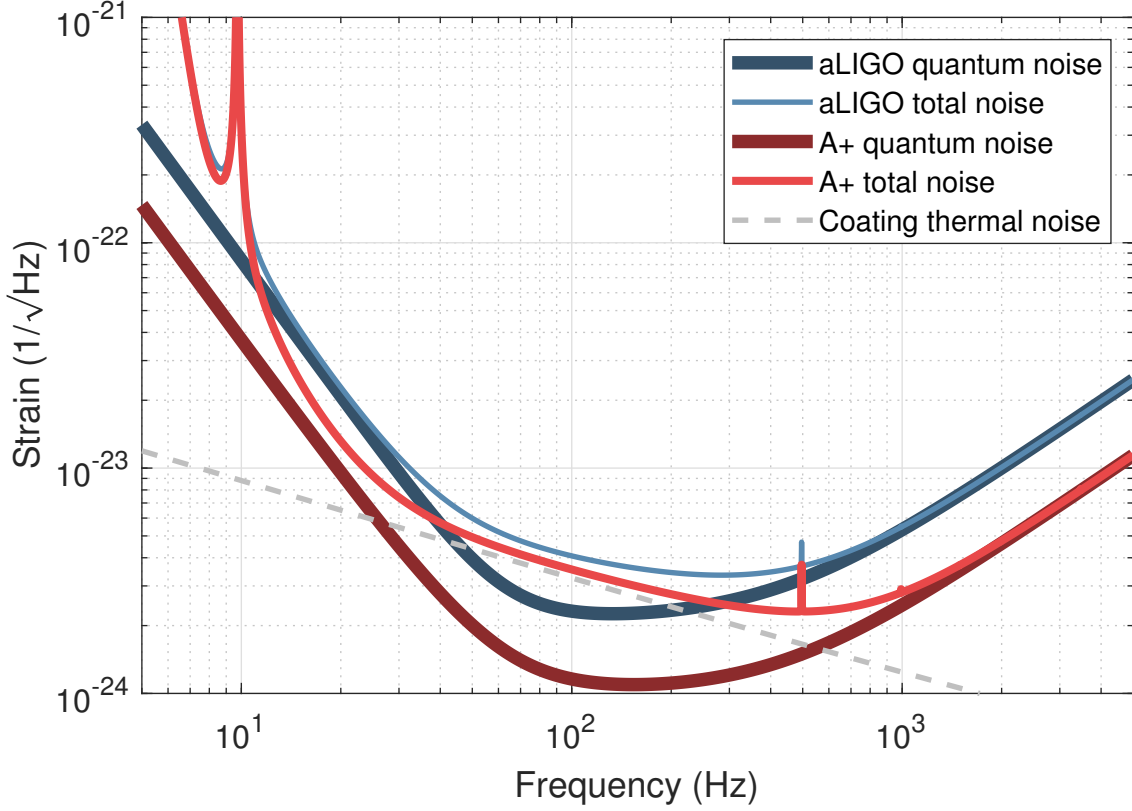


Figure 2.3: The light blue and light red lines show the design sensitivity of the aLIGO and LIGO A+ detectors. The quantum noise, noise due to radiation pressure and photon shot noise, is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise. Below 10 Hz, classical noise sources, e.g. seismic motion, limit the sensitivity of the detector. Above 100 Hz, shot noise limits the sensitivity of both detectors. Shot noise rises with f due to the arm cavities. Below 100 Hz, the quantum noise goes as $1/f$ due to radiation pressure. Coating thermal noise will limit the A+ detectors's sensitivity around ~ 100 Hz.

2.4.2 Using Power Recycling to Reduce Shot Noise

To increase the laser power that is incident upon the beam splitter, a power recycling cavity can be used [40–42]. When the interferometer is on the dark fringe, most of the light from the laser gets reflected by it. Thus, the interferometer is, effectively, a mirror. Drever [43] proposed that a *power recycling mirror* (PRM) could be placed between the laser and the interferometer to form a cavity. A sketch of this is shown in Figure 2.1. The total power impinging on the beam splitter would be increased by making wavelength-scale position changes, i.e. *tuning*, of the position of the PRM such that the light reflected from the interferometer and the light from the laser add in phase, i.e. the reflected light power would be ‘recycled’. Current laser sources that can be operated for year-long timescales are capable of producing of the order 100W, see Chapter 3, and so the power recycling factor needs to be $\sim 10^4$. This technique has been implemented at LIGO [44] and other large-scale interferometers [26, 27].

2.4.3 Using Signal Recycling to Shape Quantum noise

An interferometer’s bandwidth depends on the amount of time that signal-carrying light is in the arms for. Meers suggested that the signal light could be recycled in a similar fashion to how power is recycled [40]. This detection bandwidth can be altered by introducing a *signal recycling mirror* (SRM). A coupled cavity, known as the *signal recycling cavity* (SRC), between the SRM, ITM and ETM will be formed by placing the SRM between the photodiode and the beam splitter. A sketch of

this is shown in Figure 2.1. The finesse of the arm cavities as seen by the light emerging at the AS port depends on the tuning of the SRM. There are two common modes of operation: *resonant sideband extraction* (RSE) [45,46] and *signal recycling*. Alternatively, the SRM can be tuned so that the interferometer is optimised for a narrow band of frequencies, e.g. [47].

When in RSE mode the SRM is tuned so that the SRC is anti-resonant, thus broadening the interferometers sensitivity bucket at the expense of losing peak sensitivity. This was motivated by needing to reduce the thermal load on the beam splitter by increasing the finesse of the arm cavities [46]. However, by increasing the finesse of the arm cavity, the bandwidth of the interferometer is reduced. To retain the desired arm cavity finesse from a sensitivity point of view, the SRM can be tuned so that the finesse of the arm cavity experienced by the signal is lowered.

In signal recycling mode, the SRM is tuned so that the SRC is resonant. This deepens the sensitivity bucket at the expense of bandwidth. The trade-off between bandwidth and peak sensitivity comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detector's bandwidth and the energy within this bandwidth; so for the same amount of power at the beam splitter, this can be thought of as a fundamental limit to the interferometer's sensitivity. This is known as the Mizuno limit [32].

2.4.4 Reducing Shot Noise by Removing Junk Light with a Mode Cleaner Cavity.

All light that is detected by the output photodiode contributes to the total shot noise of the measurement; therefore, only light which carries signal should be incident on the photodiode to enhance the SNR of the measurement. In an ideal interferometer, the light emerging at the AS port will be a beam that is defined by the mode that is resonant within the arm cavity. To minimise diffraction losses and due to the mirrors' manufacturing tolerances, the light in the arm cavity will be in the *fundamental Gaussian mode*. However, due to optical imperfections, e.g. astigmatism, and the RF side-bands used to control the interferometer, the light emerging from the AS port contains light in higher order modes. This undesired light is referred to as *junk light*. A mathematical description of Gaussian beams can be found in Appendix F.

To remove junk light, mode cleaners are used (e.g. [48, 49]). A mode cleaner is a high finesse cavity that strips away higher order mode light and the control sidebands. The LIGO output mode cleaner (OMC) [48] is a monolithic, fused silica bow-tie cavity that has a round-trip of ~ 1 m. As higher order modes acquire a different amount of phase as they propagate compared to the fundamental Gaussian mode, an optical cavity can be designed such that the fundamental mode resonates while the higher order modes do not. These higher order modes will be suppressed by a factor of the cavity's finesse. However, it is possible for some higher order modes to not be suppressed if their round trip Gouy phase is nearly the same as that of the fundamental mode; thus, the cavity is designed so that only higher order modes

which carry an insignificant amount of light will do this. The control sidebands are attenuated if they are outwith the cavities linewidth. The beam transmitted by the mode cleaner will be predominantly in the fundamental mode, and so the SNR of the signal measured by the photodiode will be increased.

2.4.5 Radiation Pressure Noise and the Standard Quantum Limit

Radiation pressure noise is a quantum effect that arises due to the uncertainty in the momentum that light imparts to the test masses when it is reflected from them [37]. These quantum fluctuations will give rise to a differential force on the ETMs, meaning that this noise couples as $1/f^2$ to the measurement of the positions of the ETMs. As the force exerted by a beam is $F = \hbar\omega_{EM}N/c = P/c$, the amplitude spectral density due to the radiation pressure of the length of a Fabry-Perot cavity with mirrors of mass m is

$$\Delta x_{\text{RPN FP}} = \frac{\sqrt{\hbar\omega_{EM}P}}{2mc\pi^2 f^2}. \quad (2.13)$$

To reduce the effect of radiation pressure noise, the LIGO test masses are 40 kg. The effect of radiation pressure noise on the LIGO detectors's sensitivity curves can be seen in Figure 2.3.

As the relative shot noise decreases while radiation pressure noise increases as the laser power is increased, a balance can be struck between the two noises. For a laser

which produces a coherent state, i.e. a classical laser, the standard quantum limit (SQL) [37, 50, 51] sets an optimal laser power for a given ETM mass which minimises the total quantum noise. The SQL results in a $1/f$ locus that is determined by the mass of the ETM and the duration of the measurement.

The sensitivity curve of an interferometer with a signal recycling mirror that is tuned to either signal recycling mode or RSE mode will ‘touch’ the SQL at one frequency; however, for an arbitrary detuning of the SRM, the sensitivity curve of the interferometer will drop below the SQL at some frequencies due to non-linear optical effects [52].

2.4.6 Coating Thermal Noise and Other Noise Sources

There are many classical sources of noise to overcome in a gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of atoms in the amorphous mirror coating layers of the test masses [53]. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector, as shown in Figure 2.3. Other noise sources include seismic noise from the motion of the Earth, Newtonian noise from pressure waves in the Earth’s crust, residual gas pressure noise that alters the refractive index of the arms, and the noise associated with the interferometer’s electronics (e.g. [54, 55]).

2.5 Reducing Shot Noise with Non-Classical States of Light

The two-photon formalism developed by Caves and Schumaker [56,57] can be used to see how non-classical states of light can improve the sensitivity of an interferometer. In this framework, light is analogous to a quantum simple harmonic oscillator (SHO). A SHO can be described by two orthogonal components, such as the momentum and position quadrature; for light a convenient basis to use is the *amplitude* and *phase* quadrature. The amplitude and phase quadratures, $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively, defined in terms of annihilation and creation operators of the electromagnetic field, $\hat{a}$ and $\hat{a}^\dagger$, are

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) , \quad (2.14)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) . \quad (2.15)$$

From this formalism, an uncertainty relation between the phase and amplitude quadratures can be derived:

$$\Delta\hat{\mathbf{X}}\Delta\hat{\mathbf{Y}} \geq \frac{1}{2} \quad (2.16)$$

If the uncertainty in one quadrature is reduced, the uncertainty in the other is increased. Such a light field is *squeezed*. The phase between the squeezed field and the detected quadrature is defined as the squeezing angle.

As the gravitational wave signals will be in the phase quadrature of the light, it

would be advantageous to, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature; this can be achieved by using squeezed light. Squeezed light is created with an optical parametric amplifier; this is a device which generates correlated photons by converting single higher energy photons into two lower energy photons. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at the LIGO Livingston Observatory (LLO) [58] and GEO600 [59].

However, fluctuations in the amplitude quadrature of the light perturb the position of the test masses; thus, they couple to the signal at the AS port. During LIGO's third observation run, the squeezing angle had to be tuned to prevent radiation pressure noise from limiting the BNS range of the detector [58].

To infer the quantum state of a light field, a technique known as *balanced homodyne detection* (BHD) can be used to observe one quadrature of the light field [60]. By combining the signal light field that is to be measured with a much stronger *local oscillator* light field on a beam splitter, the quantum state of the signal light field is encoded in the combined beam's intensity. The relative phase between the signal and local oscillator determines the quadrature of the quantum state that will be observed. Chapter 5.1 explores this in more detail.

2.6 Interferometry Techniques to Beat Quantum Limits

Squeezed light and an effect known as ponderomotive squeezing [50], can be used to surpass the SQL at some frequencies. Ponderomotive squeezing arises due to the natural correlations between the phase and amplitude quadratures of the light created by the interferometer. A demonstration of the SQL being surpassed at the LIGO Livingston detector can be found in [61]. Although [61] shows the SQL was beaten at some frequencies, the noise at other frequencies was increased relative to if no squeezing was used. Although the LIGO detectors are operated for broadband gravitational wave sensitivity, sub-SQL interferometer configurations could be used if detecting certain astrophysical sources with merging frequencies around the frequency at which the SQL is beaten was of interest.

To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency dependent squeezer can be utilised [50]. The light's squeezing angle can be rotated as a function of frequency. This is done by using a high finesse detuned filter cavity. At frequencies within the linewidth of the cavity, the light reflected from the filter cavity will experience a phase shift and so the squeezing angle will be rotated; in contrast, at frequencies outside the cavity's linewidth, the light incident upon the cavity is reflected without experiencing any phase shift and so the squeezing angle is not rotated. At low frequencies, the amplitude quadrature can be squeezed and thus the interferometer's radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and thus the

interferometer's shot noise can be decreased. A frequency dependent squeezed light source that is suitable for LIGO has been demonstrated [62] and frequency dependent squeezing will be part of the A+ detector.

Variational readout is a technique where quantum noise due to radiation pressure is removed from the measurement by using a frequency dependent homodyne angle [50]. This is realised in a similar way to frequency dependent squeezing; the AS beam is coupled to a filter cavity and thus the detected quadrature of the beam emerging from this cavity is frequency dependent. This beam is then measured on photodiodes using a conventional BHD scheme. Variational readout schemes are sensitive to small amounts of loss within the filter cavity, and so this technology is not mature enough to be implemented into current gravitational wave detectors.

To beat the bandwidth limit, energy has to be injected into the system. White-light cavities as a means of increasing the detector bandwidth have been studied [63–65]. However, these interferometer configurations have yet to be experimentally demonstrated.

There are several sub-SQL schemes which avoid the quantum noise problems related to making a position measurement of the test masses. If a momentum measurement of the test masses could be made, then a random impartation of ‘position’ would not affect future measurements of the momentum [66]. This is a type of quantum non-demolition measurement. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$; this type of measurement can be made with a speed meter interferometer [67–69].

2.7 The Effect Losses Have on Measuring Quantum States of Light

Losses between the arm cavities and the readout photodiodes result in less signal as well as more noise. The decrease in signal is straight forward to understand: if there are less photons to measure, then the signal will be smaller. Since the power is lower, the signal to noise ratio will be larger since shot noise goes as $\sqrt{\text{Power}}$. As gravitational wave detectors implement squeezing to decrease the quantum noise in the detection quadrature, it is important to preserve the quantum state of the signal field by having minimal loss. Real optical components have physical attributes that result in incident light power being wasted. These mechanisms are known as losses, and examples of these include absorption and scatter. A lossy optical component can be modelled as an ideal one downstream of a beam splitter whose splitting ratio represents the loss (e.g. [38, Section 2]). The incoming light enters a beam splitter, some light is directed to the ideal component and some gets lost. In the remaining input to the beam splitter, vacuum fluctuations can enter. Some of these fluctuations will make it to the ideal component, and so the state of the light that is detected is contaminated with extra vacuum fluctuations compared to if there was no loss. The factor by which the shot noise increases, ϵ , due to a simple loss mechanism, such as a non-ideal photodiode (see Section 2.8), is given by

$$\epsilon = \sqrt{\frac{1}{\eta} - 1}, \quad (2.17)$$

where $0 < \eta < 1$ is the loss.

Mode matching losses have a greater contribution to the total shot noise compared to losses of the same size generated by a simpler mechanism [70]. This source of loss can be reduced through careful design of the optical layout. Mirrors with a tunable radius of curvature can be used to ensure that there is minimal mode matching loss. This is the focus of Chapter 6.

2.8 Photodiodes

Photodiodes are used to sense the signal-containing light from the interferometer via the OMC (see Section 2.4.4)³. A photodiode is a square-law detector that produces a current that is proportional to the power of the light it is detecting. The photodiode’s *quantum efficiency* is the fraction of photons that are absorbed by the photodiode and into a photocurrent; a quantum efficiency less than unity is a source of loss. Photodiodes are usually operated with a *reverse bias* – a positive voltage across the cathode and anode – to widen their depletion region, thus improving their quantum efficiency and bandwidth (see Section 8.2 for more detail). Due its bandgap, InGaAs is a widely used semiconductor for detecting 1-1.6 μm light, and suitable InGaAs photodiodes were available for the current generation of gravitational wave detectors. The requirements for the photodiodes used in a gravitational wave detector are discussed in Section 8.1. Usually, off-the-shelf InGaAs photodiodes are of the order 1 mm in diameter to keep their junction capacitance low enough for high-speed

³In this thesis, unless specified, I will be talking about the ‘DC’ photodiodes used to sense the gravitational wave signal. There are other types of photodiodes used in gravitational wave detectors, e.g. RF photodiodes; these are not discussed.

operation⁴ and to keep their dark current low. Photodiodes often follow a p-doped, intrinsic, n-doped (PIN) semiconductor structure (see Section 8.2) as this design can yield photodiodes with high quantum efficiency and fast response times. InGaAs PIN photodiodes can often detect up to 100 mW of light power, and they can be operated at room temperature.

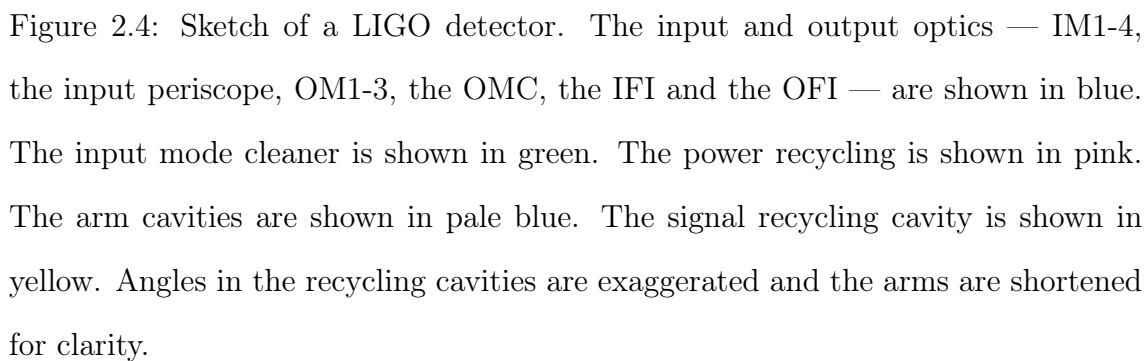
Since future gravitational wave detectors may need to operate with 2 μm light (see Section 2.10), the current photodiodes may be replaced and ‘Extended’ InGaAs photodiodes may be used instead. Extended InGaAs is an industry term for InGaAs photodiodes with cut-off wavelengths that are above 1.6 μm . Based on conversations with four photodiode manufactures and distributors [71–74], it is unlikely that commercially available extended InGaAs photodiodes for detecting 2 μm light that are suitable for a gravitational wave detector exist. As discussed in Chapter 8, it is difficult to produce defect-free extended InGaAs, and defects in extended InGaAs photodiodes can cause the noise and quantum efficiency of the photodiode to be degraded. It may be noted that previous experiments in squeezing at 2 μm have been limited by the properties of the extended InGaAs photodiode that was used [75, 76]. In [75], to minimise excess noise from the photodiode, the photodiodes were not reverse biased; this came at the expense of the photodiode’s quantum efficiency and thus the total level of observed squeezing was reduced. The characterisation of some extended InGaAs photodiodes is described in Chapter 8.

⁴High-speed operation is not needed for a photodiode in a gravitational wave detector.

2.9 Overview of the LIGO Detectors

The LIGO detectors are a dual-recycled-Fabry-Perot Michelson interferometer with 4 km arms [2]. One of these is in Livingston, Louisiana (LLO) and the other is in Hanford, Washington (LHO). The layout of the detectors is shown in Figure 2.4 [77] and its sensitivity curve is shown in Figure 2.3 [78, 79]. The term dual-recycled refers to the fact that the interferometers use both power and, the mathematical dual of power recycling, signal recycling. The interferometers are housed within vacuum systems comprised of several horizontal access modules (HAM) and beam splitter chambers (BSC). Figure 2.4 includes all the optics from the pre-stabilised-laser (PSL) to the photodiodes that the main beam interacts with.

The first cavity after the PSL is the input mode cleaner; this provides a beam with high spatial purity to the interferometer and is used in the initial locking sequence as a stable reference frequency. The power recycling cavity features two extra mirrors, known as PR2 and PR3, in addition to the PRM. The role of these mirrors is to make the round trip Gouy phase sufficient so that it is stable; the signal recycling cavity also has three mirrors for the same reason [44]. The beam passes through the output Faraday isolator (OFI), and is steered into the OMC. The gravitational wave signal is encoded in the beam transmitted by the OMC, so photodiodes monitor this light.



2.10 The Future of Ground Based Gravitational Wave Detectors

The next generation of ground-based gravitational wave detectors include the Einstein Telescope [80, 81], LIGO Voyager [82] and LIGO Cosmic Explorer [83–85]. LIGO Voyager will be a cryogenic interferometer installed in the current LIGO infrastructure and will be a factor of ~ 2 more sensitive than LIGO A+. The Einstein telescope and Cosmic explorer interferometers will be ten times larger than the current generation of interferometers and will be able to: test the no-hair theorem, test alternative theories of gravity, test models of string theory, measure objects with large redshifts to learn about the formation of cosmological structures, measure closer to the big bang than is possible with electromagnetic telescopes, detect cosmic strings, probe the physics of neutron stars and measure what happens during a supernova. As well as this, the third generation of gravitational wave detectors may measure signals from exotic sources which have not yet been theorised.

The current generation of gravitational wave detectors will be limited by the thermal noise around 100 Hz, see Figure 2.3, due to the Brownian motion of the molecules that make up the mirror coatings [86]. Since these thermally driven fluctuations are a limiting source of noise, proposed future gravitational wave detectors, such as the Einstein Telescope [80, 81] and LIGO Cosmic Explorer [83], may use cryogenically cooled mirrors.

However, the material that the test masses are made from, fused silica, has

undesirable noise properties when it is cooled [87] and so the test masses may be made of crystalline silicon; crystalline silicon is opaque at $1\text{ }\mu\text{m}$ [88] and so this also forces a change from a $1\text{ }\mu\text{m}$ laser. Additionally, cryogenically cooled mirror coatings may not be compatible with the current laser wavelength [89–91], which is $1\text{ }\mu\text{m}$, and so future gravitational wave detectors may use either a $1.5\text{ }\mu\text{m}$ or $2\text{ }\mu\text{m}$ wavelength laser [84, 85]. $2\text{ }\mu\text{m}$ is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength [92].

2.11 Conclusion

Further detections of gravitational waves will lead to a better understanding of the universe. To reach the design sensitivity of the LIGO detectors, among other upgrades, the input laser power needs increasing. To further lower the LIGO detectors noise floor, the A+ upgrade will utilise frequency dependent squeezed light. For this to be successfully implemented, the quantum state of the light field must be accurately detected with a balanced homodyne readout and losses in the readout must be minimal.

As there is potential to reduce the noise floor that will limit the A+ detector by using cryogenic mirror coatings that are compatible with $2\text{ }\mu\text{m}$ light, future detectors may use a $2\text{ }\mu\text{m}$ laser; this leads to the question: is laser technology at this wavelength sufficiently mature to enable quantum limited measurements? Balanced homodyne detection and photodiodes with high quantum efficiency and low dark noise will be

required for any future interferometer.

Chapter 3

A Prototype High Power Laser for LIGO's Fourth Observation Run

The work in this chapter is reported on in [93], and undertaken alongside Nina Bode, Matt Heintze and Andrew Spencer.

3.1 Introduction

The LIGO detectors require high input power, nominally 125 W at 1064 nm [94], to decrease the shot noise which limits sensitivity at frequencies above 100 Hz. The original laser for aLIGO cannot reach the required power without generating excess

pointing noise. In this chapter, the new laser layout is discussed. The pre mode cleaner (PMC) throughput power, beam quality, intensity noise, pointing noise and frequency noise of the prototype laser were measured. This laser was able to produce 100 W of intensity stabilised light. The pointing noise was measured to be above the requirement; however, this prototype was made in a lab which was noisier than the PSL, and the pointing noise may be improved by reducing turbulence in the water-cooling system. While the 125 W nominal power level was not obtained, this layout provides more power than the current laser. With additional work in optimising the layout, the maximum output power of this laser may be increased.

If coherent light from a laser is detected, the shot noise associated with power P is $\sigma = \sqrt{\frac{hc}{\lambda}P}$, so the signal to noise ratio at high frequencies scales with $\sqrt{P}$. Alternatively, to decrease the shot noise limit the wavelength could be decreased; however, the LIGO detectors operate at 1064 nm due to the availability of mirror coatings with low absorption and low thermal noise. Future interferometers may operate at longer wavelengths due to the potential reduction of mirror coating thermal noise that can be achieved at longer wavelengths [92].

In the field of gravitational wave detection, high power refers to lasers with powers of the order 100 W. High power is a loosely defined term as lasers ranging from 100 W to 100 kW are defined as high power in the literature [95–105]. As the power is increased, other properties of the laser tend to be degraded. It needs to operate uninterrupted on year-long timescales; this poses a major engineering challenge.

The laser for LIGO must be continuous wave, linearly polarised, have high spatial purity, low intensity noise and low frequency noise. These requirements are detailed in [94]. To overcome the technical challenges that come with the high power and low noise requirements for the LIGO laser, the master-oscillator-power-amplifier (MOPA) topology was selected. This type of laser is based on a high performance seed laser which is then amplified. As amplifiers are linear and coherent, the low noise properties of the seed laser characterises the noise of the high power beam.

LIGO's original laser used a high power oscillator (HPO) as the means for obtaining the target power. The HPO was a ring cavity containing laser crystals that was seeded by a 35 W low noise laser. This design went through extensive testing and refinement at the AEI [106] and was installed as part of LIGO's pre-stabilised-laser (PSL). However, due to pointing noise from the water cooling of the crystals within the HPO coupling to the differential arm motion signal (DARM), as well as one HPO being destroyed by accident, the HPO was bypassed and the LIGO detectors operated using the 35 W laser [107].

To reach higher powers during LIGO's third observation run, instead of the HPO, a neoVAN-4S single-pass amplifier was used. This stepped up the laser power from 35 W to 70 W [55, 108]. The next stage of development was a complete redesign of the LIGO laser based on using two upgraded versions of the neoVAN-4S. The neoVAN-4S-HP amplifier showed promising results [104, 105], so a prototype laser was constructed and tested at LIGO Livingston. This design will be installed as the new PSL for the start of LIGO's fourth observation run [107].

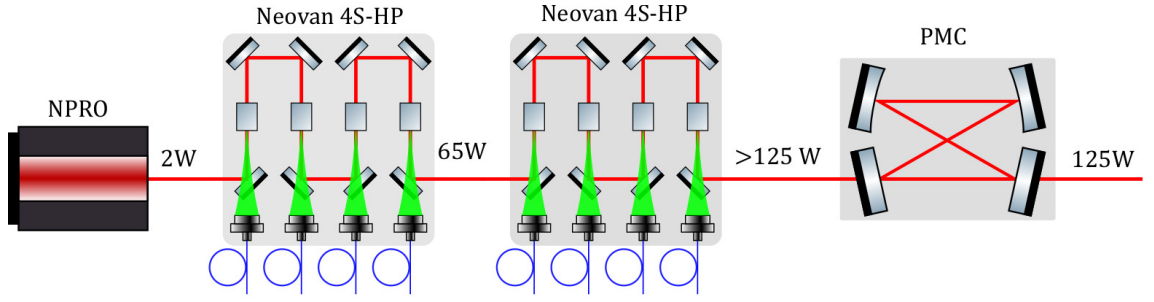


Figure 3.1: Conceptually, the PSL will be made up of four components: the NRPO, two neoVAN-4S-HP amplifiers and the PMC cavity. The NPRO provides seed light with good noise characteristics. These noise characteristics are preserved by the amplifiers, so the resulting beam will have high power and low noise. The PMC filters junk light and RF intensity noise so a spatially pure mode can be sent to the interferometer.

3.2 Concept for LIGO's New Laser

LIGO will use a MOPA laser that is seeded by a non-planar ring oscillator (NPRO). The NPRO beam will be amplified by two single-pass amplifiers in series and be followed by a mode cleaner cavity to produce a low-noise high-power laser in the fundamental mode. This concept is shown in Figure 3.1. As the NPRO has good free running noise characteristics the amplified beam will as well. Results from a setup using a single neoVAN-4S-HP showed these amplifiers can be continuously operated for many tens of days [105]. The led to the proposal to construct the PSL from two of these in series. A similar setup using a 35 W seed laser and two neoVAN-4S-HPs was able to generate 195 W of light [104] for nearly 80 days with no significant drops in power.

3.2.1 Non-Planar Ring Oscillators

The first element of the PSL is a non-planar ring oscillator [109–111]. The NPRO provides a low noise laser making it a suitable seed for a MOPA system. This is a single frequency laser constructed from one Nd:YAG crystal which acts as both the gain medium and the cavity, thus it produces a laser with a high amount of frequency stability. The NPRO used was a Mephisto, built by Coherent, and it can produce 2W of 1064 nm light.

3.2.2 The neoVAN-4S-HP Solid-State Amplifier

To amplify the NPRO beam, two neoVAN-4S-HP amplifiers will be used. The neoVAN-4S-HP is a single-pass, solid-state amplifier manufactured by Neolase. It is constructed from four Nd : YVO₄ crystals which are pumped by fibre coupled, low noise laser diodes. It should be seeded by s-polarised light at 1064 nm. These gain mediums are in contact with a metal plate which is cooled. This keeps pointing noise low as flowing water is not directly in contact with the gain medium. As these amplifiers have low noise pump-diodes and will be seeded by an NPRO, the noise of the light produced by them should be determined by the NPRO.

3.2.3 The Pre Mode Cleaner Cavity

The pre mode cleaner (PMC), see Figure 3.2, is used as the hand-off point between the PSL and the rest of the interferometer. Light transmitted by this cavity is used as the input beam for the interferometer. The PMC is a bow tie cavity with an FSR of 9 MHz, thus it passively filters RF intensity noise. Control sidebands are provided after the PMC but before the IMC, meaning that the IMC cannot be used to provide this stabilisation. The PMC is locked to the fundamental mode so the beam that is delivered to the rest of the interferometer will have a low higher order mode content. This is important as the amplifiers produce light with spatial defects. The PMC has been redesigned [112] to prevent contamination due to glue on its mirrors and to make maintaining the PMC easier. The PMC design no longer includes a vacuum enclosure to isolate it from acoustic noise as this was found to be unnecessary.

3.3 Layout of the Prototype Laser System

An optical layout based on the concept described in Section 3.2 was constructed. A sketch of the layout is shown in Figure 3.2. In addition to the main beam path, there are low power diagnostic beams after each amplifier that can be used to characterise the laser. The prototype layout did not include the tabletop frequency stabilisation servo (TTFSS) so this has not been included in the layout description. However, there is space for the TTFSS on the optical table.

The NPRO beam passes through some optics to clean up the polarisation. Transmission through a steering mirror is used to monitor the power of the NPRO. An electro-optic modulator (EOM) is used to produce 35.5 MHz sidebands for the Pound-Drever-Hall (PDH) scheme used to lock the PMC. A Faraday isolator ensures light reflected back from the amplifiers does not damage the NPRO. The beam passes through a lens to mode match the beam to the first amplifier. Two mirrors are used to align the NPRO beam to the amplifier. A half-wave plate ensures the polarisation into the first amplifier is set correctly. These optics are shown in Figure 3.3.

After the first amplifier, the beam passes through a high power attenuation stage, which consists of a half wave-plate and two thin film polarisers (TFP), to allow for safe set up and low power operation. When setting up this layout, the amplifiers must be run at maximum power since they exhibit significant thermal lensing. The high power attenuators are not envisioned being used during observation runs. The light from the laser is then directed through a Faraday isolator and wave-plate before reaching the second amplifier. The mode matching lenses for the second amplifier are on either side of the Faraday isolator. Two mirrors immediately before the second amplifier are used for its alignment. The transmission through one of the steering mirrors before the second amplifier is used for monitoring the beam produced by the first amplifier. This section of the layout is shown in Figure 3.3.

After the second amplifier, the beam goes through a high power attenuation stage before passing through an acousto-optic modulator (AOM). This AOM is for controlling the power of the laser. Downstream of the AOM, a pick-off beam is used to measure the properties of the laser produced by the second amplifier. Before

entering the PMC, the beam is transmitted through a TFP so that the beam is set to be p-polarised. This part of the layout can be seen in Figure 3.4.

The beam produced by the second amplifier is then coupled to the PMC; this requires two mode matching lenses and two steering mirrors. There are three beams transmitted by the PMC: a high power one, which would be sent to the interferometer, and two low power ones for stabilising the frequency and intensity of the laser. The low power beams contain of the order 100 mW of light. The intensity stabilisation servo (ISS) inner-loop photodiodes are in a light-tight enclosure. One of the photodiodes is used as the in-loop sensor while the other is used as the out-of-loop sensor. The PMC is shown in Figure 3.5. The reflected light is used to lock the PMC; this path can be seen in Figure 3.4.

The diagnostic bread board (DBB) [113,114] is integrated into the layout via several pick-off points and can be used to monitor the status of both amplifiers. The DBB enables measurements of intensity noise, frequency noise, beam pointing noise and higher order mode content. The DBB is shown in Figure 3.6. The DBB consists of a cavity, motorised lenses, an automatic alignment system and several photodiodes. Despite having motorised lens mounts in the DBB, static mode matching lenses are required in the diagnostic beam paths. To characterise the NPRO beam, mirrors on magnetic mounts can be inserted into the layout.

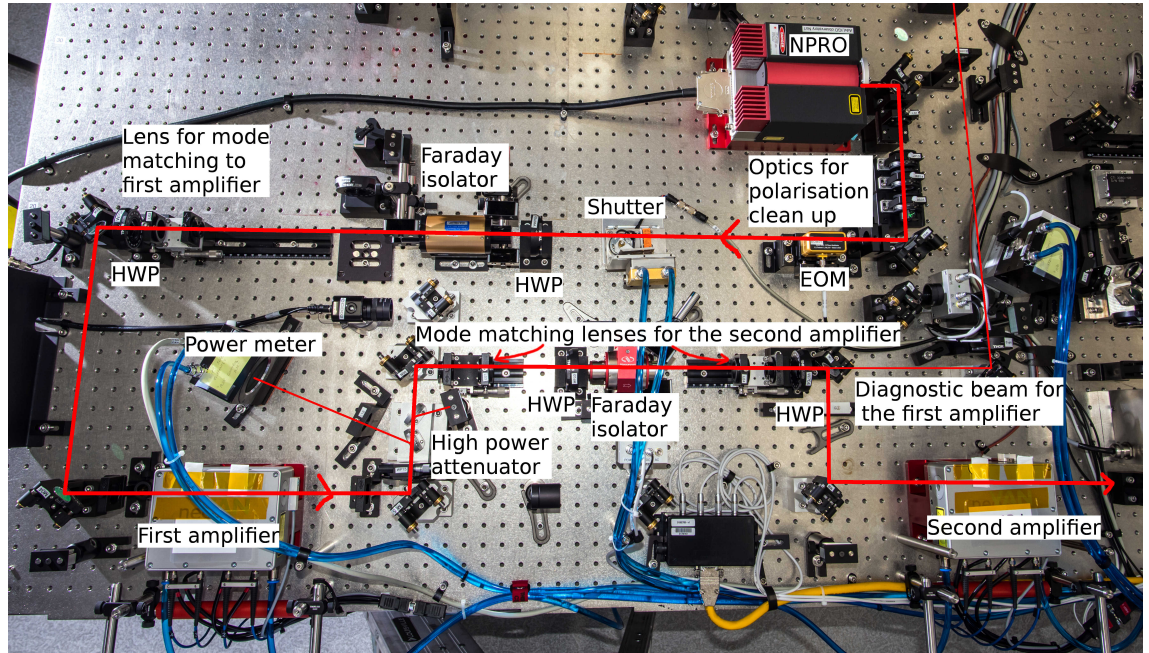


Figure 3.3: Photograph of the PSL prototype showing the beam path (red lines) from the NPRO (top right) to the second amplifier (bottom right). The beam path up to the first amplifier includes optics for setting the polarisation of the beam, an EOM, a Faraday isolator and lenses for mode matching. After the first amplifier, the beam passes through a high power attenuator, mode matching lenses and a Faraday isolator. The first amplifier (bottom left) is monitored using a low power pick-off beam (centre right).

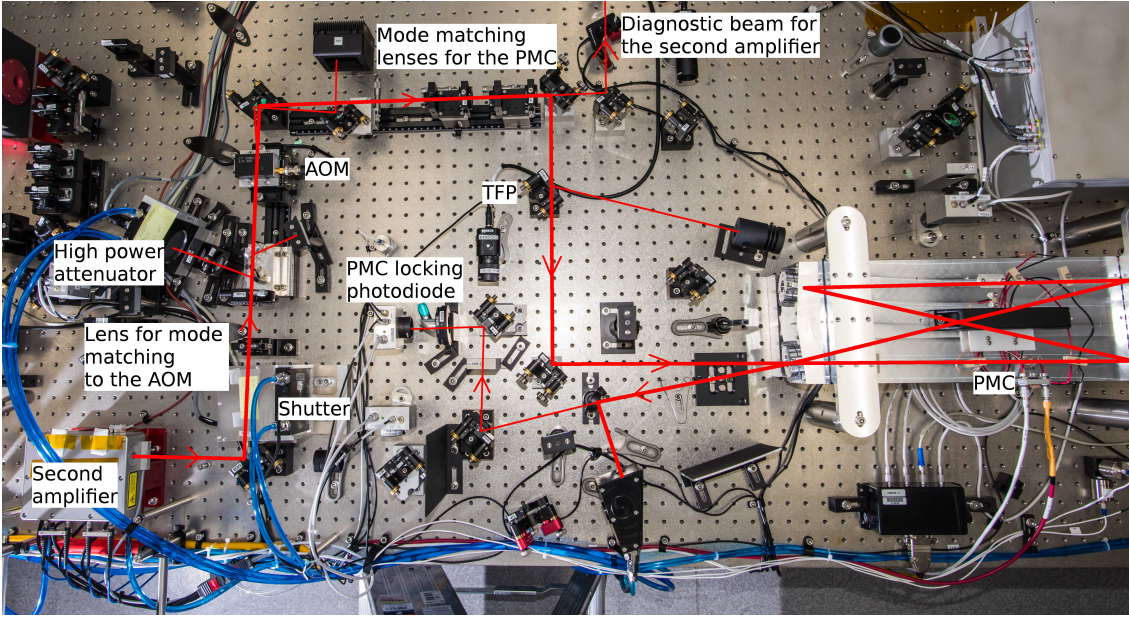


Figure 3.4: Photograph of the layout showing the beam path (red lines) from the second amplifier (bottom left) to the input of the PMC (centre right). This beam path includes a high power attenuator, an AOM, mode matching lenses for the PMC and a TFP. The PMC can be seen more clearly in Figure 3.5. The path taken by light reflected from the PMC is shown. This light is used to lock the PMC. A low power beam for monitoring the second amplifier is obtained using the transmission through a steering mirror (centre top) for the alignment to the PMC.

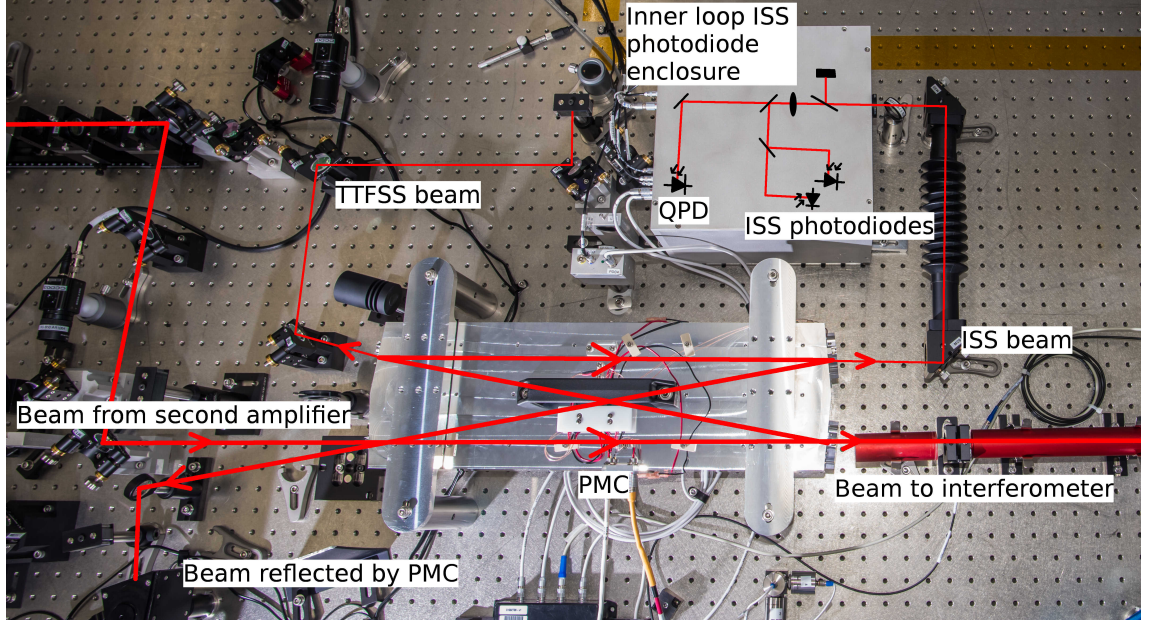


Figure 3.5: Photograph of the PMC (centre) and the inner ISS photodiode box (top right). The beam (shown in red) from the second amplifier enters the PMC. The majority of this light is sent to the interferometer. A small fraction of the light in transmission of the PMC is used to stabilise the intensity and frequency of the laser. These beams are labelled as the ISS beam and the TTFSS beam. The ISS beam is directed into a light tight enclosure. This enclosure contains two photodiodes which are for measuring the intensity noise of the laser. One of these is an in-loop sensor; the other is an out-of-loop sensor. A QPD can be used to monitor the alignment of the ISS photodiodes. The reflected beam from the PMC is made up of the junk light in the beam from the second amplifier.

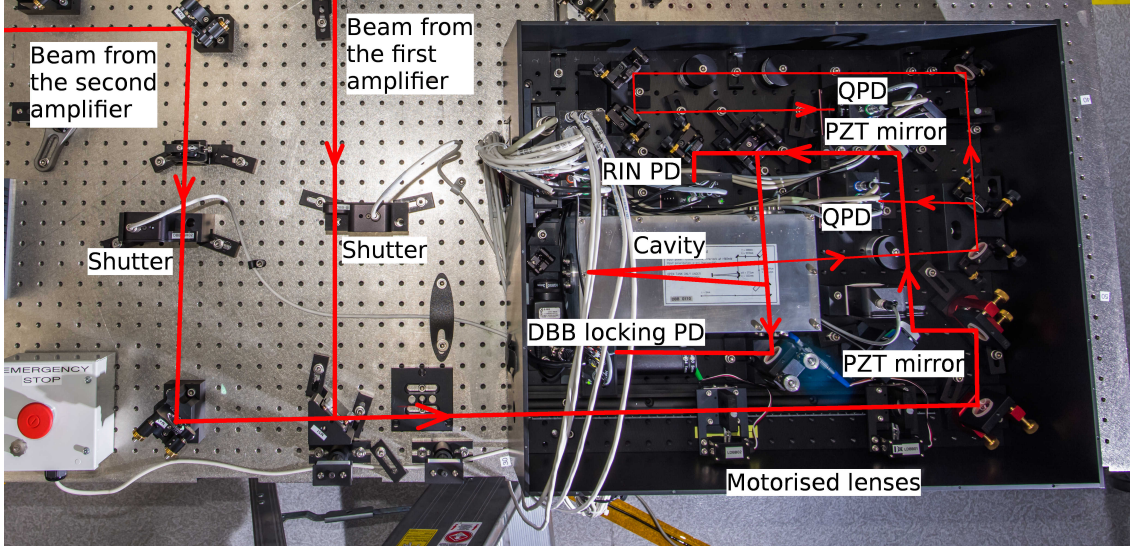


Figure 3.6: Photograph of the DBB with the beam path highlighted in red. Low power beams are used to measure properties of the laser produced by both amplifiers. The shutters control which amplifier will be characterised. The DBB contains motorised lenses and PZT mirrors so that remote mode matching and automatic alignment can be performed. The intensity noise of the beam can be measured with the RIN photodiode. The DBB locking photodiode is used to lock the cavity to the beam. This photodiode is also used during mode scans to measure the higher order content of the beam. QPDs separated by 90° of Gouy phase are used to measure the pointing noise of the incoming beam.

3.4 The Power Transmitted by the PMC

The PSL should deliver 125 W to the interferometer [94]. The power transmitted through the PMC is ultimately determined by two factors: the power in the beam generated by the amplifiers, and the distortion of this beam. The beam's distortion is affected by the amplifiers' input alignment, mode matching and thermal lensing. Increasing the pump-diode current increases the power that is produced by the amplifier as well as the thermal lensing. The flow rate of the coolant can be increased to mitigate thermal lensing; however, this can generate pointing noise.

The prototype layout was able to deliver a maximum of 111 W through the PMC during a 55 hour trial run. This is shown in Figure 3.7. The power was measured with recently calibrated power meters. The power emerging from the PMC dropped to 107 W during the course of the trial run, while the power incident onto it remained ~ 125 W. Changes in the lab temperature and humidity caused slight alignment changes. This resulted in the power decreasing. As the laser will be in a controlled environment, this effect will be less severe when it is installed.

To increase the output power, the mode matching to the PMC could be improved and the amplified beam's distortion could be reduced. Mode matching between the final amplifier and the PMC could be improved at the percent level. Increasing the flow rate would reduce the beam's distortion and may allow for higher pump-diode currents, however concerns of pointing noise meant that the flow rate was kept low. With higher flow rates, a beam with 137W of power and a higher mode content of

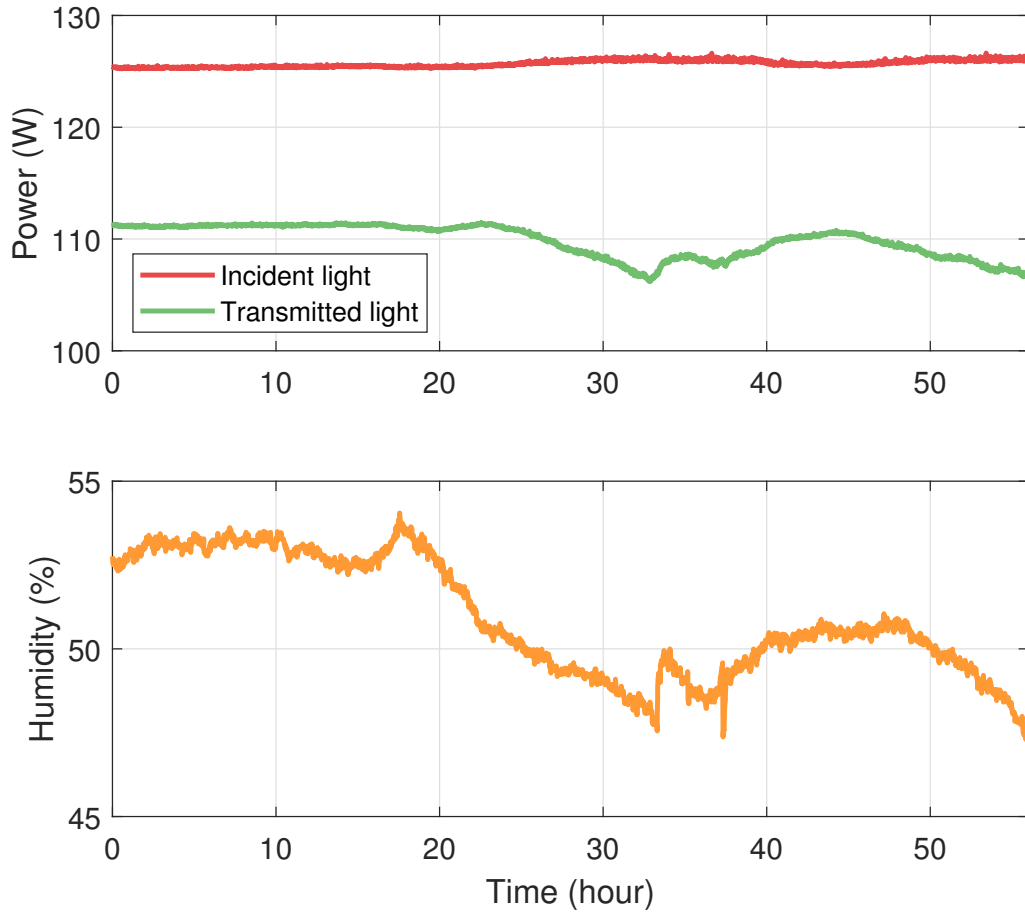


Figure 3.7: The top panel shows the power that was incident on the PMC and the power emerging from the PMC over a 55 hour trail run. The light produced by the amplifiers did not change significantly for the duration of this measurement, however the transmitted light through the PMC dropped from 111 W to, at its lowest, 107 W. This is likely due to the environmental conditions in the lab changing. The bottom panel shows the humidity of the lab. Changes in the humidity, which can be caused by a temperature change, coincide with changes in the power emerging from the PMC.

4.6% was produced by the amplifiers.

To actuate on the power of the laser, some of the incident light will be ‘dumped’ to provide headroom for the actuator to increase the beam’s power. This means that the full power will be reduced by ~ 10 W when the intensity stabilisation is implemented.

Long term measurements should be taken, however the results in [104] show these amplifiers can be run continuously for at least 80 days. Staff at LIGO Hanford measured the input laser power dropping from 55 W to 52 W over the course of one year [93]; this is likely due to the ageing of the laser diodes as well as mode mismatches resulting from changes in the thermal state of the setup.

3.4.1 The Effect of Cooling on the Amplifier’s Power and Beam Quality

As the quality of the beam emerging from an amplifier is affected by the temperature of the amplifier’s head, an experiment was performed to determine a suitable operating temperature. As the Nd : YVO₄ crystals are cooled non-uniformly, the thermal lens is non-circular. For small amounts of thermal lensing, the beam will become slightly astigmatic. Due to the pointing noise witnessed at the Hanford detector, the flow rates were set as low as possible before the power in the fundamental mode dropped significantly.

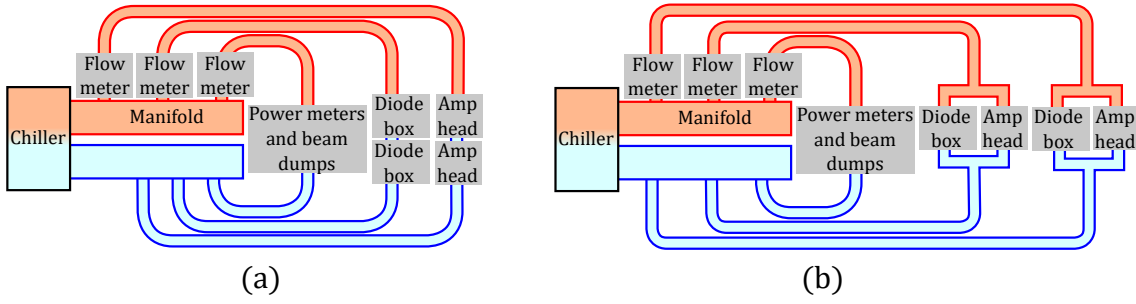


Figure 3.8: Cartoons of water cooling layout for the prototype laser. Panel (a): The laser will be run with the amplifier heads cooled in series with each other to reduce pointing noise. Panel (b): To perform the amplifier head temperature experiment, the water cooling was arranged so that the amplifiers temperature could be independently controlled.

The cooling layout that was used to find a suitable amplifier head temperature to operate with is shown in Figure 3.8. The diode box and amplifier head were connected in parallel, and each amplifier was on a separate cooling loop so that the amplifier head temperatures could be independently controlled. The flow rate to the first amplifier was kept high to produce a high quality input beam for the second amplifier. The HOM content for the beam entering the second amplifier was 3.8%. The second amplifier and diode box flow rate was decreased and the temperature of the diode head was monitored. The higher order mode content was measured with the DBB by averaging over sixty mode scans. The power was measured with a high-power power meter.

The change in beam quality and output power as a function of amplifier temperature are shown in Figure 3.9. At 22.3°C the higher order mode content was $\sim 4.5\%$ and the power in the fundamental mode was ~ 127 W. The changes in the

second order mode make up the majority of the HOM content change is shown in Figure 3.10. This is due to the beam becoming more astigmatic as the amplifier head heats up. It was decided that the diode heads would be operated such that their temperature did not rise above $\sim 27^\circ\text{C}$ to keep a high power in the fundamental mode.

To be able to keep the flow rate as low as possible on the table, the water cooling was set up so that the amplifiers heads were in series with each other. The diode boxes were on another loop to the heads. The cooling scheme that will be used is shown in Figure 3.8. Meters which monitor the flow rate of the coolant will be placed on the return side of the system rather than the supply side. By having the flow meters on the return side, they will detect a drop in flow if there is a leak, allowing for the water supply to be cut.

3.4.2 Amplifier Alignment Procedure

To obtain high beam power and low higher order mode content, the alignment of the input beam to the amplifier needs to be optimised. The output power of the amplifier may increase at the expense of higher order mode content, so the beam profile and the beam power must be monitored when aligning the amplifier. Optimising the alignment to the amplifier is an iterative process, so with each increase in power a small change in the amplifier's alignment is required.

Initially, the amplifier was switched off. This protects it against damage as

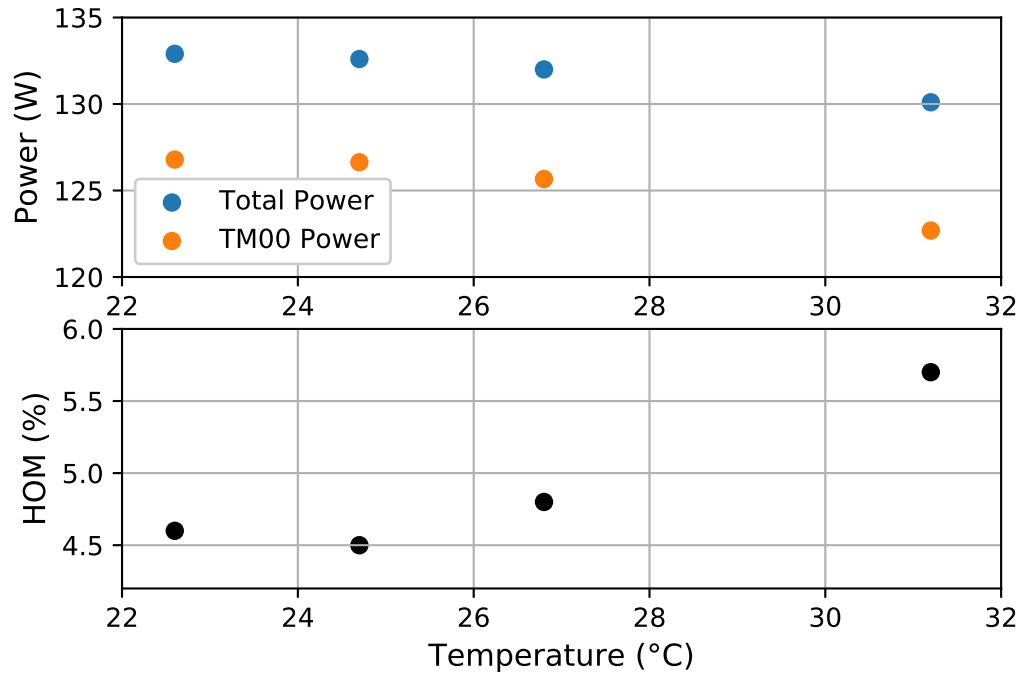


Figure 3.9: The top panel: as the second amplifier head's temperature increased, the power produced by it decreased. The bottom panel: The higher order mode content increased as the temperature of the amplifier increased. This means that the power in the fundamental mode decreased as the temperature decreased due to two effects: less overall power and worse beam quality.

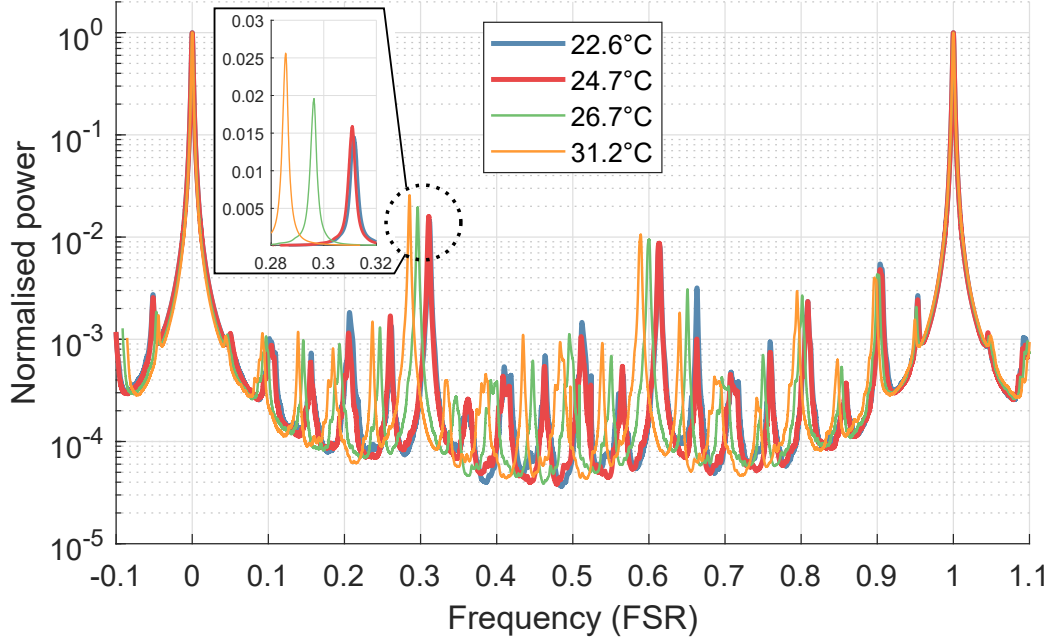


Figure 3.10: Mode scans were performed with the DBB for each amplifier head temperature. The DBB's auto-alignment system minimised the power in the first order mode. The second order mode for this cavity is located at ~ 0.3 FSR; its exact location drifts due to thermal expansion of the cavity. There is some power in the fourth order mode. As the temperature of the amplifier head changes, the second order mode content of the beam changes at the percent level. This second order mode height is shown in the inset. The second order mode changes due to the beam's astigmatism changing.

well as preventing the possibility of stray light beams with several tens of Watts of power being in the lab. The beam path between the NPRO and the exit of the first amplifier were inspected to make sure the beam path was clear and terminated with dumps.

Next, the NPRO was set to produce 1.3W of light. The power of the light was measured on the input side of the amplifier. The beam was roughly aligned so that the spot would hit the centre of the amplifier's input aperture using the two mirrors that immediately precede the amplifier. The power meter was placed on the exit side of the amplifier, and the input alignment was adjusted until the power emerging from the amplifier was the same as the input power. This indicates that the beam will be roughly aligned and some amplification will occur when the pump-diodes are turned on. The high power attenuator immediately after the amplifier was set to dump all the light. The initial mode matching was based on the manufacturers specifications, and the input light was set to s-polarisation.

The pump-diode currents were increased until the amplifier 'switched on'. This was at 2 A of current through each of the pump-diodes. Amplification begins when there is sufficient pump light to cause enough population inversion that stimulated emission occurs. At this threshold, the output power was maximised by optimising the alignment. Then, the mode matching was adjusted in tandem with correcting for mis-alignments caused by non-perfect centring of the beam on the mode matching lens.

A fast low-power power meter was used to monitor the beam transmitted through

the high power attenuator. Compared to the high-power meter, the low-power meter drifted less and responded to changes in power faster. This gave quicker feedback and allowed for easier alignment.

A Wincam was set up to monitor the beam profile. If the amplifier is significantly misaligned and mode mismatched, the beam profile will become distorted. Large misalignment will cause the beam profile to look lopsided, and smaller misalignments will cause the beam profile to have shoulders.

Once the amplifier had been optimised for a low pump-diode current, the pump-diode current was incremented in small steps of roughly 1-2A. Each time this was done, the thermal lensing increased. Due to the non-uniform way in which the heating occurred, the input alignment was re-optimised each time as was the input mode matching. By going in small steps of diode current, the optimal alignment was safely achieved. The amplifier should be operated at high diode currents, $\sim 9\text{A}$, so at low currents it was sufficient to be within 1% of optimal.

To determine when the alignment was optimal, mode scans with the DBB were performed. Due to thermal lensing, the beam will be slightly astigmatic and so optimising the mode matching from the amplifier to the DBB was difficult. When optimising the pump-diode currents, the thermal lensing changes, so the mode matching to the DBB needs to be corrected for this.

3.5 Intensity Stabilisation of a 100 W Beam

The free running-intensity noise of the laser exceeds the level needed for the interferometer to reach design sensitivity. Therefore, the laser must be stabilised. As actuators on the lasers intensity, a current shunt and an AOM were compared. Using the current shunt rather than the AOM enabled 10 W more power to be obtained in the beam emerging from the PMC, however the behaviour of the current shunt was highly dependent on the DC pump-diode current at 10 Hz where the laser noise requirement is most strict. If the amplifiers were required to operate under different circumstances during the course of an observation run, the current shunt may prove to be unreliable. In contrast, the AOM has proved reliable during previous observation runs. While the AOM causes the beam to have less DC power, this may not be a problem for O4 as it is unlikely that the full 125 W will be used due to point defects on the test masses.

3.5.1 The Concept and Requirements for Intensity Stabilisation Servo

The intensity noise requirement for LIGO's laser is found in [94] and reproduced in Figure 3.11. As an imbalance between the light powers in the arms gives rise to a laser intensity to test mass motion coupling, at low frequencies (0.1-10Hz), the requirement on intensity noise is set by the motion of the test masses due to seismic vibrations; in this frequency band, the intensity noise should be low enough that the

motion of the test masses is dominated by seismic noise. At 10 Hz, the laser must be stabilised to a level of $\sim 1 \times 10^{-9}/\sqrt{\text{Hz}}$. Above 10 Hz, the requirement on the intensity noise is relaxed as there is passive filtering of the light by the PRC. The light is passively stabilised by the PMC to reduce the intensity noise of the laser at RF frequencies.

The active intensity stabilisation of the laser is done in two stages as shown in Figure 3.12. First, the laser is stabilised by the inner loop to a level required to lock the IMC. The light used to measure the intensity noise for the inner loop is a low power beam that is transmitted by the PMC. There is $\sim 1 \text{ mW}$ of light in this beam, thus the laser cannot be stabilised beyond $\sim 2 \times 10^{-8}/\sqrt{\text{Hz}}$ due to shot noise. Once the IMC is locked, a pick-off from the beam entering the interferometer is used to measure the intensity noise of the laser. This beam has enough power, $\sim 350 \text{ mW}$, to reach the level required, $1 \times 10^{-9}/\sqrt{\text{Hz}}$, during observation mode. The noise of the beam entering the interferometer must be measured just after the IMC as this contains the intensity noise which cannot be measured by the Inner-loop diode e.g. the noise due to the periscope immediately after the PMC.

3.5.2 Free Running Noise of the Prototype Laser

To determine the open-loop gain needed for the servo to suppress the noise to the requirement, the free-running noise of the laser must be measured. The noise was measured at different points along the beam path to see the contribution that the

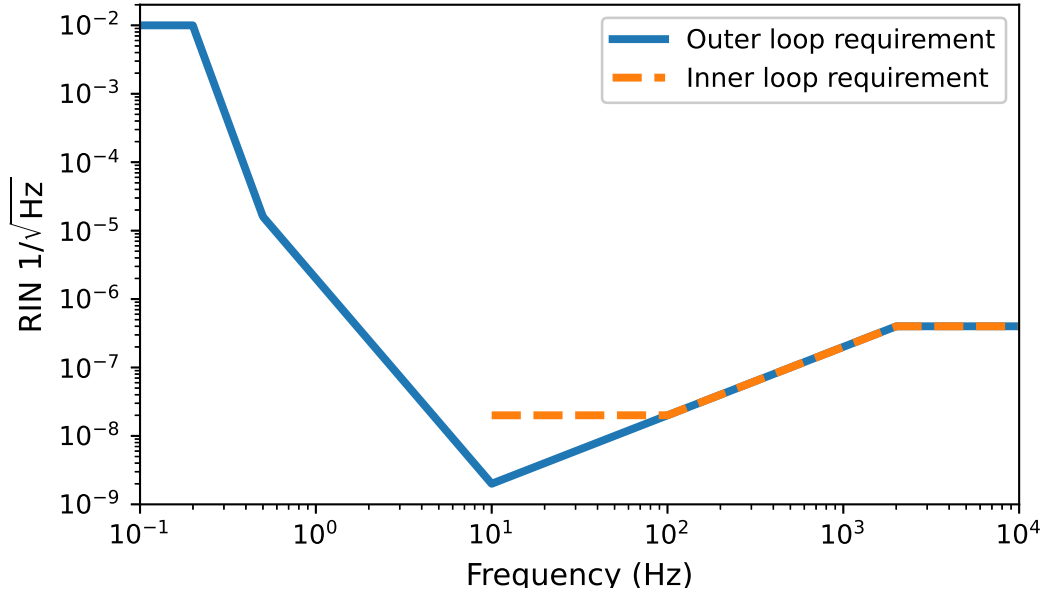


Figure 3.11: A reproduction of the intensity noise requirement from [94]. The outer loop requirement is shown in blue and the inner loop is shown by the dashed orange line. To lock the IMC, the inner loop requirement must be met. At frequencies below 10 Hz, the outer loop requirement is determined by the radiation pressure noise that is expected to arise due to imbalances in the arm power. The requirement is most stringent at 10 Hz. The requirement relaxes above 10 Hz due to the PRC cavity pole. The inner loop requirement is less extreme than the outer loop as it is only used to stabilise the laser in the initial locking sequence.

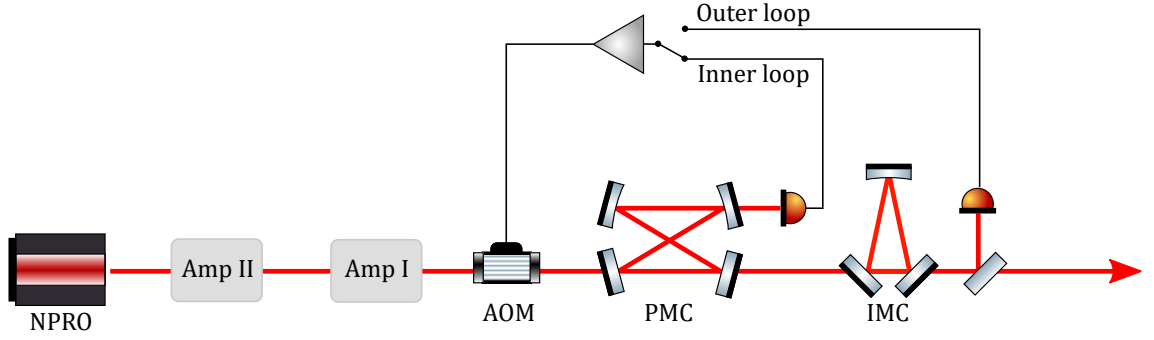


Figure 3.12: The intensity stabilisation concept involves two loops. The inner loop is responsible for stabilising the laser so that the IMC can lock. The outer loop is used to stabilise the laser for observation mode. The two loops sense the laser intensity noise at different points. The inner loop uses a low power beam, ~ 1 mW, transmitted by the PMC. The outer loop uses a higher power, ~ 350 mW, pick-off from the main beam. An AOM is used to actuate on the laser's power.

NPRO, each amplifier and the PMC made to the total noise.

The power noise of the NPRO was measured using the DBB RIN photodiode (Figure 3.13). The NPRO had a typical noise level for this type of laser [111]. This was recorded via CDS, so the measurements near 10 kHz are affected by the anti-aliasing filter. The noise between 100Hz and 1kHz had features associated with the motion of the table and mirror mounts.

The power noise of the first and second amplifiers was measured using the DBB RIN photodiode (Figure 3.13). The noise was measured with a spectrum analyser rather than CDS so that measurements up to 100kHz could be made. As the amplifiers were pumped with low noise diodes and were operating in their saturated

regimes, the RIN is expected to decrease after each amplifier. When an amplifier is saturated, the amplifier will not behave linearly and instead will add a fixed input power regardless of its seed power. As expected, at frequencies above 10 kHz the RIN after the second amplifier was less than the RIN after the first amplifier. There was a peak at 20 Hz; this can be attributed to the air conditioning. The noise was covered in a forest of lines with 4 Hz spacing, starting from 4 Hz and continuing up to high frequencies. These lines were found to originate in the Beckhoff slow-control computer which is used as an interface for the amplifiers. The laser control box needs better electrical isolation between its digital and analogue sections to remove the 4 Hz lines.

The noise of the beam emerging from the PMC was measured (Figure 3.13). The PMC introduces length to intensity noise coupling via the offset in the PDH loop. The offset needed to remove this coupling drifted due to changing levels of RF pick-up in the PMC servo electronics. The coupling from length noise of the PMC to intensity noise was decreased as much as possible.

3.5.3 Power Control Actuators

The actuator that was used to provide intensity stabilisation was an AOM. The use of a current shunt integrated into one of the pump-diodes in the second amplifier was investigated. The current shunt had the potential to reduce the number of transmissive optics in the beam path and increase the total power. Despite this, the

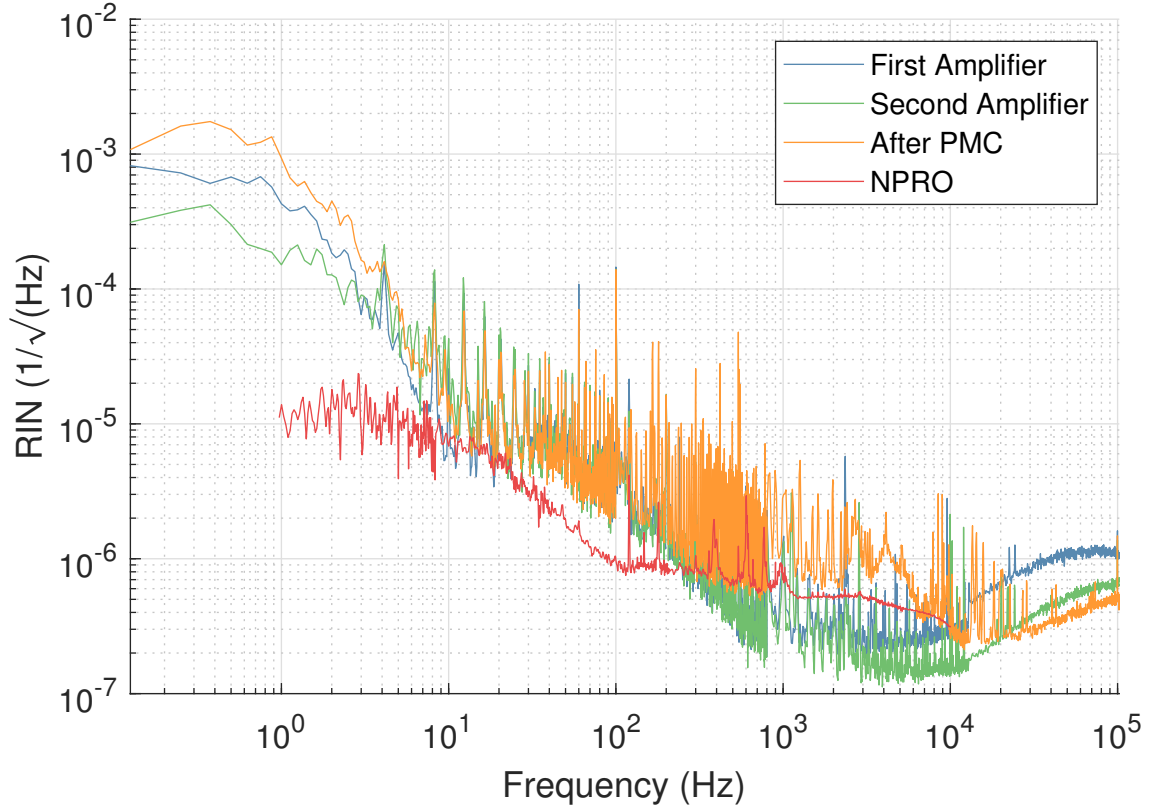


Figure 3.13: The free running noise of the NPRO and each amplifier is shown. The noise after the PMC was also measured. There is a forest of 4 Hz lines due to the Beckhoff control computer in the amplifier and PMC noise curves. Due to the non linear nature of the saturated amplifiers, at high frequencies the output beam intensity noise decreases relative to the input beam. The PMC couples length noise to intensity noise between 100 Hz and 10 kHz. This excess can be tuned away in the PDH servo.

current shunt may be an unsuitable choice for stabilising LIGO's laser. The current shunt transfer function depends on the DC current, at 10 Hz it changed by ~ 10 dB when the DC current was changed by ~ 1 A (see Figure 3.17), and so this may make the intensity stabilisation servo unreliable if the DC pump-diode currents need to be changed over the course of an observation run.

3.5.3.1 Acousto-Optic Modulator

An AOM uses acoustic waves to scatter light into higher order modes. The amount of light that is Bragg scattered depends on the intensity of the acoustic wave. The AOM used was a Crystal Technology 3080-199. An RF driver, Gooch and Housego r21080-2am-rfx, was used to control the intensity of the acoustic wave. This driver produced an 80 MHz signal whose power can be modulated. The AOM's response was linearised with an IC (AD734). By modulating the power of this signal, the power in the zeroth order beam can be controlled. The diffraction efficiency of this AOM was found to be 74%. This means that the AOM can be used for coarse and fine power adjustment. To modulate the intensity of a laser beam, some of it must always be scattered into the first order beam. To operate linearly, the DC level of the light that is dumped was ~ 10 W.

The transfer function, in relative intensity units, between the AOM control electronics to the intensity of the zeroth order beam as measured by the ISS photodiode is shown in Figure 3.15. The schematic for this measurement is shown in Figure 3.14. This transfer function is flat from 1 Hz to 100 kHz, and, as the servo's unity

gain point will be between 10-100 kHz, it is worth noting that above 10 kHz there is some phase lag.

3.5.3.2 Current Shunt

Modulating the current through a laser diode will modulate the intensity of the laser. This is a standard way of implementing intensity control at the 10W level [117–121]. Using a current shunt as an intensity actuator has been demonstrated in a similar system at the 100W level [105]. The current shunt technique allows for one to have fewer transmissive high power optics in the beam path, which potentially can reduce noise and thermal distortions of the beam. It requires a smaller DC offset compared to an AOM in order for it to work, therefore higher beam power can be achieved.

The amplifier's pump-diode current determines the power produced by the amplifier. By modulating the current through one of the pump-diodes, the power of the laser can be stabilised. The current shunt consists of a power MOSFET and a resistor. It diverts current from the pump-diode by pulling it through the resistor instead. The current that is pulled through the resistor depends on the voltage at the gate of the MOSFET (see [122, Chapters 2 and 3]).

The transfer function between the current shunt input and the ISS photodiode in transmission of the PMC was measured for a variety of different DC pump-diode currents. This is shown, in relative intensity units, in Figure 3.17. A schematic of the measurement is shown in Figure 3.16. The transfer function was found to have

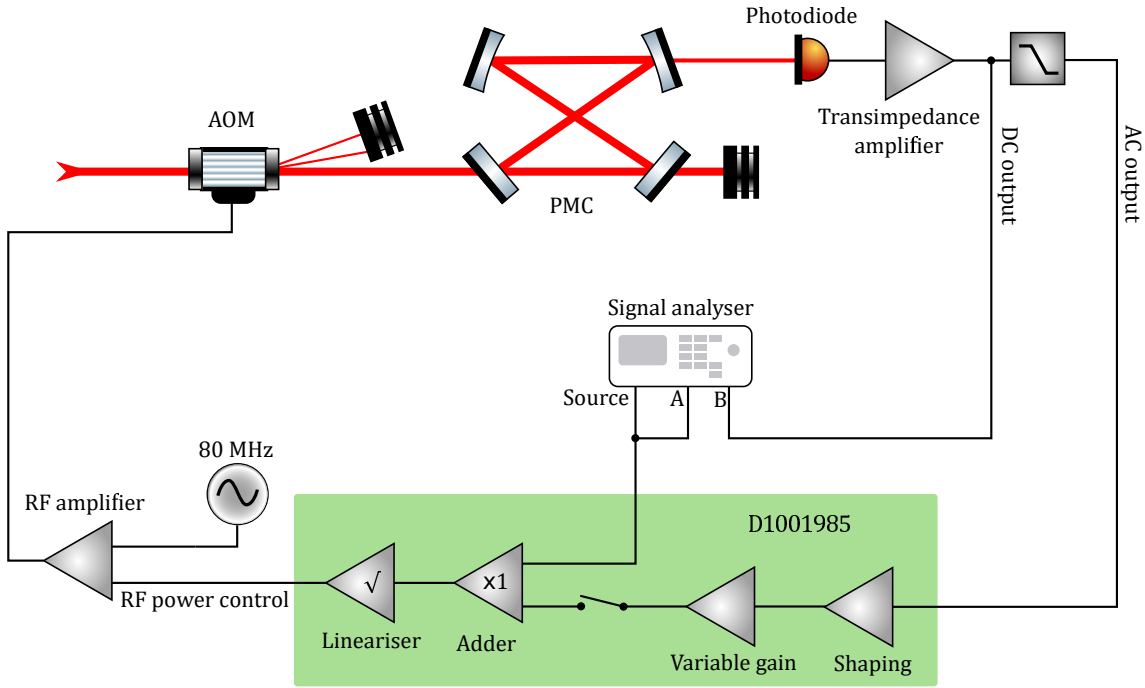


Figure 3.14: A sketch of how the AOM transfer function was measured. The components of the intensity stabilisation servo has also been shown. The PMC was locked and a signal analyser was used to perform a swept sine measurement. An IC (AD734) that performs a square root operation to an input signal was used to linearise the AOM's effect on beam power to an input signal. The transfer function was measured between the input to the lineariser and the photodiode used for the intensity stabilisation. The servo electronics are found in [115] and [116].

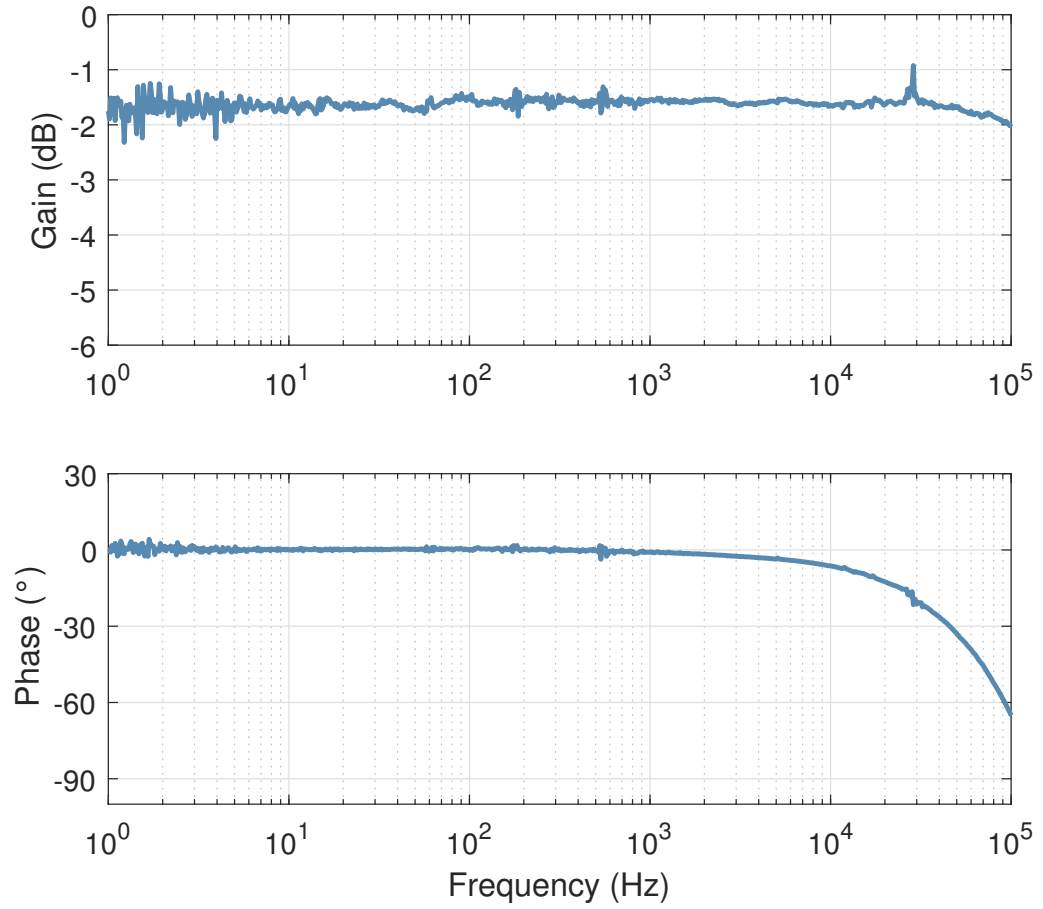


Figure 3.15: Using the layout shown in Figure 3.14, the transfer function between the input to the linearisation stage before the AOM to the light power change on the ISS photodiode was measured. The response was normalised by dividing by the DC power on the photodiode. The transfer function is flat over the region where the open loop gain needs to be greater than one (~ 10 kHz). There is some phase lag between 10-100 kHz; this is where the unity gain point of the servo will be and so this needs to be accounted for when determining if the loop will be stable.

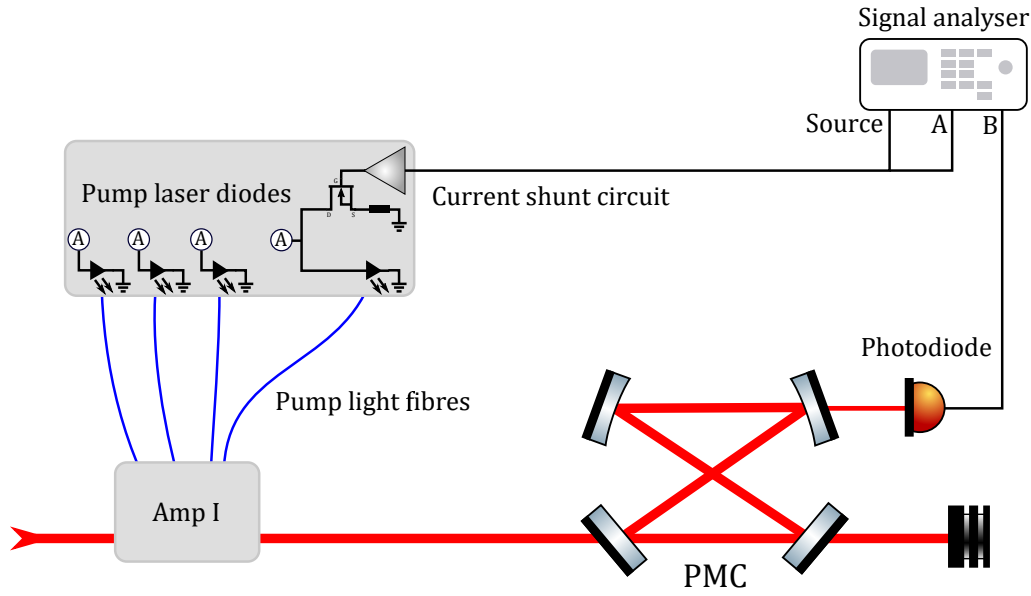


Figure 3.16: A sketch of how the current shunt transfer function was measured. The PMC was locked and a signal analyser was used to perform a swept sine measurement. The current of one of the pump-diodes in the second amplifier was modulated by applying a signal to a current shunt. The transfer function was measured between the input to the current shunt electronics and the photodiode used for the intensity stabilisation.

significant amounts of variation in gain and phase between 1 Hz-100 kHz. Above 10 kHz the transfer function did not vary much with DC pump-diode current. This is good, as the unity gain point of the intensity stabilisation servo will be between 10-100kHz. The AOM is a much stronger actuator than the current shunt; this can be compensated for by changing the servo electronics. It was found that the behaviour of the actuator at 10 Hz, where the servo requirement is hardest to meet, strongly depended on the DC pump-diode current.

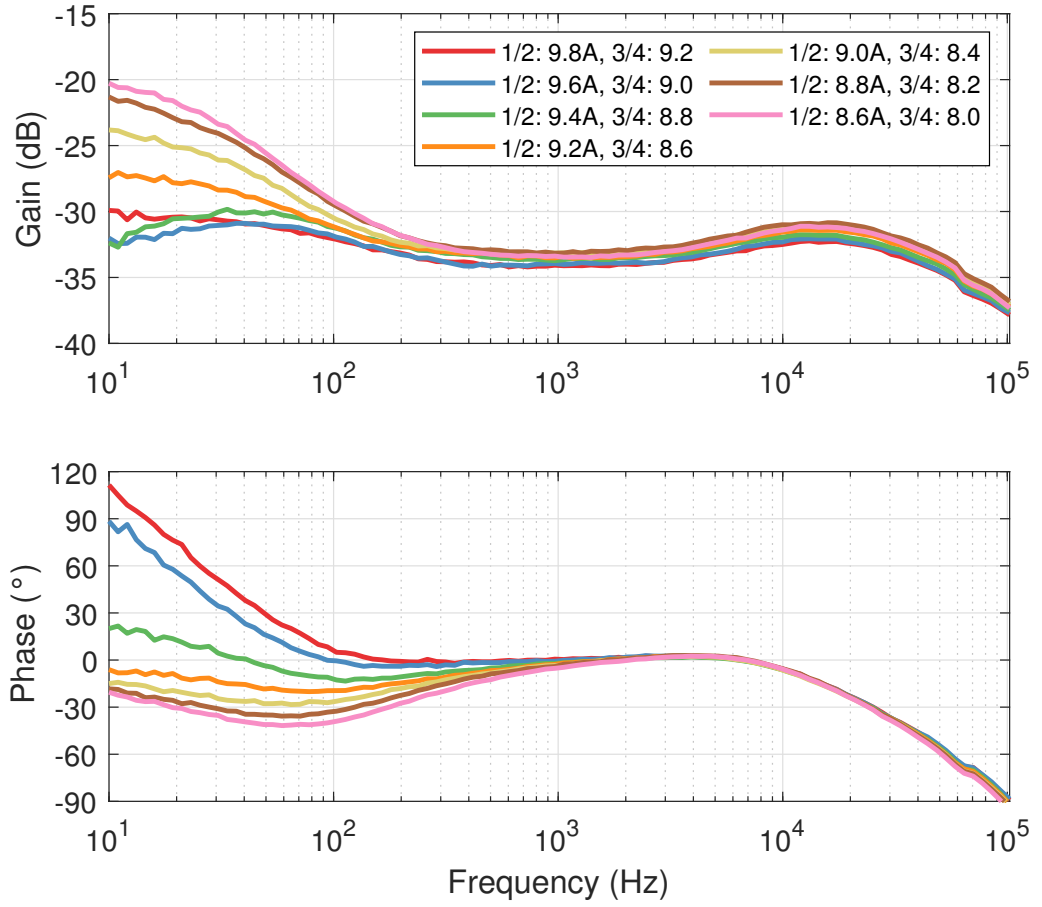


Figure 3.17: Using the layout shown in Figure 3.16, the transfer function between the current shunt input electronics and the intensity stabilisation photodiode was measured. The transfer function was normalised by dividing by the DC power on the photodiode. At 10 Hz, where the noise requirement is hardest to meet, the gain of the transfer function changes significantly depending on the pump-diode's DC current. The phase at this frequency does not matter as it is not near the unity gain point of the servo.

3.5.3.3 The Reason for Choosing the AOM Instead of the Current Shunt

While the current shunt offers some advantages over the AOM, it cannot reduce the power at the many tens of watts level. This is a functionality that LIGO would likely need as the detectors have never operated at full power, at present the LIGO detectors operates with between 25-50W, due to point absorbers on the test masses. The pump-diode DC current would have to be reduced to reduce the power by many tens of Watts. This would affect the noise of the laser and the open loop transfer function of the servo. These amplifiers work best in their saturated regime and the RIN may be increased if the pump-diode currents are decreased significantly. There may be other circumstances in which the pump-diode currents need to be altered; towards the end of their life-time the diodes need to be frequently adjusted. Alternatively, high power attenuation optics would be required. Unlike the current shunt, the AOM's response depends on the DC light level in a trivial way. The AOM causes a decrease in total power, but this downside is outweighed by the reliability of the AOM compared to the current shunt. When the LIGO detectors are able to operate at full power, then the current shunt actuator would be worth revisiting

3.5.4 Open Loop Transfer Function of the Intensity Stabilisation Servo

The free running noise of the laser transmitted by the PMC is shown in Figure 3.13 and the requirement is shown in Figure 3.11. Based on the noise and the

requirement, the unity gain point of the servo must be above 10 kHz. At 10 Hz, where the requirement is hardest to meet, the noise requirement is $2 \times 10^{-8} \frac{1}{\sqrt{\text{Hz}}}$ and the noise is $1 \times 10^{-5} - 1 \times 10^{-4} \frac{1}{\sqrt{\text{Hz}}}$, so there needs to be at least 60 dB of open loop gain at this frequency. If the noise of the laser was not dominated by the 4 Hz lines, the amplitude stabilisation could be achieved with a single pole low pass filter. However, this servo uses additional shaping to ensure that there is enough gain to suppress these peaks.

A sketch of the servo is shown in Figure 3.18. The servo features two circuits that add shaping; these are found built into the photodiode circuit [116] and within the rack electronics [115]. The servo model is described in detail in Section 3.5.4.1. The inner loop ISS features two photodiodes, one for in-loop measurements and one for out-of-loop measurements. The rack electronics allow for the servo to be controlled via CDS. This gives the user the ability to turn the servo on/off, to vary the overall gain and activate extra integrators.

A swept sine was applied at a times one adding stage of the servo to measure the open loop transfer function. The signal at the other input to this stage and the signal immediately after this stage was measured. The ratio of these two gives the open loop transfer function of the servo. For this measurement, the gain in CDS was set to 20 dB. Once the intensity stabilisation was activated, the integrators were engaged to increase the open loop gain at low frequencies. The light transmitted through the PMC was 100 W. The open loop transfer function is shown in Figure 3.19. The unity gain point is at 48 kHz with $\sim 70^\circ$ of phase margin. This servo satisfies the requirements.

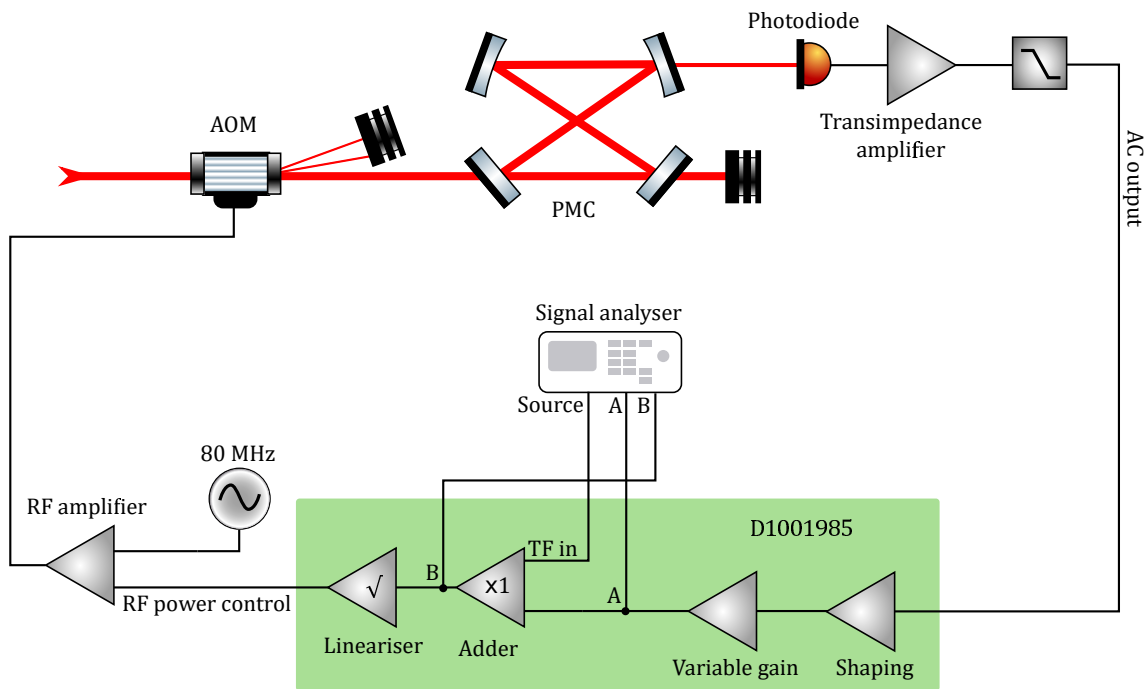


Figure 3.18: Sketch of the amplitude stabilisation servo. The open loop transfer function was measured by injecting a signal into a $\times 1$ adding stage and measuring the ratio of the signal immediately after and just before this stage.

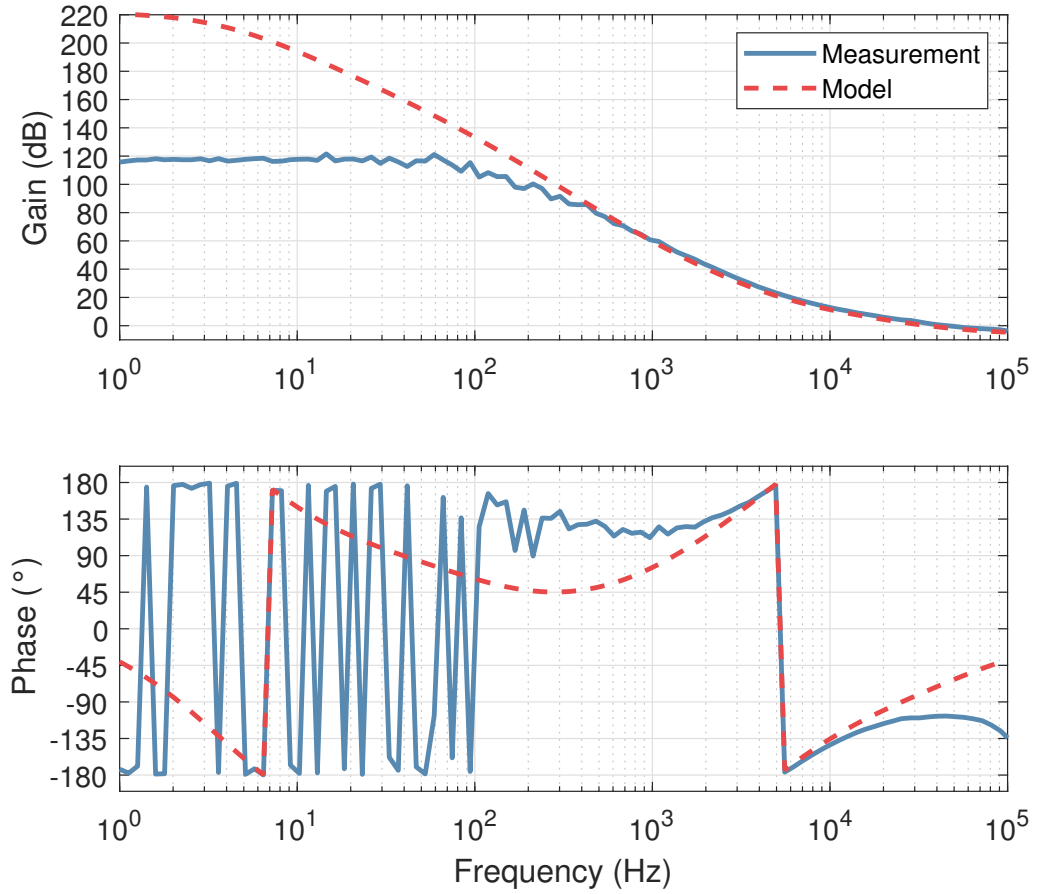


Figure 3.19: The open loop transfer function was measured using the layout shown in Figure 3.18. The unity gain point is at 48 kHz with 70° of phase margin. The shaping of the servo is $1/f$ at the unity gain point, but to achieve higher gain at low frequency, the servo has a $1/f^3$ slope below the unity gain point. The measurement saturated below 100 Hz. The model and measurement do not agree well below a few kHz due to the measurement saturating.

3.5.4.1 Model of the Open-Loop Transfer Function

To understand the behaviour of the servo at frequencies where the measurement of the open loop gain saturated, a model is required. The overall gain of the servo is determined by how much light there is on the in-loop photodiode and the setting of the variable gain. The following description puts all the gain that is distributed over all the components of the servo into one block. The photodiode transimpedance amplifier has a low-pass corner frequency of 5.9 Hz. This passes through a filter which boosts the photodiode's response to AC signals; this filter consists of two transitional differentiators with corner frequencies at 0.05 Hz and 3 Hz. The rack electronics feature two transitional integrators and one transitional differentiator. The first transitional integrator has corner frequencies at 1.06 kHz and 4.83 kHz. The second transitional integrator has corner frequencies at 1.06 kHz and 4.42 kHz. The transitional differentiator has corner frequencies at 59 kHz and 2.9 MHz. Activating the extra integrator stages removes the zeros in the transitional integrators at 1.06 kHz. The AOM introduces phase lag via a delay between the input and the output of 2.5 μ s. This model is summed up in Table 3.1 and shown in Figure 3.19.

3.5.5 Measurement of the Intensity Stabilisation of a 100 W Beam to the Inner Loop Requirement

The PMC was set to transmit 100 W of intensity stabilised light. The stabilised laser noise is shown in Figure 3.20. A noise budget was created using the signal

Component	Transfer function
Photodiode low pass	$\frac{36.945}{s+36.95}$
Photodiode filtered output	$\frac{3600(s+0.3142)^2}{(s+18.85)^2}$
Rack transitional integrator I	$\frac{0.21946(s+3.035e04)}{(s+6660)}$
Rack transitional integrator II	$\frac{0.023982(s+2.777e04)}{(s+666)}$
Rack transitional differentiator	$\frac{49.153(s+3.707e05)}{(s+1.822e07)}$
Rack additional integrators	$\frac{1 \times 10^{-8}(s+6660)^2}{(s+0.666)^2}$
AOM phase lag	$\exp(-2.5 \times 10^{-6} s)$
Overall gain	2.6775×10^{10}

Table 3.1: Model of the intensity stabilisation servo expressed in terms of the complex angular frequency, $s = \omega + i\sigma$. Note that in reality the gain is distributed amongst each block; for clarity, in this model all the components have unity gain at DC and the overall DC gain factor is given. The additional integrators have been expressed as a separate block to the transitional integrators with a pole that cancels a zero in the transitional integrators; in the transfer function of this additional integrator, the zero was chosen to be below any frequency of interest and does not exist in the servo. It was included here to make modelling the servo more clear.

measured on the in-loop photodiode and the shot noise level of the light used to stabilise the laser. This agrees with the out of loop noise measurement above 10Hz. Below 10Hz, there is excess noise in the out of loop measurement compared to the noise budget. Possible sources of this noise could be scattered light or dust falling through the ISS beam. Near the unity gain point, due to the open loop transfer function phase and gain, there is a small increase in the stabilised noise compared to the free running noise, however this is out of frequency band of the requirement, so this is not an issue. This setup will be able to deliver at least 100 W of intensity stabilised light, which will likely be more than is required at the time of writing (2020), to the interferometer.

3.6 Pointing Noise of the Beam Entering the PMC

The pointing noise witnessed at LHO is one of the reasons why single-pass amplifiers were chosen as the replacement for the HPO. Pointing noise manifests as intensity noise since a change in beam direction can be modelled as the introduction of higher order modes. Pointing is passively stabilised by the PSL via the PMC. These new amplifiers are cooled in such a way that the pointing noise on the PSL beam is reduced. The pointing noise requirement is based on the transfer function of the HG10 mode between the input of the interferometer and the dark port [123].

The pointing noise was measured using the DBB and is shown in Figure 3.21. To measure pointing noise, the laser was locked to the DBB cavity. The reflected light

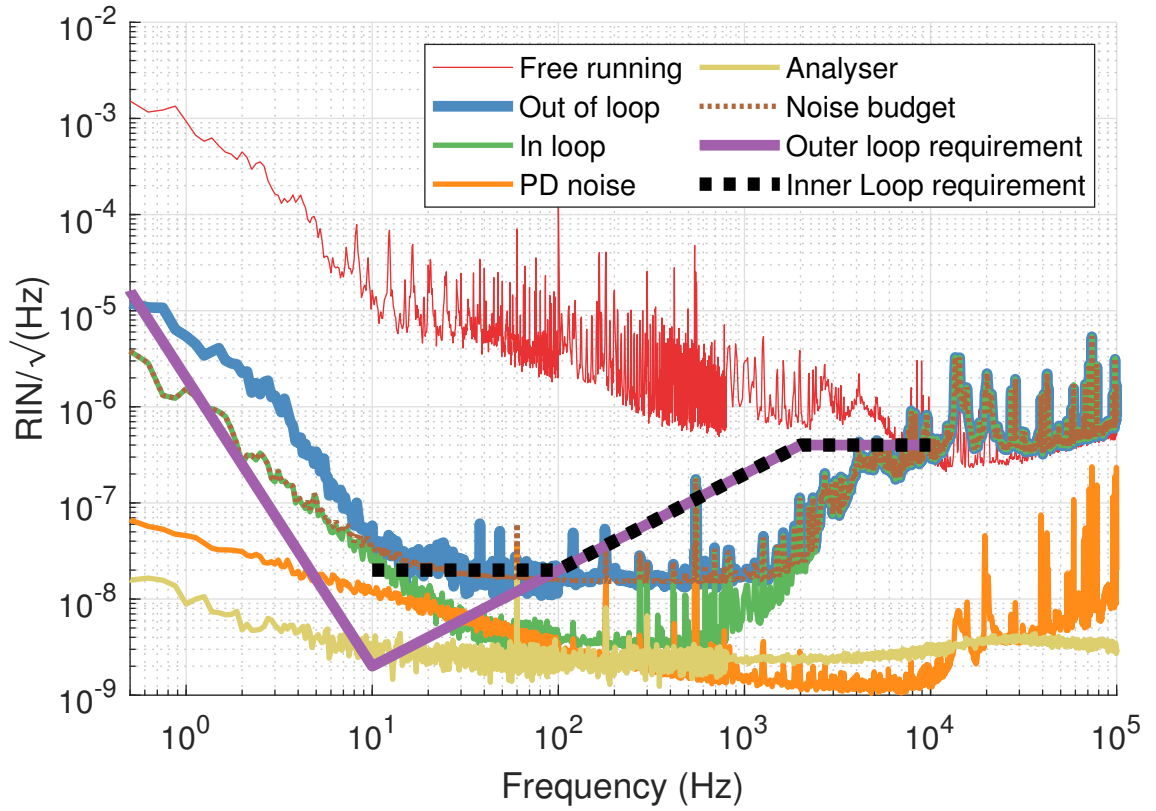


Figure 3.20: The free-running noise of the laser is shown in red. The noise of the light transmitted by the PMC is shown in blue; this meets the requirement that is shown by the dashed black line. The noise level is determined by the in loop noise, shown in green, the shot noise and other technical noise sources; the sum of these is shown by the dotted brown line. Between 10 Hz–100 Hz, the noise of the laser is limited by shot noise. There is a discrepancy between the noise budget and the measured noise. This is likely due to dust falling through the beam or scattered light. Near the unity gain point of the servo, the noise is enhanced by the servo due to gain peaking.

from the cavity was made up of the fundamental mode of the cavity (some of this mode leaks out) and the first order modes associated with misalignment. The beat between these two fields can be used to determine the pointing noise of the beam; this is known as differential wavefront sensing [124]. The electronic noise floor is at $1 \times 10^{-7}/\sqrt{\text{Hz}}$. The pointing noise is expressed in units of $p = \delta x/w_0 + i\delta\alpha/\Theta$, where Θ is half the divergence angle of the beam, α is the angle of the beam, x is the beam's position and w_0 is the beam's waist.

While the pointing noise measurement was affected by scattering below ~ 200 Hz, the pointing was measured to exceed the requirement below ~ 1 kHz. To reduce the pointing noise caused by turbulence in the cooling system, the water lines should be made to have gradual bends in them. As the LIGO PSL table is in a quieter environment than this setup, an improvement will be seen when this is constructed on the PSL table.

3.7 Free Running Frequency Noise of the Prototype Laser System

The LIGO detectors employ a multi-stage frequency stabilisation system. Ultimately, the arm cavity is used as a frequency reference since $\delta f/f = \delta l/l$, however the initial stabilisation of the laser uses an ultra low expansion fused silica reference cavity located on the laser table [125]. The free-running frequency noise must be

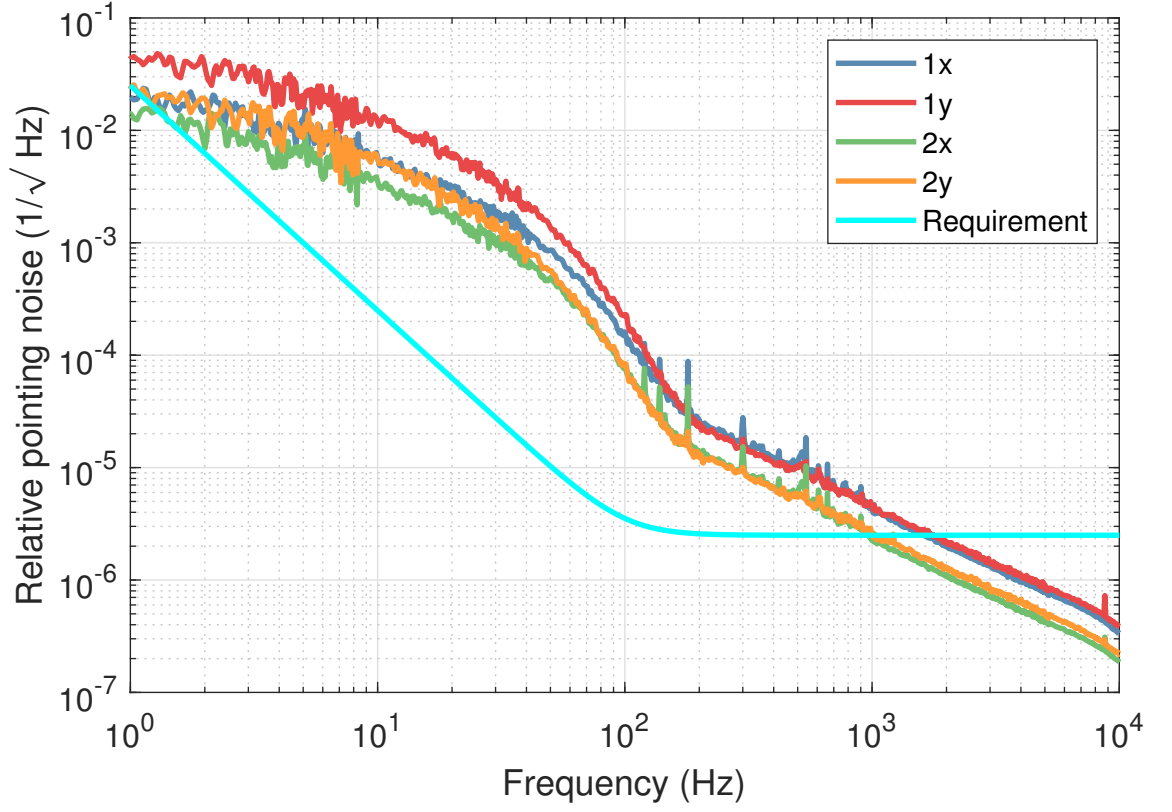


Figure 3.21: The four relative pointing degrees of freedom for the beam into the PMC was measured with the DBB. The requirement [123] is shown in light blue. The pointing noise is above the requirement, however the TnT lab has a higher amount of environmental noise than the PSL. Measurements below 200 Hz were be affected by scattering.

measured to design the frequency stabilisation servo.

The PMC can be used to measure the frequency noise of the beam emerging from the PSL. The beam's frequency will be determined by the length noise of the PMC and the frequency noise of the NPRO. The signals used to control the PMC can be converted into a measurement of the systems frequency noise. The control signal which is sent to the PZT in the PMC can give the frequency noise of the system where there is sufficient open loop gain. Around the unity gain point, the response of the servo needs to be factored in. Alternatively, the error signal at the mixer and the open loop gain can be used to measure the frequency noise of the system.

The NPRO is locked to the PMC by controlling the length of the PMC. The length of the PMC is controlled with two actuators. A PZT is attached to one of the mirrors to provide fast control, and a heater which determines the thermal expansion of the PMC provides slow control. The PDH technique is used to sense the length of the cavity. RF frequency side-bands are applied to the laser via an EOM, and the light that is reflected from the PMC is demodulated at the same frequency as the RF sidebands. When the cavity is near resonance the demodulated signal is proportional to the difference between the laser frequency and the frequency corresponding to the cavity length. A sketch of the PMC length control servo is shown in Figure 3.22.

To see how the error signal used to lock the cavity and the open-loop transfer function gives the frequency noise, consider Figure 3.23. The difference between

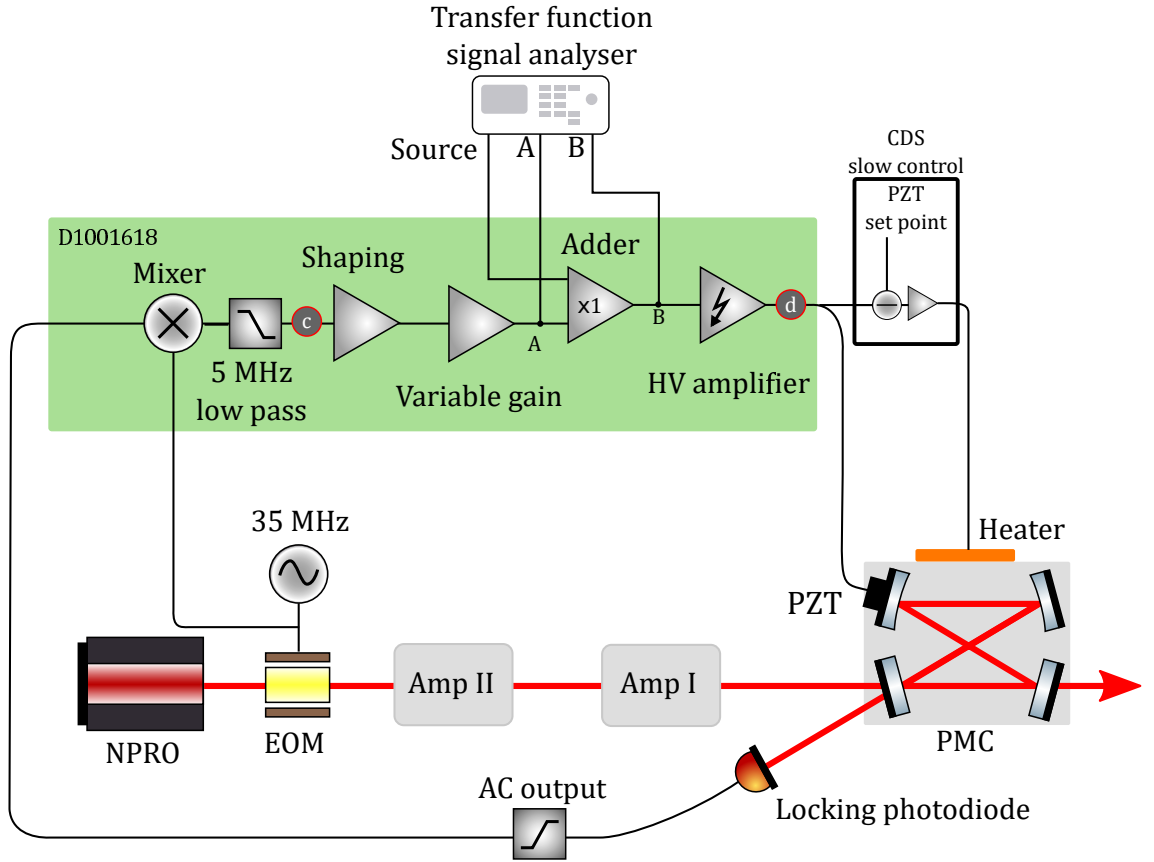


Figure 3.22: A sketch of the frequency stabilisation servo. An EOM creates 35 MHz sidebands which are used to produce a PDH locking signal. A PZT controls the length of the cavity at high frequencies. A heater and a digital loop is used to provide slow control that corrects for frequency drifts associated with change in the NPRO temperature and the room temperature. This is required as otherwise the PZT would run out of range. The open loop transfer function was measured using signals at A and B. The frequency noise was measured using the error point and control signals marked by *c* and *d*, with this notation corresponding with the notation in Figure 3.23 and Equation (3.1).

the laser frequency and the cavity frequency is obtained via the PDH technique. The PZT signal, c , is then obtained via the transfer function of the optics, i.e. the frequency to voltage transfer function corresponding to the slope of the demodulated PDH signal, and the transfer function of the electronics. Combining these transfer functions gives G . First, the error signal is defined as $c = a - b$. Next, the control signal is expressed in terms of the error signal, $cG = d$. Expressing the signal b in terms of d allows for signal a to be written in terms of the error signal c ,

$$a = c(1 + GH). \quad (3.1)$$

Equation (3.1) shows that to measure the frequency noise of the laser, a , the error signal should be, approximately, multiplied by the open loop transfer function, GH .

3.7.1 Calibrating the Mixer Signal, the PZT and Modelling the Open Loop Transfer Function for Locking the PMC

As the cavity is short compared to the frequencies of interest, the transfer function representing that converts a difference in NPRO and PMC frequency, the signal due to the PDH scheme, to a voltage is a flat gain factor. A measurement of this can be obtained by measuring the slope of the PDH signal as the frequency of the cavity passes through a resonance. By driving the PZT with a ramp and simultaneously measuring the signal before and after it is demodulated, the transfer function of the optics can be obtained. Figure 3.24 shows this measurement. The time axis can

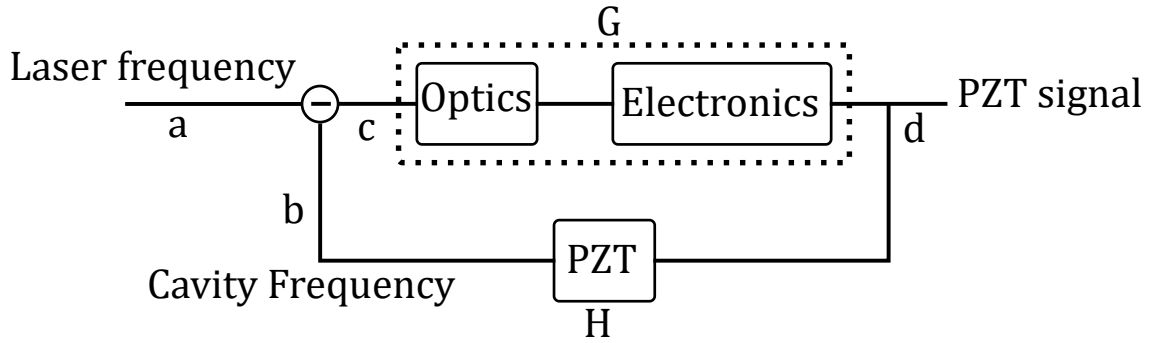


Figure 3.23: The servo can be considered as being made of three parts: the optics, the shaping electronics and the PZT. The transfer function of the optics represents how a difference in frequency between the laser and the cavity transforms into a signal after the demodulator. When at resonance and below the cavity pole, this is a flat transfer function that is proportional to the difference in frequency. Since this cavity pole is in the MHz region and this measurement will only go to the kHz region, the cavity pole can be ignored. The transfer function of the electronics includes everything after the demodulator up to and including the final high power OP amp. The PZT transfer function converts a voltage to a cavity frequency. The PZT acts as a low pass filter due to it having capacitance.

be converted to frequency as the full-width-half-maximum (FWHM) of the reflected light is determined by the optical properties of the PMC. The design value of the FWHM for the PMC is 1.19 MHz. The slope of the demodulated signal as the cavity passes through the resonance can be used to give the conversion between difference in frequency of the cavity and laser to voltage after the demodulation electronics. This factor is 0.1364 MHz/V.

A ramp was applied to the PZT and the spacing between two resonances was measured to convert the PZT voltage to the cavity frequency at low frequencies. The difference in ramp voltage between the two resonances corresponds to the FSR of the cavity. This is shown in figure 3.25. The nominal FSR for the PMC is 148.529 MHz, so the PZT to frequency transfer function is 42.2870 MHz/V. The PZT forms a low pass filter as it has capacitance; this must be included when modelling the servo. The corner frequency of the PZT was estimated to be at 1700 Hz. The corner frequency of the PZT is above the unity gain frequency of the PMC locking servo.

The measurement and model of the open loop transfer function, i.e. the transfer functions of the optics, electronics and the PZT multiplied together, is shown in Figure 3.26. The model is within 2 dB of the measurement. A sketch of how the open loop transfer function was measured is shown in Figure 3.22. The unity gain point of the servo was measured to be at approximately 800 Hz. The difference between the model and the measurement may be due to the use of nominal values for the properties of the PMC.

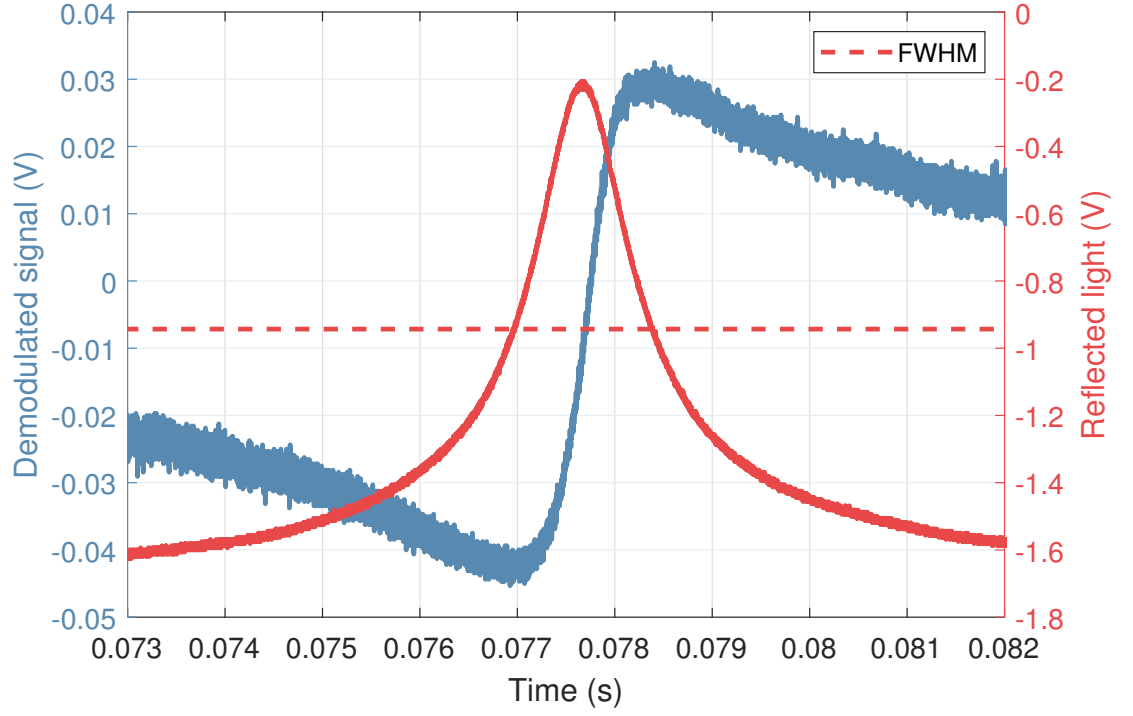


Figure 3.24: A ramp was applied to the PZT and the photodiode signal was measured before and after demodulation. The demodulated photodiode signal is shown in blue. The photodiode signal is shown in red. Around the peak of the photodiode signal, the slope of the demodulated signal gives the gain of the ‘optics’ transfer function. This can be expressed in units of MHz/V by converting the time axis to a frequency axis. This is done by using the FWHM of the photodiode signal as a reference frequency. The level of the FWHM is indicated by the dashed red line.

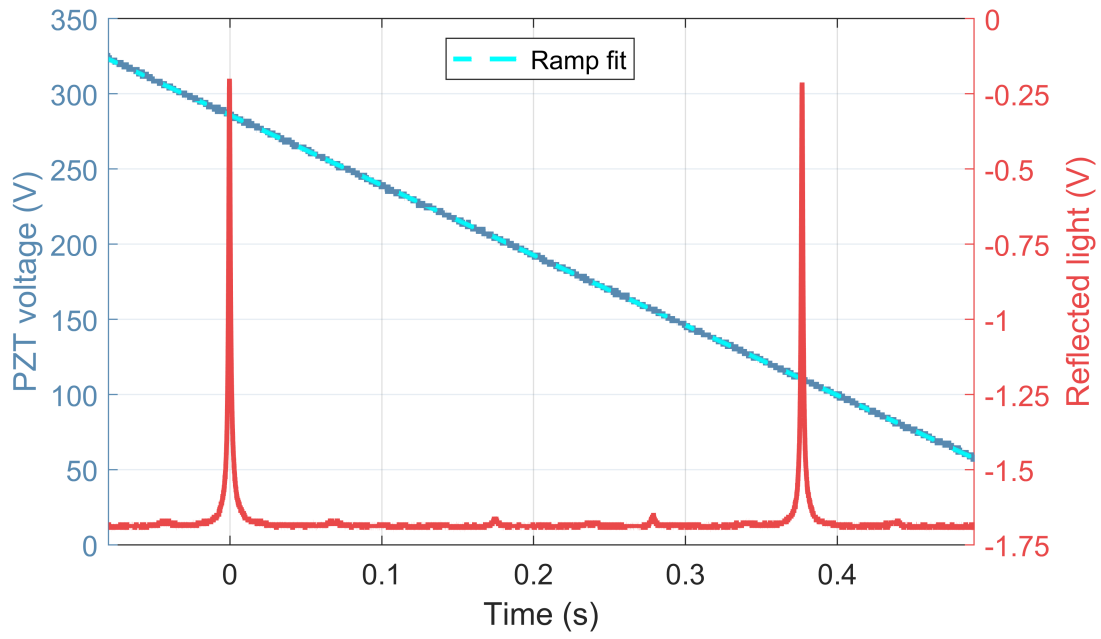


Figure 3.25: A ramp was applied to the PZT and the reflected light from the cavity was measured. The FSR of the cavity is known, so the time in this measurement can be converted to frequency. This gives the PZT transfer function at low frequencies. The ramp signal is shown in blue and the reflected light is shown in red.

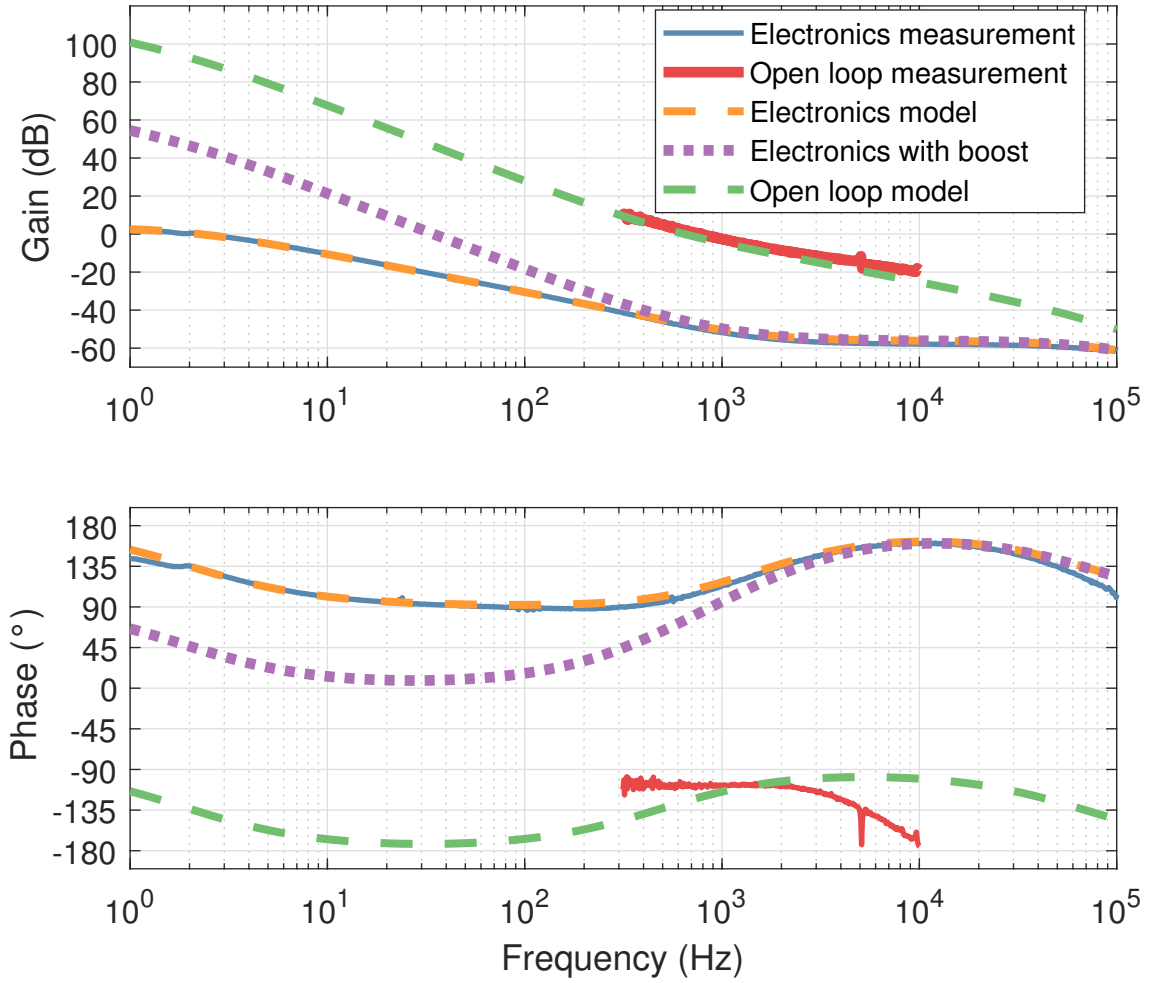


Figure 3.26: The open loop transfer function was measured using the layout shown in Figure 3.22. This is shown in red. The transfer function of the electronics without the ‘boost’ provided by an extra integrator stage was measured and is shown in blue. The model of this is shown in orange. The boosted electronics model is shown in purple. The green line shows the expected servo gain based on the boosted electronics model and the measurements shown in Figures 3.24 and 3.25. This model is within 2 dB of the measurement. The unity gain point was measured to be at ~ 800 Hz.

3.7.2 Frequency Noise Measurement

The measurement of frequency noise is shown in Figure 3.27. Despite imperfect calibration, this measurement provides useful information about the frequency noise of the laser. As the PMC was not vacuum sealed, between 100 Hz to 1 kHz, there was a bulge in the frequency noise. This was due to the air conditioning generating PMC length noise combined with non perfect offsetting in the PDH loop. Below 100 Hz and above 1 kHz, the frequency noise of the laser was determined by the NPRO. Near the UGP, there was gain peaking which will enhance the noise.

Below the unity gain point, as the PZT response is flat, the PZT signal can be converted into frequency noise. The mixer signal and the open loop transfer function $c(1 + GH)$ gives the frequency noise. These two measurements are within a factor of 2 of each other. This noise is higher than the expected level for the NPRO. There is uncertainty in the calibration as nominal values for the optical properties of the PMC were used.

At frequencies much above the unity gain point of the servo, a signal applied to the PZT will not be suppressed by the loop. Near the unity gain point, a signal applied to the PZT will be attenuated. A peak at 1111 Hz was used to check the calibration of the loop. The closed loop gain at 1111 Hz was modelled to be -5.2 dB, so the peak measured at the mixer should be reduced by a factor of $1 - 10^{-5.2/20} = 0.45$. The peak generated at the PZT was measured at the mixer to be reduced by a factor of 0.70; the difference between the prediction and measurement would be

fixed with better characterisation of the PMC.

The discrepancy between the blue and red traces in Figure 3.27 may be due to the optical values used for the PMC, as these were not measured. A frequency to amplitude conversion occurs if the cavity is detuned, however this conversion does not happen at the FSR of the cavity [126]. Thus, by measuring the transfer function between a frequency modulation generated by a broadband EOM and the amplitude of the light transmitted by the cavity, an accurate measurement of the length of the cavity can be made. To measure the finesse of the cavity, an RF amplitude modulator could be used. The transfer function between the input modulation and the transmitted light depends on the FSR and the finesse of the cavity [126]. The AOM in the layout cannot be used as its bandwidth is too low. In Figure 3.27, the red trace below the unity gain point is better calibrated since the FSR does not depend on the cleanliness of mirrors.

3.8 Conclusion

The PSL has undergone a major redesign due to the pointing noise introduced by the cooling for the HPO. The new layout consists of an NPRO and two solid state amplifiers followed by the PMC. A prototype of this new design was constructed and tested. This layout is shown in Figure 3.2. The setup procedure was established and a cooling scheme was trialled.

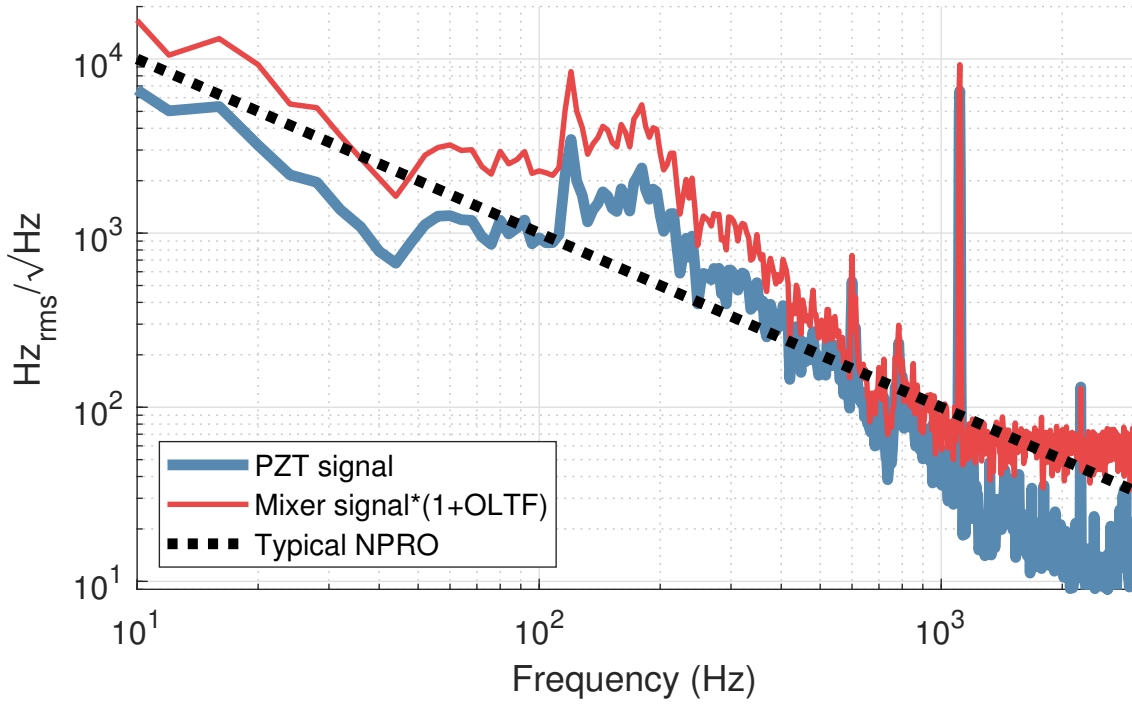


Figure 3.27: The mixer signal combined with the open loop transfer function is shown in red. This is equivalent to the frequency noise of the system. The PZT signal should agree with this measurement below the unity gain point (~ 800 Hz). A signal was applied to the PZT at 1111 Hz. This is below the corner frequency of the PZT and above the unity gain point for the servo. This should match with the mixer signal. There is a factor of two between these two measurements. This can be attributed to imperfect calibration due to nominal parameters of the PMC being used. The typical noise expected from an NPRO is shown by the dashed black line. Between 100 Hz and 1000 Hz, there is a bulge in the frequency noise. This is likely due to acoustically driven length noise of the PMC as the PMC is not vacuum sealed.

The frequency noise of the beam transmitted by the PMC was investigated using signals in the PMC control loop. The laser is dominated by NPRO noise at frequencies below $\sim 100\text{Hz}$ and above $\sim 1\text{kHz}$. As the PMC is not shielded from acoustic vibrations, the PMC introduced noise between these two frequencies. This measurement is shown in Figure 3.27.

The pointing noise was found to be exceed to the requirements (See 3.21). As the pointing noise is close to the requirement, a significant reduction may be achieved when PSL is in a quieter environment. To reduce the pointing noise further, the flow rates being reduced will help, however this reduces the output power. Active alignment could be implemented using the light that is reflected from the PMC. Turbulence generated by bends in the water cooling pipes could be reduced to lower the pointing noise.

The level of power achieved with this setup will be sufficient for O4 as the usable amount of power is limited by noise couplings which arise due to point absorbers in the test masses [55]. The output power was measured over two days and was found to drop from a peak of 111 W to 107 W. This is shown in Figure 3.7. This was due to changes the lab temperature affecting the alignment. Higher output power could be achieved if more pump-diode current and cooling was used. Additionally, to increase the power emerging from the PMC, the modematching into the PMC could be improved.

100W of intensity stabilised light was transmitted through the PMC. The free running intensity noise of the beam transmitted by the PMC was measured and

the inner loop intensity stabilisation was tested. Two actuators were considered, an AOM and a current shunt; the AOM was chosen as this offers the most flexibility in terms of loop gain and operating power. The inner loop noise requirement was met with this setup. The stabilised noise is shown in Figure 3.20.

Future iterations of the PSL may use fibre lasers and fibre amplifiers [102, 103] as there is rapid development in this technology. These have been demonstrated at the power levels required for LIGO. Coherent power addition is another technique that can be used to obtain high beam powers (e.g. [127, 128]). When long term operation has been demonstrated, these technologies may be a viable replacement for the current laser design.

Chapter 4

Junk Light Generated by the Signal Recycling Cavity

4.1 Introduction

A gravitational wave creates a differential signal in the arm cavities which is directed to the signal recycling cavity (SRC) (see Section 2.4.3), and then to the output mode cleaner (OMC). The OMC removes *junk* light from the signal beam (see Section 2.4.4), so the light emerging from the OMC will only contain the signal generated by the differential arm length change. Some of the junk light will come from mode mismatches between the arm cavities and the SRC (see Section 4.1.1), and non-optimal mode matching decreases the interferometer's sensitivity (see Sec-

tion 4.1.2). The parts of the LIGO detector discussed in this chapter are shown in Figure 4.1.

Currently, based on the measurements reported on in Section 4.2, there is significant uncertainty on the arm mode's beam parameter at the signal recycling mirror (SRM). The active wavefront control within HAM6 is designed around this uncertainty (see Chapter 6). The effect of errors in the focal length of SR3 and SR2 on the beam parameter and higher order mode content (HOM) of the arm light at the SRM are explored in this chapter. To do this, a LIGO-like interferometer, was simulated using FINESSE; this is discussed in Section 4.3 and Section 4.4.

Using this simulation, the uncertainty in the beam parameter is explored (see Section 4.5). The difference in the HOM content of the beam transmitted through the SRM between an ITM single-bounce measurement and a measurement which involves the arm cavities is discussed in Section 4.6. The effect that the interferometer being in signal recycling and resonant sideband extraction (RSE) mode had on the HOM content of the beam as the focal length of SR3 deviated from its nominal value was also discussed in Section 4.6. Section 4.7 is a summary of the results in this chapter.

4.1.1 Mode Matching the Arm Cavities, the SRC, and the OMC

Since the SNR of the gravitational wave signal is directly related to the mode matching between the arm cavities, the SRC, and the OMC, it is crucial to have accurate knowledge of the beam parameter at the SRM. The mode in the arm is well defined since the arm cavities are high finesse. Uncertainties in the RoC of optics in the SRC telescope (SR2 and SR3, see Figure 4.1) results in an uncertainty in the beam parameter of the arm mode at the SRM, as well as the eigenmode of the SRC. The SRC is a low finesse cavity, so a range of modes that do not match the SRC's eigenmode can be sustained within it.

Junk light will be generated if the arms are perfectly matched to the OMC but not to the SRC because the SRC is a low finesse cavity. If the arms are matched to the OMC but not to the SRC, then actuators for changing the mode matching between the arm and the OMC could not be used to increase the differential arm signal sensed by the photodiode after the OMC. This means the mode matching between the arm and the OMC can be optimal but the light can still contain HOM. Additionally, if the beam is astigmatic, then optimal mode matching would not correspond to perfect mode matching.

4.1.2 Losses due to Mode Mismatch

Losses degrade the sensitivity of the interferometer (see Section 2.7). It is known that $\sim 20\%$ of the light is lost between the AS port and the readout photodiodes, as the shot noise can be compared to the expected shot noise for a given fringe slope, input power and known losses. Approximately half of the loss is accounted for, e.g. some comes from the Faraday isolator, some comes from the OM3 pick off; however, the other half is not fully understood, e.g. see [129]. The level of squeezing can be used to cross check the loss [130]. Some of the loss which is unaccounted for may come from mode mismatch between the arm cavities and the OMC.

Loss due to mode mismatch between the squeezed light source, the interferometer, and the OMC acts coherently. This is known as coherent destructive modal interference [131]. Mode mismatch can be modelled as a modal beam splitter that transfers some light from the HG00 mode to the HG20+02 mode. The amount of light transferred between the HG00 and HG20+02 mode is determined by the magnitude of the mismatch, and the phase difference between the HG00 and HG20+02 mode depends on the nature of the mismatch. An error in waist size is in-phase (real), whereas an error in waist location is out-of-phase (imaginary). In general, the mismatch introduces a complex phase. A sequence of mode mismatches can be modelled in the same way as a Mach-Zehnder interferometer. The difference in phase between the HG00 and HG20+02 mode is analogous to a difference in path length in a Mach-Zehnder interferometer. It differs from a typical Mach-Zehnder interferometer as the HG00 to HG20+02 splitting ratio is not well balanced. As the

loss due to mode mismatch losses act coherently, the degradation of the squeezed state can be greater for loss due to mode matching than for other types of loss, e.g. the loss due to photodiodes with a quantum efficiency less than one. If the squeezer is perfectly matched to the OMC, then the mode mismatch between the squeezer and the interferometer is four times the loss you would expect from treating a mismatch loss as a regular loss. This is explained in detail in [131, 132].

4.2 Measurements of the Mode Matching Between the Arm Cavities and the OMC at the LLO Detector Between 2018-2019.

The mode matching between the arm cavities and the OMC, and the input light to the OMC were studied at the LLO detector between 2018-2019, and these measurements are described in the following section. Techniques for determining the mode matching between the arms and the OMC include: actuating on the waist size and waist location of the beam coming from the arm and measuring what effect this has on the mode matching to the OMC; direct beam profile measurements; the application of different DARM offsets to determine what portion of light in an OMC scan corresponds to the arm mode and what portion of the light comes from elsewhere in the interferometer (e.g. from contrast defect); and use of a beacon signal to distinguish between the signal-carrying arm cavity light and junk light.

Modelling work relies on the mode matching of the PRC and the arms to be high, and this was estimated from measurements of the interferometer's reflectivity to be $> 99\%$ [133]. High amounts of mode matching between the PRC and the arms allows for single-bounce and full interferometer measurements to be compared. The upper limit on mode matching from the arm to the OMC is 99%, and this was obtained from actuation on the beam parameter and the use of a beacon signal (meaning this must be arm cavity light). The entire collection of measurements finds the mode matching to be between 87% and 99%. Since this is a large discrepancy, the measurements are briefly discussed below.

Mode scans for an input power of 10W find the mismatch to be 11% [134]. The DARM offset was changed to determine what light was coming from the arm mode. An input power of 10W is low, the design is 125W, so the thermal state of the interferometer may be different in this measurement compared to when the interferometer is in observing mode. This method was repeated for 40W of input power [135]. In this measurement, it was found that there was 4% mode mismatch.

A direct measurement of the beam profile found 15% mode mismatch between the OMC waist and the beam transmitted through the SRM while the interferometer was in an ITM single-bounce [136,137] configuration. This measurement also shows significant astigmatism, with the waist in the x and y directions being separated by ≈ 0.5 m. The x direction data is discarded in the analysis as there was trouble orienting the beam profiler. Since there is good mode matching between the input and the arms ($> 99\%$), there should not be much difference between this beam and the beam when the interferometer is locked if the thermal lensing in the ITM

was similar to how it would be in observation mode. This type of measurement cannot be performed while the arms are locked as they require commissioners to be working near the arm cavities. This analysis is in close agreement with the mode scan measured in [134].

Altering the beam parameter using the ITM thermal lens actuator has shown that the mode matching between the arm cavity and the OMC can be improved by 1% – 2% [138]. Actuating on the beam parameter of the arm cavity with the SRM CO2 heater increased the interferometer’s optical gain [138] since the amplitude of the beacon signal increased by 10%. The ITM and the SRM are separated by sufficient Gouy phase, so actuating on them covers enough beam parameter phase space to determine if the mode matching was optimised. The interferometer was in single-bounce mode, but since the mode matching between the PRC and arm cavity is high, this is similar to a locked-arm cavity measurement. The interferometer’s input power was 25W, so thermal effects may be different between this measurement and when the interferometer is in observation mode. This type of measurement was repeated using 40W of input power [139]. Assuming the ITM thermal lens was optimal, the mode matching from the arm to the OMC was found to be optimal with no SRM heating applied. This suggests that the mode matching is $> 99\%$. A beacon signal was used to determine the mode matching between the arm to the OMC.

Modelling of the arm mode finds that a 3.5% improvement in mode matching between the arm cavity and the OMC can be made using the ITM thermal lens and the SRM thermal lens [140]. This modelling is compared to the measurement

made with data from [137]. It was found experimentally that a 4% improvement in the mode matching between the arm to OMC could be achieved for a single-bounce measurement with 25W of input power. The interpretation of the measurements in [140] contradicts that of [139], as [139] shows that the mode matching between the arm and the OMC is optimised.

Just after the interferometer had lost lock, a series of mode scans were used to create a model which predicts 92% mode matching between the beam transmitted through the SRM and the OMC [141]. The input power was 20W and the lock lasted 1 hour and 41 minutes, so the interferometers thermal state immediately after this lock loss was the same as it would be for a typical time during observing mode when the input power was 20W. The thermal lens in the ITM was measured using the Hartmann wavefront sensors, and this was used to model the interferometer at the time of unlocking. The Hartmann data is noisy, which could cause uncertainty in the mode matching prediction made by the model, although no error on the fit parameters is given in [141].

The mode matching between the squeezer and the optics within HAM6 was studied [142,143]. Measuring the beam profile between the squeezer and the OMC allows for the layout in HAM6 to be modelled. This means that there is less uncertainty in where the mismatch between the arms and the OMC originates. The mode matching between the squeezer and the OMC was measured to be 98%. The model of HAM6 that was developed from this data differs significantly from the nominal values of the optics in HAM6, a 15% error in the RoC of OM2 is required to explain the beam profile measurements. This may not be unreasonable, as accurate measurements of

focal lengths for mirrors with $f = 2\text{m}$ are challenging to make.

4.3 Modelling Interferometers With Mode Mismatches Using FINESSE

FINESSE is software for simulating optical layouts. Optical layouts are modelled with components such as mirrors, lenses, and spaces. The way these components are connected is defined by nodes. FINESSE performs a nodal analysis to calculate the steady state amplitude of light fields within the optical layout, similar to how a circuit can be analysed. PyKat is a python wrapper for FINESSE which provides many quality of life improvements such as the ability to programmatically update and re-run simulations [144].

The light fields that are calculated in a FINESSE simulation can be outputted by using detectors such as a photodiode, `pd`, or an amplitude detector, `ad`. These can be configured to detect only certain modes using the `mask` command. The `bp` detector is used for outputting the beam parameter, and the `gouy` command computes the Gouy phase accumulated along a set of spaces. These detectors were used to find the ratio of HOM to HG00 light, the beam parameter of the arm mode that is transmitted through the SRM, and the Gouy phase accumulated along the SRC.

The beam parameters at nodes in a FINESSE simulation are determined by `cav`

and **gauss** commands. The **cav** command is used for defining linear and circular cavities. The beam parameters are found by solving for the cavity eigenmode. Once these nodes have had a beam parameter defined at them, the components connected to them have their beam parameter calculated based on the set of linear equations that define the component. If there are more than one **cav** commands used in a simulation, the latter **cav** command overwrites the previous one. The **gauss** command executes after the **cav** commands and sets the beam parameter of the laser used in the simulation.

FINESSE can perform simulations using HG modes, allowing for the effects of mode mismatches to be modelled if the problem satisfies the paraxial approximation. A coupling between the HG modes happens whenever a beam described by q_1 enters a segment described by q_2 . This could happen if the **gauss** command used on the laser does not correspond to the eigenmode of a cavity defined by the **cav** command.

A laser beam can be modelled as a linear combination of Hermite Gaussian modes. Ideally, an infinite number of these modes would be used in a calculation; however, since it is not feasible to simulate an infinite number of TEM modes, the maximum number of TEM modes needs to be defined in a FINESSE simulation. This is done using the **maxtem** command. FINESSE will calculate the coupling of the light amongst these modes. If an insufficient number of TEM modes is used in the simulation, the output will be nonsense because power from the higher order TEM modes that would be required to accurately model the optical layout will be ‘aliased’ into the ones defined in the simulation. When a cavity becomes unstable, FINESSE will not generate physical results since the cavity will not be mathematically

describable.

The distance between components, L , in a FINESSE simulation is defined with a macroscopic distance, l , which is modelled with the **space** command, and a microscopic distance which is modelled with the tuning parameter, ϕ , belonging to components such as mirrors (**m**) and beam splitters (**bs**). The macroscopic distance l is always an integer number of wavelengths, whereas the microscopic tuning, ϕ , determines the position of a component to sub-wavelength accuracy. FINESSE uses the convention that one wavelength corresponds to 4π rad phase. It is useful to split the total length, $L = l + \phi$, in this way since the l determines things like Gouy phase, whereas ϕ determines whether a cavity is on resonance/which mode is resonant. When simulating mode mismatches, the command **phase 2** is required to give physically correct results.

4.3.1 Thin and Thick Beam Splitters

Thin beam splitters are made with the **bs** command. Care needs to be taken when using these components since incorrect usage can yield erroneous results. To illustrate this, two simulations of Michelson interferometers were run, one with a thin beam splitter and the other with a thick beam splitter made up of 3 thin beam splitters and two spaces with a refractive index > 1 . See Figure 4.3 for an illustration of a thick beam splitter. The FINESSE code for these simulations can be found in appendix C.3.

For energy to be conserved, each reflection from a dielectric coating where the refractive index is lower on the reflecting side than on the transmitting side causes a light field to gain 180° of phase. In a Michelson, there is one path with two of the π phase change reflections, while in the other path there is only one such reflection. This means when the arms have equal lengths, destructive interference occurs at the AS port. The intensity, I , at the AS port of the Michelson as a function of the tuning of one of the arm mirrors, $\Delta\phi$, including the FINESSE convention for phase, is

$$I = I_0 \sin^2(\Delta\phi), \quad (4.1)$$

where I_0 is the input power.

The effect of using a thin beam splitter versus a thick beam splitter in a FINESSE simulation is shown in Figure 4.2. The thin beam splitter causes there to be constructive interference at the AS port for equal arm lengths, whereas Equation (4.1) shows that it should be destructive interference. Using a thick beam splitter fixes this ‘bug’, as the phase at the beam splitter is accounted for.

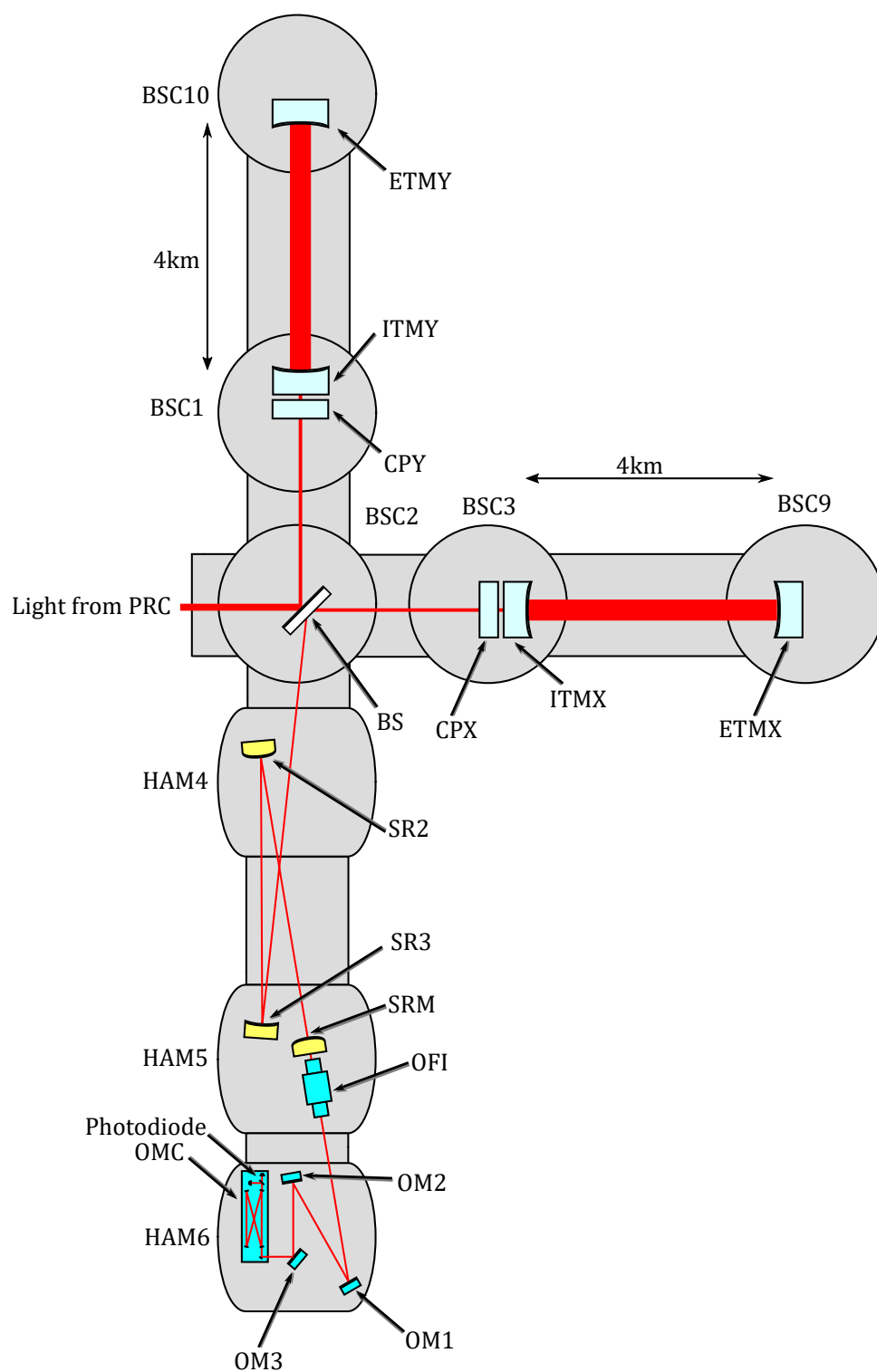


Figure 4.1: (Caption on the next page.)

Figure 4.1: (Figure on the previous page.) Sketch showing the parts of the LIGO detectors discussed in this Chapter. A more complete sketch is shown in Figure 2.4. The angles and lengths are exaggerated for clarity. Light from the arm cavities is incident upon the beam splitter (BS). Differential arm length signals encoded onto this light are directed towards SR3, SR2, and then the SRM. Between the SRM and the OMC, there are optics for steering and mode matching (OM1-3). Finally, to strip away junk light, the light enters the OMC. The light emerging from the OMC is detected by photodiodes.

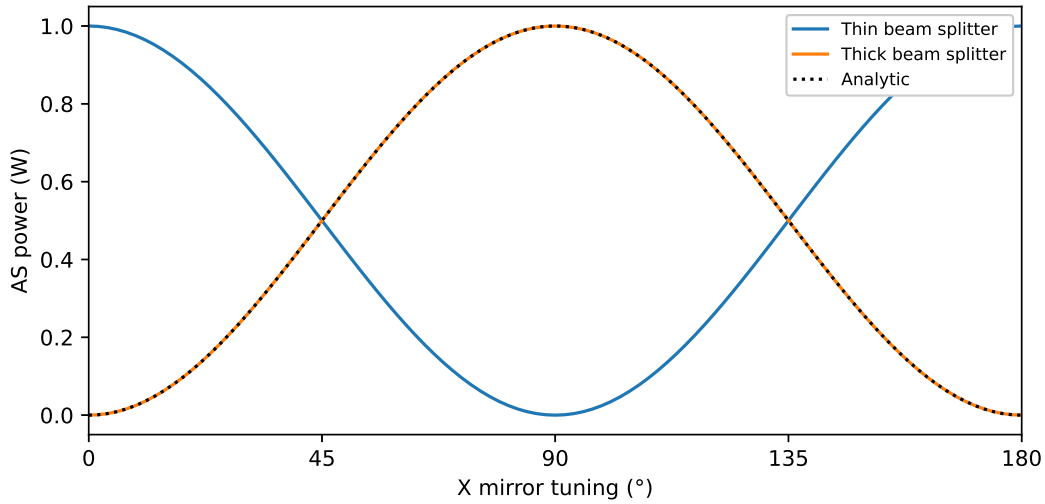


Figure 4.2: The simulation of a Michelson interferometer with thin a beam splitter is shown in blue. The same simulation was done with a thick beam splitter, this is shown in orange. The analytic behaviour of a Michelson interferometer (Equation (4.1)) is shown by the dashed black line. The tuning of one of the arm mirrors was changed and the power at the AS port was simulated. Comparing the outputs of the simulation to Equation (4.1) shows that thick beam splitters should be used.

4.4 Modelling of a Michelson Interferometer With Fabry-Perot Arm Cavities and a Signal Recycling Cavity

To investigate what effect a mode-mismatch between the SRC and the arm cavities has on the HOM content of the beam transmitted through the SRM, a ‘LIGO-like’ interferometer was simulated in FINESSE. This interferometer uses two identical, design specification arm cavities. The lengths between the optics in the SRC were obtained from the Zemax model of the LLO interferometer, and the focal lengths of the optics in the SRC are the measured values for the optics used in the LLO detector. The reason for using identical arm cavities is to eliminate contrast defect. The OMC, IMC, and PRC were not included in this simulation as these do not affect the beam in the SRC. A schematic of the simulation is shown in Figure 4.3. The default focal lengths, transmission coefficients and spacings for this simulation are listed in Table C.7.

To perform ITM single-bounce measurements, the ETMs are misaligned. This results in the ITM having an effective power reflectivity of unity. To replicate this in FINESSE, the R for each ITM was set to 1 minus the loss of the ITM HR coating. The ETM and ITM both have 10 ppm loss. Because the mode matching between the PRC and the arms is close to 100%, this is valid.

The interferometer was set up to be in DC readout mode. The tuning of the arm

order modes. There are two sorts of errors which cause mode mismatch: RoC errors and errors in the distance between each of the optics in the SRC. RoC errors are due to uncertainties in the RoC of SR3, SR2, the SRM, and unintended levels of thermal lensing in the ITM and compensation plate.

The largest geometry errors in the SRC come from RoC errors in SR3 and SR2, with both errors affecting the beam in a similar way as there is not much Gouy phase between them. Therefore, to investigate how geometry errors affect the beam transmitted through the SRM, SR3's and SR2's radii of curvature were varied.

Certain components have a small impact on the uncertainty in the beam parameter at the SRM. The thermal lensing in the ITM is assumed to be 50 km, and it would require significant deviation, in the direction of more heating, for this lens to have a large effect. The radius of curvature of SRM is well enough known that it is not the dominant cause of junk light in the SRC. The spacings between the optics in the SRC is known to a great enough precision that errors in their locations have a negligible effect.

For simplicity, no Schnupp asymmetry was included. In a real interferometer, a Schnupp asymmetry is required to make the interferometer controllable. Since spaces are by default on resonance in FINESSE, no Schnupp asymmetry is needed to control the inner Michelson length. The Rayleigh length of the beam at the ITM is of the order of several hundred metres. This is huge in comparison to the Schnupp asymmetry, which is 8 cm. The Schnupp asymmetry will have a small effect on the HOM content of the beam transmitted through the SRM, but since the dominant

sources of HOM are RoC errors in SR3 and SR2, the behaviour of the beam in the SRC can be largely understood without it.

The beam parameter of the input laser was defined so that there was no difference between the single-bounce and the locked arm simulation. To define the beam parameter, the `gauss` command was used. In this simulation, the mode matching from the laser to the arm cavities should be perfect. The amount used was 99.99999%; this was found to be sufficient. See Appendix C.1 for details on how w_0 and z were determined for the `gauss` command used on the laser. The power of the input laser was set to 5 MW.

The `cav` command was used to set up the arm cavities, although the `gauss` command used on the laser means the `cav` command is not required (on the fly testing showed this to be the case). A `cav` command was not used on the SRC since this forces the beam parameter at the SRM to be determined by the SRC geometry. An SRC cannot be defined in FINESSE using the `cav` command as one of the end mirrors cannot be properly defined.

The detectors for analysing the mode matching between the nominal arm mode (which matches the laser mode) and the SRC mode were `pd` and `ad` with various `mask` commands applied to them. To detect the total light, no `mask` is applied. To detect the higher order light, a `mask [pd] 0 0 0` command is used. This sets the sensitivity of the photodiode to the HG00 mode to 0. The mode matching is then determined as $1 - \text{HOM}/\text{total light}$. `maxtem` is set to 6. As a cross check, two amplitude detectors were used to detect the HG02 and HG20 modes:

```
ad ad_HOM_2_y 2 0 0 nSRMb
ad ad_HOM_2_x 0 2 0 nSRMb
```

When plotting the power of the HG20+HG02 mode, the square of the `ad` detector output needs to be taken. The `bp` detector was used to find the beam parameter. To find the Gouy phase accumulated in the SRC by the HG00 mode, the `gouy` command was used.

Both the signal recycling and RSE modes of operation were investigated. The tuning of the SRM determines whether the interferometer is in RSE mode or in signal recycling mode. If the interferometer has arm cavities, the SRM detuning which corresponds to RSE is 90° while the detuning which gives signal recycling is 0° . For an interferometer without arm cavities, to operate in RSE mode the SRM detuning should be set to 0° , while to be in signal recycling mode the detuning of the SRM should be set to 90° .

4.4.1 Determining Tunings for DC readout

To maintain the same SRC finesse seen by the differential mode for each simulation, the interferometer was set up so that it was on the dark fringe, with 30 mW of light being transmitted through the SRM. This was achieved by applying a differential tuning to the ETMs for simulations with the arm cavities, and a differential tuning to the ITM for single-bounce simulations. The light transmitted through the SRM

for an interferometer with arm cavities as a function of differential ETM detuning is shown in Figure 4.4. For an interferometer without arm cavities, the power transmitted through the SRM as a function of differential ITM detuning is shown in Figure 4.5. The tunings for operating with 30 mW of light being transmitted through the SRM are shown in Table 4.1 and Table 4.2.

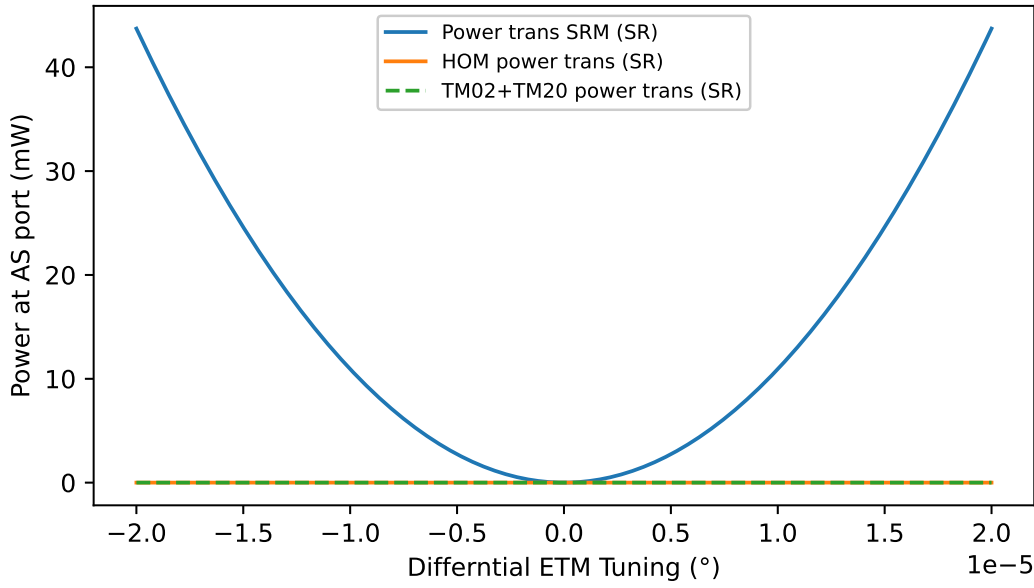


Figure 4.4: The minimum of the dark fringe is at 0° differential ETM detuning, as shown by the blue line. A slight offset causes light to be at the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange line and dashed green line show the HOM content of the beam.

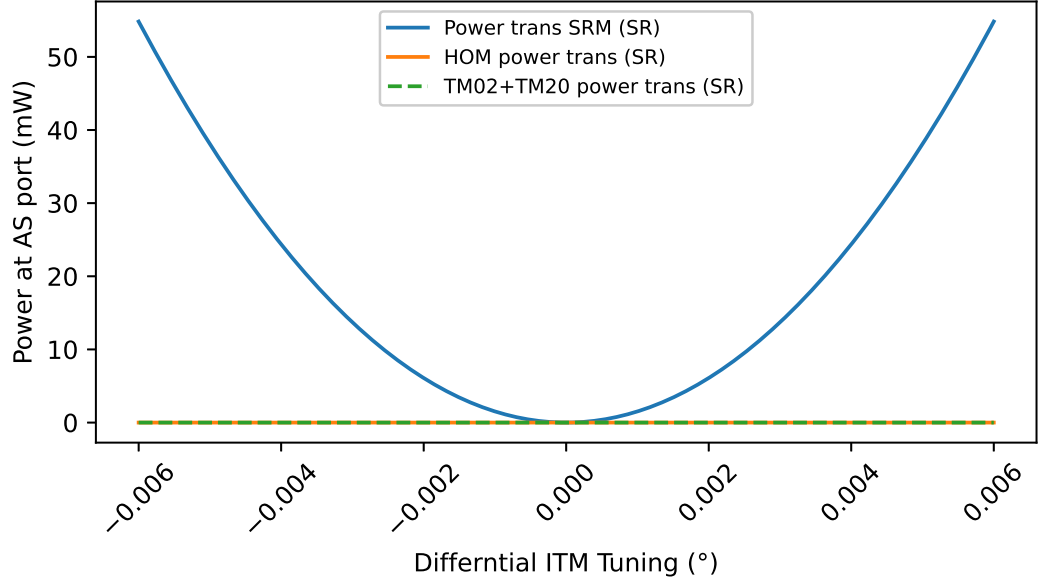


Figure 4.5: The minimum of the dark fringe is at 0° differential ITM detuning, as shown by the blue line. A slight offset causes light to be at the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange line and dashed green line show the HOM content of the beam.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ETM _x	0	$+2.76\text{e} - 06$
ETM _y	0	$-2.76\text{e} - 06$
ETM _x	90	$+2.64\text{e} - 05$
ETM _y	90	$-2.64\text{e} - 05$

Table 4.1: The tunings used for the arm-cavity simulation to get 30mW of light at the dark port.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ITM _x	0	$+0.00709$
ITM _y	0	-0.00709
ITM _x	90	$+0.00074$
ITM _y	90	-0.00074

Table 4.2: The tunings used for the no-arm-cavity simulation to get 30mW of light at the dark port.

4.5 The Beam Parameter of the Arm Mode at the SRM

The mode defined by the arm was simulated at the SRM. The beam parameter at the SRM depends on macroscopic quantities such as the distances between mirrors and the RoC of the optics, rather than microscopic tunings. Therefore, the beam parameter is not affected by the interferometer being in RSE or signal recycling mode. The beam parameter depends only on the geometry of the cavities. The focal lengths of SR3 and SR2 were varied around three standard deviations of their measured values, and the effect this had on the beam transmitted through the SRM was determined with a `bp` detector.

The effect of a focal length error in SR3 and SR2 on the width and RoC of the arm mode at the SRM is shown in Figure 4.6 and Figure 4.7. These figures show that the beam parameter depends mainly on the focal length of SR3, although SR2 has a noticeable effect. When the focal lengths of SR3 and SR2 are at their nominal values, the simulated beam has a width of 2.1 mm, a RoC of -4.0 m, and the Gouy phase accumulated between the ITM and the SRM is 18.7° . There is a slight difference between the RoC found here and the design value of -3.88 m; this difference is due to the focal lengths of the optics used in the SRC at LLO being different from the design values.

Measurements of the single-bounce beam profile [137] suggests that the beam is much larger at the SRM than the design value. This would correspond to a positive

error in the focal lengths of SR3, as shown in Figure 4.6. If the beam were this large, it is unlikely that the SRC would be stable, as shown in Figure 4.10. The measurement of the beam width found in [137] was obtained under fairly artificial conditions, thus it may not represent the beam parameter of the arm mode when the interferometer is in observing mode.

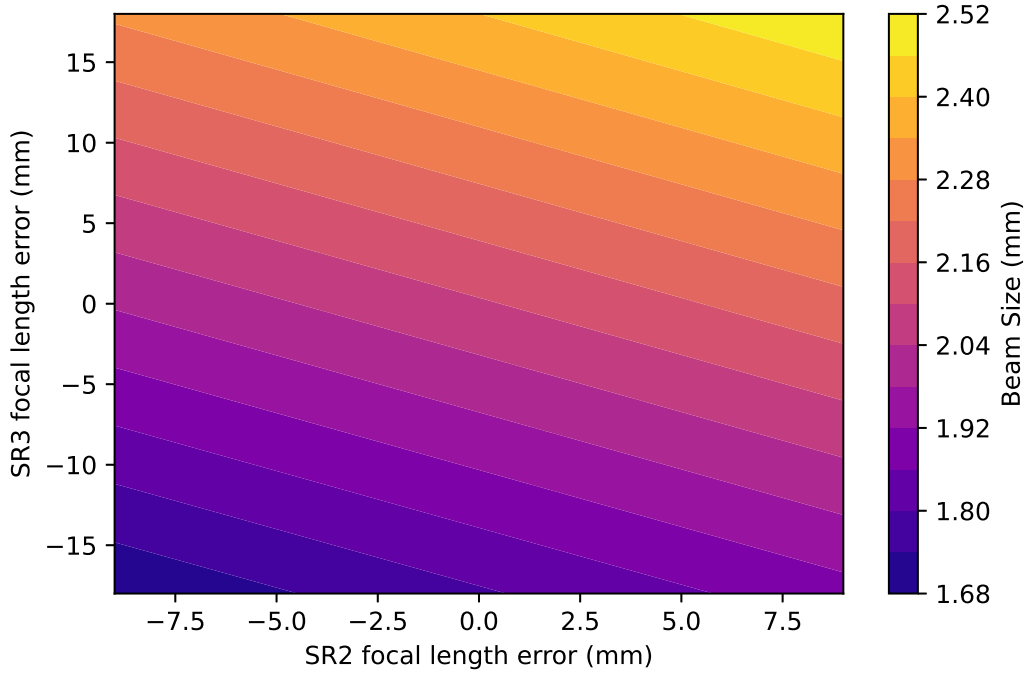


Figure 4.6: The Gaussian width of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. The Gaussian beam width is strongly dependent on SR3’s focal length error, whereas SR2’s focal length error is less important. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see Figure 4.10), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

Information about the beam parameter of the arm mode can be inferred from the Gouy phase accumulated by the arm mode in the SRC. The Gouy phase as a function of focal length error in SR3 and SR2 is shown in Figure 4.8. Measurements of Gouy phase of the SRC find it to be 22.81° [145] and 21.3° [146]. These measurements of Gouy phase suggests that the beam at the SRM is smaller than the design value and has a lower curvature. This Gouy phase also corresponds to the same similar amount of HG00 loss that is measured and simulated (Figure 4.13). The Gouy phase accumulated in the SRC is almost entirely dependent on the RoC of SR3 and SR2, positioning errors of up to 1 cm of SR3 and SR2 do not affect the Gouy phase significantly. Figure 4.8 shows that it is unlikely that the Gouy phase could be 22.81° in a state where the arm has greater than 90% mode matching to the SRC, as the RoC of both SR3 and SR2 would have needed to have been three standard deviations from the measured value. This could be accounted for by a much shorter thermal lens in the ITM (see Figure 4.9). The measurement of 21.3° of Gouy phase is plausible with the 50 km thermal lens. The single pass thermal lens was measured to be 38.92 uD, i.e. ≈ 25.7 km [141], which corresponds to the design value and the value used in this simulation.

The focal lengths of SR3 and SR2 can be estimated from the Gouy phase measurement. A normal probability density function was used to model the uncertainties in the focal lengths of SR3 and SR2. This was combined into a 2-D normal distribution. Combinations of SR3 and SR2 that result in 21.3° of Gouy phase were found and the value of the 2-D normal distribution for those combinations was calculated. The maximum value of the resulting probability density function (which is normal) corresponds to the most likely combination of focal lengths for SR3 and SR2, and one

standard deviation corresponds to the error. Figure 4.11 shows the that focal length of SR2 to be -3.205 ± 0.003 m and the focal length of SR3 to be 17.971 ± 0.002 m.

The beam parameter at the SRM is determined by the focal length of SR3 and SR2, and the focal lengths of SR3 and SR2 have a probability density associated with them (see Figure 4.11). Thus, each beam width and RoC pair that was computed in this simulation also had a probability density associated with it. Based on the measurement of Gouy phase, the beam is likely to have a width of 1.84 mm and a defocus of -2.80 Diopetre. This is shown in Figure 4.12. This is a slight simplification as the beam parameter cannot be such that the Gouy phase of the cavity becomes too small ($< \sim 17.5^\circ$). The cavity will be unstable if the Gouy phase is too small, so beam widths of more than 2.25 mm are unlikely. The shape of the distribution of the arm mode beam parameter at the SRM is driven mostly by the focal length error of SR3.

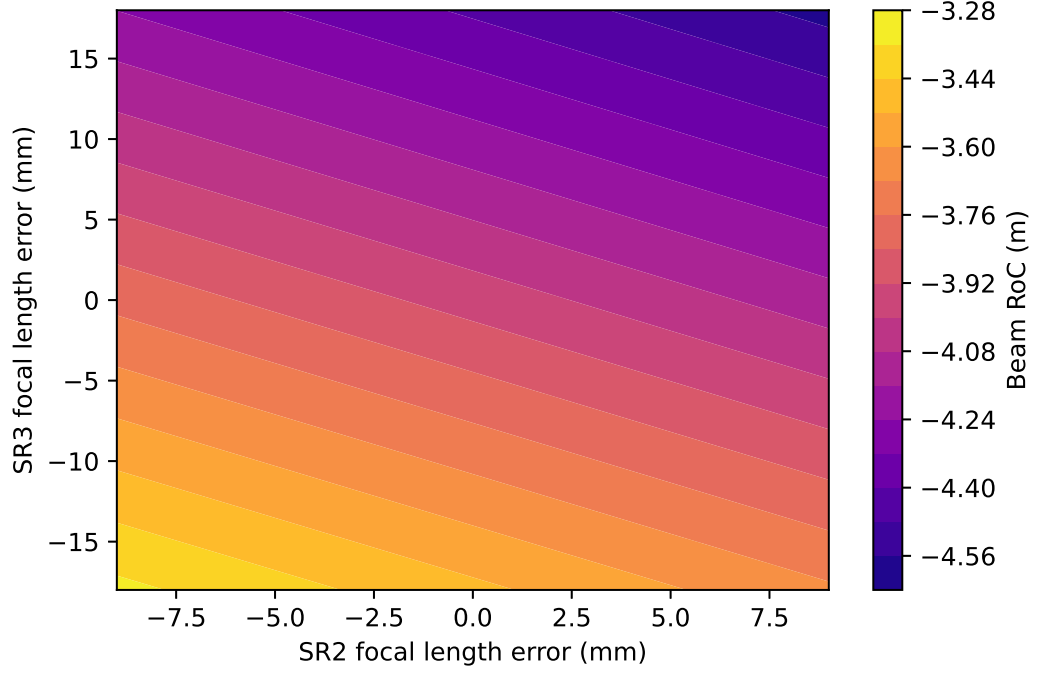


Figure 4.7: The RoC of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. If the arm cavities and the SRC are well mode matched, then the RoC of the beam transmitted through the SRM would be $\frac{5.67\text{m}}{1.45} = 3.91\text{m}$. When the SR3 focal length error $> 6\text{mm}$ and SR2 error is zero the SRC cavity is unstable (see Figure 4.10), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

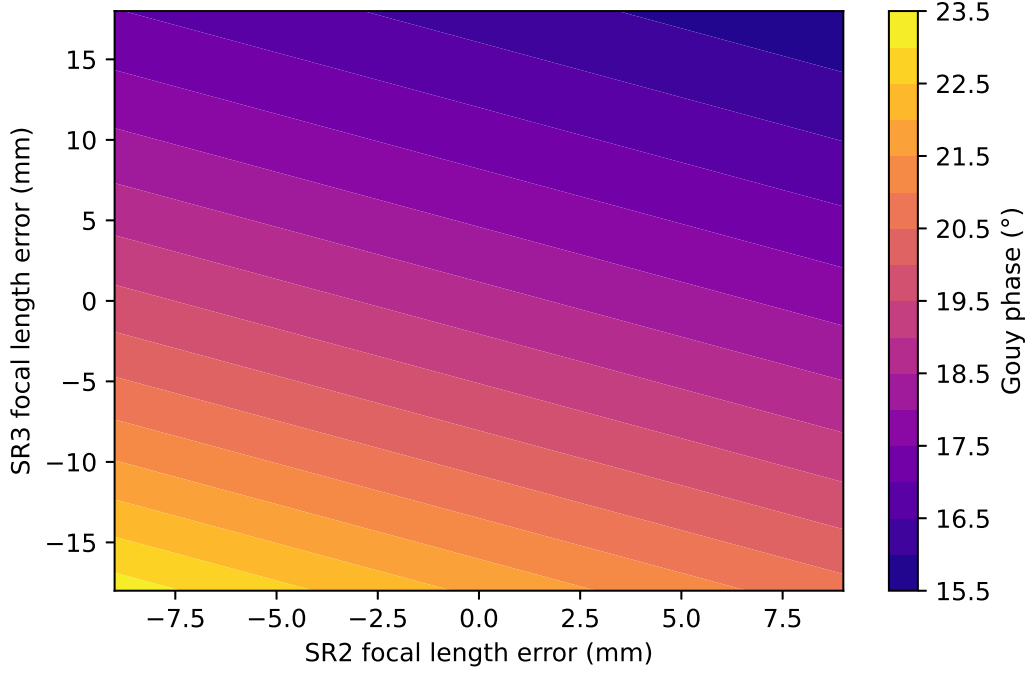


Figure 4.8: The Gouy phase between the ITMs and the SRM as a function of the error ($\pm 3\sigma$) in SR3 and SR2. When the SR3 focal length error > 6 mm and SR2 error is zero, the SRC cavity is unstable (see Figure 4.10), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

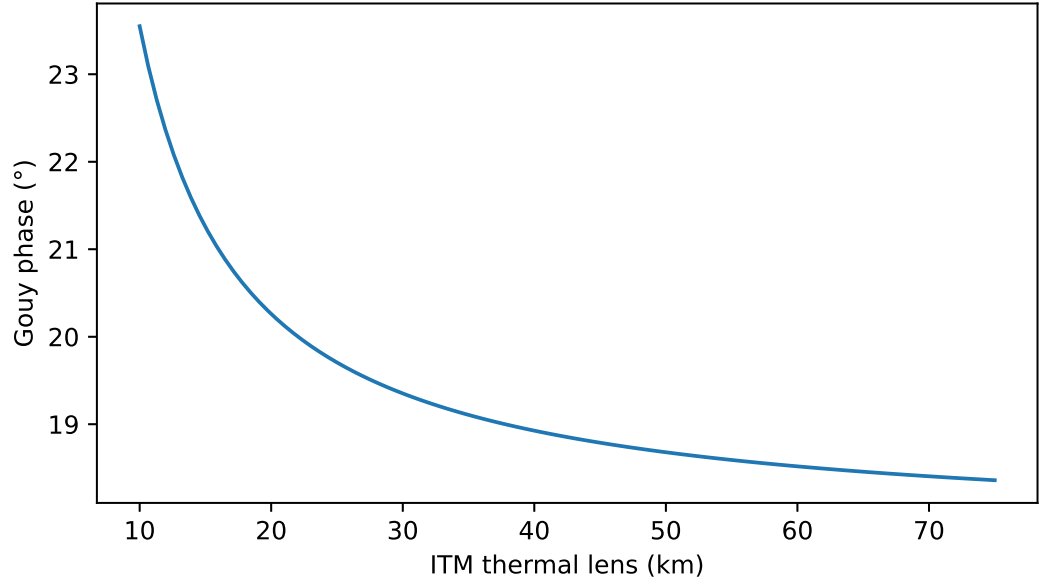


Figure 4.9: The Gouy phase between the ITMs and the SRM as a function of the thermal lens in the ITMs for the default values of SR3 and SR2. The nominal ITM thermal lens focal lengths are 50 km, and this corresponds to about 19°. More heating results in a greater thermal lens, however the LIGO detectors compensate for this with the thermal compensation system (TCS).

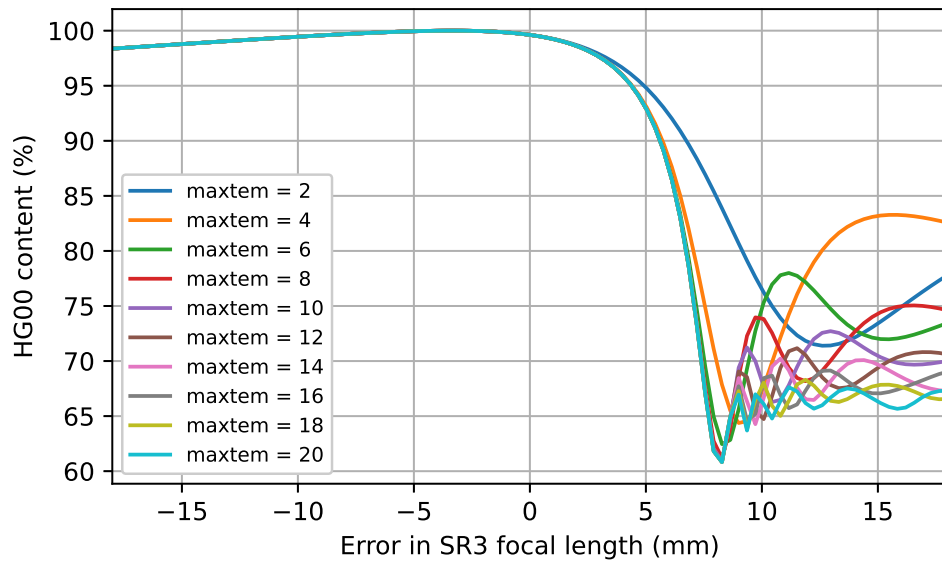


Figure 4.10: The percentage of HG00 light transmitted through the SRM as a function of SR3 focal length for the Michelson interferometer with Fabry-Perot cavities in its arms was simulated. Each simulation used a different `maxtem` command. The focal length of SR3 was changed. For focal lengths with a positive error greater than 7.5 mm, the cavity becomes unstable and FINESSE does not produce reliable results. For a stable SRC, the minimum number of TEM modes for sub 1% accuracy is six.

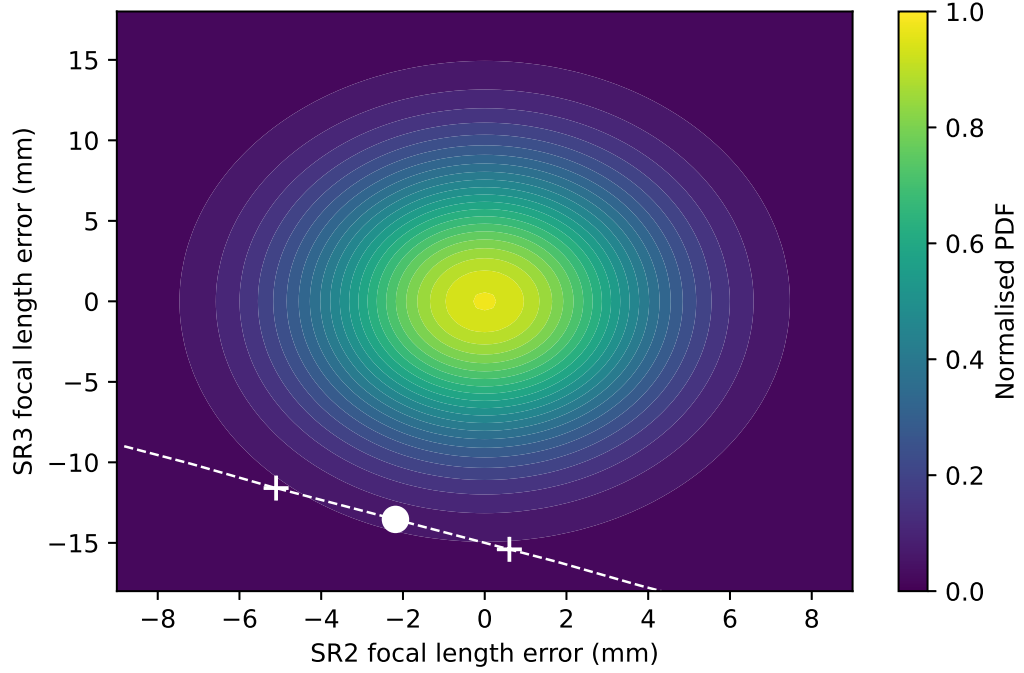


Figure 4.11: The measurement error of SR3 and SR2 follows a normal distribution. These were combined to make a 2-D normal distribution. Dark blue corresponds to the least likely combinations of SR3 and SR2, and bright yellow corresponds to the most likely. The measured Gouy phase was 21.3° [146]; this is indicated by the white dashed line. The white circle marks the most likely combination of SR3 and SR2 which gives 21.3° of Gouy phase, and the white crosses show one standard deviation from this. The white circle and crosses indicate that the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m.

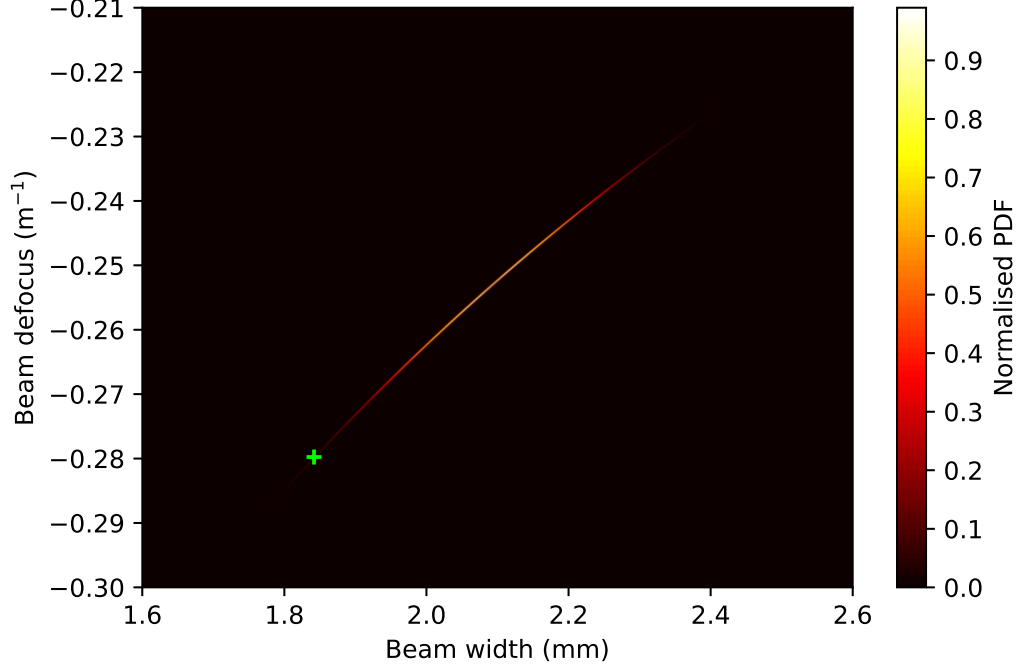


Figure 4.12: The beam parameter at the SRM was calculated for combinations of SR2 and SR3 focal lengths. The associated probability density for these combinations can be calculated (see Figure 4.11), so the probability of the beam parameter having a certain width and defocus can be calculated. The probability density for each pair of beam width and defocus is indicated with the colour bar. The measured Gouy phase of 21.3° [146] is indicated with a green cross (1.84 mm, -2.80 Dioptre). This is a simplified approach since the beam parameter cannot be such that the SRC is unstable, so certain combinations of beam width and beam defocus are not possible. Beam parameters with a width $> \sim 2.25$ mm result in an unstable SRC, therefore these beams are unlikely to be present at the SRM at LLO.

4.6 Power Transmitted Through the SRM for a Mode Mismatch Between the Arm Cavity and the SRC

The HOM content of the beam as a function of the error in focal length in SR3 was investigated. As the arm cavities become more mismatched to the SRC, less HG00 mode is transmitted. The mode mismatch can be seen in Figure 4.7. As the focal length of SR3 becomes further from its nominal value, the radius of curvature of the beam at the SRM stops matching the radius of curvature of the SRM. Since the SRC is a low finesse cavity, a significant mismatch can be tolerated before there is a large drop in the HG00 power transmitted through the SRM. This is shown in Figure 4.13. In Figure 4.13 and Figure 4.14, it is shown that the nominal value of the focal length SR3, 17.985 m, is longer than optimal. When the SRC is on resonance, the peak in transmission is -3 mm from the nominal focal length of SR3. This is shown by the blue and green lines in Figure 4.14.

The power of the HG00 light transmitted through the SRM as a function focal length error is shown in Figure 4.14. In signal recycling mode, shown as blue and green in Figure 4.14, the transmitted power has a peak shape. This is because an error in the focal length of SR3 moves the cavity off resonance. In RSE mode, shown as red and orange in Figure 4.14, a change in focal length of SR3 does not have much effect on the amount of HG00 transmitted by the SRM since an anti-resonance is a much broader feature than a resonance.

In a single-bounce measurement, the beam incident onto the ITM from outside

the cavity is reflected straight back. This is equivalent to the reflectivity of the ITM $R = 1$. When the light is resonant in the arm cavity, the ITM acts as a highly reflective mirror which applies a frequency-dependent phase shift to the light. Thus, there is no difference between a single-bounce measurement and a measurement with arm cavities at frequencies within the linewidth of the cavity. This is shown in Figure 4.13. Higher order mode beams generated by errors in the SRC are not suppressed by the arm cavity as they are reflected by the ITM.

The ratio of the light in the HG00 mode versus the total light depends on the detuning of the SRM, rather than the presence of cavities in the arm. This is shown in Figure 4.13. When the light enters the arm cavities, it is reflected with an extra π rad phase; however, the HOMs do not enter the cavity, so they do not acquire this phase. Since the HOMs only interact with the SRC, the ratio of HOM to HG00 depends on the detuning of the SRM rather than whether the interferometer is in signal recycling or RSE mode. The amount of higher order mode generated in the SRC is proportional to the HG00 power in the SRC, so the percentage of HOM beam is dependent only on the tuning of the SRM and the focal length error of SR3.

As the SRC gets closer to being unstable, more TEM modes are required to simulate the behaviour of the cavity. The effect of using more TEM modes in the simulation is shown in Figure 4.10. For positive errors above 6 mm in the focal length of SR3, the cavity becomes unstable. It was found empirically that six TEM modes is sufficient to simulate the interferometer while it is stable. The hatched regions of Figure 4.13 and Figure 4.14 show where the simulation is not accurate.

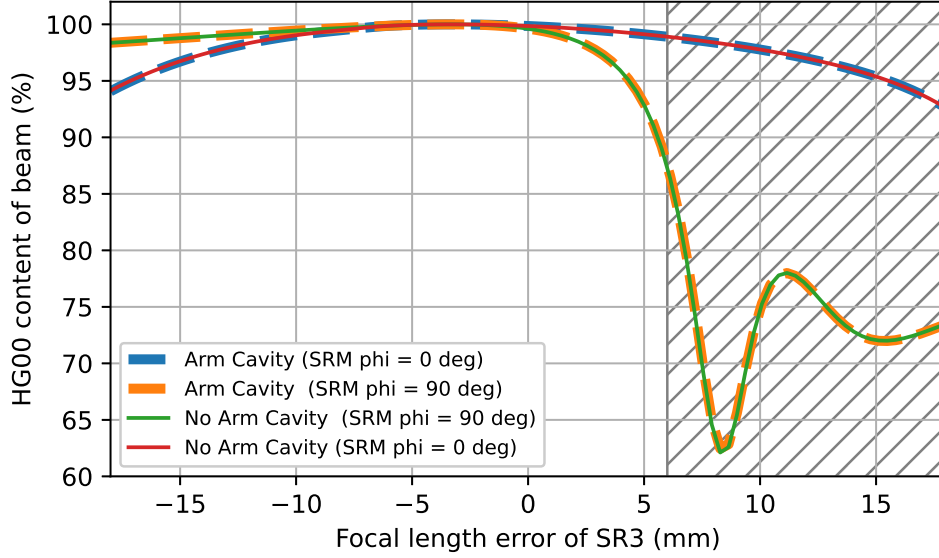


Figure 4.13: The percentage of light which is in the HG00 mode as the focal length of SR3 changes compared to the total light power does not depend on whether there are arm cavities. It is determined by the detuning of the SRM. The blue dashed line shows the behaviour of the beam with arm cavities present and with the SRC in signal recycling mode (SRM $\phi = 0^\circ$). The red line shows how the beam changes with focal length error for a single-bounce measurement and with the SRC in RSE mode (SRM $\phi = 90^\circ$). The orange dashed line corresponds to the interferometer having arm cavities and being in RSE mode (SRM $\phi = 90^\circ$). The green line shows the behaviour of the beam when the interferometer is configured for a single-bounce measurement and is in signal recycling mode. The hatched region indicates where the SRC becomes unstable (see Figure 4.10), so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

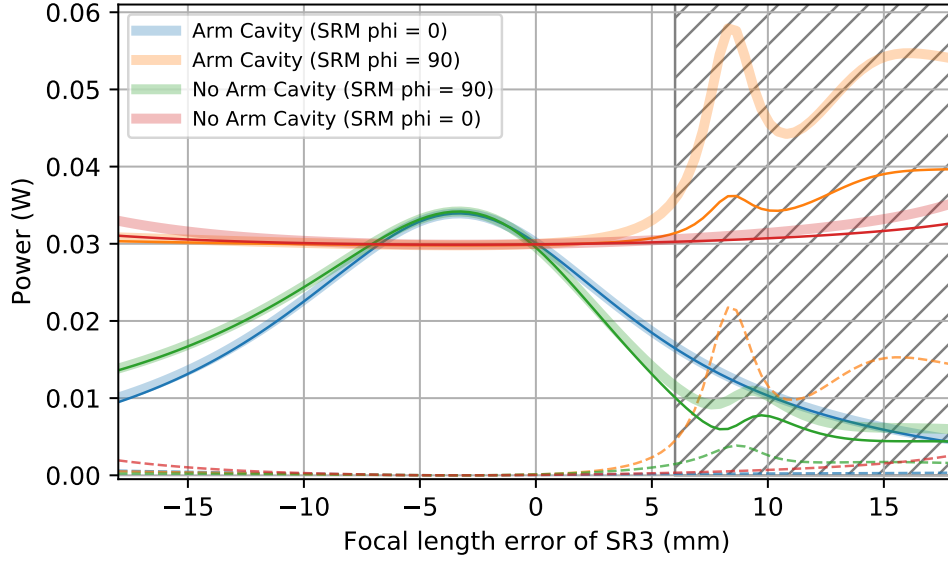


Figure 4.14: The total power transmitted through the SRM for each simulation is shown by the thick pale lines. The thin solid lines correspond to the power in the HG00 mode, and the thin dashed lines correspond to the power in the higher order modes. Blue lines show the behaviour when the interferometer has arm cavities and is in signal recycling mode, and the orange lines shows when it is in RSE mode. Green lines show how the power changes when the interferometer is configured for a single-bounce measurement and is in signal recycling mode, and the red line shows when its in RSE mode. The hatched region indicates where the SRC is becomes unstable, and so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

4.6.1 Change of finesse of SRC

As the transmission of the SRM decreases, the finesse of the cavity increases. A FINESSE simulation was run to see the effect of increasing the power transmission of the SRM. The HG00 mode power decreases with lower SRM transmission slower than the power transmitted in the HG20+02 mode. This is shown in Figure 4.15.

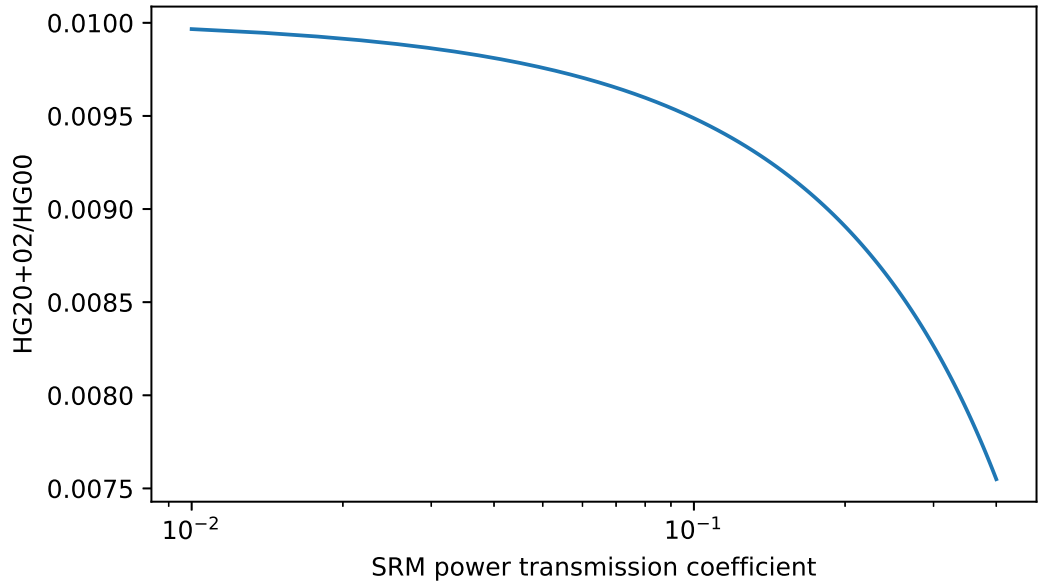


Figure 4.15: Ratio of HG20+02 to HG00 as the transmission of the SRM is increased for a mode mismatch between the arm cavity and the SRC. The interferometer is in RSE mode. A mode mismatch was introduced by setting the focal length of SR3 to be 12 mm shorter than the default amount.

4.7 Conclusion

The mode mismatch between the arm cavities and the SRC results in a loss mechanism whereby light is scattered from the HG00 mode into higher order modes. Loss in interferometer usually reduces the SNR by $\sqrt{\text{Power}}$, however since there are multiple points between the squeezer and the OMC where mode mismatches can happen, the quantum noise is enhanced by more than the usual amount for a loss. There exists a collection of measurements which shows the arms are matched between 92% and 98% to the OMC.

FINESSE simulations were used to model mode mismatches between the arm and the SRC caused by RoC errors in SR3 and SR2. The simulation used as-built parameters for the SRC at the LLO site and design specification arm cavities. This allows us to study the effect of non-ideal mode matching in a close to real life situation. The interferometer was modelled in signal recycling mode and RSE mode for locked-arm-cavities and single-bounce configurations.

It was found that the dependence of the ratio of HG00 light to the total light depends mainly on the focal length of SR3 and is not affected by the presence of arm cavities. This is because the arm cavities are well matched to the input laser. When the arm cavities are on resonance, the ITMs act as highly reflective mirrors with frequency dependent behaviour. This means if the interferometer's thermal state remains the same, there is no difference in HOM content of the beam transmitted through the SRM between a single-bounce measurement and an arm-

cavity measurement.

The ratio of HOM to HG00 for the beam emerging from the SRM depends on the detuning of the SRM, although it does not depend on whether the interferometer is in signal recycling mode or RSE mode. As HOMs do not enter the arm cavity, they cannot distinguish between the signal recycling and RSE modes of operation. When the SRC is on resonance, it is much more sensitive to an error in SR3's focal length than when the SRC is anti-resonant. The light is less sensitive to an error in the focal length of SR2.

The uncertainty in the beam parameter of the arm mode at the SRM and the eigenmode of the SRC is driven by the uncertainty in SR3. This is shown in Figure 4.6 and Figure 4.7. The single pass Gouy phase of the SRC for different combinations of SR3 and SR2 was simulated; this is shown in Figure 4.8. The Gouy phase acquired in a single pass of the SRC was measured to be $21.3 \pm 0.3^\circ$, so it can be inferred that the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m. This is shown in Figure 4.11. These values of focal length would result in the arm mode having a width of 1.8 mm and a defocus of $-0.28, \text{m}^{-1}$ immediately after it is transmitted through the SRM, and there would be a few percent HOM in OMC scans (see Figure 4.12).

Future work could involve using a heater to change the focal length of SR3 and measuring the effect this has on the Gouy phase accumulated in the SRC. This measurement could be used to validate the model of the geometric properties of the SRC. As the lengths between all the optics is well known, and the focal length

of the ITMs is well known, the squeezer beam could be used to gain information about the radii of curvature of SR3 and SR2. This would be done by setting up a single-bounce measurement. Having greater confidence in the beam parameter of the arm mode at the SRM would be of great benefit for the design of future upgrades, such as balanced homodyne detection (see Chapter 5 and Chapter 5), to the LIGO detectors.

Chapter 5

Balanced Homodyne Detection for LIGO A+

A major change between aLIGO and LIGO A+ will be the replacement of DC readout (see Section 5.1.1) with balanced homodyne detection (BHD). The motivations for implementing BHD into the LIGO detectors are described in Section 5.1. The primary reason can be summarised as: if the detectors are to reach their design sensitivity, they require BHD. In contrast to DC readout, the local oscillator (LO) and signal do not follow the same optical path in a BHD scheme; the consequence of this is the local oscillator path length must be stable enough to not contaminate the signal with noise. Figure 5.2 is a sketch of the local oscillators beam path. This is discussed in Section 5.2. The proposed BHD layout is discussed in Chapter 6.

5.1 The Principles of and Motivations for Balanced Homodyne Detection

BHD is a method of measuring the information encoded in any quadrature of a signal beam (see e.g. [38, Section 2]); thus, BHD forms a key part of many quantum optics experiments (e.g. [60, 61, 147]). In a BHD readout scheme, the signal beam is set up to interfere with a LO beam with the same optical frequency on a beam splitter, and the interference is measured with photodiodes at the beam splitter's output ports. A sketch of a BHD readout is shown in Figure 5.1. The LO beam has a larger power than the signal beam, and the power of the LO is chosen to bring the quantum noise above the noise floor of the photodiode and its preamplifier. An arbitrary quadrature of the signal beam can be measured by correctly setting the relative phase between the LO and signal beams. This phase is known as the *homodyne angle*. If the two photocurrents are balanced, subtracting them results in a measurement which is proportional to the selected quadrature of the signal beam. Additionally, the subtraction results in the removal of the classical intensity noise of the LO from the measurement. BHD is mathematically described in Section 5.1.2).

Without BHD, the LIGO detectors will not be able to reach the design sensitivity for A+. A squeezed light field can be used to surpass the standard quantum limit (see Chapter 2). To reach the level of frequency dependent squeezing needed to meet the A+ design sensitivity, the homodyne angle must be 90° . There is a low amount of tolerance on the homodyne angle before the quantum noise is sig-

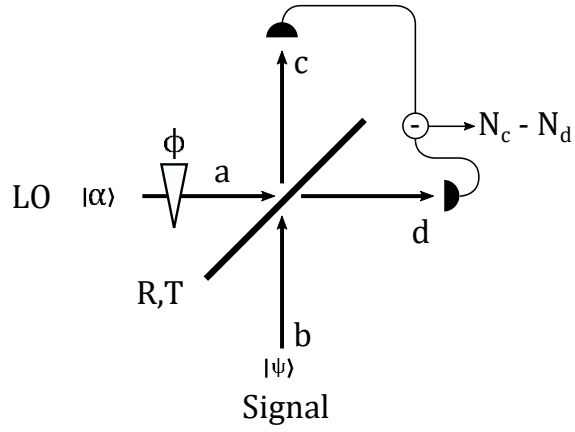


Figure 5.1: A coherent state $|\alpha\rangle$, enters port a of a beam splitter. A signal $|\Psi\rangle$ enters port b . These two beams interfere, resulting in photons being detected at the c and d ports. The two photocurrents are subtracted, and the signal that is produced is proportional to an arbitrary quadrature of the state $|\Psi\rangle$; the difference in the signals at the c and d ports is given by Equation (5.8). The quadrature of the state $|\Psi\rangle$ that is observed depends on the homodyne angle, ϕ .

nificantly degraded [148], and the current readout method (see Section 5.1.1) does not offer sufficient control over the homodyne angle. Currently, the squeezing angle is set to be at the same angle as the readout quadrature; the preference would be to set the readout quadrature to the squeezing angle [58]. The effect that controlling the squeezing angle has on the quantum noise floor has been demonstrated at LLO [61]. However, [61] used a diagnostic BHD setup which was too noisy to make gravitational wave detections.

5.1.1 Alternatives to BHD

The current detection scheme is known as DC readout [31]; this is a form of homodyne detection. A differential offset in the arm lengths produces the LO in DC readout; the size of the LO can be set with the differential offset in the length of the arms, and the homodyne angle can be set with the differential arm power [149]. DC readout has some key positive outcomes, such as there being no phase noise between the LO and the signal, and the LO and signal being perfectly aligned and mode matched. However, while theoretically the homodyne angle can be set, due to different amounts of loss in each arm, the homodyne angle is not a free parameter; for LLO, the homodyne angle during O3 was $\sim 11^\circ$ [150]. The inability to tune the homodyne angle increased the level of the detector’s quantum noise ¹.

The first generation of detectors used heterodyne readout (e.g. [151]). Hetero-

¹This increase was buried under the classical noise floor; however, the LIGO detectors will be quantum noise limited in the future

dyne readout involves a modulation-demodulation scheme whereby the signal is beat with a radio-frequency (RF) LO to a frequency where technical noise is no longer a concern. It is then demodulated at the same modulation frequency to recover the original signal. This method comes with the draw back that the shot noise is enhanced by the modulation-demodulation process [152]. Heterodyne detection is described in more detail in the final paragraph of Section 8.1.

5.1.2 Quantum Mechanical Description of BHD

To understand how BHD can be used to detect quantum states of light, it is useful to mathematically express light and interference in terms of quantum mechanics. From the two-photon formalism of light (see Section 2.5), the general quadrature operator, $\hat{\mathbf{X}}_\phi$ can be obtained,

$$\hat{\mathbf{X}}_\phi = \hat{\mathbf{X}} \cos \phi + \hat{\mathbf{Y}} \sin \phi, \quad (5.1)$$

where ϕ is the homodyne angle. By setting the homodyne angle, one can sense an arbitrary combination of the phase and amplitude of a field.

Consider the interference of a LO field, $\hat{a}$, and a signal field, $\hat{b}$, on a 50:50 beam splitter. The resulting fields, $\hat{c}$ and $\hat{d}$, are detected on two photodiodes. This is

shown in Figure 5.1. The photocurrents are proportional to

$$\hat{\mathbf{N}}_c = \hat{c}^\dagger \hat{c} = \left(\sqrt{R} \hat{a}^\dagger + \sqrt{T} \hat{b}^\dagger \right) \left(\sqrt{R} \hat{a} + \sqrt{T} \hat{b} \right) \quad (5.2)$$

$$= R \hat{a}^\dagger \hat{a} + T \hat{b}^\dagger \hat{b} + \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}) \quad (5.3)$$

$$\hat{\mathbf{N}}_d = \hat{d}^\dagger \hat{d} = T \hat{a}^\dagger \hat{a} + R \hat{b}^\dagger \hat{b} - \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}). \quad (5.4)$$

As the beam splitter is 50:50, $R = T = 1/2$. This results in the difference in the photocurrents being proportional to

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = \hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}. \quad (5.5)$$

The operators $\hat{a}$ and $\hat{b}$ can be ‘linearised’, $\hat{a} \rightarrow \alpha + \delta a$ to separate the quantum behaviour and classical behaviour. Furthermore, the phase between the signal field and the LO field can be expressed in terms of the homodyne angle with $\hat{b} \rightarrow \hat{b} \exp(i\phi)$. By neglecting terms of the order $\delta a \delta b$, this allows one to rewrite Equation (5.5) as

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b + 2\beta\delta \mathbf{X}_\phi^a. \quad (5.6)$$

The variance, $\sqrt{V(X)} = \Delta X$, of this measurement would then be

$$V(\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d) = 4\alpha^2 V(\delta \mathbf{X}_{-\phi}^b) + 4\beta^2 V(\delta \mathbf{X}_\phi^a). \quad (5.7)$$

If the LO is much stronger than the signal field, $\alpha \gg \beta$, then the measurement is

$$\hat{\mathbf{N}}_c - \hat{\mathbf{N}}_d = 2\alpha\beta \cos \phi + 2\alpha\delta \mathbf{X}_{-\phi}^b. \quad (5.8)$$

5.1.3 Other Motivations for Implementing BHD in LIGO A+

Different gravitational wave sources carry their information in different frequency bands, and the detector may be optimised to reflect this [50, 70, 153, 154]. For instance, BNS signals continue to higher frequencies than BBH signals, so it may be of interest to optimise the detector around these frequencies to improve our understanding of the properties of BNS systems. To optimise the sensitivity of the interferometer to different sources, the detectors noise would need shaping. To do this, BHD can be used alongside other upgrades (e.g [19, 50]).

More advanced upgrades (see Chapter 2.6 or [50]) could alter the detector’s quantum noise substantially, and they rely on the implementation of BHD. These changes to the interferometer are non-trivial, and would require significant redesign of many parts of the layout, so the implementation of BHD for A+ does not necessarily correspond to a suitable BHD system for one of these more advanced interferometer configurations. However, BHD for A+ would provide experience with implementing BHD on a large-scale interferometer, which would help when considering how to implement these more novel interferometer configurations.

BHD has some benefits over DC readout relating to the reduction of the DC light emerging from the AS port [149]. With a lower amount of DC light at the AS port, the coupling between the SRC length and the gravitational wave signal will be reduced². Additionally, there will be less scattering from the septum window

²The dominant coupling due to the power entering the SRC can be intuitively understood by considering a Fabry-Perot cavity with a slight tuning offset. A DC offset will give rise to linear

between HAM5 and HAM6³.

5.2 The Local Oscillator Path Length Stability Requirement for A+

Phase noise arises due to phase fluctuations in the LO beam being indistinguishable from phase fluctuations in the signal beam. A gravitational wave produces a phase modulated signal in the signal beam, and the LO field phase is modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be low enough for the signal generated by the interference of the LO and signal beams to be due to the phase modulation of the signal beam.

This section is structured as follows: first, the LO path is described. Then, a linear response function between a differential strain and signal at the AS port is obtained. Next, the sensitivity curve for A+ (see Figure 2.3) is expressed in units of $\frac{W}{\sqrt{\text{Hz}}}$ by deconvolving it with the linear response function. This light noise curve

coupling between the cavity length and the power transmitted by the cavity. This coupling scales with the size of the DC offset. As this noise is in the amplitude quadrature of the signal beam, unlike the gravitational wave signal which is in the phase quadrature, it will not be amplified by the LO if the homodyne angle is set to readout from the phase quadrature. The second route by which SRC length noise may couple to the signal is via optical springs between the SRC and arm cavities [155]. Due to this optical spring effect, a change in length of the SRC will cause the arm cavity lengths to change as well.

³Strictly speaking, the scattering from the septum window is not a property of DC readout as the window could be removed. However, this window is crucial for the maintenance of the interferometer.

was then used to find the maximum allowable intensity fluctuations due to phase noise between the LO and the signal beams. Finally, this phase noise requirement is converted into an optical path length requirement.

5.2.1 Local Oscillator Path

The layout for the optics which combine the LO and signal beam is shown in Figure 5.2. The LO will be picked off from the power recycling cavity and guided by one steering mirror, BHDM1, to the BHD combining beam splitter, BHDBS2. The signal beam will exit the OFI and be steered by one mirror, OM0, onto BHDBS2. OM0 and BHDM1 will be used to align the beams so they overlap on BHDBS2. The angle of BHDBS2 will be used to make the LO and signal beams overlap in the far field. Two lenses are used for mode matching the LO beam to the signal beam.

The LO could have been derived from a separate laser, but to acquire laser light with low frequency and intensity noise the LO will be picked off from a point within the interferometer. The two ways of obtaining a LO beam considered were a beam picked off from the main beam splitter's AR coating (BSAR) or the beam transmitted through PR2 (POP) [156]. The determining factor in choice of pick-off point was the effect of backscatter. The POP port was chosen as it is far superior to the BSAR choice at all frequencies and scatter phases. This is because backscattered light from the BSAR enters only one arm, thus making it a differential effect. Any backscatter that enters the POP port is common-mode, and therefore DARM is far

less sensitive to this source of backscatter.

5.2.2 Calculation of the Local Oscillator Path Length Stability Requirement

The first step is to express the A+ sensitivity curve in units of $\text{rad}/\sqrt{\text{Hz}}$. The signal generated by the differential motion of the ETMs is low-passed by the coupled SRC-arm cavity, so the linear response function can be written as

$$H(f) = \frac{H_0}{1 - if/f_c}, \quad (5.9)$$

where H_0 is a gain factor, f is the frequency and f_c is the corner frequency of the SRC-arm cavity. For the LIGO detectors, $f_c = 380 \text{ Hz}$ [155]. This linear response function does not include optical spring effects, but these are small and not important for this calculation. H_0 can be calculated by dividing the level of phase noise of the signal beam, $\Delta\phi_{\text{sig}}$ by the product of the A+ sensitivity curve (see Figure 2.3) and $1/(1 - if/f_c)$ at a frequency where the detector will be shot noise limited. Care needs to be taken to avoid double counting the squeezing; the signal beam will be squeezed in the phase quadrature, but because the A+ sensitivity curve includes squeezing, the squeezing factor can be omitted from the phase noise associated with the signal beam. From Equation (2.9), the relationship between the phase of the

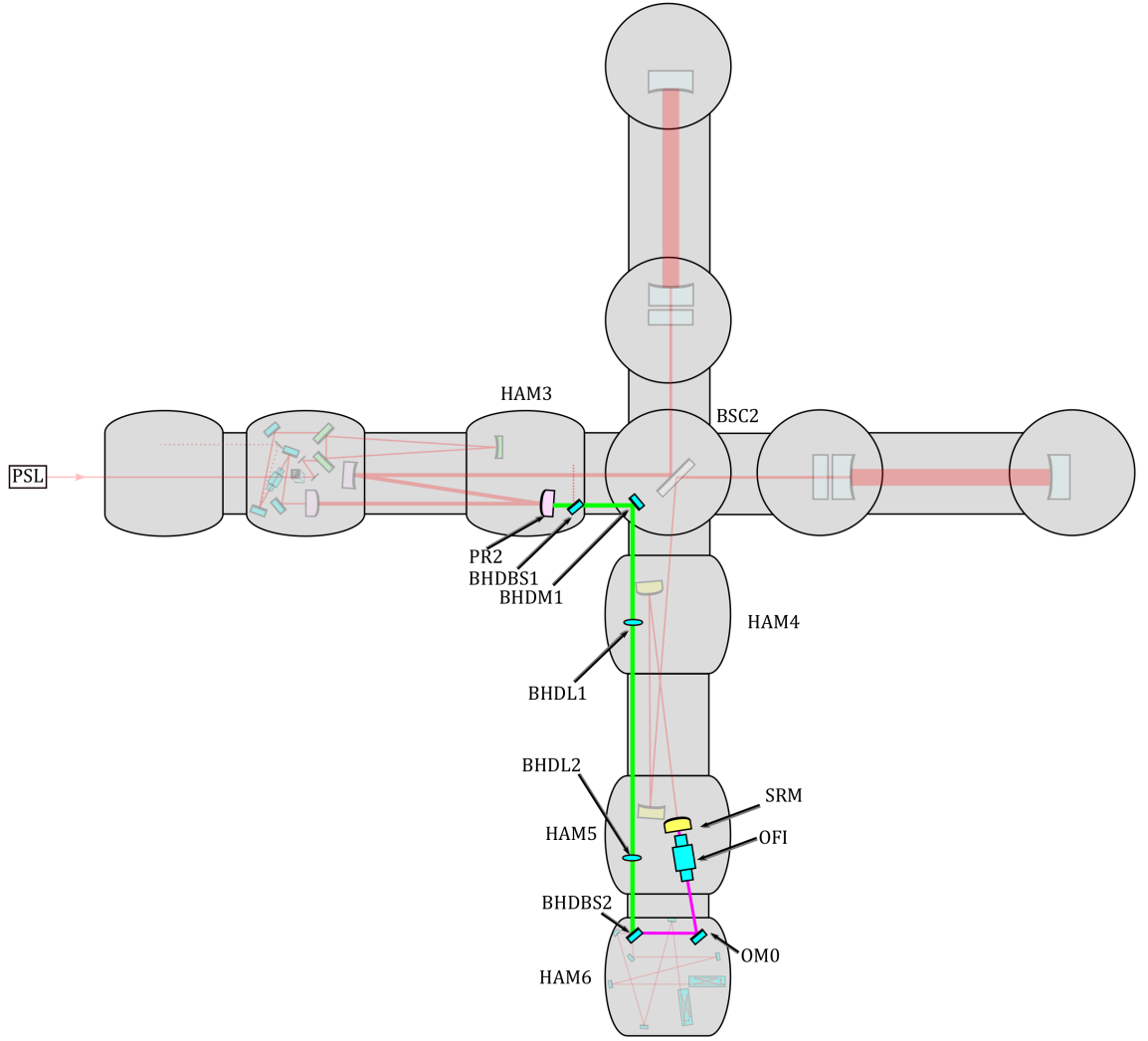


Figure 5.2: Sketch of the LO (green) and signal (pink) beam paths in relation to other parts of the interferometer (see Figure 2.4 and Section 2.9). The LO beam emerges from PR2. Some of this light is used to control the interferometer, so BHDBS1 is needed to split the light. BHDM1 directs the LO onto BHDBS2. The signal beam emerges from the SRM. It passes through the OFI and is reflected by OM0. The signal combines with the LO on BHDBS2. The layout of the optics in HAM6 is shown in Figure 6.1.

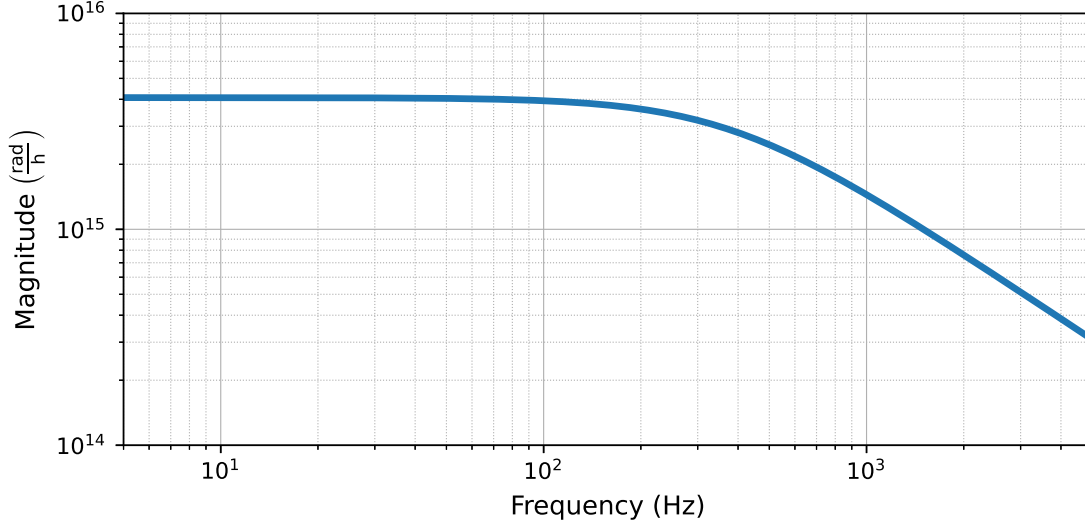


Figure 5.3: The linear response function between strain and light power, expressed in units of radians/gravitational wave strain, at the AS port. The corner frequency at ≈ 380 Hz comes from the pole of the coupled SRC-arm cavity.

signal beam ϕ and its power P_{sig} can be written as

$$\Delta\phi_{\text{sig}} = \sqrt{\frac{h\nu}{4P_{\text{sig}}}} \quad (5.10)$$

where h is Planck's constant and ν is the frequency of the light. For A+, $P_{\text{sig}} = 5\text{mW}$. The linear response function is shown in Figure 5.3. To express the A+ sensitivity curve in units of $\text{rad}/\sqrt{\text{Hz}}$, the sensitivity curve should be divided by the linear response function shown in Figure 5.3.

The next step is to use the equations for interference at a beam splitter to convert a change in power into a change in phase. When the LO and signal beams interfere

at BHSBS2, the exiting beams have power P_c and P_d . These are given by

$$P_c = \left| \frac{1}{\sqrt{2}} E_{\text{sig}} e^{i\omega t} + \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_{\text{sig}}|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} + |E_{\text{sig}} E_{\text{LO}}^\dagger| \cos(\phi) \quad (5.11)$$

and

$$P_d = \left| \frac{1}{\sqrt{2}} E_{\text{sig}} e^{i\omega t} - \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_{\text{sig}}|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} - |E_{\text{sig}} E_{\text{LO}}^\dagger| \cos(\phi), \quad (5.12)$$

where E_{sig} and E_{LO} are the complex field amplitudes of the signal and LO beams; ω is the laser frequency; t is time and ϕ is the homodyne angle. The power of local oscillator beam will be $P_{\text{LO}} = 100 \text{ mW}$. The difference in P_c and P_d signals gives the BHD signal,

$$P = 2|E_{\text{sig}} E_{\text{LO}}^\dagger| \cos(\phi). \quad (5.13)$$

Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_{\text{sig}} E_{\text{LO}}^\dagger| \sin(\phi) d\phi. \quad (5.14)$$

Since the homodyne angle is 90° , this becomes

$$dP = -2|E_{\text{sig}} E_{\text{LO}}^\dagger| d\phi. \quad (5.15)$$

Using Equation (5.15), the phase noise requirement for the local oscillator can be expressed in units of $\text{rad}_{\text{rms}}/\sqrt{\text{Hz}}$. This is shown in Figure 5.4. This requirement was also derived in [156, 157] and [158]; however, these calculations all differ from each other by factors of ~ 1 . The result in Figure 5.4 is more pessimistic than the

results in [156, 157] and [158] as a signal beam of 5 mW was assumed.

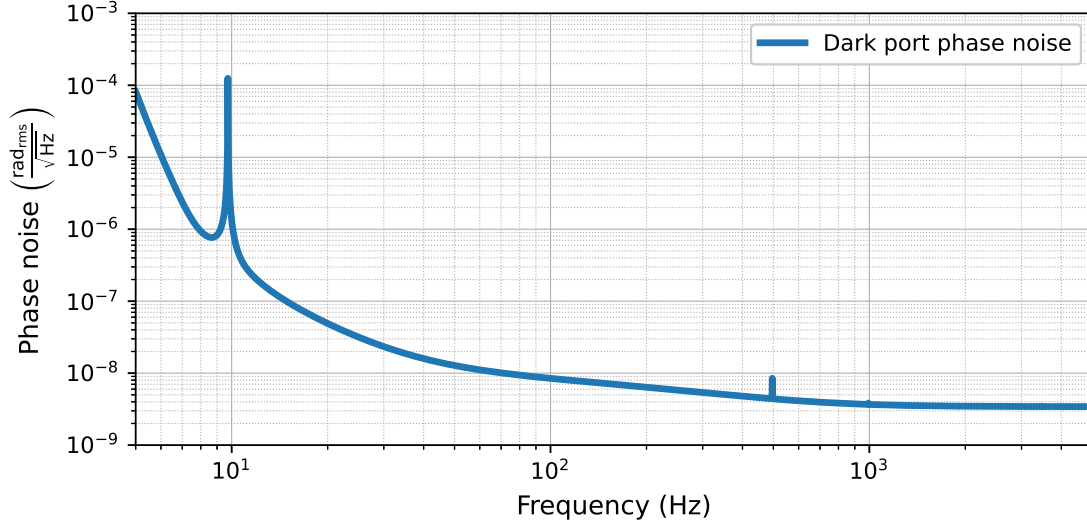


Figure 5.4: The sensitivity curve for the A+ detectors expressed in units of $\text{rad}_{\text{rms}}/\sqrt{\text{Hz}}$. At frequencies above 1 kHz the noise is flat ($3.4 \times 10^{-5} \text{ rad}_{\text{rms}}/\sqrt{\text{Hz}}$) as it is dominated by shot noise. At lower frequencies, the noise is increased relative to shot noise due to radiation pressure, coating thermal noise and other noise sources. At 5 Hz, the noise is $8 \times 10^{-5} \text{ rad}_{\text{rms}}/\sqrt{\text{Hz}}$.

To express phase in terms of optical path length, the phase needs to be multiplied by $\lambda_{\text{EM}}/(2\pi)$. The relationship between the optical path length, n_p , and the motion requirement for the optics, n_o , assumes that there will be two mirrors at 45° which are used to steer the AS and LO beams. As the optics reflect at 45° , the path length change due to the optic's motion is multiplied by $\sqrt{2}$. The beams are combined on a beam splitter, and the motion of this beam splitter needs to be counted twice. Figure 5.6 illustrates this. The relationship between optical path length and the motion of the optics is $10 \times \sqrt{(\sqrt{2}n_o)^2 + (\sqrt{2}n_o)^2 + 2 \times (\sqrt{2}n_o)^2} = n_p$. Since there

are many uncorrelated sources of noise in the LIGO detectors, a factor of 10 is used as a safety margin. Multiplying the phase noise curve shown in Figure 5.4 by $\lambda_{\text{EM}}/(2\pi \times 10\sqrt{8})$ gives the requirement for the longitudinal motion of the optics; this is shown in Figure 5.5. To determine the amount of isolation required, the typical motion of the optical table needs to be compared to the motion requirement for the optics; see references [159, 160]. This led to triple suspensions, known as HAM Relay-optics Triple Suspensions (HRTS), being used to passively isolate the optics in the local oscillator and signal beam paths.

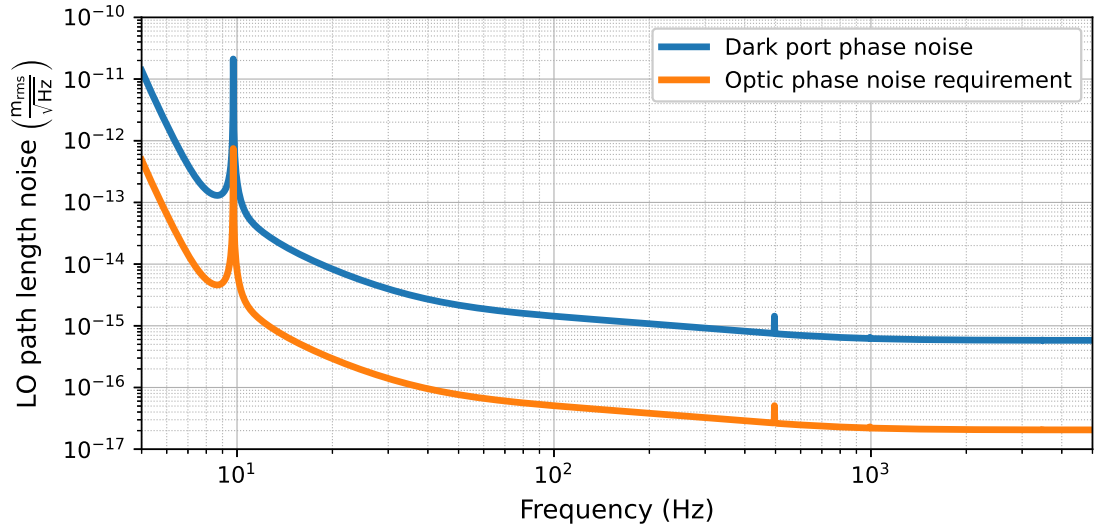


Figure 5.5: The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve in units of path length noise between the AS and LO beams, and the orange line shows the longitudinal motion requirement for the optics which combine the LO and signal beam. At 5 Hz, the longitudinal motion requirement for the optics is $4.9 \times 10^{-13} \text{ m}_{\text{rms}}/\sqrt{\text{Hz}}$. The red line shows the level of shot noise; this is $2.1 \times 10^{-17} \text{ m}_{\text{rms}}/\sqrt{\text{Hz}}$.

5.3 Conclusion

One of the major changes to LIGO for A+ will be the implementation of BHD because BHD is required for the LIGO detectors to meet the sensitivity requirement for A+ as frequency dependent squeezing requires the homodyne angle to be a free parameter. Details of the LO path were discussed and analysed. The LO will be derived from a beam picked off from the POP port, and a sketch of how the signal and LO will be combined is shown in Figure 5.2. We derived the relative path length stability requirement for the LO and signal beams by; the requirement is shown in 5.5. The local oscillator's phase noise requirement led to HRTS being chosen to steer the LO and signal beams onto the combining beam splitter, BHDBS2.

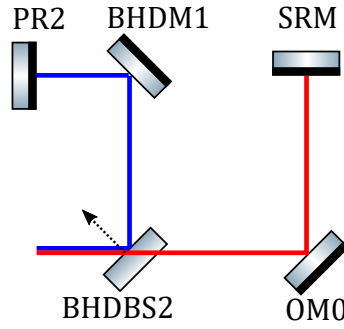


Figure 5.6: A sketch of the LO and signal beam paths. The LO beam path is shown in blue, and the signal beam is shown in red. A dashed arrow indicates the direction that the recombining beam splitter, BHDBS2, may move in. If BHDBS2 moves in this way, the phase of the signal beam has increased while the phase of the LO beam has decreased at the point where the signal and LO beams combine. This means that the recombination beam splitter motion needs to be counted twice in calculations of the phase noise requirement.

Chapter 6

Design of the LIGO A+ HAM6

Layout

The LIGO A+ HAM6 vacuum chamber will contain the readout photodiodes, the output mode cleaners (OMC)¹ and their associated steering optics. See Figure 2.4 for HAM6's location with respect to the rest of the detector. Fundamentally, the optical layout in HAM6 must combine the signal beam and the LO beam and sense the combined beams to form a balanced homodyne detector (BHD). The implementation of BHD for LIGO A+ requires a complete redesign of the HAM6 layout. As losses in the interferometer's readout will degrade its sensitivity, it is vital that the active wavefront control (AWC) is designed in such a way as to minimise this.

¹The OMC assembly includes the readout photodiodes.

In Section 6.1, the optical properties of the proposed HAM6 layout are examined. Section 6.2 is primarily a discussion of the design challenges associated AWC in HAM6. Section 6.3 is an analysis of how mode mismatches between the interferometer and the OMC can be corrected by the AWC. Section 6.4 is the conclusion for this chapter. This work was undertaken by a team made up of myself, Ken Strain, Russell Jones, Stephen Webster, Mark Barton and Eric Oelker.

6.1 The Chosen HAM6 Optical Layout

The design that is being taken forward at the time of writing (January 2021) is shown in Figure 6.1. Optically, there are many layouts which could have satisfied the requirements and constraints laid out in Section 6.2 . The choice of layout for HAM6 was ultimately determined by mechanical factors such as table space, number of suspensions and the preferred design for the Balanced Homodyne Single Suspension (BHSS). The optical properties of this layout are summarised in Table 6.1. The mirrors OMA1, OMB1, OMA2, OMB2 etc. will be referred to as OMx1 and OMx2 as it is not important to make a distinction between the A and B paths when considering mode-matching.

The beam size and Gouy phase as a function of position along the optical axis is shown in Figure 6.3. Significant points along the optical axis are also shown in this figure. The size of the beam at the final mode-matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirror's surface.

The beam is always less than 1.2 mm within HAM6, an important fact to remember when considering how close structures can be to the beam path ².

As astigmatism degrades the mode-matching, it can contribute to the loss in the readout of the detector. Non-zero angles of incidence (AOI) of non-astigmatic beams onto curved optics cause the reflected beam to become astigmatic (for a comprehensive description of astigmatism see [161]). For this layout, the astigmatic loss was calculated to be $< 0.05\%$; see Appendix E for detail in this type of calculation. The astigmatic loss is plotted as a function of angle in Figure 6.2; as the beam will have low curvature, the loss becomes significant when the AOI is above 10° . The compromise that is made when choosing a larger AOI is between the potential for larger Gouy phase separation between OMx1 and OMx2 at the expense of loss due to astigmatism. In reality the IFO and OMC will also have a small amount astigmatism so this calculation will not be entirely correct; however, as some of the losses in the readout train of the interferometer are expected to be at the many percent level, losses of $< 0.05\%$ are considered unimportant.

The focal lengths of the OFI lens, OMx1 and OMx2 were optimised so that there was as close to 45° Gouy phase separation between OMx1 and OMx2 as possible as this gives orthogonal control over the mode-matching between the arm mode and the OMC. This is shown in Figure 6.4. With the layout shown in Figure 6.1, it is possible to achieve 42° Gouy phase separation between OMx1 and OMx2. Other layouts could provide exactly 45° Gouy phase separation; however, this layout offers

²The minimise scattering, the clear aperture size for the beam is defined to be 10 times larger than the Gaussian beam width.

many practical benefits, mainly in relation to the BHSS, over the other considered designs that outweigh an additional 3° of Gouy phase separation between OMx1 and OMx2. This layout also provides good Gouy phase separation for alignment control as there is $> 80^\circ$ between OMxS and OMx2.

In conclusion, the proposed HAM6 layout, shown in Figure 6.1, fulfils the criteria set in Section 6.2. Near optimal mode-matching and alignment control can be achieved with this layout, and a summary of the properties of the layout can be found in Table 6.1. The next stage of development for this layout involves the design and prototyping of parts and the consideration of fine details such as cable routing.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.066	-15.068	-
OMx1	-0.350	-2.855	5.709 (concave)
OMx2	-0.847	-1.180	2.360 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	500
OM0 to BHDBS2	946
BHDBS2 to OMxS	Path A: 225, Path B: 315
OMxS to OMx1	Path A: 1240, Path B: 1150
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \mu\text{m}$
OMx1 to OMx2 Gouy phase	42°
OMxS to OMx2 Gouy phase	Path A: 85° , Path B: 81°
OMx1 and OMx2 AOI	A: 9.7° , B: 4.9°

Table 6.1: Mechanical and optical properties of the LIGO A+ HAM6 layout.

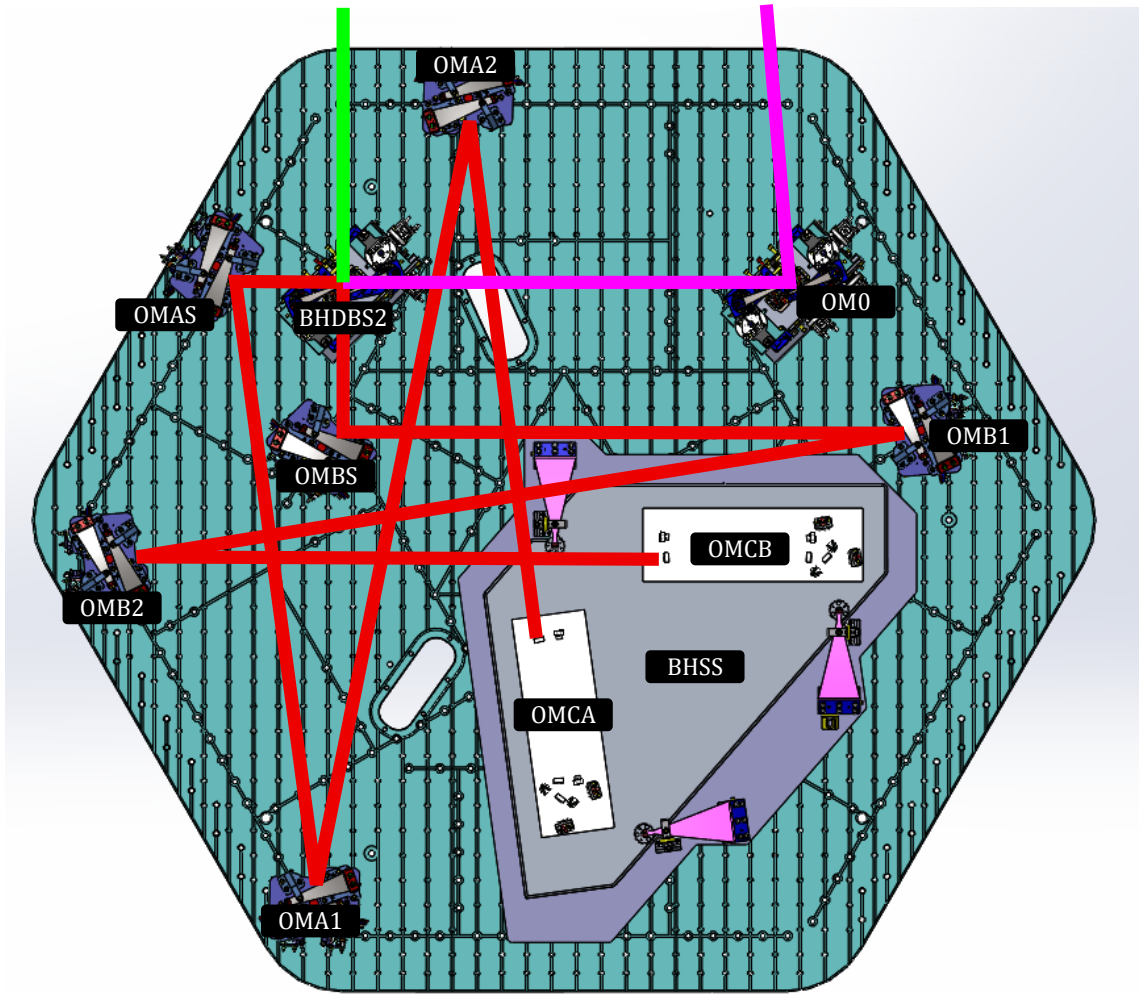


Figure 6.1: The LO beam enters HAM6 directly above BHDBS2 (north side), and it is highlighted in green. The signal beam enters HAM6 directly above OM0 (north side), and it is highlighted in pink. The signal beam is steered by OM0 onto BHDBS2 where it is combined with the LO. The two combined beams are shown in red. These are incident upon a mirror used for steering (OMxS), and then upon two mirrors curved mirrors (OMx1 and OMx2). The beams enter the OMCs, which are seismically isolated by the BHSS (the large triangular component). The spacings for these components are given in Table 6.1.

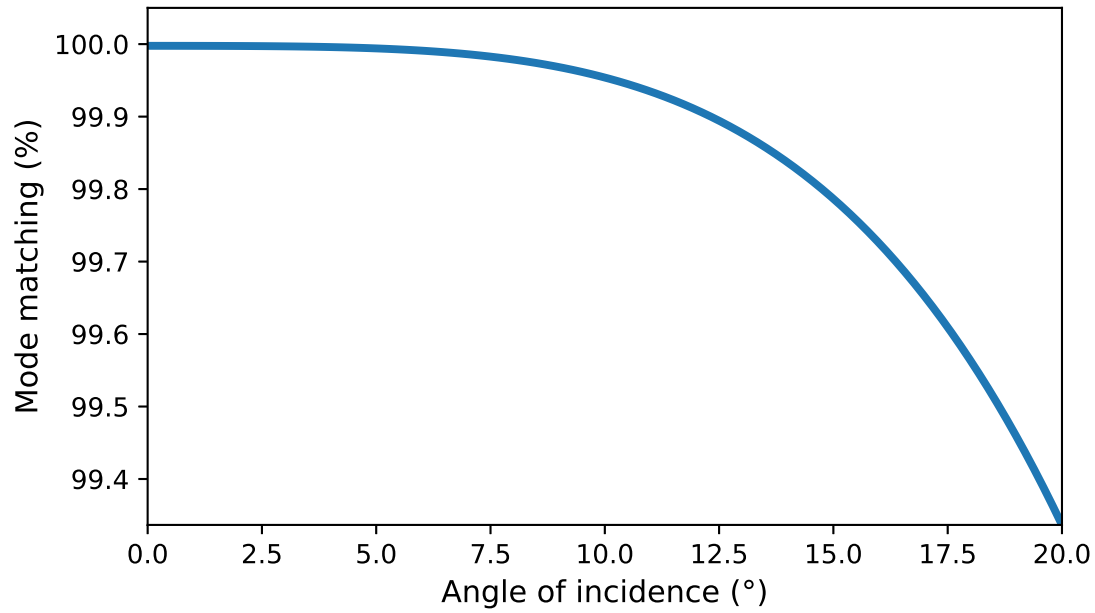


Figure 6.2: Astigmatism is a source of loss. When designing the layout shown in Figure 6.1, the AOI for the curved mirrors was altered to explore how sensitive it was to astigmatism. Using Equation (6.2), the loss was computed. This effect is negligible up to 10° after which it starts to become significant compared to the losses summarised in Table 6.2.

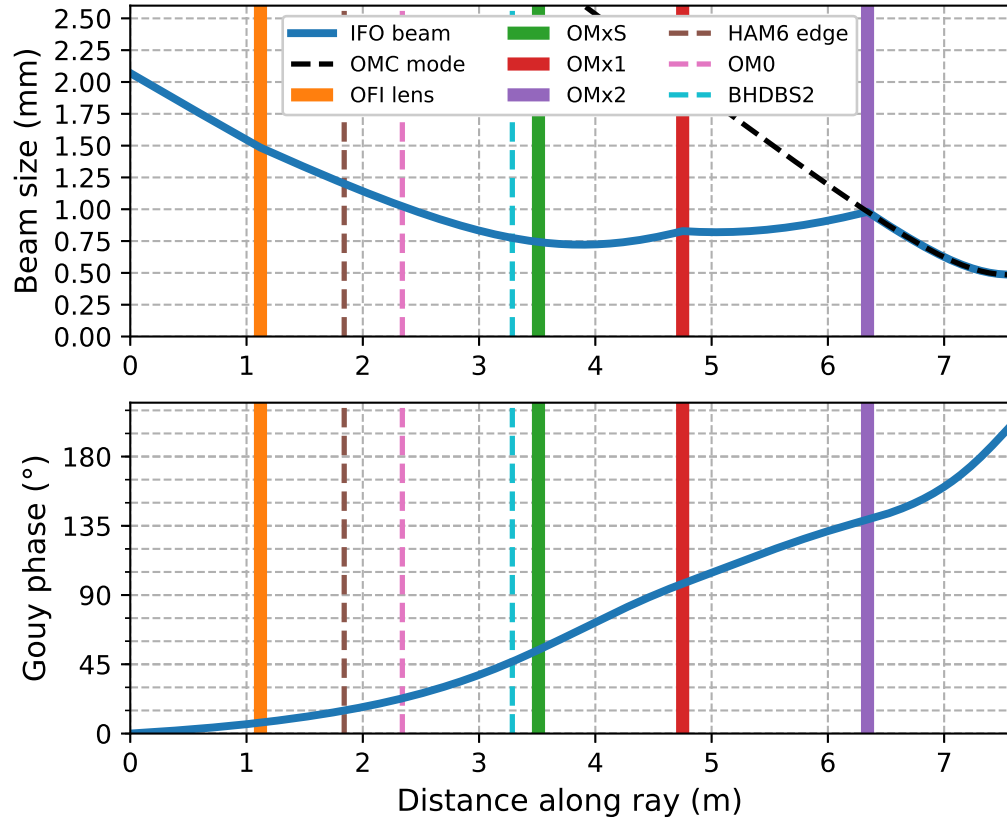


Figure 6.3: The top panel shows the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful, reference points are marked by dashed vertical lines. The bottom panel shows the Gouy phase shift accumulated by the beam as it propagates along the optical axis.

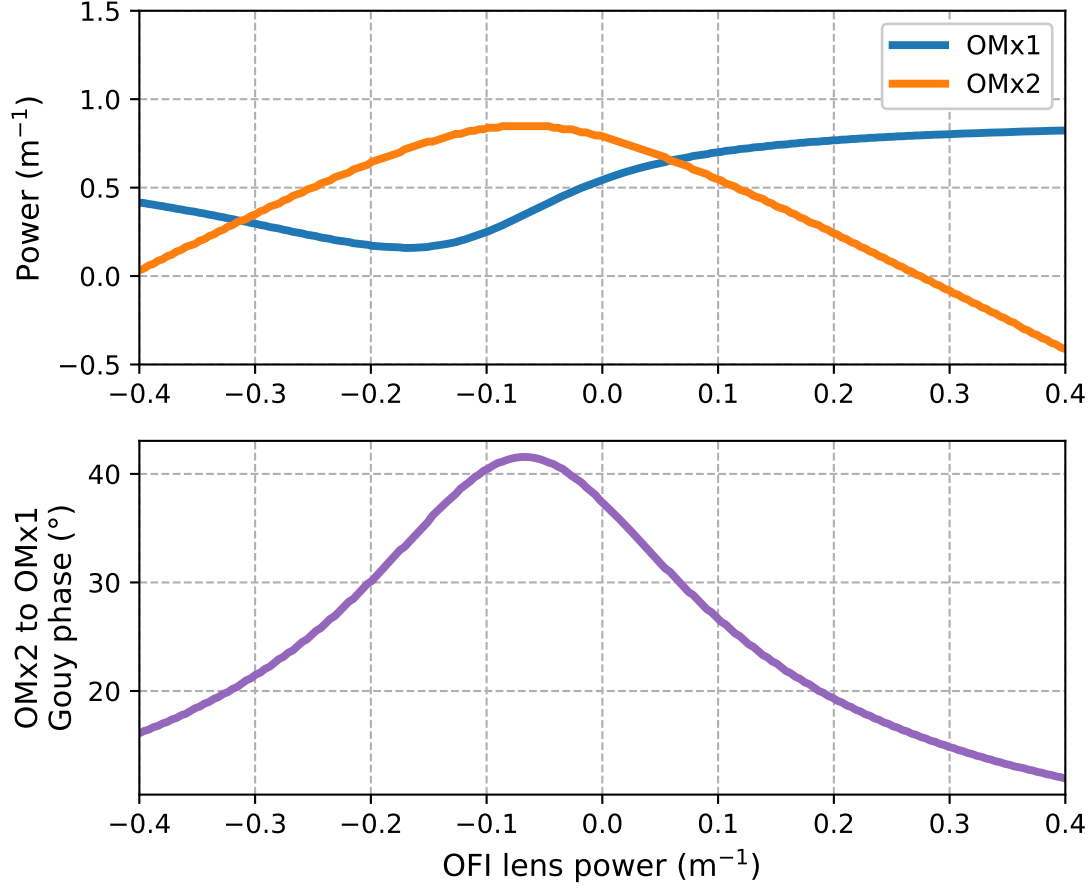


Figure 6.4: Top panel: The choice of OFI lens power affects the combination of OMx1 and OMx2 power that is required for 100% mode-matching. Bottom panel: The best choice for of OFI lens that corresponds to obtaining orthogonal control of the beam parameter in HAM6 is where the Gouy phase separation between the pair of OMx1 and OMx2 is closest to 45° . The most Gouy phase separation between OMx1 and OMx2 achievable by this layout is 42° .

6.2 The Requirements for the Optical Layout Within HAM6

The BHD optical layout in HAM6 is required to provide a readout of the gravitational wave signal. To meet the design sensitivity of A+, HAM6 must house a layout that is robust to mode mismatch as there is significant uncertainty in the beam parameter of the arm cavity mode at the SRM. Having control over the two degrees of freedom associated with mode-matching – the size and location of the beam’s waist – between the OMC and the beam exiting the SRM was challenging as this requires a lot of space relative to the size of the optical table. In addition to this, the HAM6 layout needs to leave space for the other sensors and beams used for controlling the interferometer. All proposed designs used most of the table space. This section focusses on the design considerations relating to mode-matching; other aspects of the functionality required from HAM6 is summarised in [162].

To reach the desired level of squeezing for A+, the goal for mode-matching between the arm mode and the OMC mode is $> 98\%$; the loss contribution of the components in the readout train of the interferometer are summarised in Table 6.2 [163]. The loss in sensitivity from mode mismatch can be four times greater than a simpler source of loss (e.g. photodiode quantum efficiency) due to coherent destructive modal interference [131] (see Section 4.1.2). Active mirrors, called Suspended Active Matching Stages (SAMS), will be used to correct for mode-mismatch; they will provide at least ± 50 mD of optical power tuning range [164].

The AWC in HAM6 has two functions: it will be used to correct for thermal effects in other parts of the interferometer [162] such that the mode-matching between the arm mode and the OMC is maximised, and it will be used to correct for RoC errors in the interferometer after the main beamsplitter.

Thermal changes in the arm cavity mode may be corrected for by the thermal compensation system (TCS) [165]. However, in practice operators and commissioners use each tunable aspect of the interferometer to improve its sensitivity to gravitational waves. For instance, at LLO it has been found that TCS can be used to decrease laser intensity noise coupling to DARM [166]. As each interferometer subsystem may not be used for its intended purpose, having redundancy in the design of the interferometer is desirable if unexpected noise couplings are discovered. As TCS is used to decrease a noise coupling, it may not be able to correct thermally-driven errors in the arm mode. The optical layout and AWC functionality in HAM6 were designed with this flexibility in mind.

Two curved mirrors (OMx1 and OMx2) are required to control the two degrees of freedom associated with mode-matching. To gain flexibility in the design, a lens will be used to provide further Gouy phase separation between the two active mode-matching mirrors in each path after BHDBS2. There is space for this lens on the OFI assembly; this is vibrationally isolated and is a useful distance from the mode-matching optics in HAM6. An active lens was considered here, but concerns of OFI assembly heating [167] and the unavailability of suitable flint glass meant that this idea was discarded.

To provide orthogonal mode-matching control, the two SAMS need to be separated by 45° of Gouy phase. Due to the design of the OMC, the Rayleigh range of the beams in HAM6 are of the order of one meter. As most of the Gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel a significant distance across the table before it acquires sufficient Gouy phase. This Gouy phase separation requirement gave us a method of generating candidate designs as this requirement can be interpreted as needing the mode-matching mirrors to be as far apart as possible from each other on the HAM table. Losses due to astigmatism prevents one from using a large AOI to increase the separation between optics in HAM6.

Orthogonal alignment control is obtained by having two mirrors separated by 90° of Gouy phase. For all designs, the two mirrors for alignment were immediately after BHDBS2 and immediately before the OMC as this gave the best Gouy phase separation between them. It was found that placing the first mirror close to BHDBS2 often provided good layouts. References [168, 169] describes how the longitudinal and pointing degrees of freedom associated with BHD may be sensed and controlled.

The OMCs are expected to be the largest contributor of back-scatter from HAM6 to the interferometer [162, 170]; as described in [170], backscatter can degrade the sensitivity of the interferometer. A single stage suspended platform (BHSS) is required to prevent backscatter noise from reducing the interferometer's sensitivity. Both OMCs will need to be on this platform and so all designs need to have the OMCs fairly close to each other. A large BHSS is mechanically infeasible as it will reduce space for the other components on the HAM6 table.

6.2.1 Fixed properties of the HAM6 Optical Layout

There are a few properties of the HAM6 layout that will not be changed. These include: the entry point of beams into the HAM6 chamber, the size of the nominal mode entering HAM6 and the size of the OMC waist. Table extensions can be made in places where they would not impede access, however the size of the HAM optical table does restrict the design. It was decided that the two beam paths after BHDBS2, the optic on which the LO and signal is combined, would be as similar as possible as both beams need to be matched to OMCs of the same design.

The nominal beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$ [171], so the design of the HAM6 mode-matching optics is based on this quantity. As discussed in Chapter 4, the SRM beam parameter is expected to differ from its nominal value. It is described in Chapter 4 how geometry errors in the SRM cause the arm cavity and the SRC to be mismatched. This means that the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM, and the beam parameter at the SRM that should be considered corresponds to the arm cavity mode. This set of likely modes, shown in Figure 4.12, was taken into consideration when choosing a layout for HAM6 since different candidate layouts may have been more or less robust to beam parameter errors at the SRM.

The OMC design developed for aLIGO [170], with trivial modifications, will be used to remove junk light from the signal beam and so there is no choice in target waist size. The nominal OMC waist size is $485 \mu\text{m}$. The beam enters the OMC

through a flat mirror, and a waist is located 140 mm from this mirror. There is a negligible amount of astigmatism inherent in the design of the OMC. Mode-matching optics therefore have to match the nominal SRM and OMC modes.

HAM6 loss Source	Target for O5
OMC throughput loss	< 2%
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	< 2% (as low as possible)
Any other optical loss	< 1%

Table 6.2: Target for losses in HAM6. Reproduction of [163]

6.3 Visualisations of the Robustness of the A+ HAM6 Layout to Mode Mismatch

As described in Section 6.2 and Chapter 4, it is expected that there will be some mismatch between the arm mode and the OMC. To visualise the effectiveness of the AWC provided by the layout’s active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode-matched from by the AWC and a distribution of modes that those optics may have to correct for can be compared to gain insight into the effectiveness of the layout.

The proposed layout in Section 6.1 can mode-match the majority of beam parameters one would expect based on uncertainties in the RoC of the optics within the SRC. Despite this good coverage, measurements from LLO show that the beam

parameter may fall outside the region of WS which the layout can compensate for as the LLO SRC differs significantly from the nominal design. Alternative optical powers for the OFI lens, OMx1 and OMx2 are given in Section 6.3.5; this change would sacrifice orthogonality of the mode-matching and alignment optics in favour of robustness to RoC errors in the SRC.

6.3.1 A Description of Mode-Matching in Terms of HG Modes

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, a combination of misalignment and differing beam shape can be considered to be a mode-mismatch. However, mode-matching typically refers to the overlap of the two modes if they were co-aligned, thus it is a measure of how similar in shape the two modes are. Mode-matching between two modes is usually, to first approximation, thought of as a difference in their waist size and waist location.

Small mode-mismatches can be modelled as light scattering from the beam's, or cavity's, fundamental mode into its $HG_{20} + HG_{02}$ modes. If a cavity is designed such that the fundamental mode will be resonant while the $HG_{20} + HG_{02}$ modes will be reflected, the effect of mode-mismatch between the cavity and a signal carrying beam is to decrease the signal light that is transmitted by the cavity since some of the signal carrying light will be scattered into the reflected $HG_{20} + HG_{02}$ modes. Sensing the $HG_{20} + HG_{02}$ modes therefore gives a measure of the mode-matching

between two modes.

A change in waist size is equivalent to an in-phase coupling between the fundamental mode and the $\text{HG}_{20} + \text{HG}_{02}$ modes, whereas a change in waist location is a coupling to the $\text{HG}_{20} + \text{HG}_{02}$ modes which is 90° out of phase with the fundamental mode [124]. As the $\text{HG}_{20} + \text{HG}_{02}$ modes acquire 2ϕ compared to the fundamental mode, where ϕ is the Gouy phase acquired by the fundamental mode, spacing beam curvature actuators at 45° will give an orthogonal basis for control over mode-matching. Orthogonal control results in the beam size and beam waist, or two combinations of them that are orthogonal, can be independently controlled.

6.3.2 Calculating Mode-Matching Using The Gaussian Beam Parameter

Two perfectly mode-matched beams will have identical beam parameters at the same point along the optical axis. The overlap integral between two fields, E_1 and E_2 ,

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}, \quad (6.1)$$

can be used to determine the amount of mode-matching, $\mathcal{M}$, between two astigmatic modes. Solving this integral gives

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*||q_{y2} - q_{y1}^*|}, \quad (6.2)$$

where $q_{1,2}$ is the beam parameter of the electric field $E_{1,2}$. This is derived in Appendix E. The solution to Equation (6.2) is always $0 < \mathcal{M} \leq 1$. Alternatively, mode-matching may be computed in terms of the amplitude of the higher order modes relative to the fundamental mode [172]. The mode-matching can be calculated at any point along the optical axis of the layout as mode-matching must be preserved throughout it; however, it makes sense to calculate the mode-matching in a physically significant plane such as the input plane, a plane containing a curved optic or the plane containing the waist of the target beam.

6.3.3 WS Space Visualisations

A Gaussian beam can be described by its beam size, w , and defocus, S [173]. A WS space plots a Gaussian beam in terms of these two parameters. The effect of focussing elements on Gaussian beams can be clearly seen in this representation of them. Because a curved mirror can only change the curvature of a beam, at the mirror's surface the beams before and after reflection must have the same size. The process of a Gaussian beam, described by $q_1(S_1, w_1)$, being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i\frac{\lambda}{\pi w_1^2}. \quad (6.3)$$

From Equation (6.3) it can be seen that a mode moves in the S direction when it reflects from a curved mirror. This representation of Gaussian beams has the advantage of showing the beam in terms of intuitive quantities. The downside to

this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. Since q is a complex number, it could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and S produces plots where the effect of focussing elements and propagation are easy to visualise [174, p.680]. Curved mirrors and lenses result in a change in defocus, and propagated modes follow circular paths. As these figures are being used to visualise mode-matching and the effect of actuators on the beam size (rather than $\lambda/\pi w^2$) and defocus, the WS representation was chosen. One could express the beam parameter in terms of natural units by dividing the beam size by w_0 and the defocus by $1/z_R$.

To illustrate how actuators move a mode in WS space, consider two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will be orthogonally controllable. In a WS space, the orthogonality of two actuators can be seen by the angle between the two vectors that represent each actuation. If the two lenses are orthogonal then the actuation vectors will be orthogonal. Figure 6.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase.

The mode-matching requirement between the interferometer and the OMC for

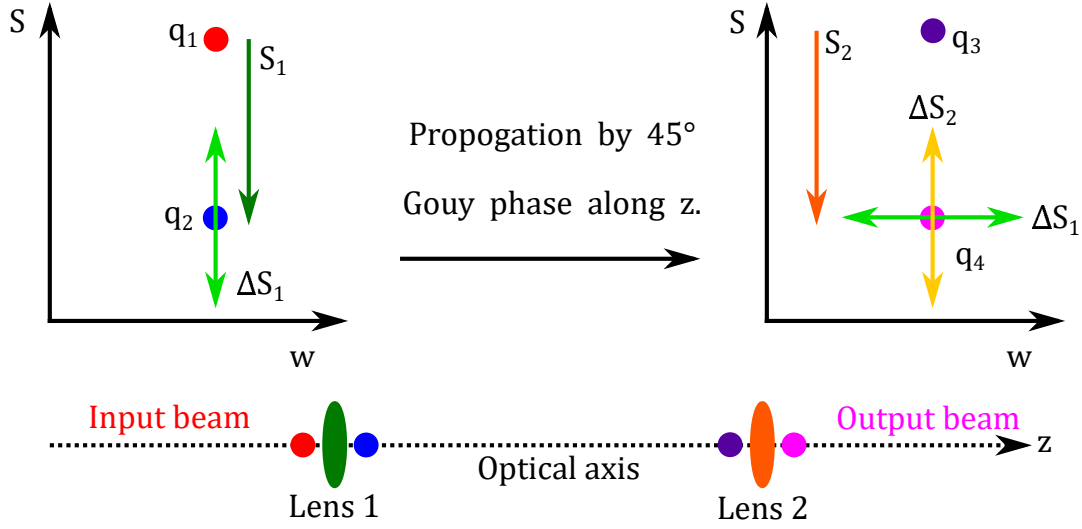


Figure 6.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} \pm \Delta S_{1,2}$. The left hand axes shows the WS plane at Lens 1. The input beam, q_1 , is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 which is shown as a blue dot. A light green arrow indicates the actuation of Lens 1. The right hand axes shows the WS plane at Lens 2. The input mode at Lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of Lens 2 is indicated by the orange and yellow lines. As Lens 1 and Lens 2 are separated by 45° of Gouy phase, the actuation provided by Lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by Lens 2, provided that the actuations are small enough. Beneath the two WS space diagrams is a sketch of the optical axis and the location of the modes.

LIGO A+ is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode; the mode-matching between two arbitrary modes can not be deduced from these contours [173]. For example, two modes may be mismatched by the same amount from the target in orthogonal directions in WS space. They cannot be 100% mode-matched as these two modes are different from each other.

6.3.4 Visualising how Active Mode-Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS space at the SRM from which the target mode-matching space can be achieved was computed. To do this, first the ring of modes in WS space that is 98% mode-matched to the OMC after reflection from the second active mirror, OMx2, was considered. If OMx2 had a static radius of curvature then the effect of OMx2 would be to move the ring vertically in the WS plane. However, the ring can also be stretched in the S direction as OMx2's radius of curvature can be changed. This can be seen, crudely, by applying $S_{\text{OMx2}} \pm \Delta S$ to the original $> 98\%$ mode-matching ring to produce two new rings separated by $2\Delta S$.

To generate the region from which the target region can be reached by OMx2, the target region should be cut by a line which intersects the ring at the extrema of the beam size. To maximise the amount of WS space covered by the actuator, the upper target region should have $+\Delta S$ applied to it whereas for the lower region the actuation should be $-\Delta S$. This results in a region of WS space which is larger than the initial target region. All modes in the new region can be at least 98% mode-matched to the OMC. The new region was then propagated to the first mirror, OMx1, where the same procedure can be applied to further expand the region. This process is shown in Figure 6.6.

The modes from the SRM that can be matched to the target are shown in Figure 6.7. To visualise how successful the layout is at handling the likely modes from the interferometer (this set of modes was investigated in Chapter 4), the interferometer modes were compared to the region of WS space containing modes which can be corrected such that they meet the $> 98\%$ mode-matching requirement. It was found that majority of likely modes will be covered by the layout as the modes near the edge of the interferometer mode continuum are more unlikely to occur than the ones near the centre; however, measurements of the SRC's Gouy phase at LLO can be used to infer the beam parameter at the SRM, and this mode would fall outside of the region of WS space which can be compensated for by the AWC. Its worth noting that the modes at the yellow end of the continuum shown in Figure 6.7 may not be physical as they correspond to an unstable SRC.

The shape of the pink region in Figure 6.7 is nominal; when the HAM6 layout is constructed the pink region's shape will be altered by geometric errors in OMx1

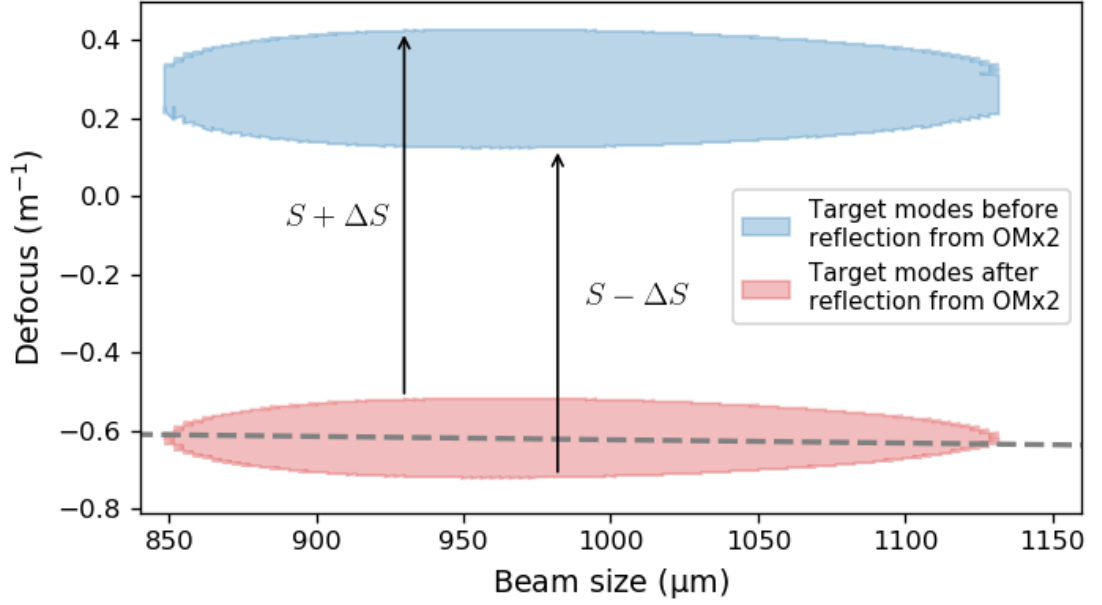


Figure 6.6: The WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode-matched to the OMC as there are no further active mode-matching elements in the layout of HAM6. Modes with at least 98% mode-matching are shown by the red region. The blue region represents modes which can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, the upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

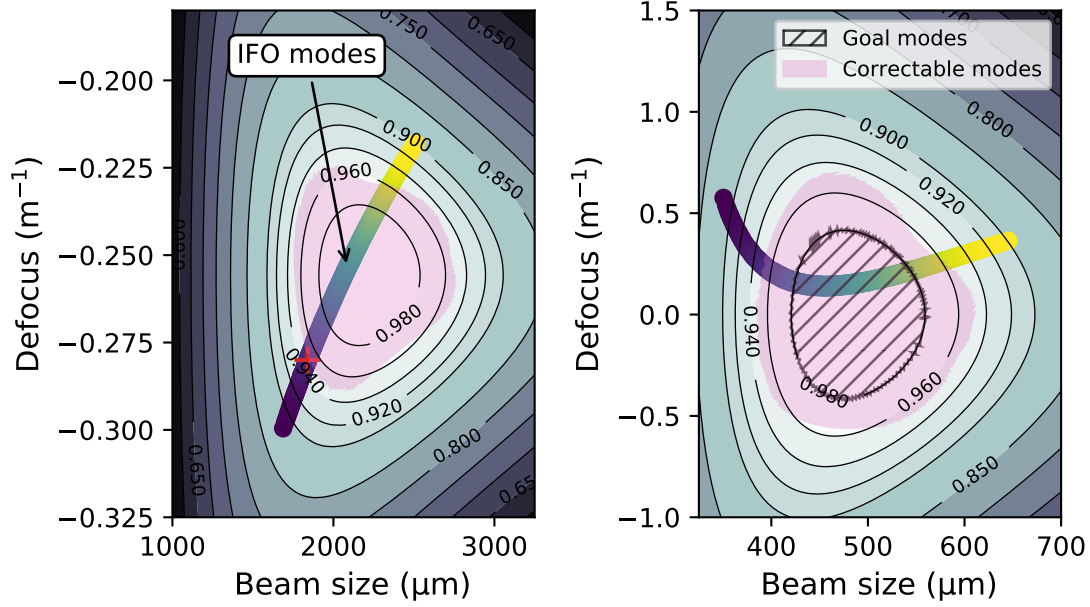


Figure 6.7: The region of WS space covered by the AWC in HAM6 if OMx1 and OMx2 for the layout discussed in Section 6.1. Left panel: the set of modes that is expected to be present at the SRM, as described in Chapter 4, is shown by the purple-blue-green-yellow curve. The pink region shows which modes that can be corrected for so that 98% mode-matching is achieved. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. The red cross indicates the mode size that can be inferred from measurements of the SRC's Gouy phase at LLO. Right panel: this shows the same modes as the left panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

and OMx2. Nevertheless, Figure 6.7 is encouraging since it shows that the actuation provided by OMx1 and OMx2 is highly tolerant to mode-matching errors generated by errors in the SRC.

6.3.5 Optimising WS Space Coverage in Favour of Expected Mode Errors

While 45° of Gouy phase separation between the mode-matching optics in HAM6 is a desirable quality for the layout to have, with an actuation range ± 50 mD, the layout can be optimised to correct for mode-mismatches between the OMC and the continuum of expected beams exiting the interferometer. The majority of the uncertainty in the interferometer's beam parameter comes from the uncertainty in SR3's RoC (see Figures 4.6 and 4.7). Ideally, one would correct for this by actuating on the RoC of SR3, however the RoC of SR3 cannot be strongly actuated on. If a mirror was placed 180° away from SR3, it could be used to compensate for the uncertainty in the RoC of SR3 i.e. one could configure the mode-matching optics within HAM6 such that they are biased for correcting the errors in the interferometer beam parameter caused by the error in SR3.

By reducing the orthogonality of the AWC actuators, more range can be obtained for one of the beam's property e.g. the size of a beam's waist. A Gouy phase separation ($\sim 30^\circ$) offers a compromise between orthogonality versus actuation range for errors in one of the beam parameters dimensions. Figure 6.8 shows how the region

of WS space covered by the HAM6 AWC can be biased in favour of the expected set of modes that may emerge from the SRM. The physical positions of the optics are the same as before (see Table 6.1), however the optical powers of the OFI lens, OMx1 and OMx2 was altered (-0.2, 1.67, 0.631 respectively) to change the Gouy phase separation between OMx1 and OMx2 to 30° . However, the Gouy phase for alignment control would be significantly reduced; this is shown in Figure 6.9.

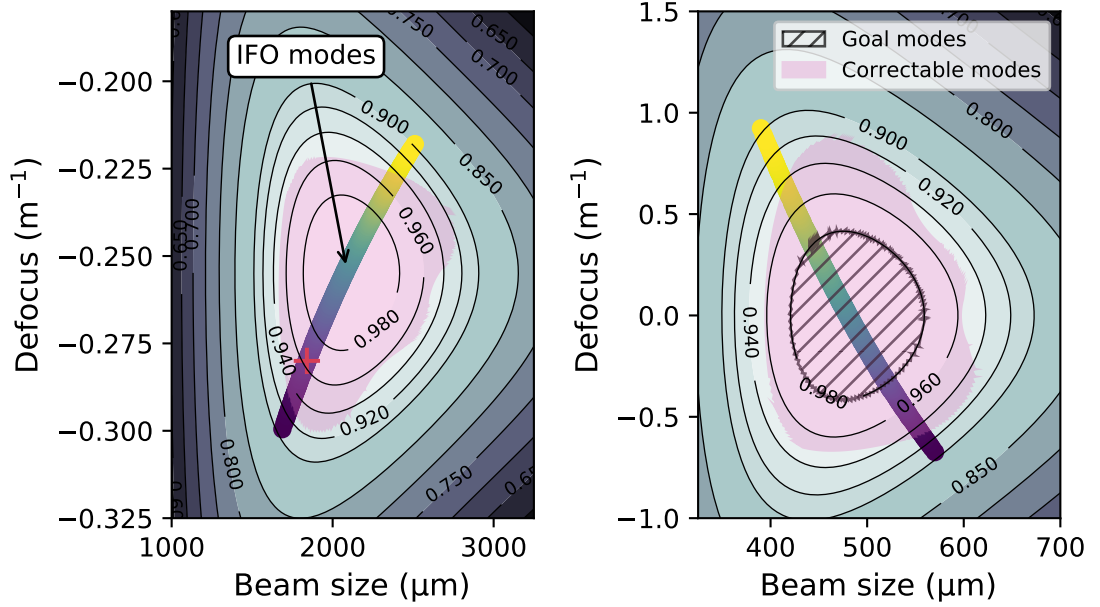


Figure 6.8: At the expense of orthogonality, the region of WS space covered by the AWC in HAM6 can be optimised by changing the Gouy phase separation between OMx1 and OMx2. Left panel: the set of modes that could be emerge from at the SRM, as described in Chapter 4, is shown by the purple-blue-green-yellow curve. The pink region shows which modes can reach 98% mode-matching by using the AWC in HAM6. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the left panel; however, they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region (hatched area).

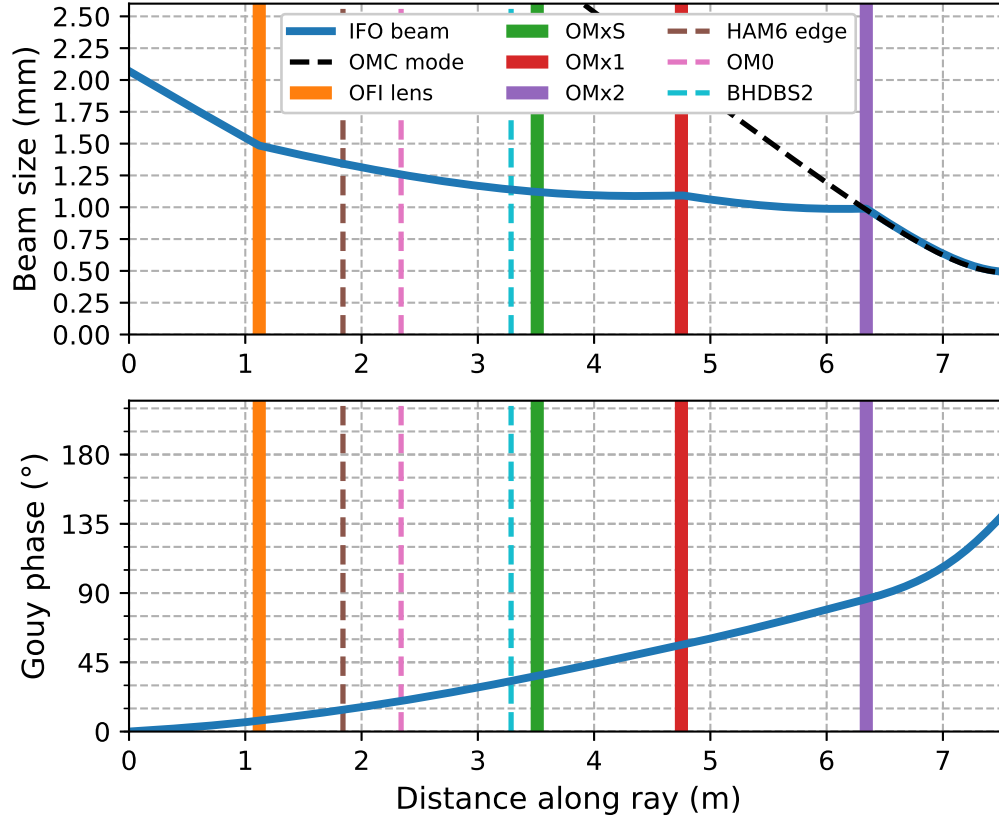


Figure 6.9: Top panel: the beam profile for the alternative layout of HAM6 suggested in Section 6.3.5. Bottom panel: The Gouy phase accumulated along the optical axis for the alternative layout. While alignment Gouy phase is degraded, the mode-matching Gouy phase was changed to bias the optical layout for correcting errors in the RoC of SR3.

6.4 Conclusion

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements was presented in Figure 6.1. The optical properties of this layout were investigated. By selection of the right combination of mode-matching optics, near orthogonal control of the signal-LO mode-matching and beam alignment to the OMC was achieved.

The effect that the AWC had on the region of WS space that contains modes which satisfy the $> 98\%$ mode-matching requirement space was investigated. WS diagrams were used to represent how robust the layout is to initial mode-mismatch. Figure 6.7 shows the amount of WS space that the AWC could correct for. It would be desirable to shape this region to improve the chances of being able to mode-match the interferometer to the OMC. At the expense of AWC and alignment orthogonality, the mode-matching optics in HAM6 could be changed from their nominal values to improve the likelihood that the AWC can correct for mode mismatches as described in Section 6.3.5.

Chapter 7

Experimental MZ

An interferometer converts a change in optical path length into a change in intensity of light. One of the fundamental limits to how precisely the path light takes within an interferometer is the shot noise level of the light. In other words, as shot noise manifests as intensity fluctuations of the light, this is indistinguishable from fluctuations in the differential phase acquired by two interfering beams. In this chapter, the calibration between power measured on a photodiode and the displacement noise of a mach-zehnder interferometer is demonstrated. This links the shot noise that is measured with this photodiode circuit to the noise within an interferometer. The results from this experiment serve as an analogy to a BHD readout schemes: as long as the the local oscillators power is much greater than the signal beams power, the shot noise limit of an interferometer with a BHD readout and the the local oscillators path length stability requirement depends only on the power in the signal beam.

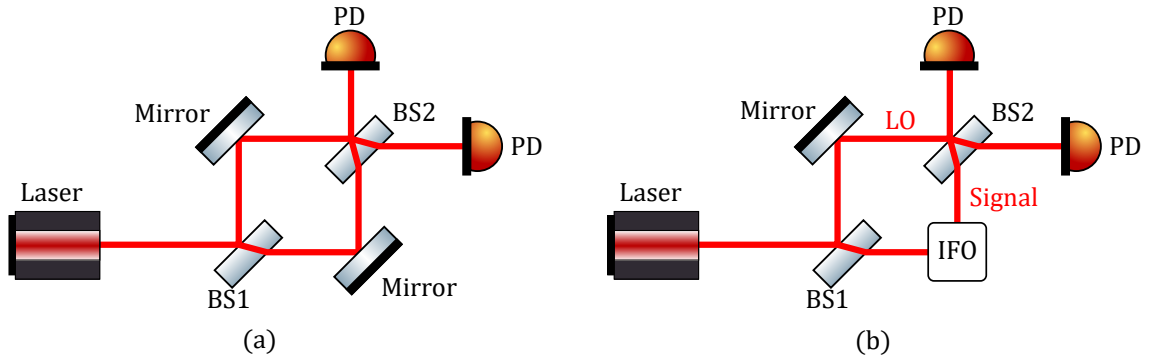


Figure 7.1

7.1 Mach-Zehnder Interferometers as an Analogy to an Interferometer with a Balanced Homodyne Readout

A sketch of a Mach-Zehnder interferometer is shown in figure ???. At the second beam splitter, two beams are combined to produce an interference pattern that carries information about the relative path length difference that the two beams experienced. This is analogous to an interferometer that uses a BHD readout. In an interferometer with a BHD readout, the two beams are called the signal and local oscillator. The transfer function between the signal beam's phase and the strain induced by a gravitational wave depends on the interferometers optical design, but at the AS port, the signal beam is simply an electric field with some phase information encoded on it and so by studying the equations of a simple mach-zehnder interferometer, conclusions about more complex interferometers can be made.

The power of the light on each photodiode, as the reflectivity, R and transitivity, T , of the beamsplitter were measured to both be 0.5 ± 0.01 , is

$$P_A = \frac{P_{\text{LO}}}{2} + \frac{P_{\text{sig}}}{2} + \sqrt{P_{\text{sig}}P_{\text{LO}}} \cos(\phi), \quad (7.1)$$

$$P_B = \frac{P_{\text{LO}}}{2} + \frac{P_{\text{sig}}}{2} + \sqrt{P_{\text{sig}}P_{\text{LO}}} \cos(\phi + \pi), \quad (7.2)$$

where P_{sig} and P_{LO} are the powers of the local oscillator and signal beams and ϕ is the phase difference between the local oscillator and signal. Subtracting the two photodiode signals gives

$$S = P_B - P_A = 2\sqrt{P_{\text{sig}}P_{\text{LO}}} \cos(\phi). \quad (7.3)$$

This signal is analogous to the one obtained from a balanced homodyne readout attached to a Michelson interferometer. In another interferometer configuration whose transfer function is $H(f)$, P_{sig} would be replaced by $P_{\text{sig}}/H(f)$.

The relationship between optical path length change, Δx and a change in power, ΔP , measured by subtracting the Mach-Zehnder's photodiode signals is

$$\Delta z = \frac{\lambda}{2\pi} \frac{1}{2\sqrt{P_{\text{sig}}P_{\text{LO}}}} \Delta S. \quad (7.4)$$

The shot noise of a beam with power P is $\Delta S = \sqrt{2\hbar\omega_{\text{EM}}P}$. The level of shot noise comes from the total power on each photodiode. If the signal beam's power is much

lower than the local oscillator's power, then the shot noise can be expressed as

$$\Delta z = \sqrt{\frac{2hc\lambda}{16\pi^2 P_{\text{sig}}}}. \quad (7.5)$$

This means that the local oscillators path length stability depends only on the power of the signal beam and the sensitivity of the interferometer will not be affected by the local oscillators power.

7.1.1 Apparatus for the In-Air Mach-Zehnder.

The in-air mach-zehnder is shown in Figure 7.3 and Figure 7.4. It was constructed from highly stable rigid mounts to reduce the amount of mount-vibration noise. The MZ was mounted on a thick aluminium base-plate which was isolated from the optical table by rubber dampeners. The MZ was acoustically shielded with a plastic box of a few mm thickness. As the beam size within the MZ was sufficiently small, the light power in one arm was attenuated with an ND filter as surface roughness of the ND filter did not matter. One of the corner mirrors in the MZ was glued to a PZT to allow for the length of one of the paths to be controlled. The light that emerged from the MZ was measured with the photodiode circuit shown in Appendix ???. The response of this photodiode circuit is flat at all frequencies of interest and is 390 V/W at DC.

A half wave-plate and polarised beam splitter were used to control the power entering the MZ and the polarisation of the light entering the MZ. The input laser

power was monitored with a photodiode. A microphone were used to monitor the environmental noise sources.

The photodiode signals were recorded with CDS (see Appendix ??). To send signals long distances across the lab, differential send-recieves circuits, constructed with the THAT range of IC, were used to remove common-mode noise that may be picked up along cable. The overall factor which converts the recorded digital signal into a voltage is determined by the number of bits used in the ADC (16), the peak-to-peak voltage that the ADC can record, 20 V, and the factor of 2 that comes from the differential send-receive circuit: the factor is therefore $20/(2 \times 65536)$.

To keep the measured signal above the digitisation noise inherent to the analogue to digital converters used in CDS, whitening filters are required. This is a circuit which boosts the AC components of a signal while keeping the DC level of the signal small enough as to not saturate the ADC. A dewhitening filter is a digital filter whose transfer function is the reciprocal of the whitening filter's. In addition to this, CDS features an anti-aliasing filter which will effect measurements that are made near the Nyquist frequency of the ADC. This response of the AA filter can be seen in figure 7.2. CDS noise is shown in figure 7.2. For this photodiode circuit, the level of shot noise due to 20 mW of light is $\sim 3 \times 10^{-8} V_{\text{rms}}/\sqrt{\text{Hz}}$ and so the whitening filter needs to boost the photodiode signal by a factor of 100.

To keep the path length difference between the two arms fixed, a control signal was applied to the PZT. This was implemented with the circuit shown in Figure ??. The servo shaping was a low pass filter with a corner frequency of 1 Hz. The PZT

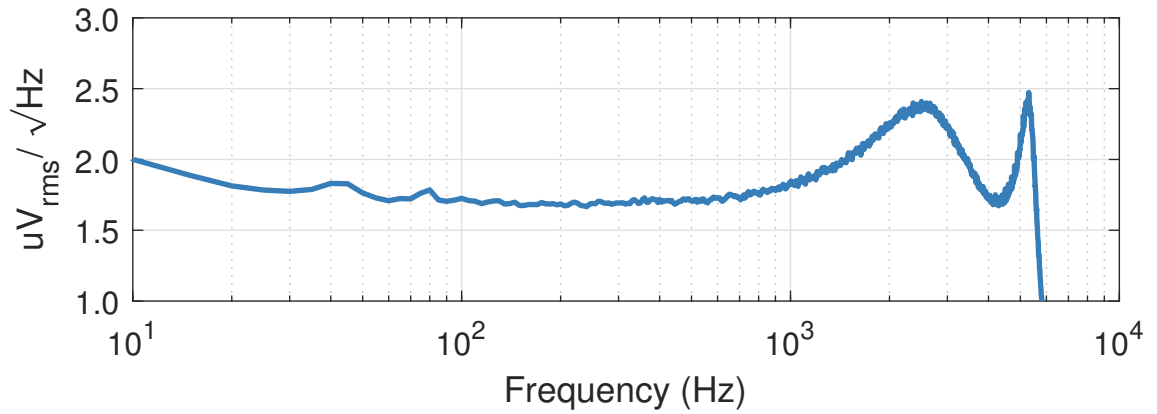


Figure 7.2

acts as a low pass filter with a corner frequency that is determined by its capacitance. To prevent oscillations due to the servo being unstable, the overall gain of the servo was controlled with a potentiometer.

7.1.2 Calibration of In-Air Mach-Zehnder

The shot noise level can be found by summing P_A and P_B . The power was measured to be 11 mW, so shot noise is... ?? (labbook post with data).

The Mach-Zehnder can be calibrated by measuring P_{sig} and P_{LO} . The power in the LO path was measured to be 20 mW (LO and Sig power measurement labbook post), and the power in the lower path before the beam passed through the ND filter was measured to be 17 mW. After the ND filter (ND filter labbook post), the power would be 0.1 mW. This is shown in figure 7.5.

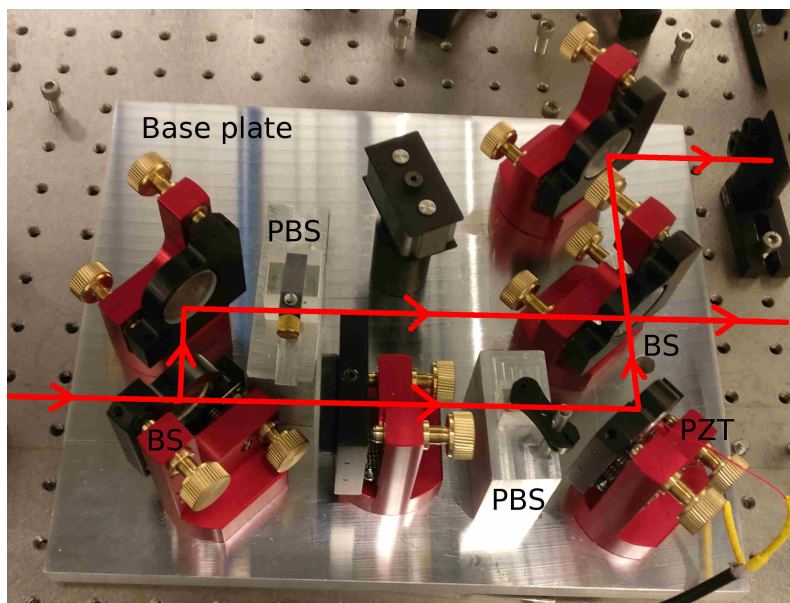


Figure 7.3: Photograph of the Mach-Zehnder base plate. The polarising beam splitters were part of an earlier design; they were removed when the measurements with this apparatus were made.

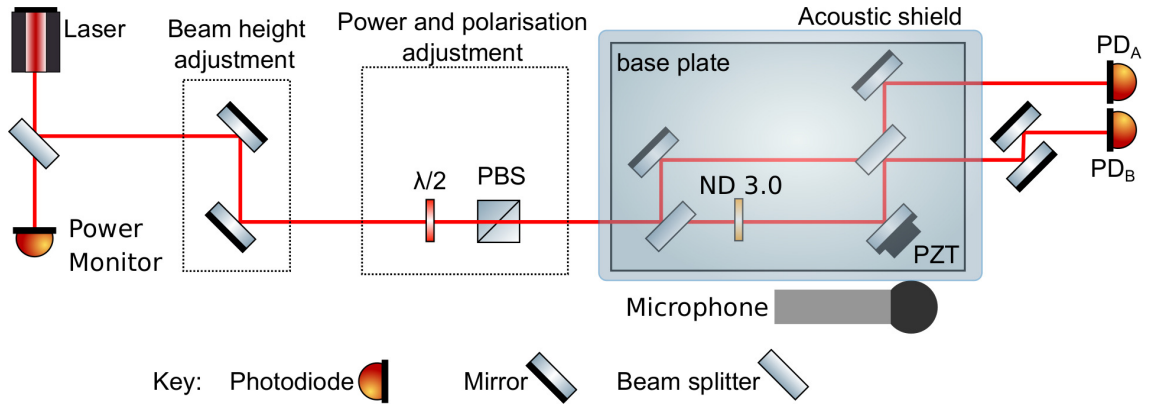


Figure 7.4: Sketch of the optical layout. The laser was a farady-isolated NPRO beam. The power of the laser was monitored using a photodiode immediately after the laser. Beam power and polarisation control was situated immediately upstream of the Mach-zehnder. The . A microphone and a humidity sensor where placed near the MZ.

Alternatively, the Mach-Zehnder can be calibrated by measuring $\Delta S/\Delta z$ by applying a ramp signal to the PZT. The ramp signal and the photodiode signals were recorded on an oscilloscope. The photodiode signals can were then plotted against the measured ramp signal, and data was fit with Equation 7.3. This is shown in figure 7.5.

Although there is a 20% discrepancy between the two calibrations, both methods of calibrating the response of the Mach-Zehnder give consistent results within margin of error determined by the quantisation noise associated with the oscilloscope and the uncertainty of the signal beam's power.

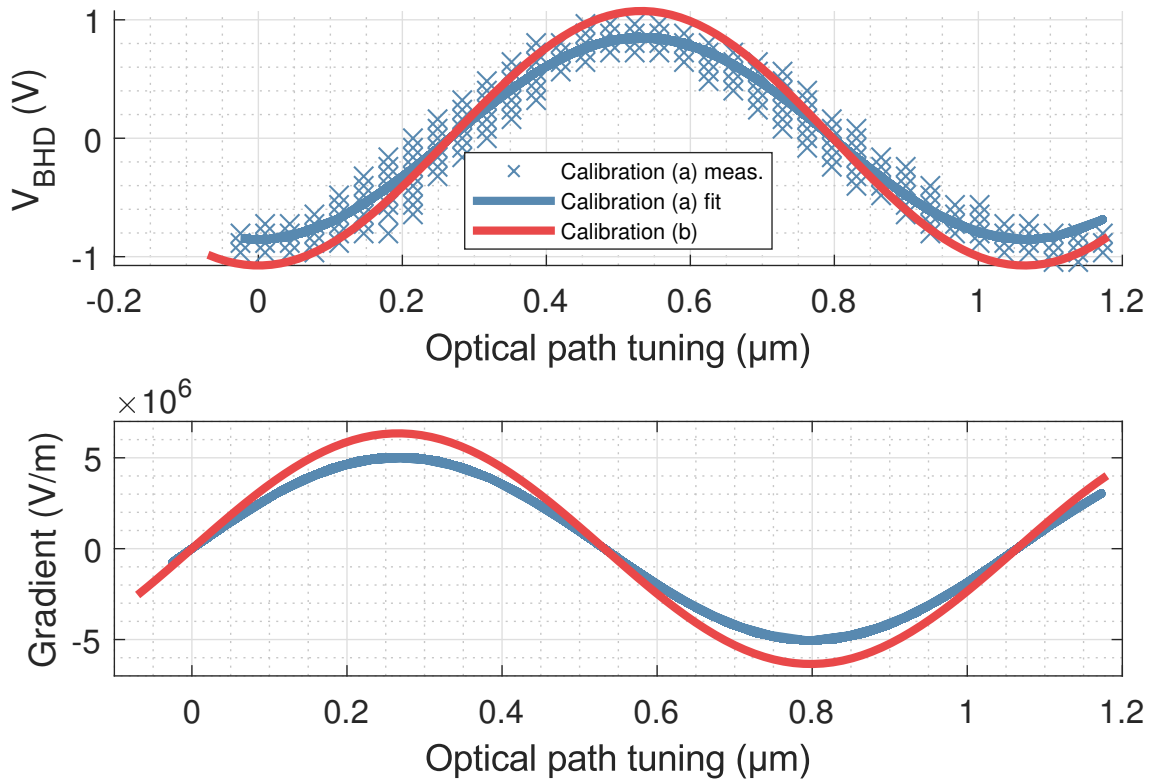


Figure 7.5: There are two ways of calibrating the photodiode signal that will be generated by a change in optical path length between the two paths in the Mach-Zehnder. Method (a) is to apply a ramp signal to the PZT while simultaneously measuring the photodiode signals. The measurement of this is shown by the blue crosses, and the fit to this data is shown by the blue line. Method (b) is to measure the power in the local oscillator and signal beams when they are not interfering. From this, Equation 7.3 can be used to get the response of the Mach-Zehnder. This is shown by the red line. The blue and red data are independent of each other, within 20% of each other and are in agreement with each other. The gradient of V_{BHD} is used to get the conversion factor for beam power to path length change.

Parameter	Value
Nyquist Frequency	32768 Hz
Frequency Resolution	10 Hz
Effective Noise bandwidth	15 Hz
Number of Averages	459

Table 7.1: A Hanning window was used to average the data and generate the amplitude spectral density.

7.1.3 Measurement of Shot Noise

The MZ was locked and the noise measured by both photodiodes was recorded via CDS. The measured signal corresponds to the level of shot noise predicted by equation 7.5 at frequencies where classical noise sources no longer dominate the signal. To demonstrate that the signal depends on the differential path length of the MZ, a modulation at 6033 Hz was applied to the PZT. This can be seen in the spectrum shown in figure 7.6. By analogy between PZT motion and noise level, for a 100 Hz measurement bandwidth, differential path length changes smaller than $5 \times 10^{-14} \text{ m}_{\text{rms}}$ cannot be measured as the measurement will be limited by the power of the signal beam.

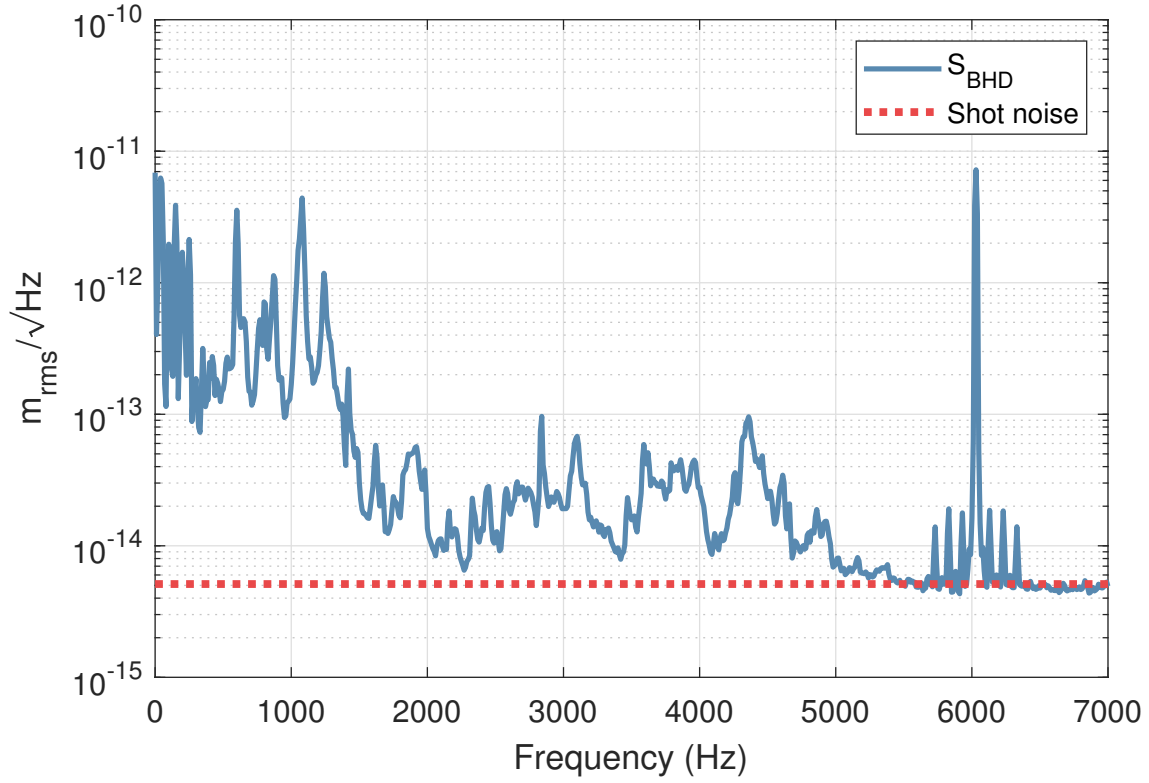


Figure 7.6: Shot noise was measured at frequencies above 5500 Hz. Below 5500 Hz, acoustic noise limited the MZ sensitivity. The peak at 6033 Hz was injected to demonstrate the effect of a differential arm length modulation.

7.2 Noise in an In-Vacuum Mach-Zehnder Interferometer and the Effect of Misalignment the Local Oscillator and Signal Beam

To measure shot noise at frequencies around 100 Hz, the frequency band of interest in a gravitational wave detector, the acoustic and seismic noise which shake the mirrors need to be reduced. This can be achieved by using suspended optics and performing the experiment in-vacuum. There was an opportunity to perform this experiment in-vacuum with left-over apparatus from a previous experiment. This proved challenging for as the interferometer was hard to lock due to large amounts of seismic motion coupling to the differential arm length. Ultimately, scattered light interfering with the differential arm signal meant that shot noise could not be measured at any frequency. However, this apparatus could be used to explore how misalignment will cause the signal to size to drop.

7.2.1 Apparatus for the In-Vacuum Mach-Zehnder

A mach zehnder was constructed from suspended optics and placed in vacuum; this is shown in Figure ?? . The suspensions used were double pendulums of with masses of 75g separated by whiles of lengths 100 mm (top to middle mass) and 150 mm (middle to bottom mass), resulting in them having longitudinal resonances around ~ 1 [?]. The balanced homodyne readout platform (BHRP) was a suspended

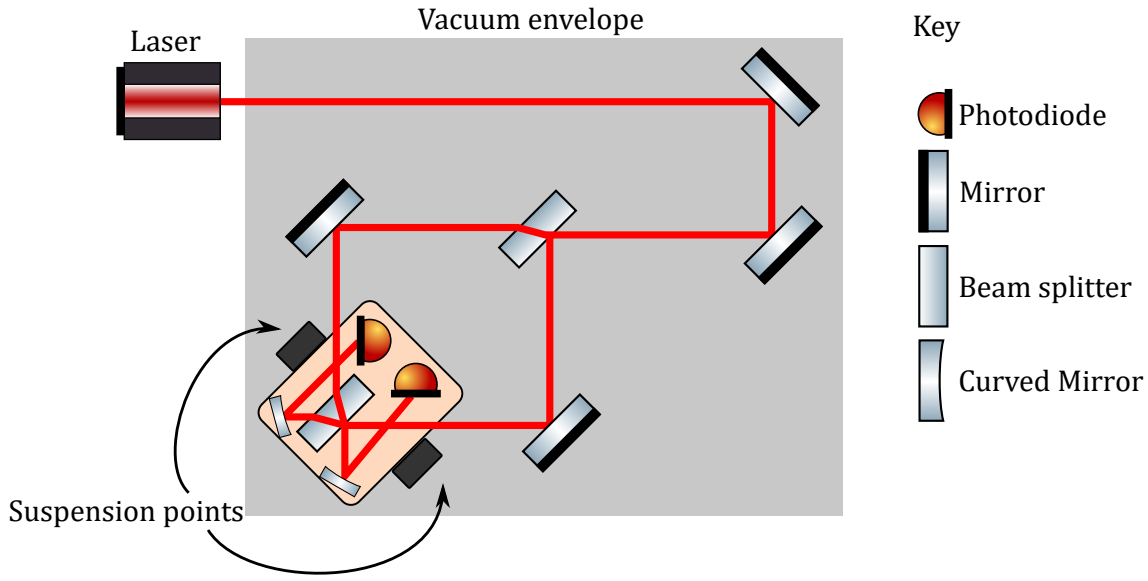
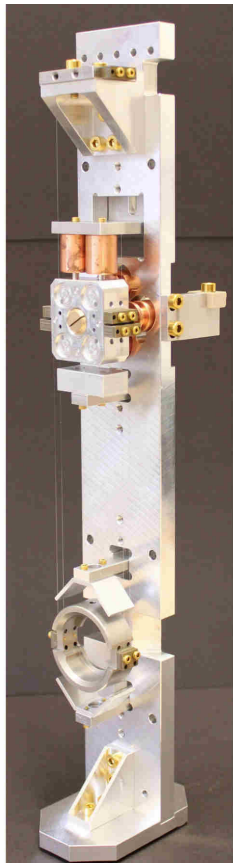


Figure 7.7

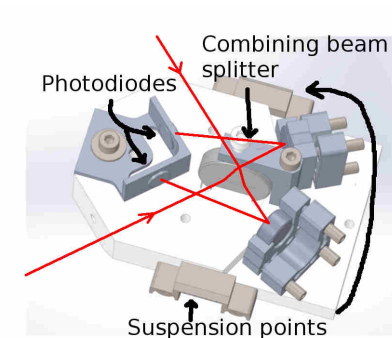
assembly that included the recombination beam splitter, steering mirrors for the recombined beams and photodiode. This photodiode circuit was the same as the one described in Section ???. The vacuum system was pumped down to 1,mbar ???. An NPRO was used to provide the laser light.

7.2.2 Calibration and Locking of the In-Vacuum Mach-Zehnder

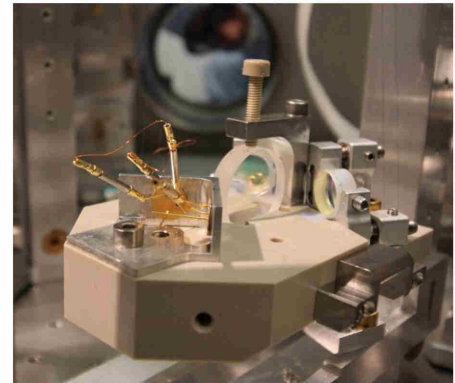
The differential arm length needed controlling below the suspension main resonance, ~ 1 Hz as the optics moved on the order of one wavelength at these frequencies. Due to the design of the of the balanced homodyne readout, there are lots of suspension-related modes that couple to the longitudinal signal at around 30 Hz. Additionally, the optical bench moved around a large amount due to poor seismic isolation. This



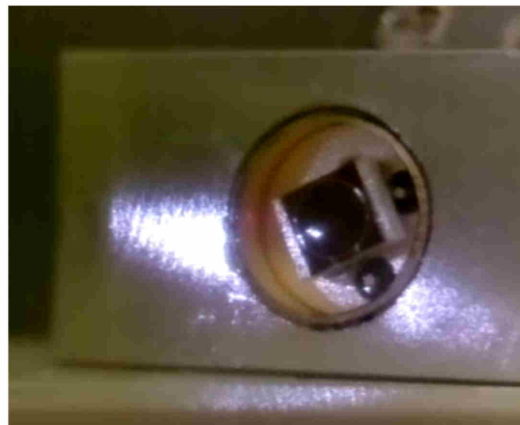
Suspended mirror



BHD PlatformCAD



BHD Platform



Scattered light on photodiode

Figure 7.8

meant that the differential arm length noise was at a similar level at $\sim 30\text{ Hz}$ as it was at $\sim 1\text{ Hz}$.

The suspended mirror was actuated on with a magnet+coil to lock the MZ arm length. The coil acted on a magnet glued to the bottom stage of the pendulum meaning that its strength fell as $1/f^2$ above the suspensions resonance. Thus, the coil would need to drive a large signal at 1 Hz , as well as one that was 1000 times greater at $\sim 30\text{ Hz}$. The magnet+coil strength was limited by geometry, e.g. the size of the mirror, size of the magnet, size of the coil, amount of current that the coil could have etc., and so was not strong enough to do this. Therefore, the unity gain point needed a large gain at $\sim 1\text{ Hz}$ while having a unity gain point limited to being below and 10 Hz by the suspension/coil driver actuator. To get the required gain, resonant gain filters were implemented in CDS, and to compensate for the phase problems these caused, phase compensation filters were made. [more detail?](#)

The transfer function of the servo is shown in Figure 7.9 and a sketch of the servo is shown in Figure 7.9. The servo consists of five logical blocks: the coil-driven suspended mirror, the ‘optical transfer function’ that represents a change in differential arm length and the power of the recombined beam, the photodiode circuit, the analogue to digital/digital to analogue converters, and the digital filtering. The dewhitening filters were inverses of the whitening filter, and so they have no effect on the shape of the loop. As the transfer function of the suspension was simulated using the model developed in ?? and the digital filters just add shaping, to calibrate the loop, the conversion factors for the electronics and optics needed to be measured; this is summarised in table 7.2.

Component	Transfer function	unit
CDS DAC/ADC converters	0.62	mV/count
Coil driver	1.0	mA/V
Coil	10	mN/A
Suspension	$H(f)$	N/m
MZ	665000	m/ λ
Fringe to counts	79000	λ /count
Digital filtering	$G(f)$	count/count

Table 7.2: Elements of the servo and their transfer functions. The suspension transfer function, $H(f)$, was obtained by simulation, the digital filter’s transfer function is trivial, the coil’s force per current was calculated using an existing script, the rest of the transfer functions were directly measured. These values were used to produce the servo model shown in Figure 7.9.

By injecting and measuring a signal across a ‘times one’ block in the digital part of the servo, as the ratio of the two signals sizes gives the loop gain, the model’s calibration was independently verified . This is shown in figure 7.10. This calibration ties up to within 0.1 dB. Therefore, the unity gain point of the servo was at 5.5 Hz with 15° of phase margin.

7.2.3 Measurement of the Differential Arm Signal

With the servo described in section ??, the in-vacuum MZ was locked. The differential arm motion spectrum is shown in figure 7.11. The curved steering mirrors used to guide the beam onto the photodiodes did not have an AR coating on their back face. This caused a large amount of scattered light. Scattered light generates

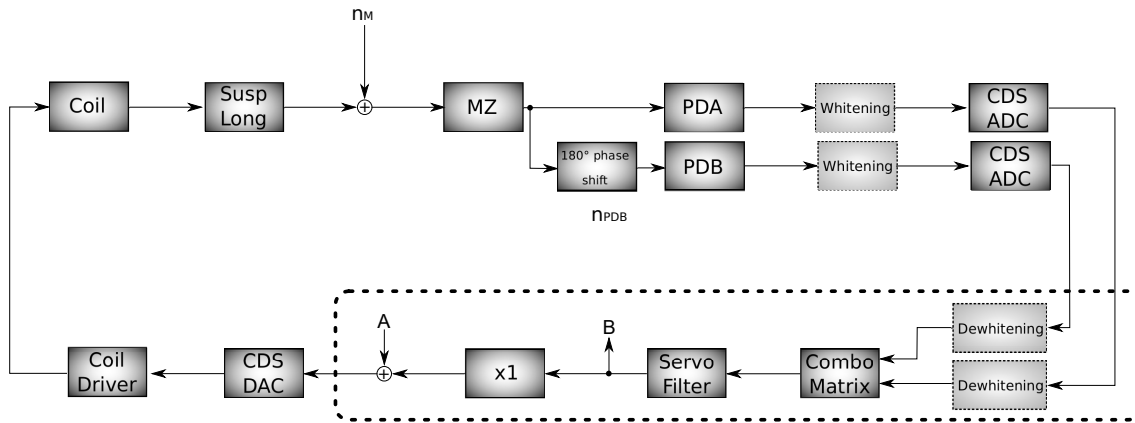


Figure 7.9: Flow diagram showing the each block of the servo used to lock the in-vacuum MZ. The digital part of the servo is shown in the dashed area. The injection points A and B were used to calibrate the servo (see Figure 7.10 are around a times one block. The mirrors motion that the servo must correct for is represented by the signal n_M . The information about the MZ differential arm length is encoded by two signals that are 180° out of phase. The whitening and dewhitening filters are paler as these do not affect the gain of the loop.

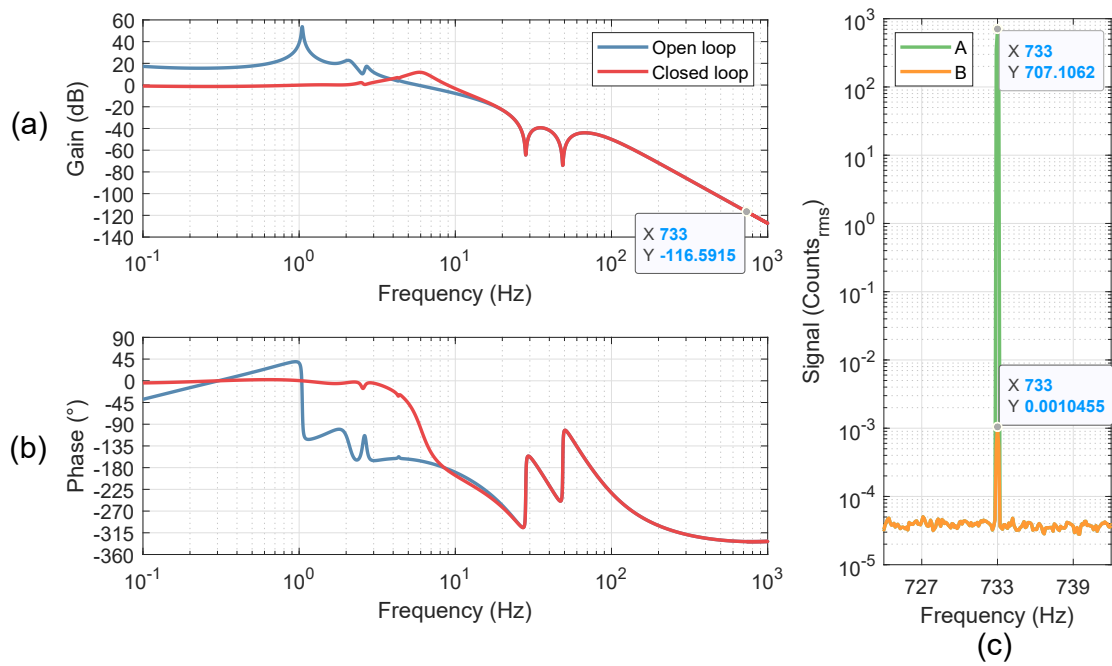


Figure 7.10: Panel (a) and Panel (b) is the Bode diagram of the Servo's open and closed loop transfer function. This was calibrated using the factors from 7.2. Panel (c) shows a measurement where a signal was injected at point A in the servo and measured at point B (see figure 7.9); these two points are separated by a times one block, and so the ratio of these signals represents the closed loop transfer function of the servo. These measurements are in agreement with the model to within 0.1 dB.

a bilinear angle-to-intensity coupling. In an interferometer, this causes excess noise due to scattered light interfering with interferometers differential light mode. One of the great engineering achievements demonstrated at LIGO is the reduction of scattered light as this is currently a limiting source of noise at low frequencies.

7.2.4 The Effect of Misalignment on the Size of a Signal in an Interferometer

This apparatus can be used to verify that an angular misalignment, θ , will cause the signal, S , to drop from its maximum value, S_0 , as

$$S_{\text{sig}} = S_0 e^{-((\theta - \theta_0)/\theta_c)^2}, \quad (7.6)$$

where $\theta_c = \sqrt{2}\lambda_{\text{EM}}/\pi w_0$ is the characteristic angle of the a beam of waist w_0 . The counts to angle was measured by applying the largest possible angle offset that CDS could provide and measuring the distance the beam moved, a few millimeter, over a distance of the order a metre. The beam's waist is know to be $\sim 460\mu\text{m}$. A differential arm signal was generated and the resulting photodiode signal was measured for a range of angular misalignments. This result is shown in Figure 7.12. To prevent noise at nearby frequencies from leaking into the measurement, a narrow frequency resolution was required (0.2 Hz). Roughly 10 averages were made used to measure the signal size at each misalignment. The variance of these averages was large, this is represented by the error bars shown in figure 7.12. A weighted non-linear least squares regression function (MATLAB's `nlinfit`) was used to fit

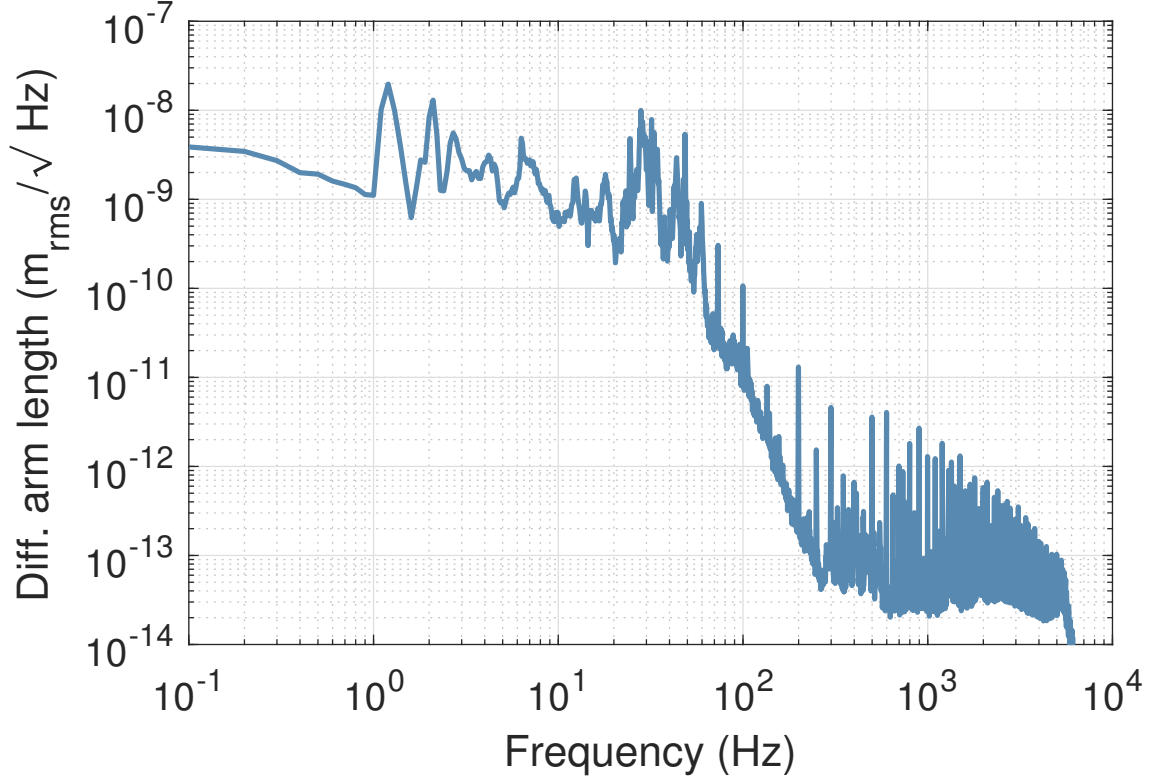


Figure 7.11: Large amounts of scattering caused there to be a high amount of noise in the differential arm length signal's noise floor, this can be seen to limit the noise above ~ 200 Hz. The effect of the pendulums resonance at ~ 1 Hz was mostly removed by the servo but it can still be seen in the noise. Many suspension related peaks can be seen around 30 Hz, these are due to the design of the BHD suspension. The $1/f^4$ slope is due to the suspension having two stages. For this measurement, shot noise would have been at $\sim 10^{-15} \text{ m}/\sqrt{\text{Hz}}$.

the data to Equation 7.6, with each point weighted with $1/\text{error}^2$. The resulting fit parameters were $S_0 = 6.9 \pm 0.3 \times 10^{-3}$ counts and $\theta_0 = 0.46 \pm 0.02$ mrad. The covariance between S_0 and θ_0 was $\sigma_{S_0, \theta_0} = 2.4 \times 10^{-9}$ and the reduced χ^2 statistic was 3.07.

7.3 Conclusion

A small, rigid mach zender was constructed to experimentally demonstrate the relationship between the local oscillators path length stability (see Equation 7.5). This interferometer was calibrated and shot noise was measured. An interferometer converts a change in optical path length into a change in intensity of light. One of the fundamental limits to how precisely the path light takes within an interferometer is the shot noise level of the light. In other words, as shot noise manifests as intensity fluctuations of the light, this is indistinguishable from fluctuations in the differential phase acquired by two interfering beams...

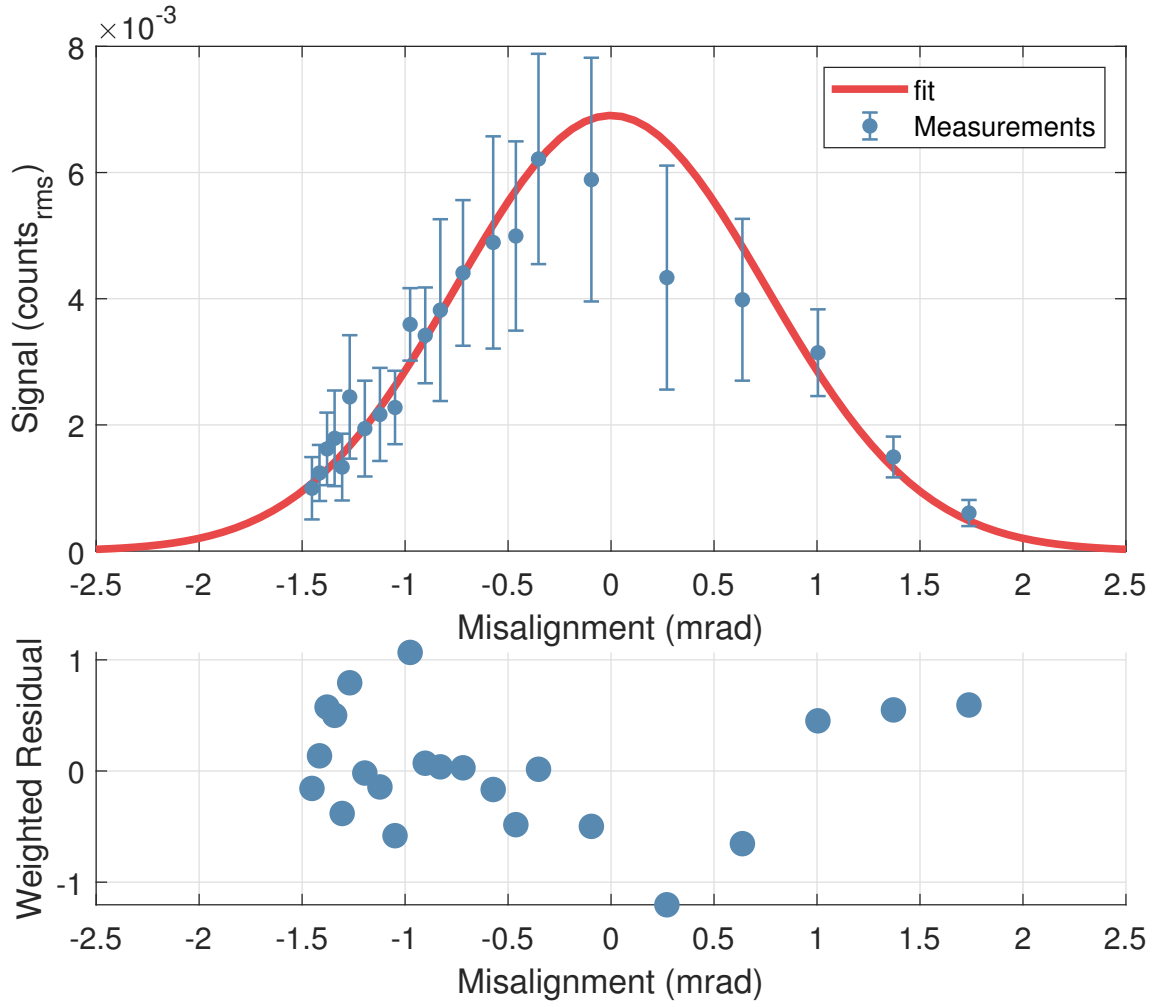


Figure 7.12: Top panel: The in-vacuum MZ was misalignment. As the MZ was misaligned, there was less power in the HG00 mode. The decrease in power for a misalignment θ is

Chapter 8

Characterisation of InGaAs

Photodiodes for detecting $2\mu\text{m}$ Light

As discussed in Section 2.8, a new photodiode technology may be needed for the third generation of gravitational wave detectors. Quantum efficiency, power handling and dark noise are three important properties for a photodiode that would be used in a gravitational wave detector, so the requirements for such a photodiode are explored in Section 8.1. Extended InGaAs photodiodes can be sensitive to $2\mu\text{m}$ light; however, for the reasons explained in Section 8.2, fabrication of an extended InGaAs photodiode that meets the requirements set out in Section 8.1 is challenging. The properties of off-the-shelf extended InGaAs photodiodes were measured under different reverse biases to investigate whether they met the requirements. As well as extended InGaAs, InSb is sensitive to $2\mu\text{m}$ light, so the properties of an InSb

detector are explored in Section 8.8.

8.1 Quantum Efficiency, Power Handling and Dark Noise Requirements for Photodiodes in a Gravitational Wave detector

The photodiodes used in a gravitational wave interferometer should have a quantum efficiency of at least 99%, as described in [163]. Loss within the interferometer decreases its SNR, as it diminishes the level to which the signal light is squeezed (see Equation 2.17). High quantum efficiency is required to detect the target of ~ 10 dB of squeezed light that future gravitational wave detectors will use [81, 84, 85].

In some applications, if a bigger signal is required, the gain can be increased by using a larger resistor in the feedback path of the transimpedance amplifier, or by using an avalanche photodiode. However, neither of these methods would work in a gravitational wave detector as both would boost the noise and the signal in proportion.

Current gravitational wave detectors run with light in the ~ 10 mW at the output (e.g. [49]) to ensure that the junk light from other modes is a small portion of the total signal (see Section 2.4.4). For the remainder of this chapter, as $q\lambda_{\text{EM}}/hc$ is of the order unity, the requirement for a photodiode is that it should be able to

produce ~ 10 mA of photocurrent without saturating.

Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that the interferometers are limited by quantum noise in their detection bands. Third generation ground based gravitational wave detectors are designed to be quantum noise limited down to about 10 Hz [80, 81, 85], so the $1/f$ noise must be sufficiently far below the shot noise of the photocurrent above 10 Hz. For a 10 mA photocurrent, the shot noise is $6 \times 10^{-11} \text{ A}_{\text{rms}}/\sqrt{\text{Hz}}$. Below 10 Hz, other rapidly rising sources of noise are predicted to limit the sensitivity of the Einstein Telescope and Cosmic Explorer.

The $1/f$ noise could, in principle, be avoided by using heterodyne detection. However, a conventional heterodyne detection scheme would increase the shot noise level of the signal [152, 175, 176]. Although an optimal choice of modulation-demodulation waveform can recover the SNR by increasing the signal [152], the benefits of using squeezed light would be lost because the shot noise would be enhanced compared to homodyne readout because vacuum noise at frequencies that do not contain signal would enter into the measurement. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by using a broadband squeezer [177].

8.2 What is an Extended InGaAs Photodiode?

The performance of an extended InGaAs photodiode depends on the number of defects within it. A basic explanation of how photodiodes work, how these defects arise, and how these defects affect the photodiode is given in this section. More details on the semiconductor physics of diodes and photodiodes can be found in many textbooks e.g. [178, 179].

The basic physics of a PIN photodiode can be qualitatively understood by considering a simpler pn junction. A pn junction consists of two types of semiconductor: p-doped and n-doped. These regions have ions which are in fixed positions and charge carriers, known as majority charge carriers, which are free to move. Holes are the majority charge carriers in the p-doped region, and the ions are negatively charged (acceptors). In the n-doped region, electrons are free to move, and the ions are positively charged (donors). In equilibrium, the number of holes is equal to the number of electrons. The free charge carriers will diffuse across the interface between the p-doped and n-doped region. This causes the p-doped and n-doped regions to become charged. This results in a region of the pn junction where there are no free electrons and holes. This is known as the depletion region.

If a photon is absorbed by a pn junction, a free electron-hole pair is generated. A photon may be absorbed by the pn junction if its energy is large enough to move an electron from the valence band into the conduction band. If the photon is absorbed within the depletion region of the photodiode, the electron and hole are separated

due to the electric field, and a current is generated in the circuit connected to the photodiode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will have a large chance of recombining and not contributing to the current in the circuit. The relationship between quantum efficiency, η , light power, P_{EM} , and photocurrent, I_p is given by

$$I_p = \eta \frac{q\lambda_{\text{EM}}}{hc} P_{\text{EM}}, \quad (8.1)$$

where λ_{EM} is the wavelength of the light, h is Planck's constant and c is the speed of light.

To widen the depletion region, an external voltage can be used. When a reverse bias is applied to the photodiode, there is a larger electric field difference at the interface between the p-doped and n-doped region; thus, the depletion region is widened i.e. applying a reverse bias to a photodiode increases the volume of the photodiode in which absorbed photons will generate photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p-doped and n-doped regions. An intrinsic semiconductor has no doping, and the number of electrons equals the number of holes within it. As the intrinsic region physically separates p-doped and n-doped regions, a larger volume of the photodiode will be depleted of charge carriers due to the electric field within the photodiode. Additionally, the capacitance of the photodiode will be decreased.

InGaAs is a widely used semiconductor for making photodiodes. InGaAs needs to be grown on a substrate. To combine two types of semiconductor, they must have similar lattice constants and share the same lattice structure; this is known as lattice matching. Typically, the indium content of InGaAs is around 53%, this is signified with the chemical formula $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$, so that the InGaAs can be lattice matched to an InP substrate¹. With this amount of indium, the InGaAs photodiode will suffer from few lattice mismatch defects, and the photodiodes created with can be of high quality.

Normally, InGaAs photodiodes are sensitive up to wavelengths around 1.6 μm , but longer wavelength sensitivity can be achieved by increasing the indium content of the photodiode (e.g. [180–182]). As mentioned in Section 2.8, these photodiodes are often called extended, and they can be sensitive for wavelengths up to 3 μm . To absorb 2 μm photons, the indium content needs to be at least 70%. One of the key issues faced when creating an extended InGaAs photodiode is the strain induced by the lattice mismatch between the InGaAs and the InP substrate [183] as this introduces defects.

Defects caused by the strain between the InGaAs and InP layers of the photodiode generate excess noise, dark current and reduce its quantum efficiency [184]. The photodiode can be made thinner to reduce this strain. The critical thickness, i.e. the thickness above which strain degrades the crystal, for extended InGaAs is of the order 0.1 nm [183] whereas the absorption length (of the order of 10^4cm^{-1} [181])

¹Ge is sometimes used as a substrate. This will be lattice matched for InGaAs with an In content around 10%; this corresponds to the bandgap energy being similar to the energy of a 1 μm photon.

requires the photodiode to be several microns thick for it to have the high quantum efficiency that is required for the detection of gravitational waves (see Section 8.1). Therefore, it is highly likely that such a thick extended InGaAs photodiode will have defects. Additionally, because InGaAs is not a good conductor of heat, a thicker photodiode can show reduced power handling [185].

The defects mentioned above manifest as abrupt changes in the crystal structure known as misfit dislocations (e.g. [186, 187]). The resulting undesirable states in the bandgap that charge carriers can occupy, known as trap states, give rise to a generation-recombination dark current [188]; thus, defects are a source of $1/f$ noise in extended InGaAs (see [189] for a description of $1/f$ noise in semiconductors). Defects cause there to be more free charge carriers in the InGaAs which leads to the depletion region becoming smaller, thus causing a decrease in quantum efficiency [190]. For indium doping levels other than 53%, buffer layers of InGaAs between the absorbing layer and the substrate can be used to mitigate the lattice mismatch [188]. The number of defects depends on the growth conditions of the photodiode, and by improving them, the dark noise of a photodiode made from extended InGaAs can be reduced by orders of magnitude [182].

An anti-reflection coating can be used to increase the quantum efficiency of a photodiode. For example, $\text{In}_{0.7}\text{Ga}_{0.3}\text{As}$ has a bandgap which is equal to the energy of a $2\mu\text{m}$ photon, and it has a refractive index, n , of approximately 3.7 at this wavelength. Assuming a single quarter-wave layer was to be applied, the film would need a refractive index of $n_f = \sqrt{3.7}$ and be $\sim 250\text{nm}$ thick. The idea of using this anti reflective coating as a passivation layer – a passive material designed for

protecting the semiconductor from the environment – is presented in [191]. Some manufacturers create a range of photodiodes with different spectral responses by changing the anti-reflective coating rather than by changing the photodiode’s indium content [71].

The dark noise and dark current of a photodiode have several contributors. In low bias regimes, the dark current is driven by the diffusion of minority carriers [190, 192]. However, the thermally driven generation-recombination of electron/hole pairs drives the $1/f$ dark noise at low frequencies and at low bias [189]. Therefore, the dark current does not predict the dark noise; however, as both increase with higher levels of defects, a photodiode with a high dark current will tend to have a high dark noise. When the bias exceeds a certain threshold, electrons can tunnel across the junction, leading to a substantial increase in dark current; this bias is typically of the order 1 V in extended InGaAs photodiodes.

The noise of a photocurrent can be modelled as

$$i_{\text{rms}} = \sqrt{2qi + \frac{4k_B T}{R_{\text{PD}}} + \left(\frac{i_c}{f^\beta}\right)^2}. \quad (8.2)$$

The first term under the square-root, $2qi$, is the term associated with the shot noise of a current i . The second term, $\frac{4k_B T}{R_{\text{PD}}}$, is the term related to the Johnson-Nyquist noise due to the shunt resistance of the photodiode, R_{PD} ; this term can become significant when the shunt resistance is low, as it is in some extended InGaAs photodiodes, and microwatt levels of light are being detected. The final term is an empirical relationship known as Hooge’s law [189]; i_c is the characteristic noise level and β is

usually 0.5 ± 0.1 .

8.3 Selection of Off-the-shelf Extended InGaAs Photodiodes

The dark current and dark noise was characterised in a selection of commercially available photodiodes. The photodiodes that were tested and their properties according to their datasheets² are listed in Table 8.1. The quantum efficiency as a function of bias was measured for one of the photodiodes, the FD10D, and the saturation limits of the FD10s, IG22x1000S4i, IG24x500S4i and IG26x500S4i photodiodes were investigated. The dark current and dark noise was measured for each model of photodiode.

With no reverse bias³, the typical quantum efficiency of these photodiodes is 80%; this is far below the target of greater than 99%. However, the quantum efficiency of a photodiode can be improved by increasing the bias voltage (see Section 8.4).

²Datasheets often quote the noise equivalent power (NEP) of a photodiode. The NEP is the noise spectral density of the dark current generated by the photodiode divided by its responsivity. For the g12183 series photodiodes, the responsivity is $\sim 1 \text{ A/W}$ and the NEP is $\sim 10^{-12} \text{ W}/\sqrt{\text{Hz}}$, so the noise, as claimed by the datasheet, is $\sim 10^{-12} \text{ A}/\sqrt{\text{Hz}}$. This would seem sufficiently low to be shot noise limited down to $\sim 1 \mu\text{W}$. However, this number does not capture the behaviour of the photodiode at illumination levels of several milliwatts where a bias is needed to ensure the photodiode does not saturate. As the noise of these photodiodes is not flat (see Appendix A), NEP should be specified as a function of frequency. The noise and responsivity are both functions of reverse bias, so the NEP is a function of bias. Another quantity related to the NEP is the specific detectivity; this incorporates the area and bandwidth of the photodiode with the NEP.

³Usually, the operating conditions of the photodiode is stated in the photodiode's datasheet when a parameter is given; however, there is no information on what reverse bias gives the responsivity in the FD10D's datasheet.

Photodiode	Peak quantum efficiency (%)	Cut-off wavelength ^a (μm)	Diameter (mm)	Datasheet
FD10D	80 ^b	2.5	1	[193]
G12183-010k	77 ^b	2.5	1	[194]
G12183-005k	77 ^b	2.5	0.5	[194]
IG26x500S4i	86 ^c	2.5	0.5	[195]
IG24x500S4i	84 ^c	2.4	0.5	[196]
IG22x1000S4i	91 ^d	2.2	1.0	[197]
G12182-005k	84 ^e	2.1	0.5	[198]
G12182-010k	84 ^e	2.1	1.0	[198]

^a λ_{EM} at which the quantum efficiency = 50% ^b $\lambda_{\text{EM}} = 1.9\mu\text{m}$ ^c $\lambda_{\text{EM}} = 2.0\mu\text{m}$

^d $\lambda_{\text{EM}} = 1.7\mu\text{m}$ ^e $\lambda_{\text{EM}} = 1.6\mu\text{m}$

Table 8.1: Information from the datasheets of the photodiodes that were tested. Note that none of the photodiodes has a quantum efficiency of at least 99%, however measurements of dark noise and dark current and investigations into their optimal operating conditions are still of interest. This choice of photodiodes covers a range of manufacturers, sizes and cut-off wavelengths.

The photodiodes listed in Table 8.1 are sold with protective windows, and these windows reflect a significant amount of light. These windows should be removed in experiments requiring high quantum efficiency.

8.4 The Effect of Reverse Bias on the Quantum Efficiency of an Extended InGaAs Photodiode

The relationship between the reverse bias and quantum efficiency of an extended InGaAs photodiode, the FD10D, was investigated. The two types of measurement

used are described in Section 8.4.1. Figure 8.2 shows a clear motivation for increasing the reverse bias of an extended InGaAs photodiode.

8.4.1 Method

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. It is straightforward to determine the change in quantum efficiency by measuring the photocurrent as a function of reverse bias. Alternatively, one can modulate the bias and measure the modulation in the photocurrent. If the quantum efficiency, η , depends on the reverse bias of the photodiode, modulating the reverse bias, V_B , at a frequency f will result in η modulating at f and so the measurement will be of $\Delta\eta/\Delta V_B$. The input modulation $\Delta V_{\text{in}} = \Delta V_B$ and the output modulation is $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$, where $V_{\text{TIA}}(V_B)$ is the DC signal from the photodiode and θ is the phase between the input and output modulation. The quantum efficiency should be independent of frequency if the photodiode is operating in a time-independent, linear fashion and so the two types of measurement should yield the same outcome.

To prove that the increase in signal was a result of the photodiode's quantum efficiency increasing and there were no non-linear mechanisms by which photocurrent was generated, the shot noise level of the photocurrent was measured. This measurement required a sufficiently quiet photodiode circuit and an amplitude stabilised laser.

To measure photocurrent with sufficient SNR, a transimpedance amplifier with electronic noise 25 times below the shot noise level of the expected signal (10 mA) photocurrent was used. This circuit is shown in Appendix B.2. The circuit allowed for the reverse bias of the photodiode to be set externally via a low-passed bench-top power supply. The low-pass filter had two poles at 16 Hz.

The optical layout is shown in Figure 8.1. A laser with a power of 50 mW and a wavelength of 1.55 μm (Pure Photonics PPCL300) was used to illuminate the FD10D, the device under test. The light from the laser was split by a 50:50 beam splitter so that a suitable amount could be used to amplitude stabilise the laser. The in-loop photodiode used to sense the laser's power was a high-performance InGaAs photodiode (C3061GH). The laser was amplitude stabilised by actuating on the current of the laser's pump-diode. The amplitude stabilisation servo had sufficient gain so that the in-loop noise was below the level of shot noise of a 10 mA current, so this could be used to measure shot noise on the out-of-loop photodiode. With this apparatus, shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to adjust the power level of the light incident on the FD10D, and a lens was used to set the size of the beam.

As the FD10D has a diameter of 1 mm, the laser beam could not have a width larger than 330 μm for there to be $< 1\%$ loss due to clipping. For the reasons stated in Section 8.1, the FD10D was tested at a power level of ~ 10 mW. The methods in this section were used to measure relative increases in quantum efficiency; measurements of absolute quantum efficiency require a calibrated power meter. A method of measuring the absolute quantum efficiency of a photodiode with squeezed

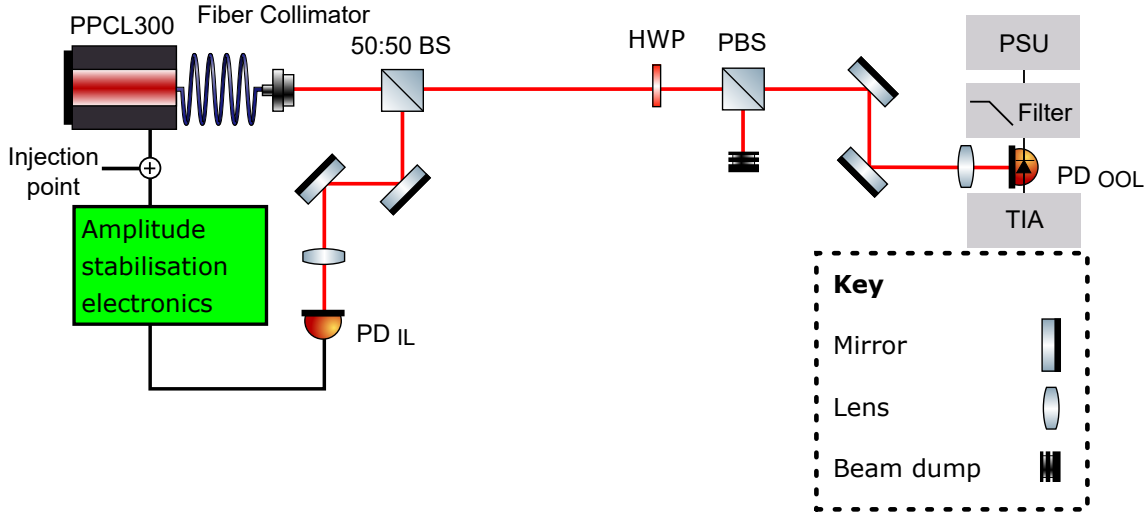


Figure 8.1: Optical layout used for measuring the effect that increasing the reverse bias of a photodiode had on its quantum efficiency and saturation limit. The out-of-loop photodiode, PD_{OOL} , was the photodiode under test (FD10D). A $1.55\ \mu\text{m}$ laser was used to assess the optical properties of the photodiode. A combination of half wave plate (HWP) and polarised beam splitter (PBS) allowed for the power of the beam to be set, and a lens was used to set the size of the beam on the photodiode under test. Amplitude stabilised light was needed to ensure the increase in photocurrent seen when the bias was increased was due to an improvement in quantum efficiency. The photodiode circuit (TIA) (see Appendix B.2) allowed for shot noise limited measurements of current around 10 mA and for the bias to be selected. The calibration signal was added onto the amplitude stabilisation error signal.

light is presented in [199].

The photodiode was biased such that the dark noise was sufficiently low (see Section 8.5 for details), and the noise of the photocurrent was compared to the level expected for shot noise. The expected level of noise, n , in units of $A_{\text{rms}}/\sqrt{\text{Hz}}$, is given by

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2}. \quad (8.3)$$

The first two terms under the square root represent the shot noise of the current produced by the in-loop and out-of-loop photodiodes, $n_{\text{OOL/IL dark}}$ is the dark noise of the out-of-loop photodiode and in-loop photodiode, n_{loop} is the electronic noise of the loop and n_{analyser} is the noise of the analyser.

The light from the laser was amplitude modulated at 1111 Hz by adding a signal to the laser's pump-diode error signal (see Figure 8.1). This modulation was used to calibrate the measurements of noise as the relative intensity of this peak was known and could be checked against the signal measured by the high-performing in-loop photodiode. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the saturation limit of the photodiode was investigated. A similar type of measurement is reported in [200].

8.4.2 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 8.2. For all beam intensities which the photodiode was exposed to, the increase in quantum efficiency was $\sim 15\%$ as the reverse bias went from 0 V to 1.8 V, the stated maximum reverse bias in the FD10D's datasheet. There may be a saturation related effect at the 1% level. Beams with a lower intensity experienced a greater increase in η than beams with a higher intensity. This could be explained by the quantum efficiency not being uniform over the photodiode – the spatial dependence of η can be crater-like with η being greatest at the edges of the photodiode. The difference between the DC and AC response is described by a non-linear process. These effects are small, and Figure 8.2 gives a clear incentive to increase the bias voltage to improve the quantum efficiency of the FD10D.

Figure 8.3 shows the noise of the photocurrent when the photodiode was biased with 0.6V. As Equation 8.3 matches the measured noise, the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency.

At 0.6 V reverse bias, for DC photocurrent ranging from 7.5 mA to 15.5 mA, the calibration peak and the noise level of the photocurrent were measured to ensure the photodiode was not saturating. Over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 8.4. The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak and the level of shot noise, so there was no sign of the photodiode

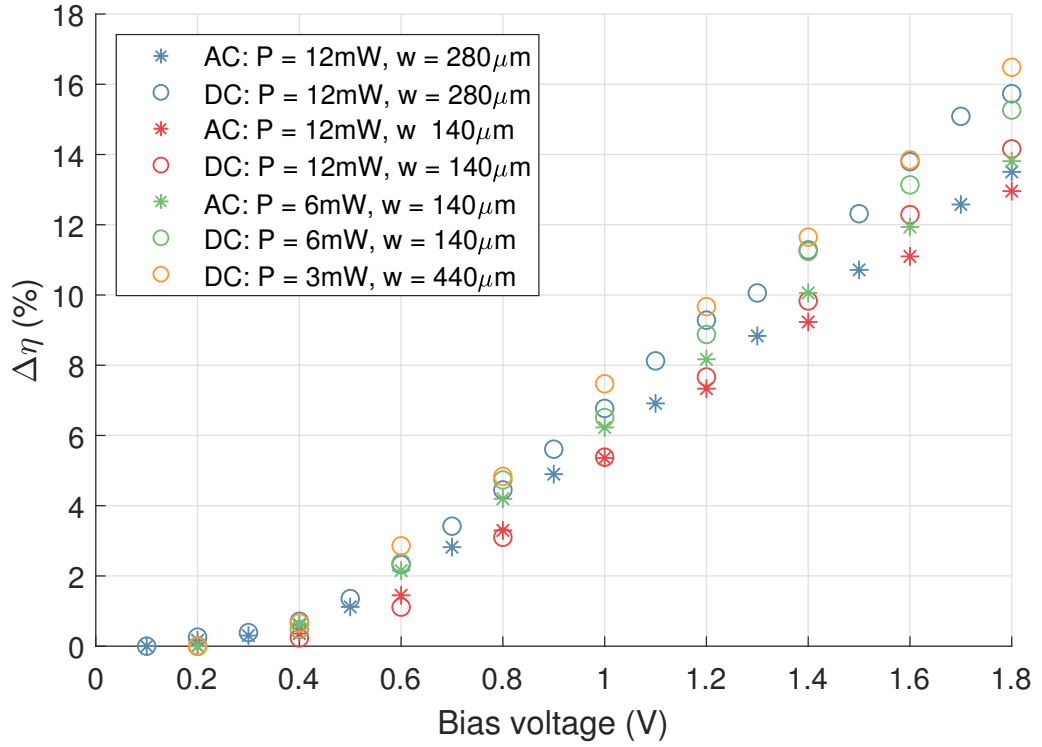


Figure 8.2: The FD10D’s quantum efficiency depends on its reverse bias; the increase in quantum efficiency was $15 \pm 1\%$ when the bias was increased to the maximum value specified in its datasheet. Two types of measurement were used to determine the increase in quantum efficiency as the bias was increased. The first method was to measure the photocurrent with a voltmeter as the bias was increased, and the corresponding results are shown as circles (‘DC’). The second method involved modulating the bias and measuring the resulting modulation in the photocurrent, i.e. an AC measurement, and these are shown by the stars. Each set of measurements was done with a different intensity by changing either the beam’s width or power. Further detail on both methods is given in Section 8.4.1.

saturating.

By measuring the photocurrent and light reflected by the FD10D's window, at 0.6 V reverse bias, its quantum efficiency was measured to be 70%. This was calibrated using two independent power meters. There is a significant amount of light reflected by the FD10D's window; if the window was not removed, at 0.6 V reverse bias the quantum efficiency would be 60%.

8.5 Dark Current and Dark Noise in Extended InGaAs Photodiodes

Following the observation that both dark noise and quantum efficiency both increase with reverse bias, there is a compromise to make between quantum efficiency and dark noise. A maximum reverse bias voltage that balanced quantum efficiency and noise was found for each photodiode listed in Table 8.1. See Section 8.4.2 for details on the dependence of quantum efficiency on reverse bias. The dark noise of the photodiode should be below the shot noise level of the detected light. For the reasons discussed in Section 8.1, the shot noise of a 10 mA photocurrent, $6 \times 10^{-11} \text{ A}_{\text{rms}}/\sqrt{\text{Hz}}$, was used as a benchmark for the photodiodes' dark noise at a frequency of 10 Hz.

Low dark currents were expected, so a transimpedance amplifier with higher gain than in Section 8.4 was used. This circuit, shown in Appendix B.1, could measure

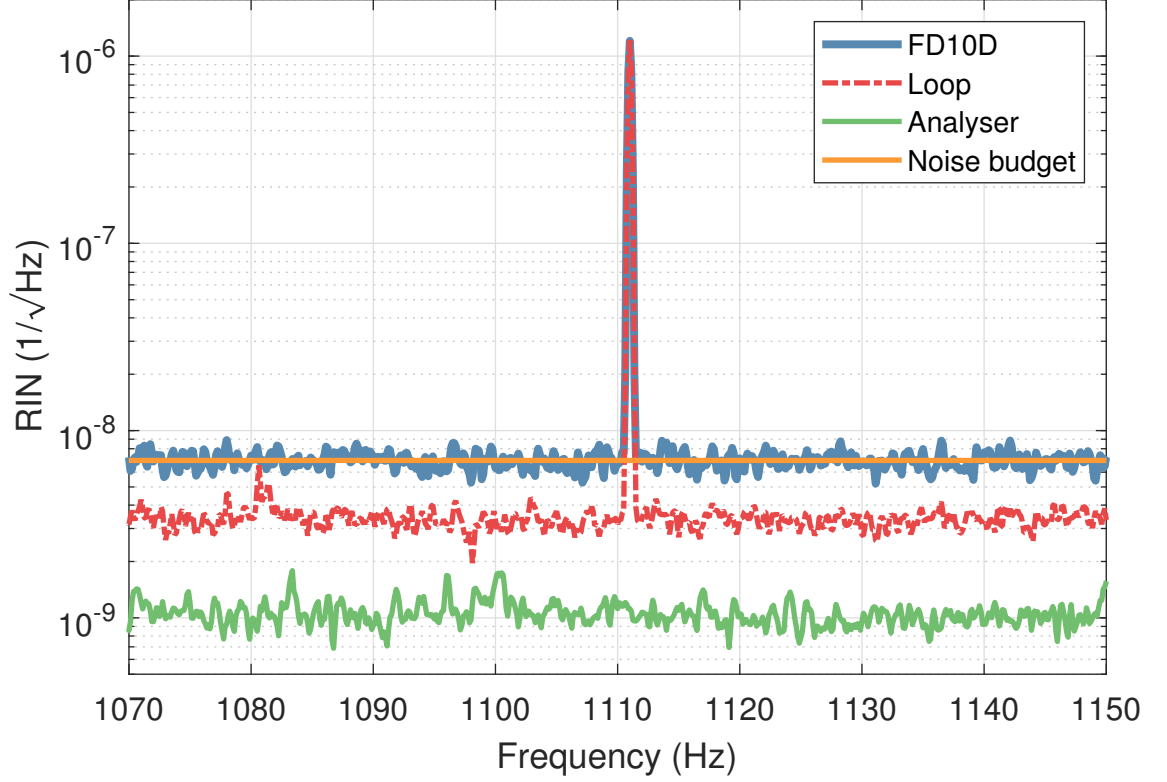


Figure 8.3: The bias of the FD10D was set to 0.6 V, and it was exposed to 15 mW of 1.55 μm light. 15 mA of photocurrent was generated. The noise of the photocurrent produced by the FD10D (blue line) corresponds to the level predicted by Equation 8.3 (orange line) meaning that the increase in photocurrent can only be explained by an increase in quantum efficiency. The 1111 Hz signal was used to calibrate the measurement as this modulation also could be measured on the high-performance in-loop photodiode.

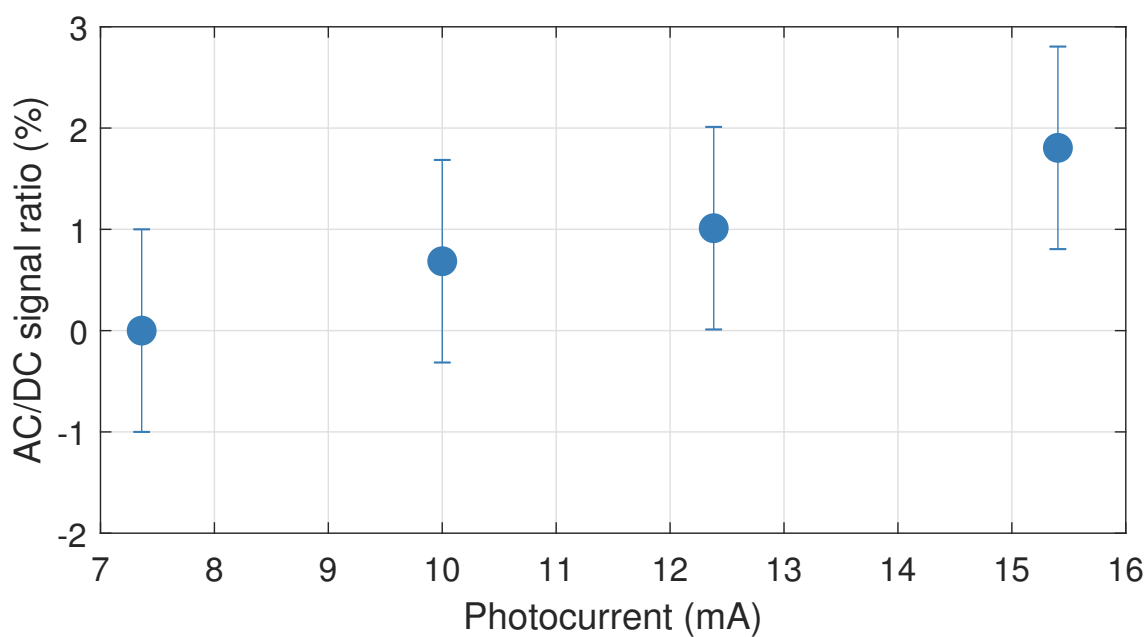


Figure 8.4: The bias of the FD10D was set to 0.6 V, and it was exposed to a series of different power levels to check at what photocurrent the photodiode would saturate at. This was done by measuring the height of the calibration peak; if this peak decreased in size relative to the DC signal then this would indicate the photodiode was saturating. This peak could only be measured to within 1%, and so the FD10D showed no sign of saturation under these conditions.

current noise of at least $0.2 \text{ pA}_{\text{rms}}/\sqrt{\text{Hz}}$ between 10 Hz–1000 Hz.

The photodiodes were placed in a light tight box while the measurements were being made. Measurements of the dark noise of photodiodes at room temperature are reported in Section 8.5.1. As the dark noise of a photodiode depends on its temperature, an experiment was performed to explore this (see Section 8.5.2).

8.5.1 The Dependence of Dark Noise and Dark Current on Reverse Bias in Extended InGaAs Photodiodes

The dark noise of the FD10D as a function of reverse bias was measured in a lab with a stable temperature of $\sim 21^\circ\text{C}$. The dark noise of the FD10D is shown in Figure 8.5. The dark noise for the rest of the photodiodes listed in Table 8.1 can be found in Appendix A.1. The dark noise at 10 Hz and the dark current as a function of bias for each photodiode is shown in Figure 8.6. The noise at 10 Hz was either determined by reading the value from the corresponding graph in Appendix A.1 or by fitting a $1/f^\beta$ slope to the data and inferring the noise from the fitted line. Not all photodiodes had a simple noise spectrum; however, usually, the spectrum would fall as $1/f^{0.5}$.

When there were duplicate photodiodes to choose from, the superior photodiode was selected for characterisation. There can be a significant variation in quality within a batch of photodiodes. Figure 8.7 shows there was a difference of two orders

of magnitude in the dark noise between two nominally identical photodiodes.

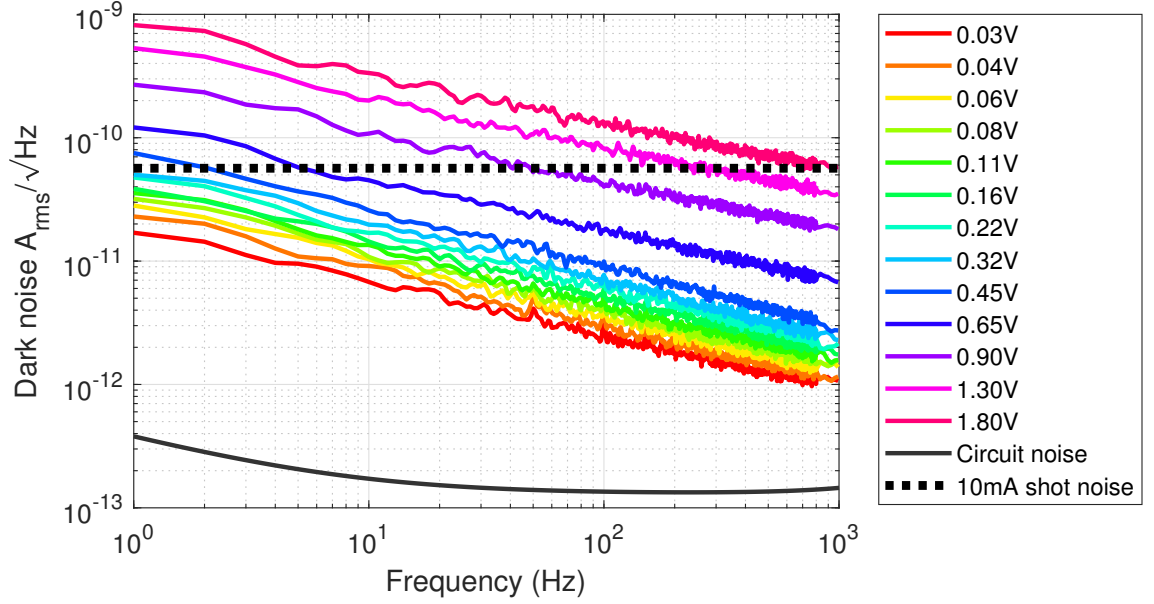


Figure 8.5: The dark noise of the FD10D increases as the reverse bias is increased. Above 0.6 V, the dark noise of the FD10D would limit the noise of a 10 mA current above 10 Hz. As the noise spectral densities have no knees, the function i_c/f^β can be fitted to the data. See appendix A.1 for the dark noise spectra of other photodiodes. Below 0.9 V reverse bias, the spectra fall as $f^{-0.45}$ and above 0.9 V bias the spectra fall as $f^{-0.4}$. The values for β are typical for ‘1/f’ noise [189]. The dark noise at 10 Hz for each reverse bias is shown in Figure 8.6 alongside the results for the other photodiodes that were tested.

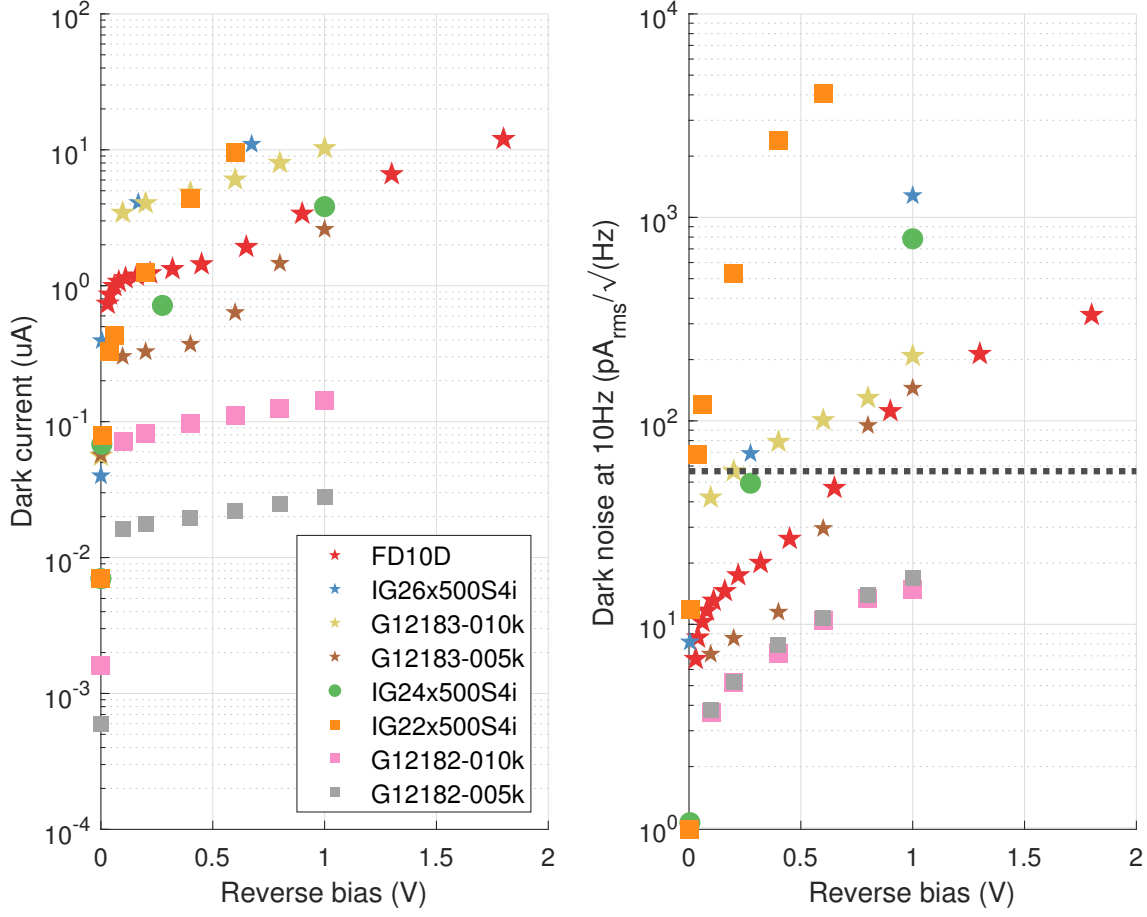


Figure 8.6: Left panel: the dark current for each photodiode was measured. Photodiodes of a similar cut-off wavelength share the same marker shape. The 1 mm photodiodes have larger markers than the 0.5 mm photodiodes. See Table 8.1 for information about the size and cut-off wavelength for each photodiode. Right panel: the dark noise at 10 Hz for each photodiode. The dashed line indicates the level of shot noise for a 10 mA current.

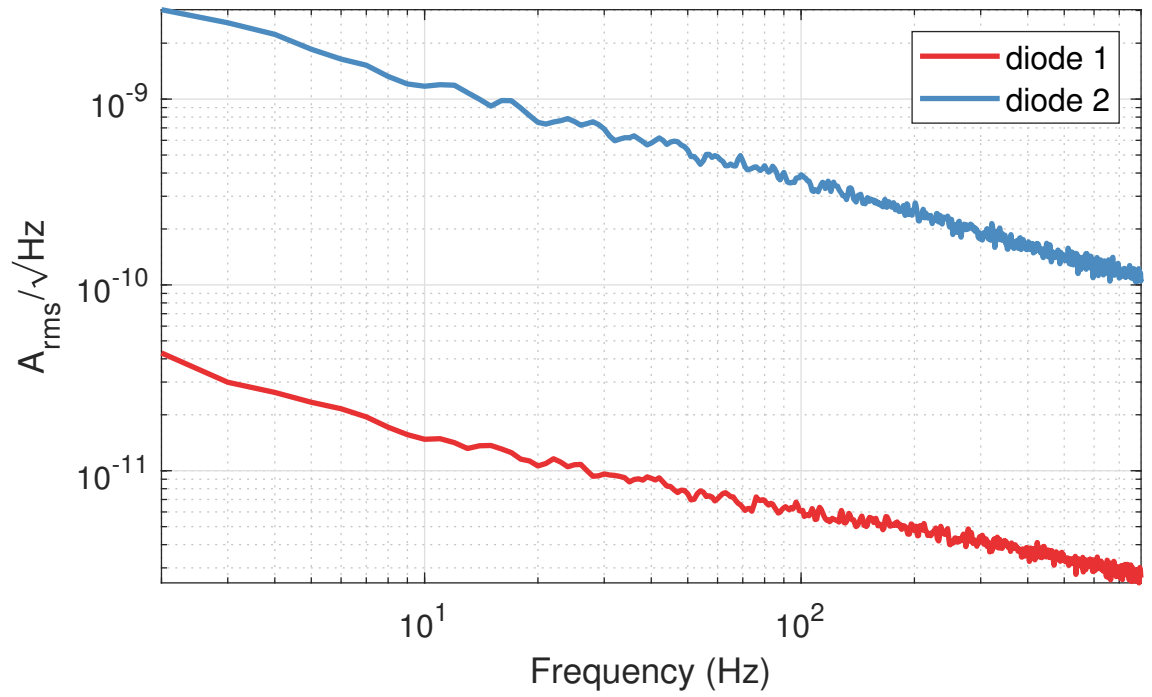


Figure 8.7: The difference in dark noise between two nominally identical photodiodes (g12182-010k) under the same operating conditions (room temperature at 1 V bias). Photodiode 2 performs nearly 100 times better than photodiode 1.

8.5.2 The Dependence of Dark Noise and Dark Current on Temperature in Extended InGaAs Photodiodes

The typical low-temperature damage threshold for extended InGaAs photodiodes is -40°C , so the dark noise of the photodiodes listed in Table 8.1 was measured at a range of temperatures from -40°C to 30°C at the maximum reverse bias specified in their datasheets. The photodiodes were glued to an aluminium block with ICEOxford cryo-varnish. The temperature of the block was controlled with power resistors and an copper rod inserted into a liquid nitrogen bath. The block is shown in Figure 8.8. A computer-controlled power supply (Tenma 72-2540) was connected to the power resistors. A pt1000 temperature sensor and a multimeter (Keithley 2000), configured to make a four-wire resistance measurement, were used to measure the temperature of the block. LabVIEW was used to perform the PID calculation. This program measured the temperature of the block and calculated the required voltage at which to set the power supply for the block to reach the requested temperature.

The dark noise at 10 Hz and the dark current for each photodiode is shown in Figure 8.9. The individual dark noise spectra for each photodiode at each temperature is shown in Appendix A.2. Except for the IG24x500S4i, the dark noise and dark current for each photodiode fell exponentially, as expected [192], as the photodiode was cooled.

The IG24x500S4i had an unexpected knee-like feature in its spectrum. Above

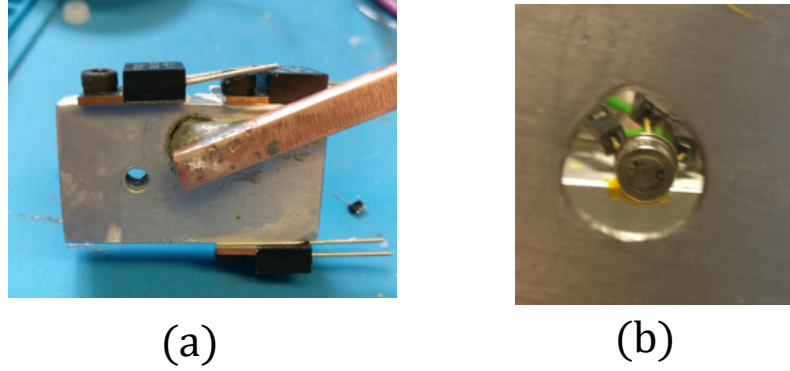


Figure 8.8: (a) The photodiode was affixed to this aluminium block. The power resistors provided heating to the block and the copper rod provided a thermal link between the liquid nitrogen bath and the block. (b) A photodiode glued with ICEOxford cryo-varnish to the block such that a considerable proportion of the photodiode was in contact with it.

10 Hz, the dark noise fell as the IG24x500S4i was cooled, however below 10 Hz the noise spectra were bunched together and had a weak dependence on temperature.

8.6 The Saturation Limit of the IG22x1000S4i, IG24x500S4i and IG26x500S4i

The IG22x1000S4i, IG24x500 and IG26x500 were found to saturate at low power if they were biased with low enough voltage such that their dark noise was below the target. The change in the height of the calibration peak relative to the DC photocurrent was measured (see Section 8.4.1 for detail); this is shown in Figure 8.10. If the

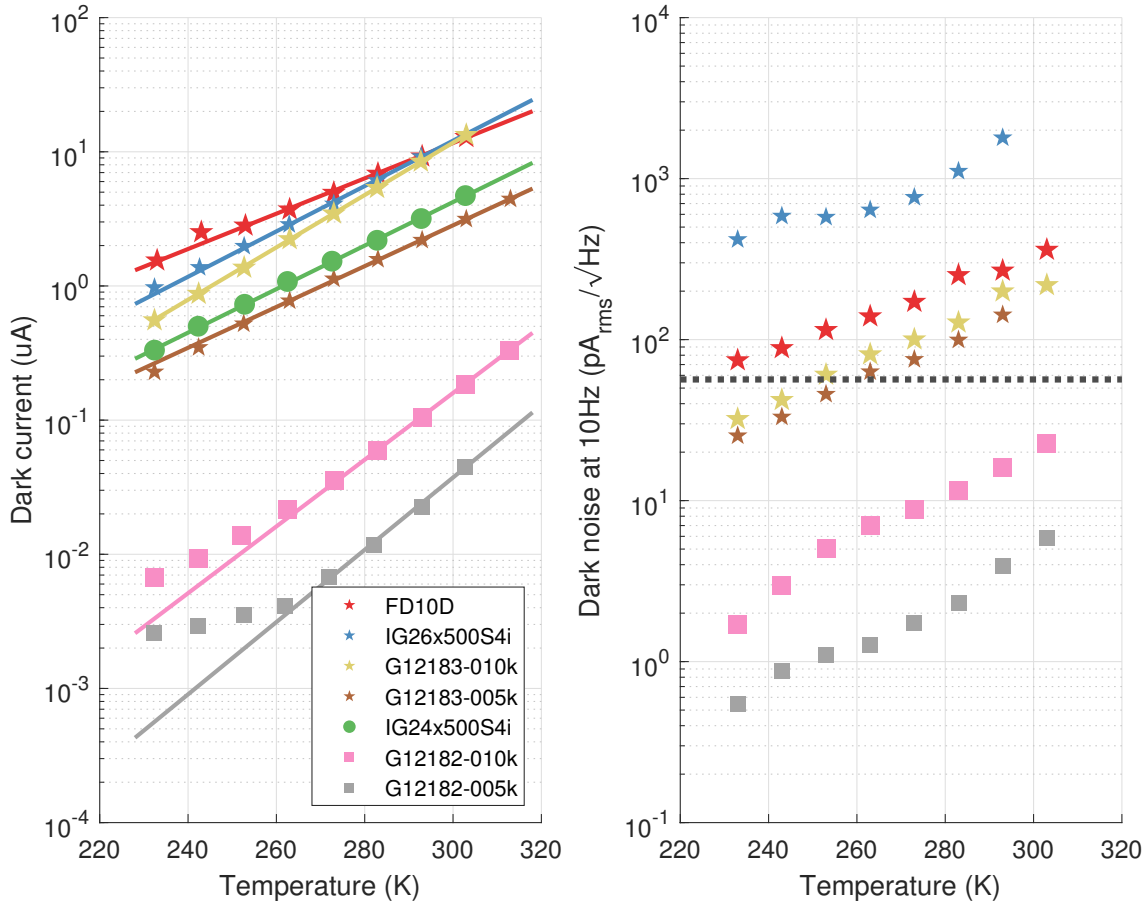


Figure 8.9: Left panel: the dark current for each photodiode at their maximum datasheet-specified reverse bias as a function of temperature. The measurement of the DC dark current approached the resolution of the multimeter around 1 nA. Exponentials were fitted to the data. Photodiodes with similar cut-off wavelengths share marker shapes, and 1 mm photodiodes have larger markers than the 0.5 mm ones. See Table 8.1 for information about the size and cut-off wavelength for each photodiode. Right panel: the dark noise of each photodiode was measured as a function of temperature. The black dashed line indicates the shot noise level for a 10 mA current. At 10 Hz, the measurements for the IG24x500S4i did not change much as the temperature changed because there was a knee around this frequency (see Figure A.13); the noise of the IG24x500S4i was $600 - 700 \text{ pA}/\sqrt{\text{Hz}}$ at 10 Hz.

ratio between the photocurrent and the size of the calibration modulation decreases as the photocurrent increases, this indicates that the photodiode is saturating.

8.7 Excess Noise in the Photocurrent made by the IG24x500S4i

The noise of a 1.3 mA photocurrent produced by the IG24x500S4i exceeded shot noise over a 100 Hz bandwidth around the 1111 kHz calibration modulation by a factor of 1.8. This is shown in Figure 8.11. Although 0.25 mA is already far below the photocurrent requirement described in Section 8.1, the noise was measured at 0.25 mA, and it exceeded the predicted amount by a factor of 1.2. The apparatus used and calibration method are described in Section 8.4.

Since excess broadband noise was observed, there was concern of bias voltage noise coupling to the photocurrent via the photodiode's impedance. Thus, a noise projection measurement was made. For a description of noise projection measurements, see [201]. The transfer function between the voltage at the anode of the photodiode and the output of the transimpedance amplifier was measured while 1 mA of photocurrent was being generated by the photodiode. A calibration modulation was applied to the bias voltage, and the noise at the anode of the photodiode and the noise at the output of the transimpedance amplifier were simultaneously measured. Using this method, 42.5 dB of bias voltage noise was subtracted from

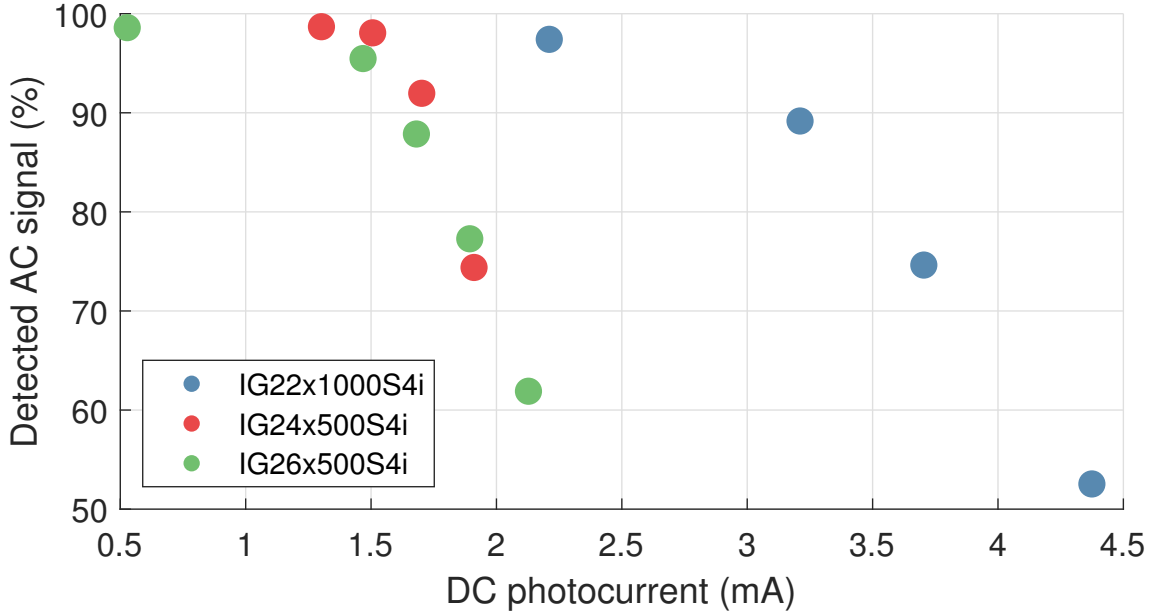


Figure 8.10: The saturation limit of the IG22x1000S4i, IG24x500S4i and IG26x500S4i photodiodes was measured. The photodiodes were biased so that their noise at 10 Hz was equal to the shot noise of a 10 mA current. The IG24x500S4i and IG26x500S4i were exposed to a beam with a 3σ width of 220 μm and the IG22x1000S4i was exposed to a beam with a 3σ width of 1000 μm . The size of the light modulation of known size was tracked as the DC photocurrent was increased. As the photodiodes saturated, the relative size of the AC signal decreased. The uncertainty in these measurements was $\pm 1\%$. These photodiodes had low saturation limits compared to the FD10D, which showed no signs of saturation up to 16.5 mA (see Figure 8.4).

the measurement; however, this did not remove the excess broadband noise. This is shown in Figure 8.12.

The equation of a diode contains a variable called the *ideality factor*. If the ideality factor differs significantly from unity, additional behaviour that was not described in Section 8.2 may be occurring within the photodiode. Thus, the ideality factor of the IG24x500S4i was investigated. The equation of a photodiode is

$$I = I_p + I_s \left(\exp \left(-\frac{qV_b}{nk_B T} \right) - 1 \right), \quad (8.4)$$

where I_p is the photocurrent, I_s is the saturation current of the photodiode, q is the charge of the electron, V_b is the bias of the photodiode, n is the ideality factor, k_B is Boltzmann's constant and T is the temperature of the photodiode. The current flowing through the photodiode as a function of forward bias was measured and is shown in Figure 8.13. The SciPy python function `scipy.optimize.curve_fit` was used to do an unweighted fit of the data to Equation 8.4. The photodiode was found to have $n \simeq 1$. The I_p term in Equation 8.4 was checked by shining light onto the photodiode. This is shown in Figure 8.14.

The source of this excess noise is still unclear: the photodiode was not saturating as the calibration signal was the right level relative to the DC photocurrent, a noise projection measurement showed that the bias voltage noise was not the source of the excess photocurrent noise, and a measurement of the ideality factor was used to determine if the photodiode behaved as an ideal photodiode would. The level of noise cannot be explained by the Johnson-Nyquist noise of the shunt resistance of

the photodiode.

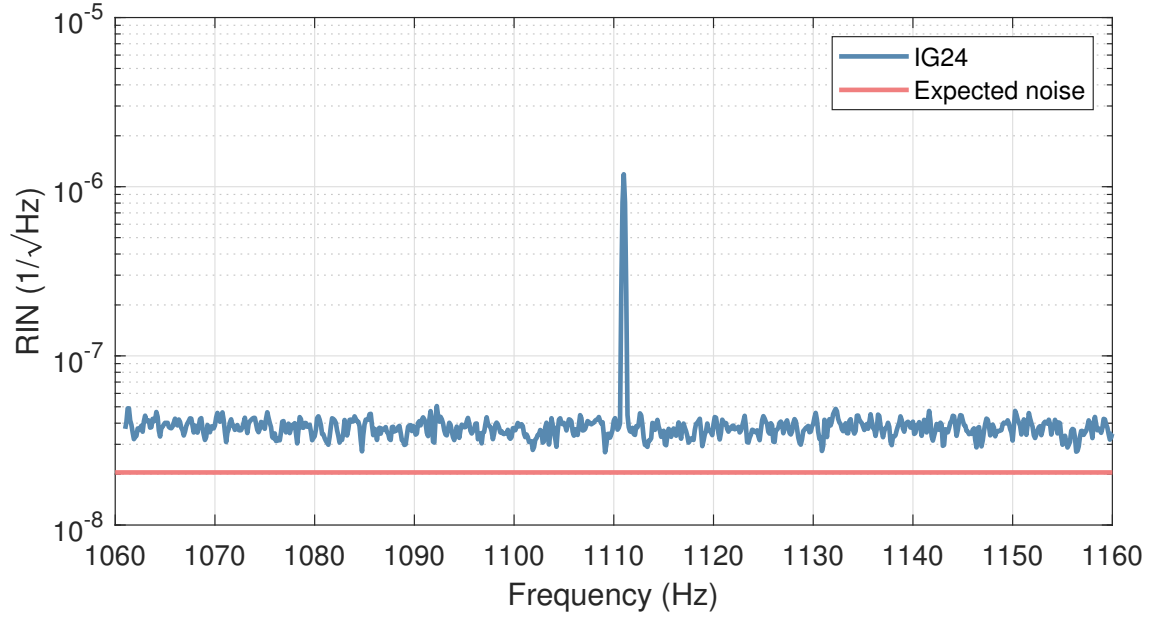


Figure 8.11: The noise of a 1.3 mA photocurrent made by the IG24x500S4i (blue) exceeded the expected amount (red) based on Equation 8.3 by a factor of 1.84. Note that the height of the 1111 Hz peak cannot be expressed in terms of amplitude spectral density units; however, this peak was measured to be the right level, so this beam was not saturating the photodiode.

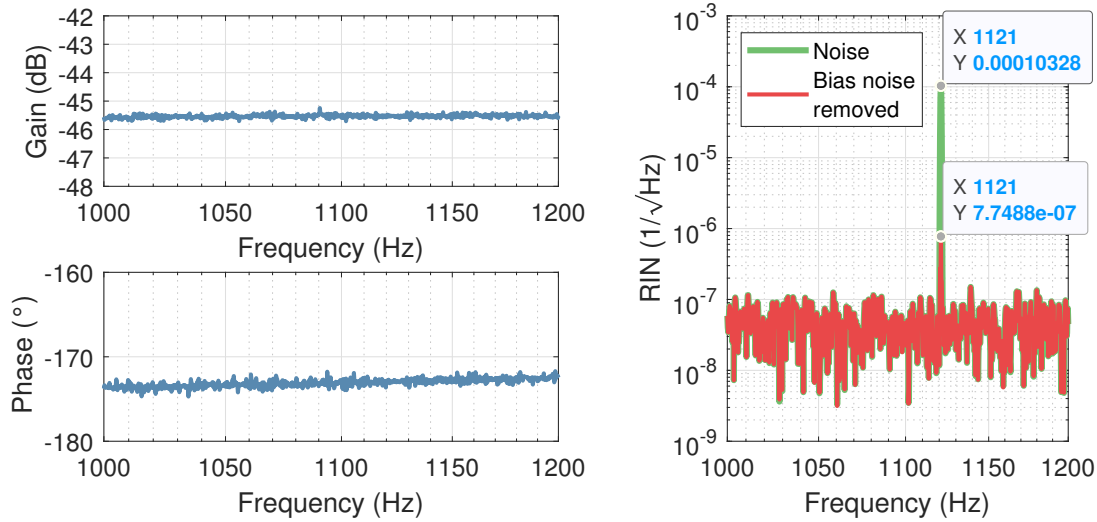


Figure 8.12: A noise projection measurement was performed to determine if the excess noise in the photocurrent produced by the IG24x500S4i (see Figure 8.11) was due to bias noise. The left hand panels show the Bode plot of the transfer function between the photodiode's cathode and the output of the transimpedance amplifier when the photodiode was producing 1.3 mA of photocurrent. The right hand panel shows the effect of subtracting bias noise from the signal at the output of the transimpedance amplifier. The peak at 1121 Hz was a modulation applied to the photodiode's cathode; 42.5 dB of this signal was removed by using the noise projection technique, and this is approximately the amount expected from the measured transfer function (45.5 dB at 1121 Hz). However, the level of the broadband noise remained unchanged thus indicating that the excess noise witnessed in the photocurrent produced by the IG24x500S4i was not due to bias noise.

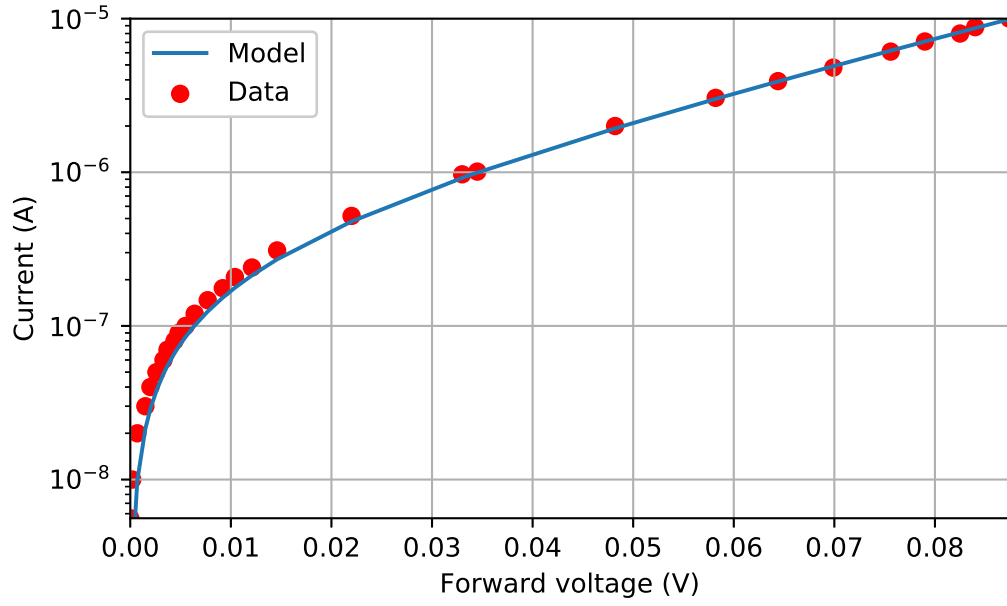


Figure 8.13: To investigate reasons why there was excess noise in the photocurrent produced by the IG24x500S4i, its ideality factor was measured (red points). Equation 8.4 was used to do an unweighted fit to the data ($I_0 = (360 \pm 10) \text{ nA}$ and $n = 1.02 \pm 0.01$). The random error in this measurement $\sim 1\%$, however a systematic error due to contact potential affected the data below 0.02 V . The IG24x500S4i was found to be close to ideal because n was close to unity and so this does not provide clues to the source of excess noise in the photocurrent.

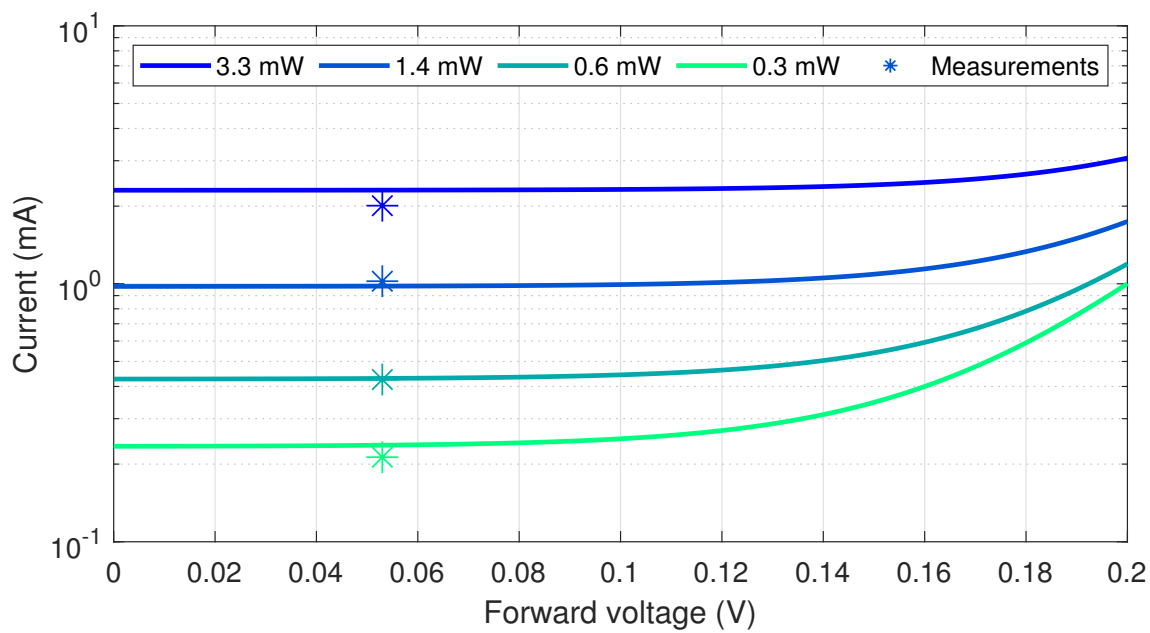


Figure 8.14: Light was shone onto the photodiode and the current produced at 0.05 V forward bias was recorded. These measurements are shown by the star shaped markers. Based on the model obtained from the data shown in Figure 8.13, the current flowing through the photodiode was predicted (shown by the lines). The model and the measurements agree with each other, although the photodiode may have been saturating for the 3.3 mW measurement.

8.8 Noise Properties of an InSb Detector

InSb photodetectors are sensitive to light with wavelengths between $1\text{ }\mu\text{m} - 5\text{ }\mu\text{m}$, and they have been used by the infra-red astronomy community (e.g. [202]). InSb detectors have near unity quantum efficiency above $2.5\text{ }\mu\text{m}$, and below $2.5\text{ }\mu\text{m}$, their quantum efficiency can exceed unity due to multi-electron processes [203]. Because InSb detectors are sensitive to light with wavelengths up to $5\text{ }\mu\text{m}$, they may need shielding from heat sources if they were used in a gravitational wave interferometer. While a quantum efficiency greater than unity does not necessarily correspond to worse noise performance, multi-electron processes can convert some of the detected photon energy to heat i.e. introduce a loss mechanism.

The noise of the photocurrent produced by a Hamamatsu P5968 detector was measured. The transimpedance amplifier used to measure the photocurrent is shown in Section B.1. InSb detectors must be operated at low temperatures, so liquid nitrogen was used to cool the photodiode to 77 K . The detector was characterised by covering its aperture with a low emissivity material (aluminium foil) which was at room temperature. The P5968's window lets light between $1.5\text{ }\mu\text{m} - 5.3\text{ }\mu\text{m}$ pass. The viewing angle of the detector was 60° ; based on the solid angle covered by the material as seen by the detector, the expected level of thermal radiation that would be detected could be calculated. The expected photocurrent due to the aluminium foil was $5.5\text{ }\mu\text{A}$. The DC current from the device was measured to be $5.3\text{ }\mu\text{A}$, so the photocurrent can be explained by the thermal radiation of the aluminium foil. The results from this test are shown in Figure 8.15. This corresponds to the device

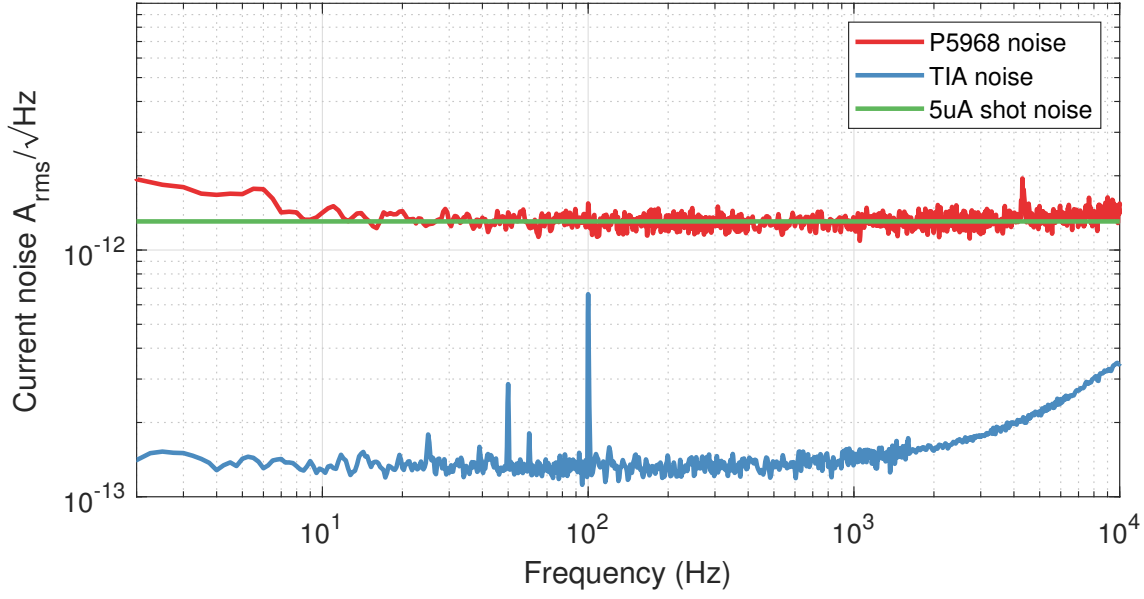


Figure 8.15: The P5968 InSb detector had its cathode grounded, and the noise of the photocurrent due to the blackbody radiation produced by room temperature aluminium was measured (red). The expected level of photocurrent was calculated to be $5.5\text{ }\mu\text{A}$ and the measured photocurrent was $5.3\text{ }\mu\text{A}$. The noise level of the photocurrent corresponds to shot noise (green). The noise of the transimpedance amplifier (blue) is well below the shot noise of the photocurrent.

having a quantum efficiency of $\sim 90\%$.

The behaviour of the P5968 under a reverse bias was investigated as the quantum efficiency of InSb photodetectors depends strongly on the reverse bias for wavelengths of $1.55\text{ }\mu\text{m}$ [204]. A reverse bias was applied to the device, and the P5968 exhibited $1/f$ noise. The full data are shown in Section A.3. Above 0.2 V reverse bias, the noise of the photocurrent became susceptible to bias voltage as shown in Figure 8.16.

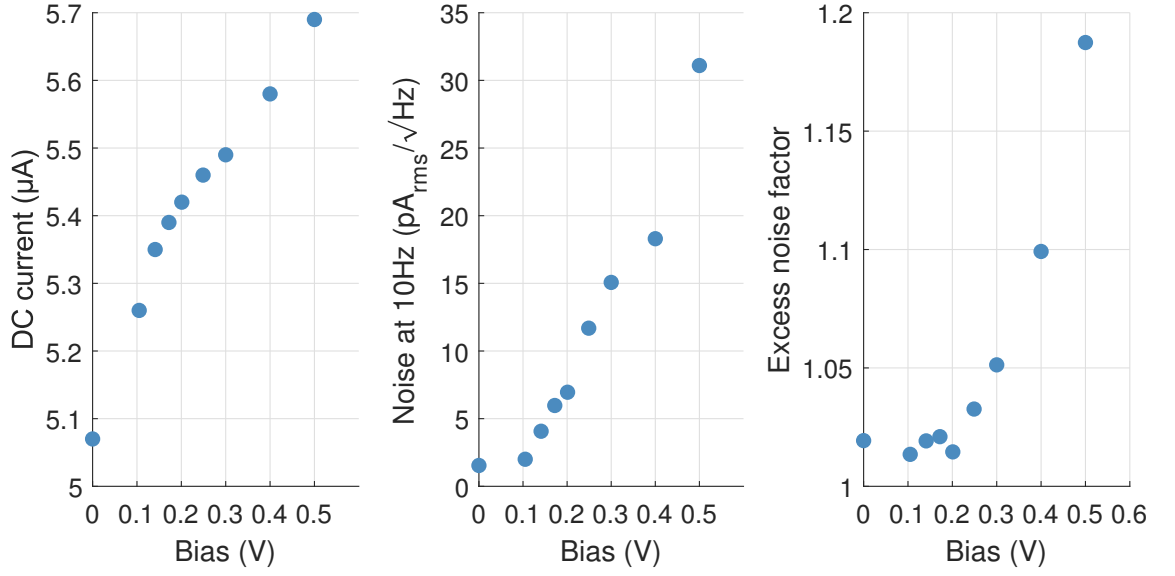


Figure 8.16: Left panel: the DC dark current of the P5968 as a function of bias. Middle panel: $1/f$ noise rose as the bias of the P5968 was increased (see Figure A.15). Right panel: At a frequency where $1/f$ was not present, the noise of the photocurrent was measured. The excess noise factor is the multiplicative factor that the shot noise corresponding to the DC photocurrent needs to be multiplied by to get the noise level that was measured.

8.9 Discussion

None of the photodiodes tested in this chapter would meet the requirements described in Section 8.1 because the quantum efficiency is specified in all of their datasheets to be below 99%. Communication with four commercial manufacturers of extended InGaAs photodiodes suggest that there is no off-the-shelf photodiode capable of meeting the requirements set out in Section 8.1 [71–74]; the results in Section 8.4 and Section 8.5 support this and give insights into the best possible operating conditions for an off-the-shelf extended InGaAs photodiode. As discussed in Section 8.2, high quality extended InGaAs photodiodes are difficult to fabricate because of the lattice mismatch between the InGaAs and the InP substrate.

As only two copies of most photodiodes were available for testing, batch-level statistics cannot be obtained from these results. For instance, the FD10D reported on in this chapter has lower dark noise than the G12183-010k; however, this does not tell one if FD10Ds are generally better than G12183-010ks. Due to the nature of semiconductors with high amounts of strain, inter and intra wafer variability is expected. For example, there was a factor of ~ 100 between the dark noise of the two G12182-010ks that were tested (see Figure 8.7). Whether this is due to their manufacturing or how they were handled prior to purchase is unclear, however this result shows that it may be worth buying several extended InGaAs photodiodes and selecting the best ones from them. Additionally, the results only cover seven different models of extended InGaAs photodiode, and it is possible that another photodiode manufacturer will have better performing photodiodes than the ones tested in this

chapter.

Section 8.4 shows the effect that biasing an extended InGaAs photodiode can have on its quantum efficiency. The quantum efficiency of the FD10D was increased by 15% when its bias was set to its maximum value. This increase in quantum efficiency occurs because the depletion region of the FD10D was made larger by applying a reverse bias, and because the electron-hole pairs that are excited by the incoming photons spend less time within the depletion region when the bias is larger, thus they have a smaller chance of recombining with another hole/electron. This motivates using as large a bias as possible as the quantum efficiency can be greatly increased.

The high dark noise associated with the bias found in the IG24x500S4i and IG26x500S4i photodiodes (see Section 8.6) meant that they had to be operated with low biases if they were to produce shot noise limited photocurrent at frequencies above 10 Hz. This resulted in them having a low saturation level, see Figure 8.10. To avoid this saturation limit, one could split the beam and use an array of photodiodes to detect the light. However, due to each photodiode in the array contributing the same amount of dark noise, each photodiode would need to have less dark noise compared to if it was detected on one photodiode⁴. Instead, better performing photodiodes could be used; for example, the FD10D showed no signs of saturation, see Figure 8.4, when operating with a bias that fulfilled the noise criteria.

⁴Splitting the light onto an array of k photodiodes makes the SNR a factor of $\sqrt{k}$ worse. As electronic noise adds in quadrature, the SNR would be $S/\sqrt{kn}$, where n is the noise of an individual photodiode.

Applying a reverse bias to a photodiode reduces its capacitance, however the few hundred picofarads of capacitance that photodiodes have at low bias voltage is a manageable amount to design a transimpedance amplifier around for a DC readout or BHD (see Chapter 7) scheme. For instance, the photodiode circuit used in Section 8.4 (see Appendix B.2) has a large enough bandwidth for a photodiode used in gravitational wave interferometer and was quiet enough to be limited by the shot noise of a 10 mA photocurrent over a bandwidth relevant in ground-based gravitational wave detection.

While the dark current does not depend strongly on the reverse bias when the bias is below the tunnelling threshold, the dark noise of every photodiode increased as its reverse bias increased. Although photodiodes with higher dark current tended to have higher dark noise, the dark noise could not be predicted from the dark current. For instance, the FD10D had a larger dark current but lower noise than the IG24x500S4i. The dark current tended to be greater in larger photodiodes; this is due to larger photodiodes being more likely to have a greater number of defects as well than having more edge-related current. However, the IG24x500S4i and IG26x500S4i photodiodes had higher dark noise.

As the dark current and dark noise of a photodiode increases exponentially with temperature, the dark noise of a photodiode can be significantly reduced by cooling it. Extended InGaAs photodiodes can be purchased with thermoelectric coolers. Once the quantum efficiency and dark noise problems surrounding extended InGaAs photodiodes have been solved, the integration of such a photodiode into a vacuum system could be investigated.

8.10 Conclusion

Extended InGaAs photodiodes are widely available detectors that can be used to sense $2\text{ }\mu\text{m}$ light; however, off-the-shelf detectors do not fulfil the requirements of a photodiode that can be used in a gravitational wave detector. In particular, this is due to them not having a large enough quantum efficiency, and this leads to a relative increase in the shot noise of the detected light.

It is technically challenging to create extended InGaAs photodiodes with high quantum efficiency due to the lattice mismatch between the InP substrate and the InGaAs. While the depletion region of a photodiode can be made larger by running the photodiode with a reverse bias, the dark noise of an extended InGaAs photodiode increases when reverse biased. Measurements of the dark noise as a function of reverse bias (see Figure 8.6) and temperature (see Figure 8.9) were made to find the operating conditions which balances quantum efficiency and dark noise.

To overcome the manufacturing challenges associated with creating extended InGaAs photodiodes, a wafer would need to be commissioned. Increases in quantum efficiency can be obtained by coating the photodiodes in an anti-reflective film; however, the primary challenge of creating high quantum efficiency photodiodes is making them thick enough while keeping defect level low. Reference [205] shows results from a photodiode with 95% quantum efficiency and a cut-off wavelength of $2.4\text{ }\mu\text{m}$, however due to the exponential dependence of absorption on thickness, it is still a significant challenge to reach 99% quantum efficiency. Future work may

include investigating the growth conditions for extended InGaAs photodiodes.

Alternative photodiode materials could be used to detect $2\text{ }\mu\text{m}$ light, and a range of cryogenic photodetectors have been used in infrared telescopes. We investigated InSb detectors (see Section 8.8) as they have near unity quantum efficiency above $2.5\text{ }\mu\text{m}$, however its characteristics at $2\text{ }\mu\text{m}$ are still unclear. To investigate these, the device's window would need removing and a $2\text{ }\mu\text{m}$ laser with several milliwatts of power would be needed because the device was not sufficiently sensitive to the laser that was available. HgCdTe is another material used to detect infrared light, and HgCdTe detectors under development by colleagues [206] could be used to make a photodiode with at least 99% quantum efficiency at $2\text{ }\mu\text{m}$ as HgCdTe does not suffer from strain related issues; however, the optimal growth conditions for these devices has not been established, and the linearity and dark noise of such HgCdTe photodiodes is unknown.

Chapter 9

Conclusion

Appendix A

Extended InGaAs Diodes Noise Spectra

A.1 Bias Section

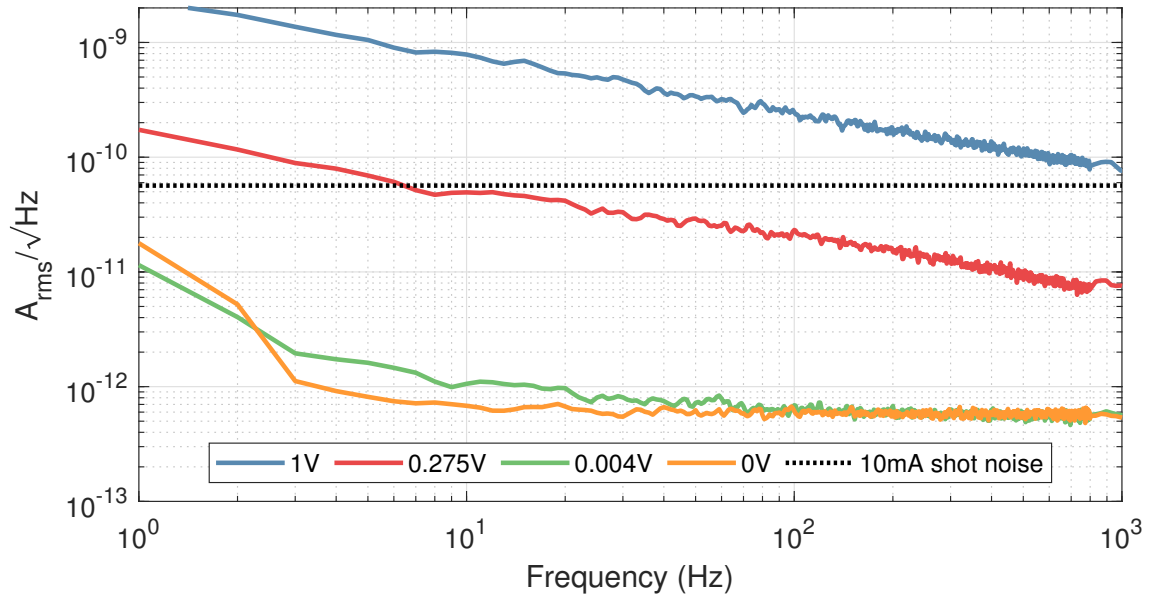


Figure A.1: IG24

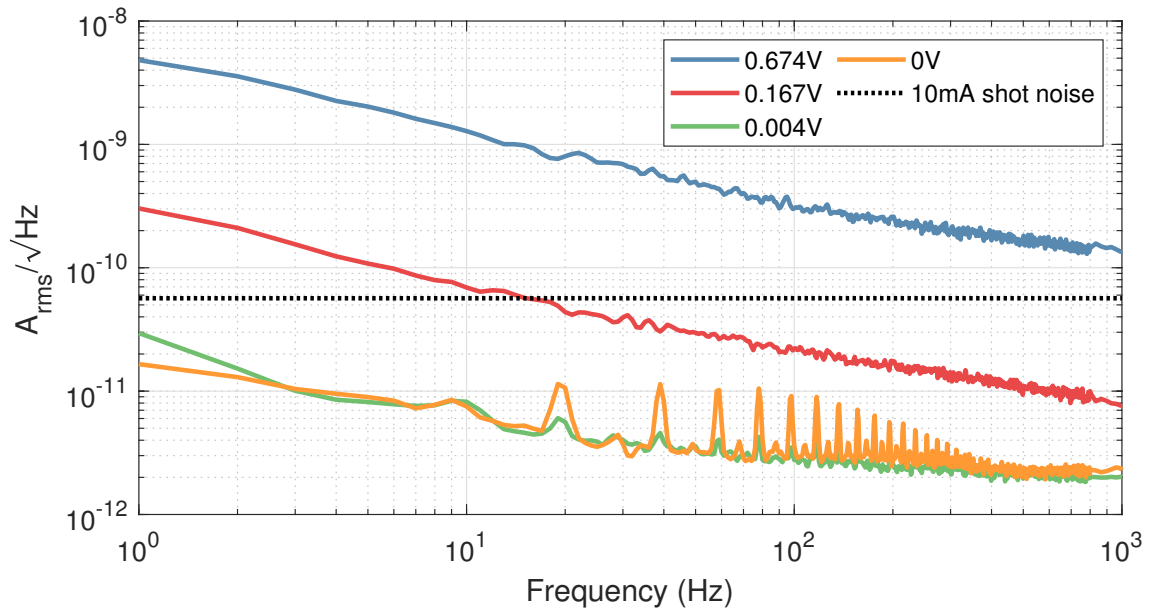


Figure A.2: IG26

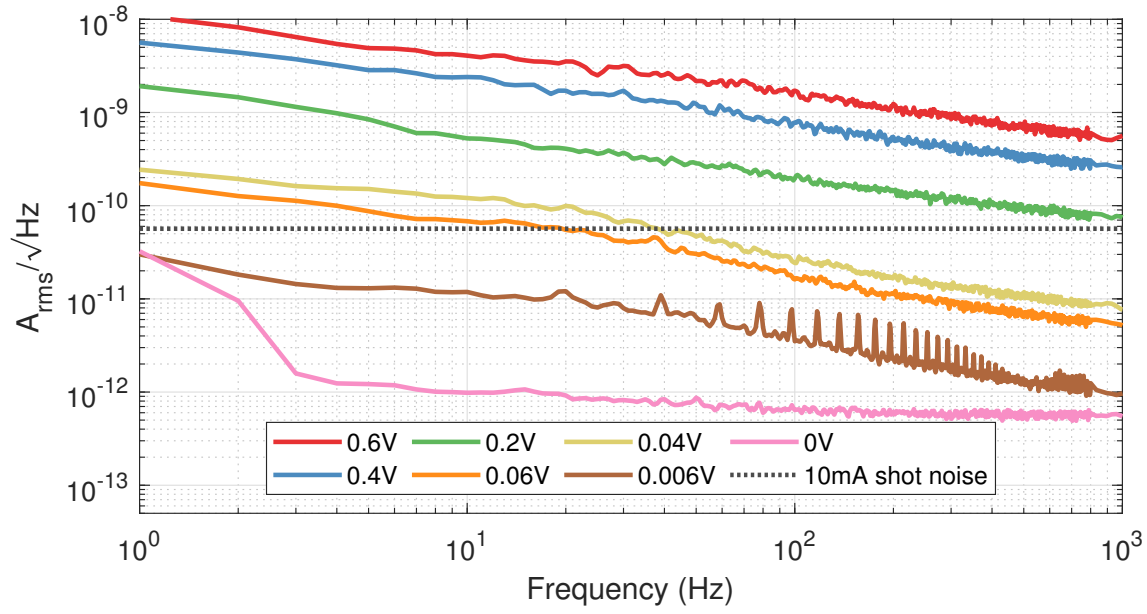


Figure A.3: IG22

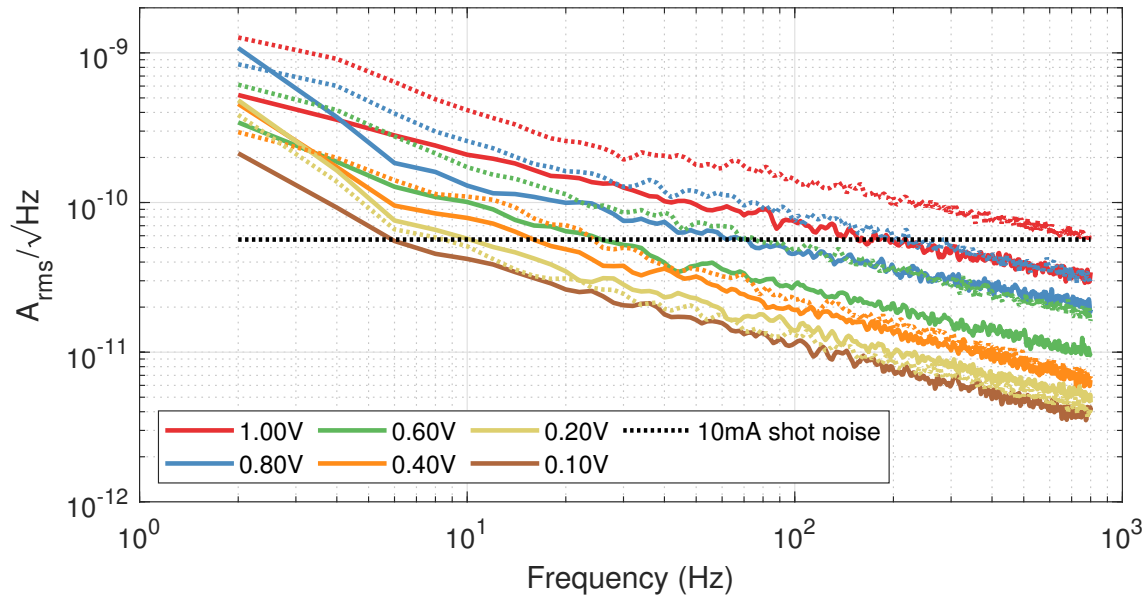


Figure A.4: G12183-010k

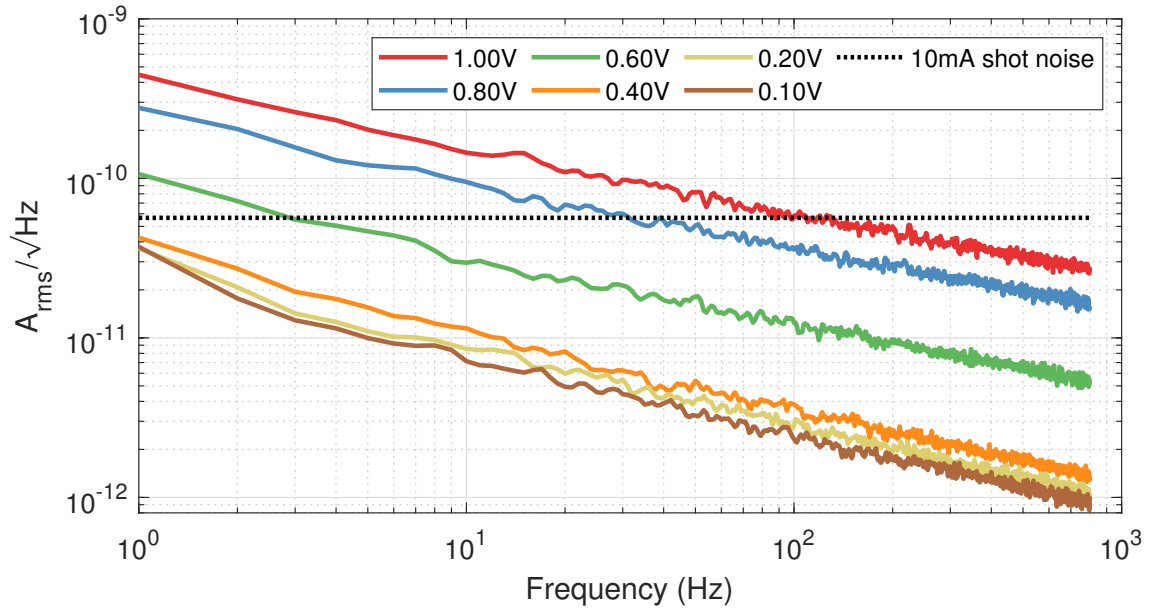


Figure A.5: G12183-005k

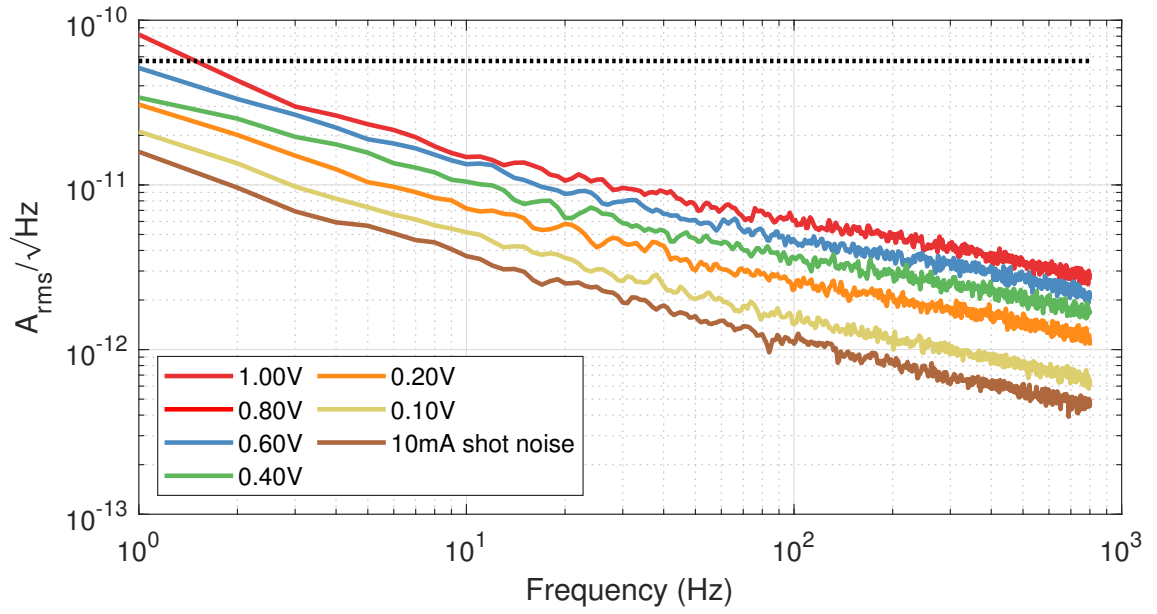


Figure A.6: G12182-010k

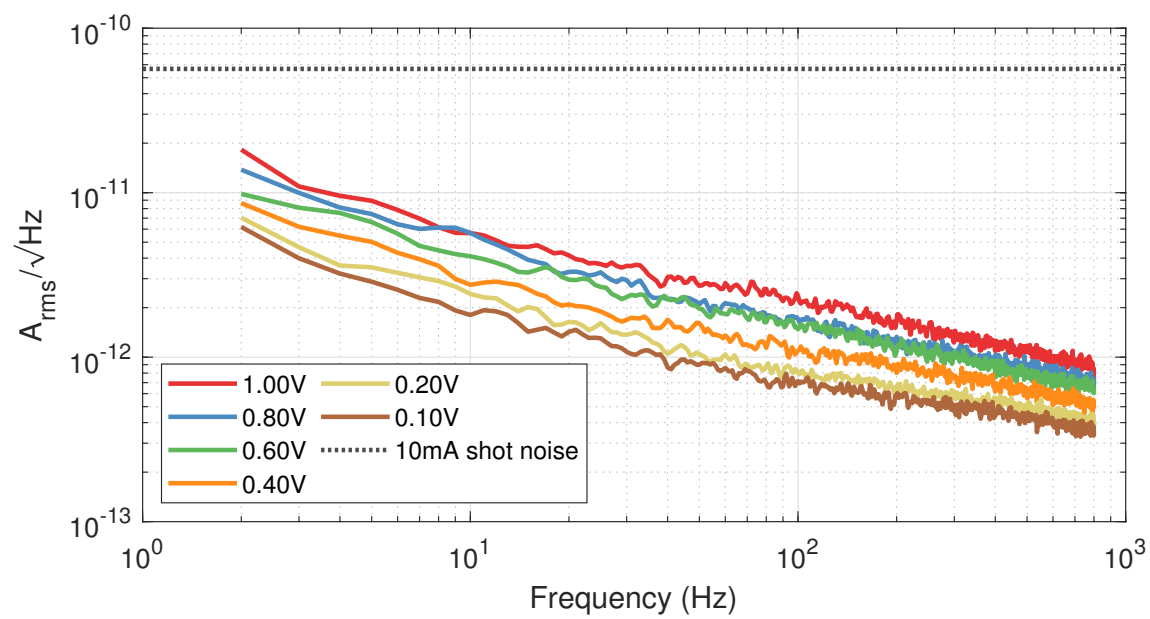


Figure A.7: G12182-005k

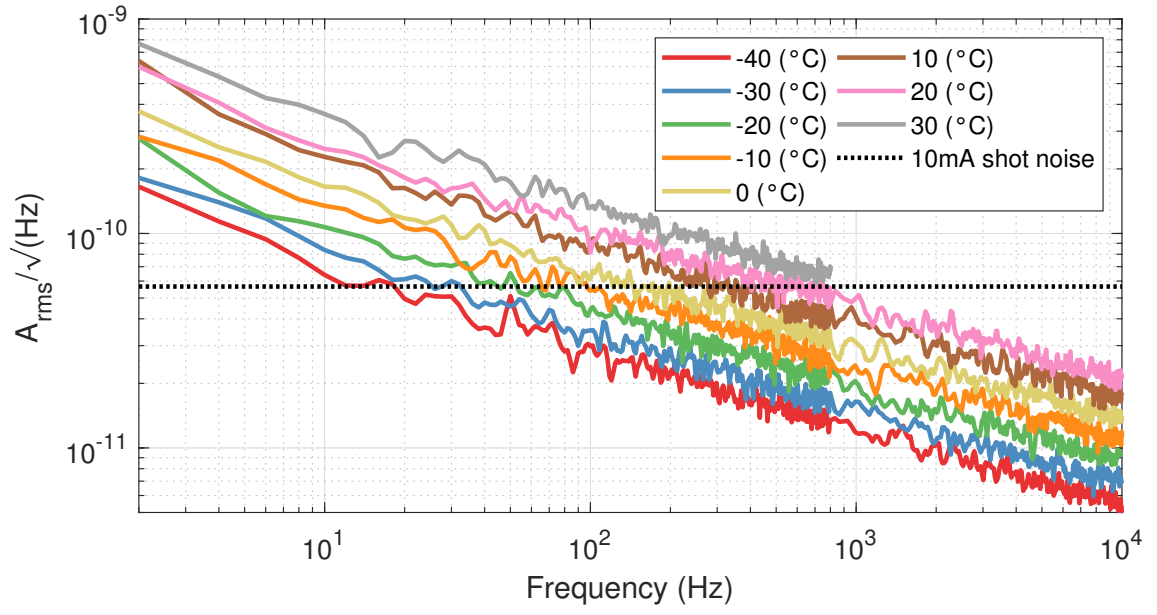


Figure A.8: FD10D 1.8V reverse bias

A.2 Temperature Section

A.3 InSb

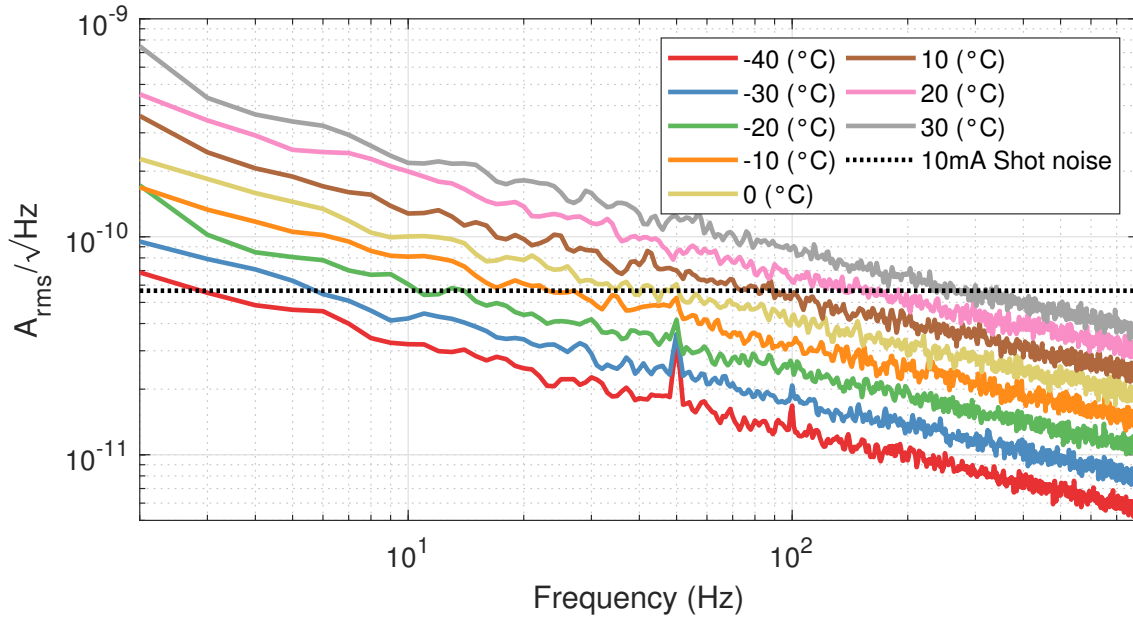


Figure A.9: G12183-010k 1.0V reverse bias

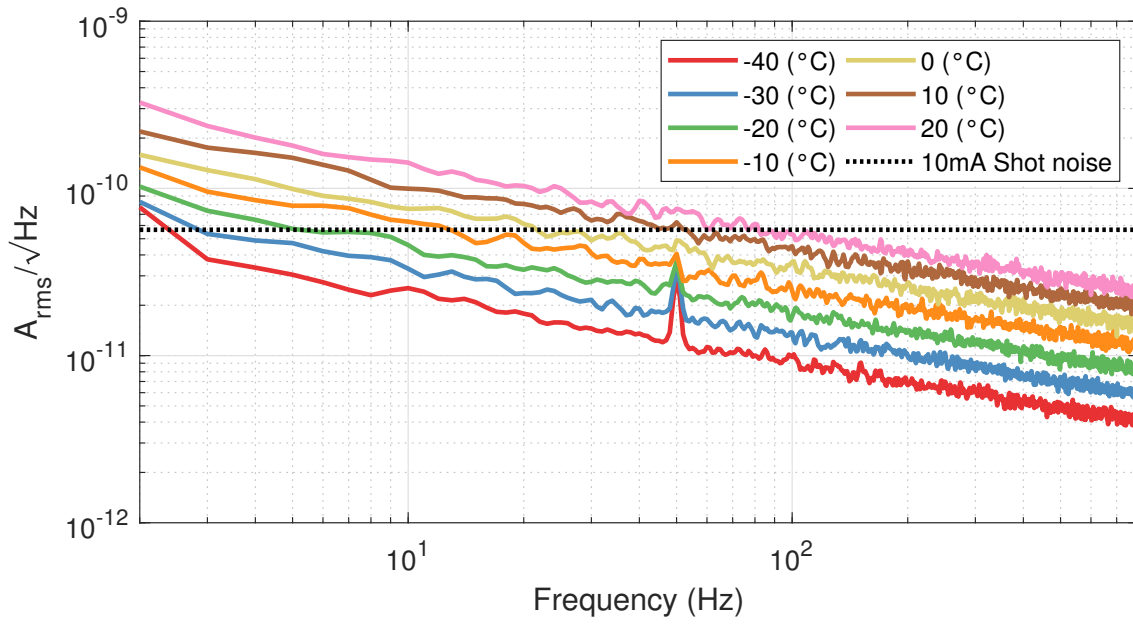


Figure A.10: G12183-005k 1.0V reverse bias

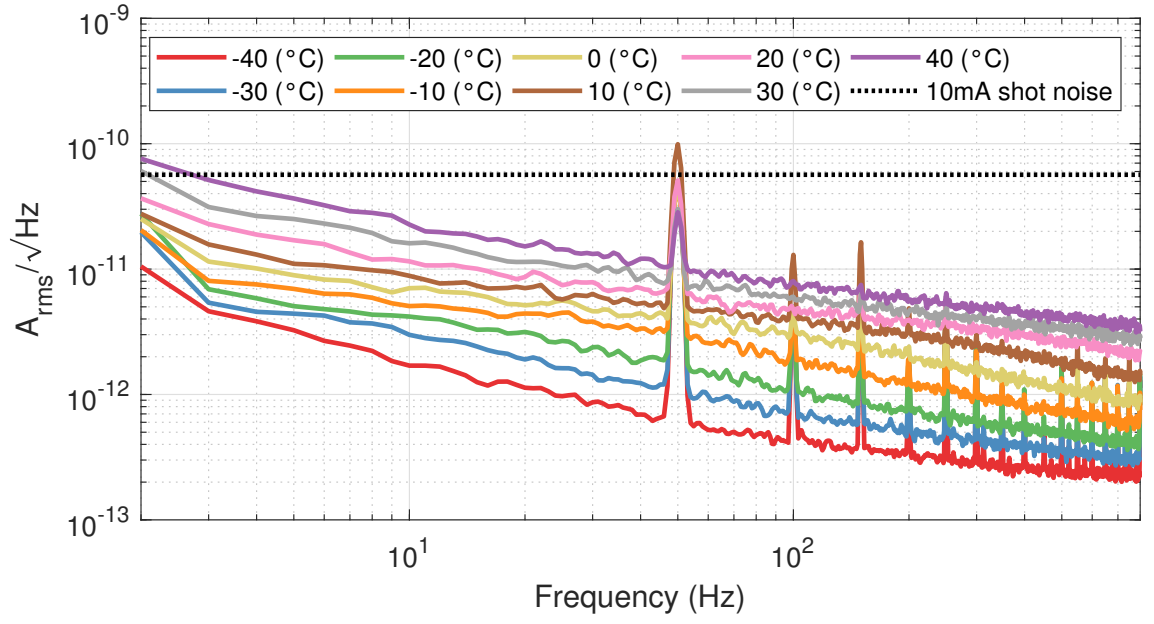


Figure A.11: G12182-010k 1.0V reverse bias

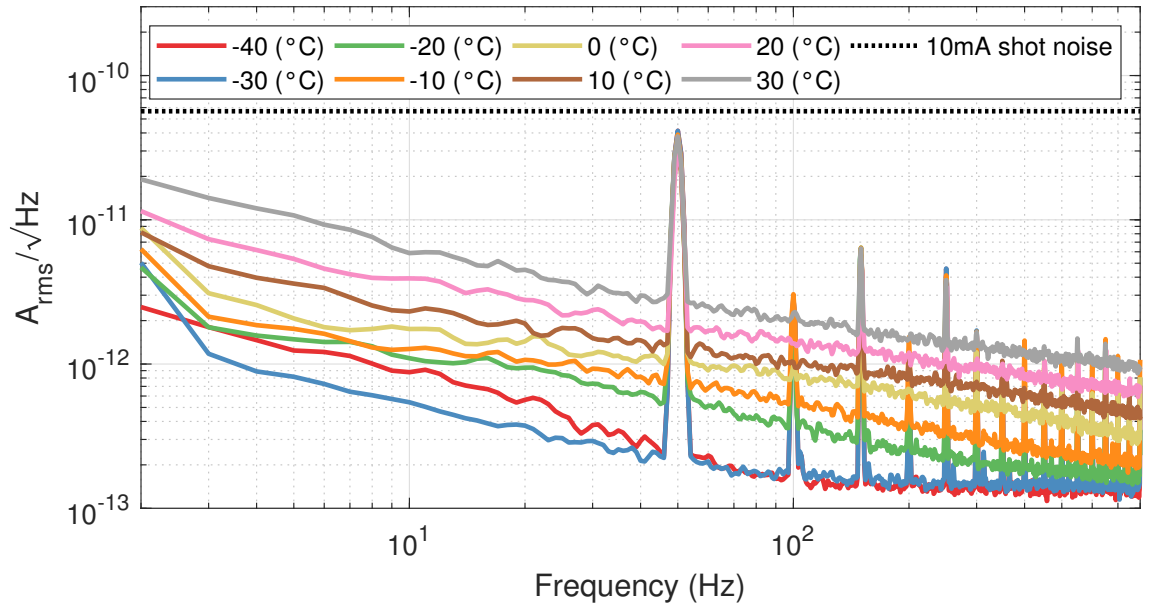


Figure A.12: G12182-005k 1.0V reverse bias

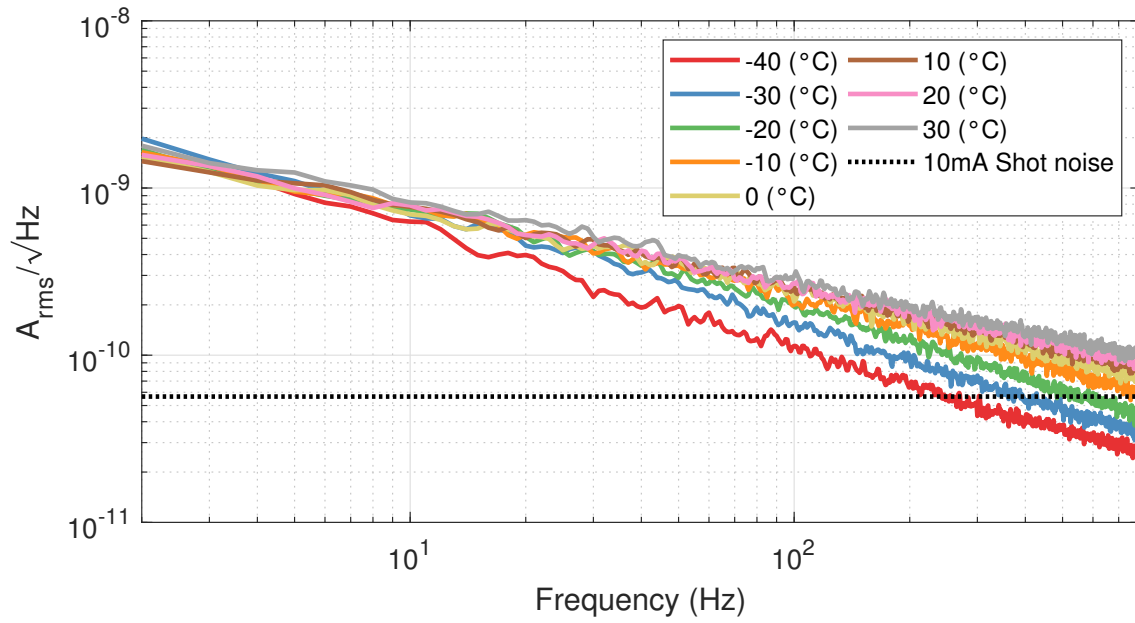


Figure A.13: IG24x500S4i 1.0V reverse bias

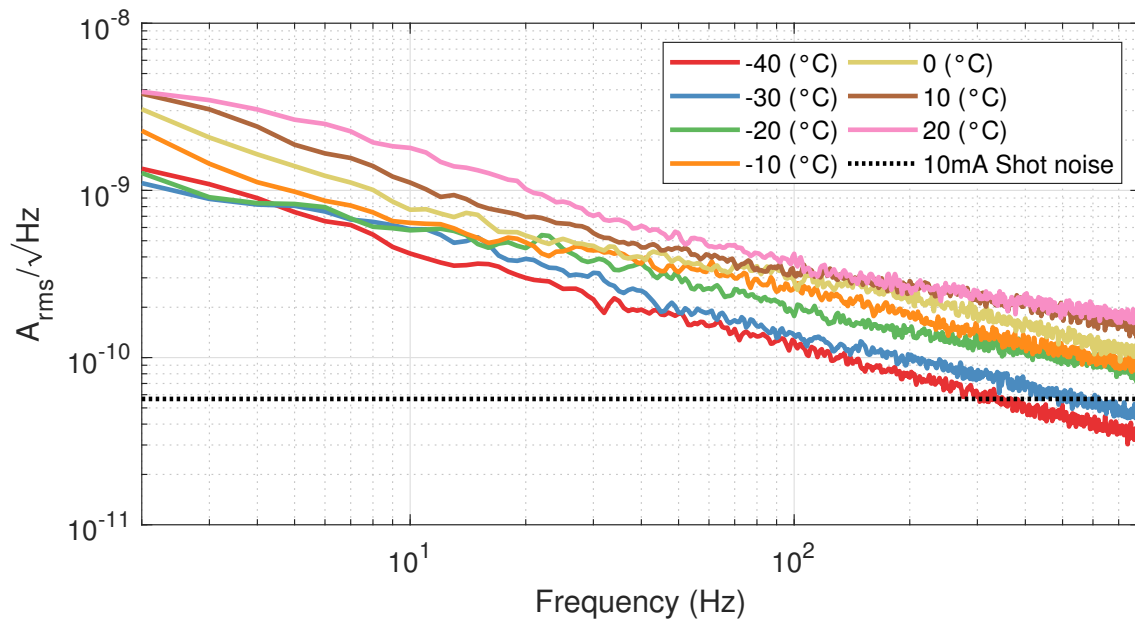


Figure A.14: IG26x500S4i 1.0V reverse bias

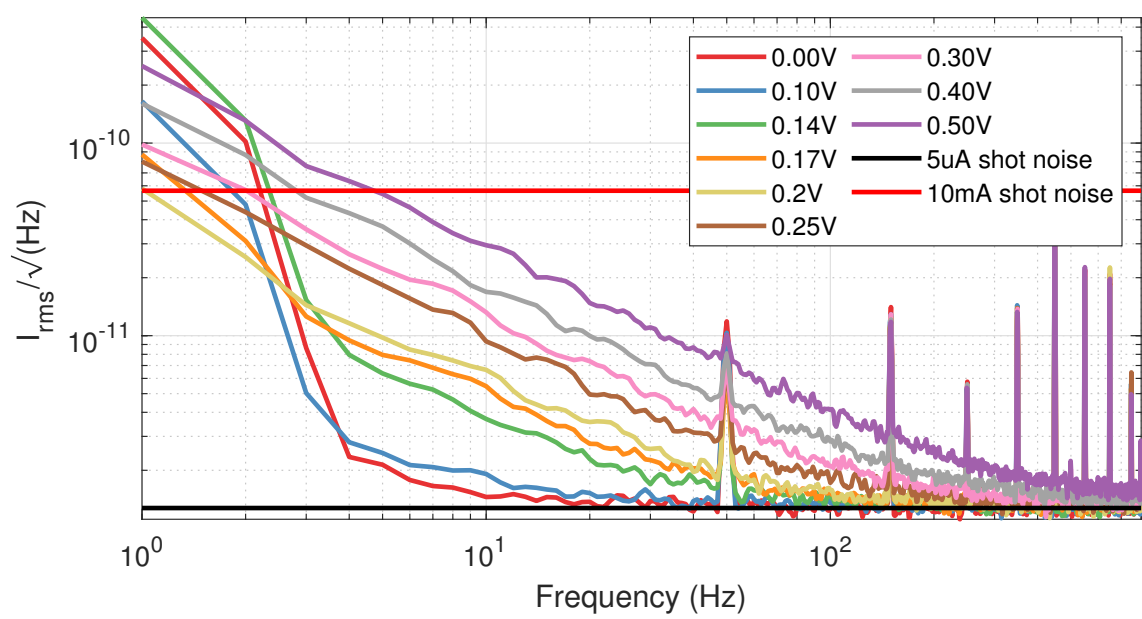


Figure A.15

Appendix B

Photodiode Circuits

B.1 Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes

A circuit which allowed for the reverse bias to be set and for the easy exchange of photodiodes was constructed. To measure small amounts of current noise, a transimpedance amplifier with a large gain is needed. As the dark current for these diodes is of the order of 1mA, the gain of the transimpedance amplifier was chosen to be 1×10^6 so that signals of the order 1 V would be produced, and so a 1 M Ω resistor was used as in the feedback path.

Operation amplifiers have current noise at their inputs, voltage noise at the their outputs and the feedback resistor will generate Johnson-Nyquist noise (e.g. [122, Figure 8.58]). Designers of OP amps, generally, improve the OP amps output voltage noise at the expense of the input current noise or vice versa and so there is not one-size-fits-all op amp. In this application, as the current noise that is to be measured is small and the feedback resistor is large, a TL071, a JFET style op amp with low input current noise, was selected. As the photodiode has capacitance and the op amp has finite bandwidth, a 2.2pF capacitor in parallel to keep the circuit stable.

Several methods of providing the reverse bias were tested. The first of these derived the bias from a LM7815 regulator in combination with a variable resistor configured a potential divider; this circuit is shown in Figure B.1. A similar circuit which was used in later experiments is shown in Figure B.2.

To verify that this was a sufficiently quiet bias voltage source that does not contaminate the measurements with noise, an ultra-quiet bias reference was used to cross check a measurement; this circuit is shown in Figure B.2. By filtering the voltage made a reference voltage IC (AD587JNZ), this circuit provided a bias with $5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz. The bias of the FD10D (see Section ??) was set to 1 V in both circuits. Figure B.4 shows that the regulator-potentiometer bias voltage source is sufficiently quiet.

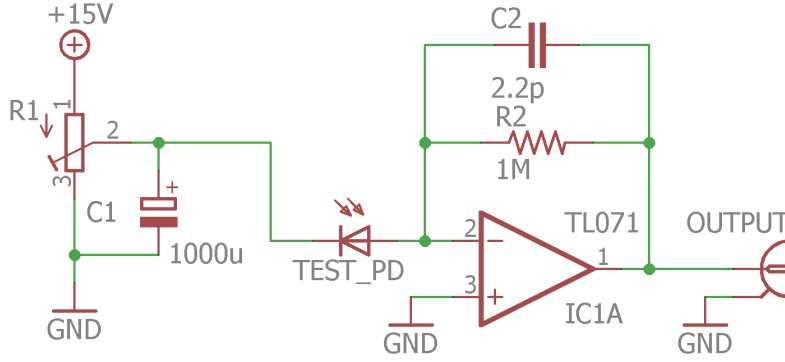


Figure B.1: Circuit used to measure the dark noise of photodiodes, represented by TEST_PD, under different reverse biases. The transimpedance amplifier has a $1\text{M}\Omega$ (R2) in parallel with a 2.2pF capacitor (C2) in the feedback path of the TL071 op amp (TL071P). The reverse bias was generated by a potentiometer (R1) connected to an LM7815 voltage regulator. This voltage was low passed using C1

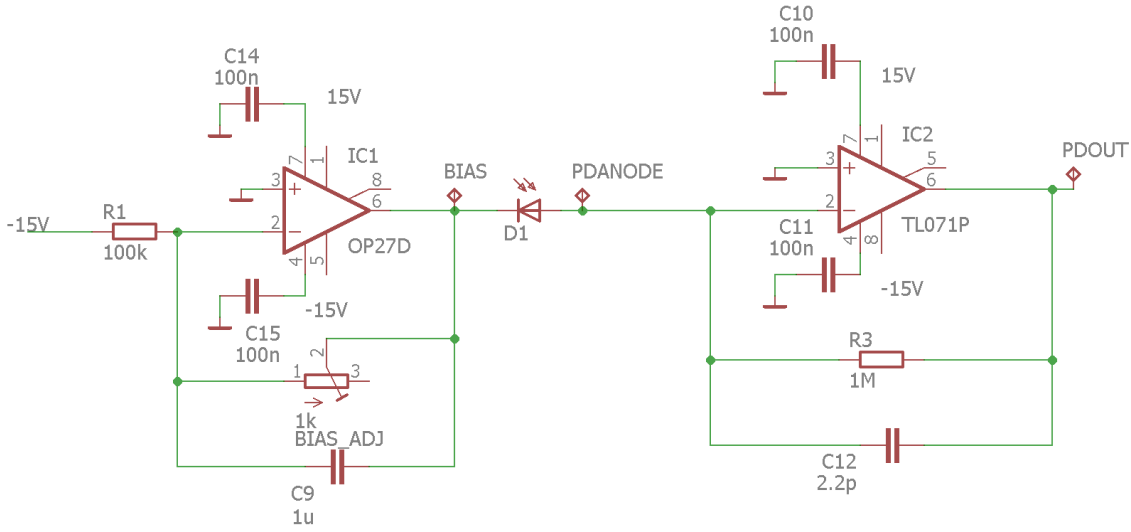


Figure B.2: Circuit used to measure the dark noise of photodiodes. The transimpedance amplifier used is identical to the one shown in Figure B.1. The bias voltage was generated using an OP27 op amp (OP27D) configured with so that the gain between the inverting input and the output of OP27D was > 1 .

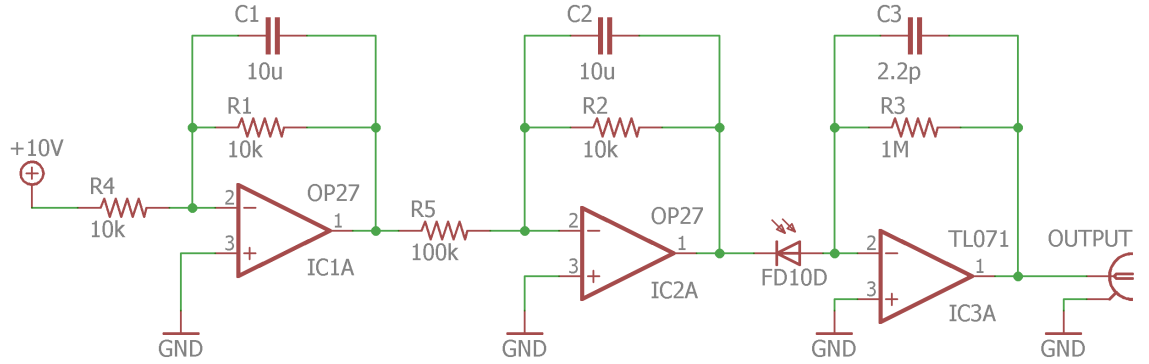


Figure B.3: Circuit used to check that bias voltage noise was not contaminating dark noise measurements. To reach $5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz, a 10V low noise reference IC (AD587JNZ) was used to generate the reverse bias. The output of this IC was filtered by two low pass filters (IC1A and IC2A). The transimpedance amplifier used is identical to the one shown in Figure B.1

B.1.1 Transfer function

The transfer function of the transimpedance amplifiers shown in Figure ?? was measured under a variety of different physical constructions. This circuit's behaviour can be affected by stray capacitance. When constructed from strip board and with the photodiode anode and cathode cables very near each other, there was of the order 10 pF of stray capacitance i.e. it was as if the circuit had an extra 10 pF capacitor between the inverting input of the op amp and ground. By making this circuit on a prototyping board with less stray capacitance between the strips and by keeping the photodiode anode and cathode wires from forming a capacitor, the stray capacitance was reduced. The measurement is in agreement with the LISO simulation of this circuit. This was cross checked using the method described in

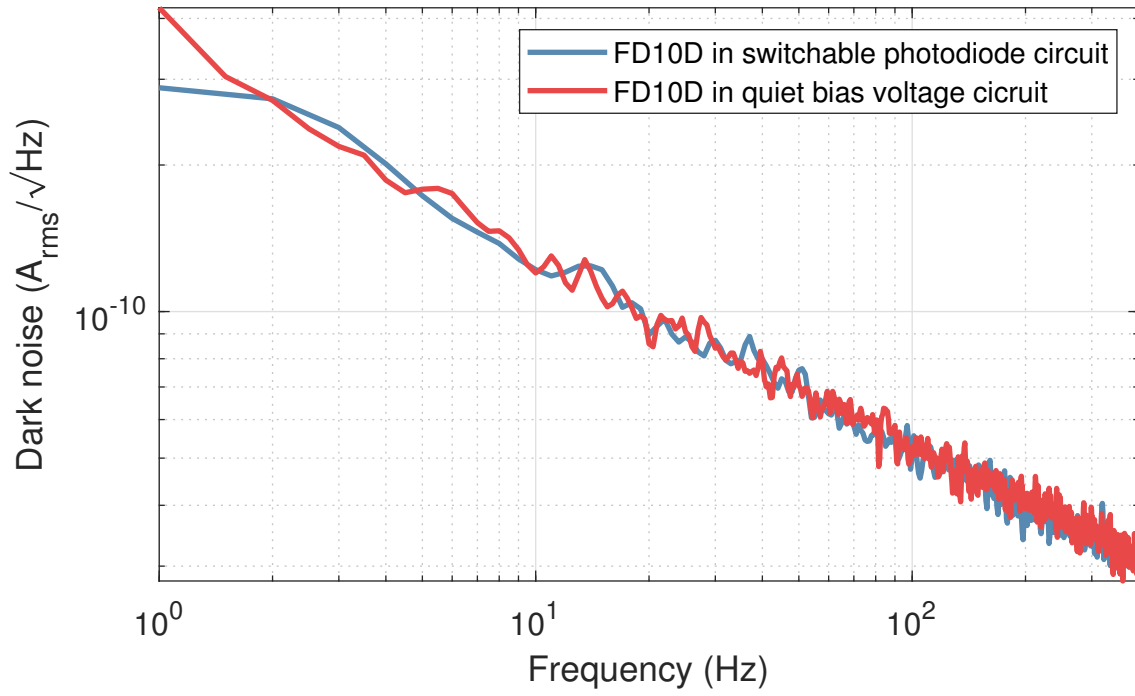


Figure B.4: The FD10D was biased with 1 V in the circuits shown in Figure B.1 and Figure B.2. The low noise fixed bias circuit and the circuit with a settable bias gave the same results, so we can be confident that the circuits shown in Figures B.1 and B.3 will produce measurements that are not contaminated by bias noise.

Section B.4.

B.1.2 Noise measurement

The dark noise of the circuits described in Section B.1 is shown in Figure B.6. The shot noise level of a typical photocurrent in a gravitational wave detector is several hundred times larger than the noise of this circuit. To measure noise of the transimpedance amplifier, instead of a photodiode, a 470 pF capacitor was connected between the inverting input of the op amp and ground.

B.2 Transimpedance Amplifier for Making Shot Noise limited Measurements of 10 mA Photocurrents

A photodiode circuit with electronic noise ~ 25 times lower than the shot noise of a 10 mA current at frequencies between 100 Hz to several 1 MHz is shown in Figure B.7. Below 100 Hz, the noise rises as $1/f$, and it equals the shot noise of a 10 mA current at 0.1 Hz. The noise for this circuit is shown in Figure ?? and the transfer function is shown in Figure ?. While Figure B.7 is specific to the experiment described in Section 8.4, the design of the transimpedance part of this circuit (the AD797 and accompanying parts) was also used Chapter 7.

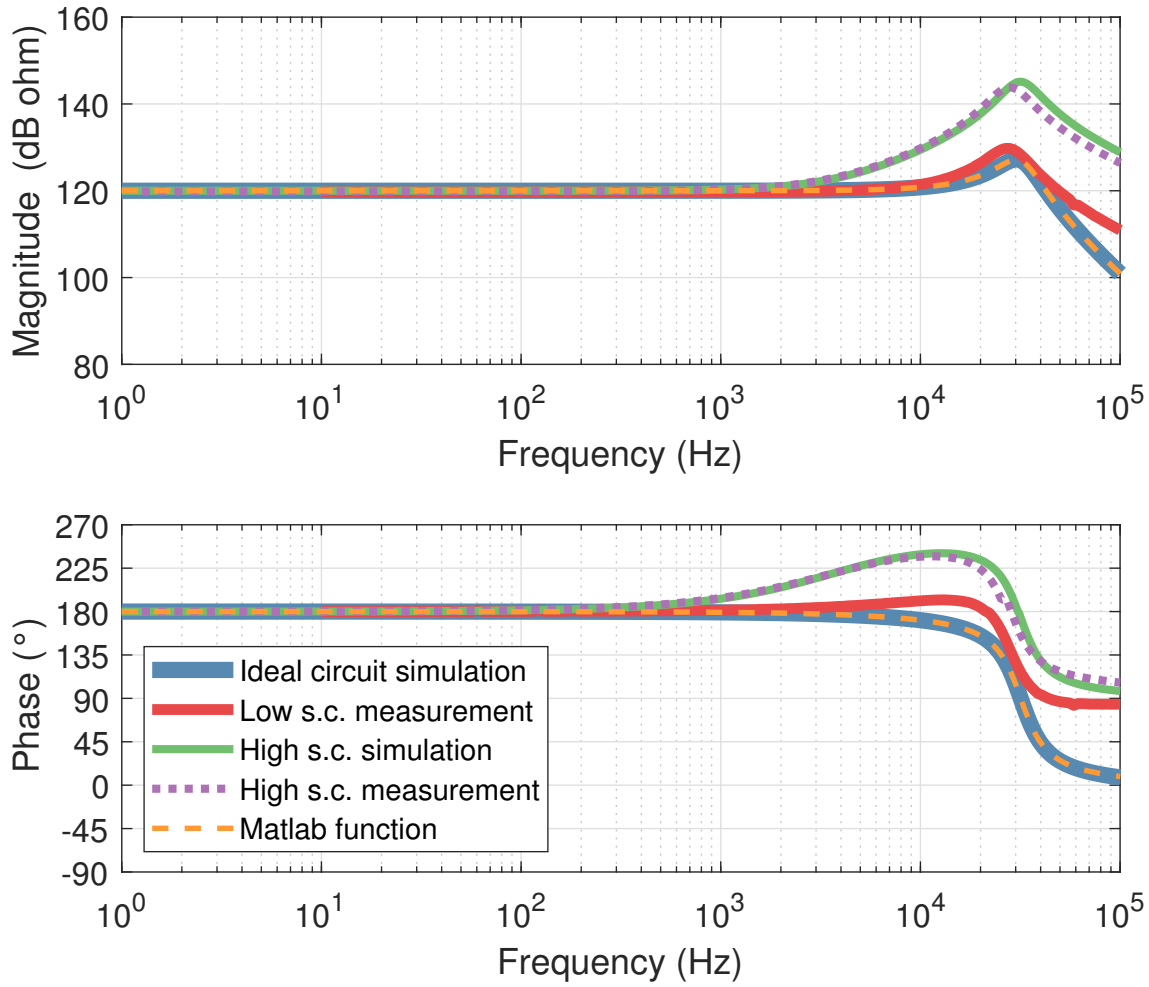


Figure B.5: The transfer function between a current into the inverting input of the transimpedance amplifiers designs shown in Figures B.1B.3 and B.2 was measured and simulated using LISO. A MATLAB function (see Section B.4) was used to check the measurement and the simulation. This circuit is sensitive to stray capacitance (s.c.) at the inverting input of the TL071. The stray capacitance was reduced by changing the layout of the circuit.

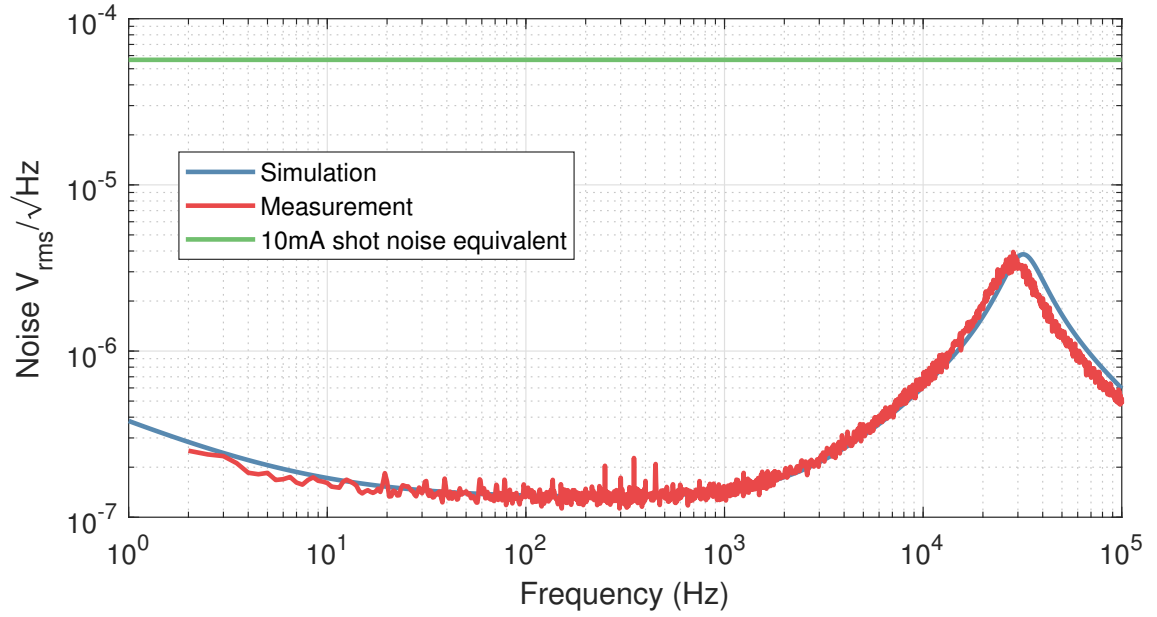


Figure B.6: At low frequencies, the noise of the circuit design shown in Figures B.1B.3 and B.2 is dominated by the low frequency voltage noise of the TL071. Above 10 Hz, the noise of the circuit is dominated by the johnson-Nyquist noise of the feedback resistor. At 30 kHz, the impedance of the feedback network causes the TL071's voltage noise to rise above the resistors noise.

To avoid oscillations due to unusually large output impedance of the AD797, in similar circuits to one shown in Figure B.7, the feedback capacitor may need a resistor in series with it (see [207, p.14]); in this circuit, one was not required. Additionally, if the photodiode is not biased or has a large area, this circuit may oscillate at a frequency between 10 MHz-100 MHz due to the capacitance of the photodiode. Every time a version of this circuit was made, an oscilloscope was used to check for such oscillations as LISO does not model this behaviour.

For convenience, the circuit shown in Figure B.7 allowed for the bias to be externally set. A DC bias was created by low-passed bench-top lab power supply. This low pass filter has two poles at 16 Hz. An AC component could be added onto the bias via a summing OP-amp.

B.3 Simulations with LISO

The behaviour of circuits was simulated using LISO. This software performs a nodal analysis to calculate the voltages at each point within a circuit. LISO can calculate the noise and transfer function of circuits that include op amps. LISO has a wide variety of op amps in its op amp library and this can be easily expanded by specifying the noise properties, open loop gain and gain-bandwidth frequency.

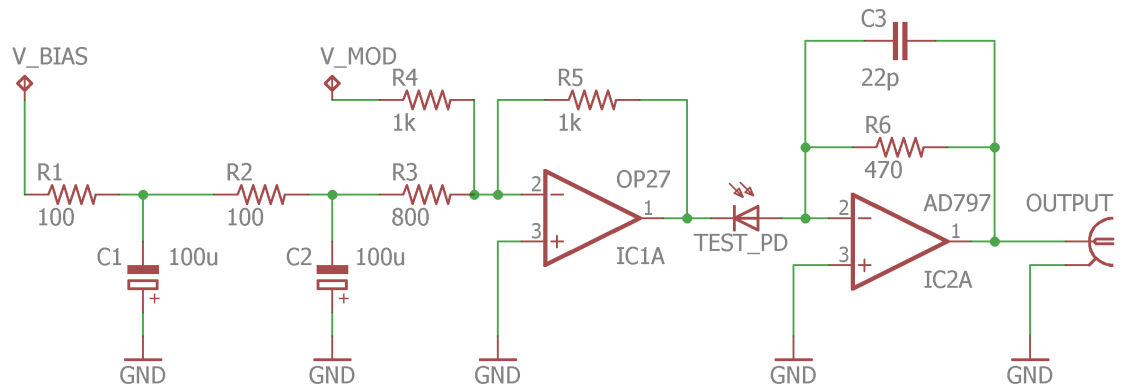


Figure B.7: The AD797 stage of the circuit (right) is a transimpedance amplifier for measuring the shot noise of 10 mA photocurrents. The left hand stage of the circuit is used to provide the bias for the photodiode. One input of this stage is a two pole low-pass filter. A bench-top power supply would be connected to the input of this (V_{BIAS}). The other input, (V_{MOD}), was used to provide a modulation to the bias voltage.

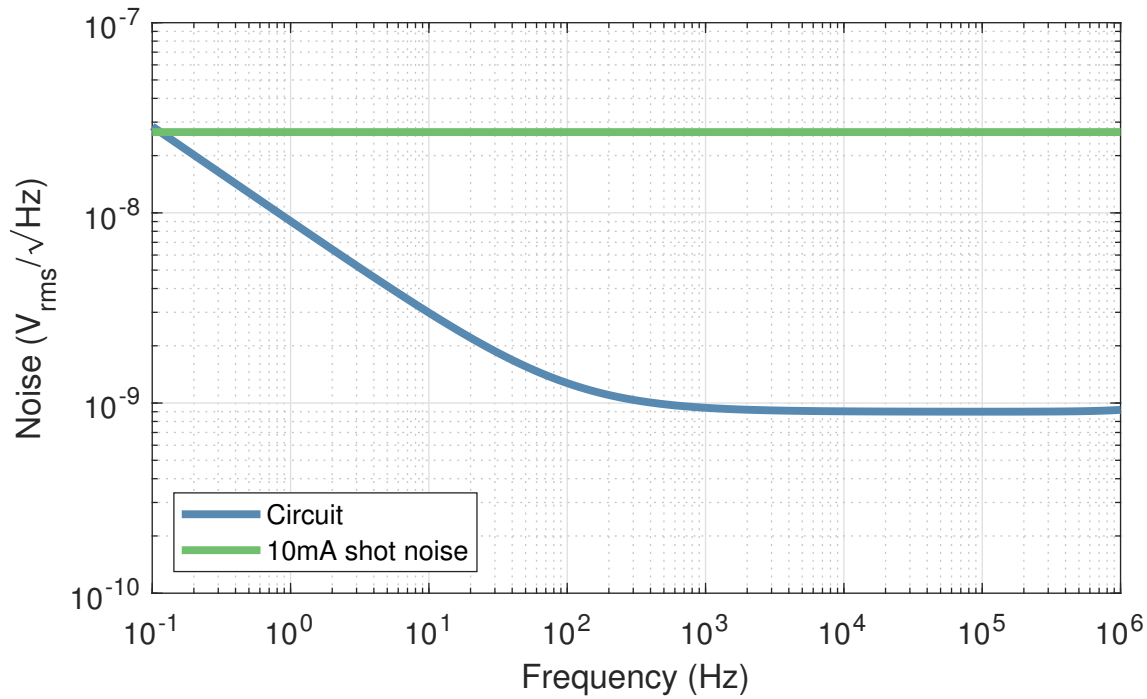


Figure B.8: LISO simulation of the noise of the circuit shown in Figure B.7.

B.4 Transfer Function Calculation With MATLAB

The open loop transfer function of an op amp is

$$A(s) = \frac{A_0}{\frac{A_0}{2\pi f_{\text{GWB}}}s + 1}. \quad (\text{B.1})$$

The transfer function between the input current to the inverting input and the voltage at the output pin for an op amp made into a transimpedance amplifier is

$$H(s) = Z_{\text{FB}} \frac{A(s)}{1 + A(s)B(s)}, \quad (\text{B.2})$$

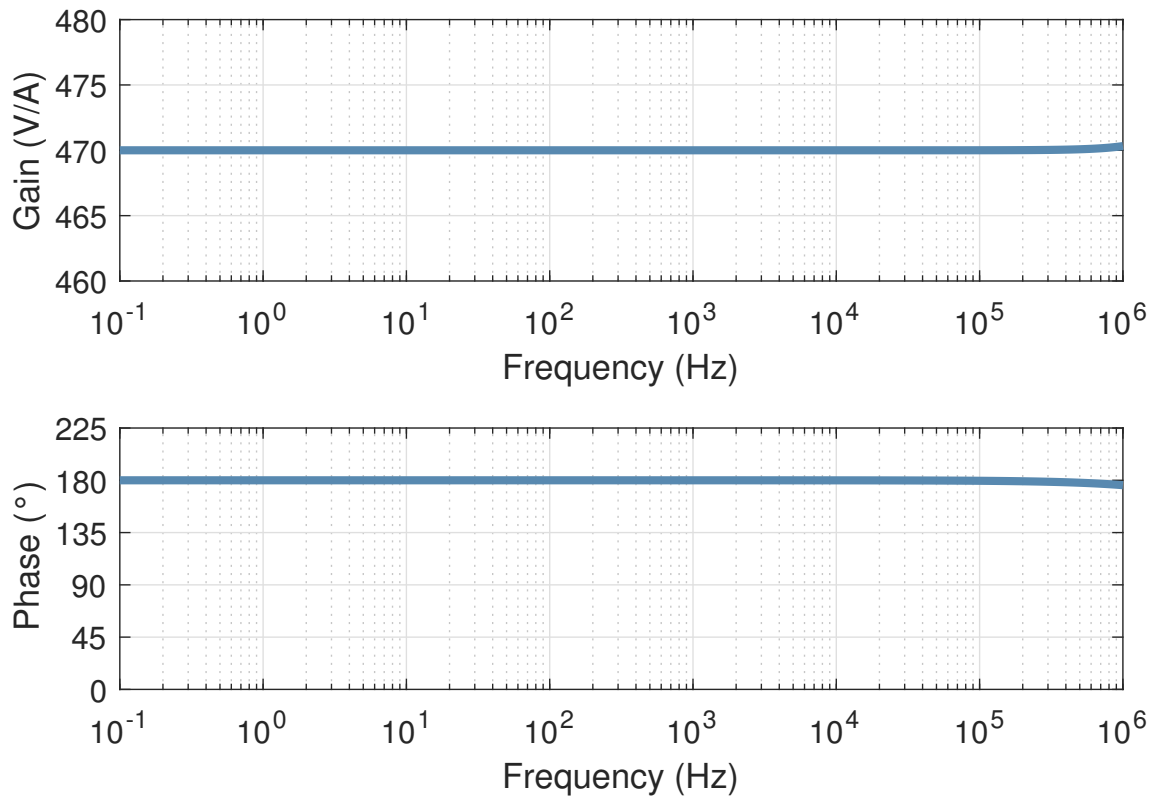


Figure B.9: LISO simulation of the transfer function of the circuit shown in Figure B.7.

where $B(s)$ is the transfer function of the components between the output pin of the op amp and the inverting input of the op amp and Z_{FB} is the impedance seen by the current flowing from the output pin to the inverting input. Without a capacitor in parallel with the feedback resistor $B(s) = \frac{1}{R_{FB}C_{PD}s+1}$. This can be problematic as the transfer function of the circuit may have a phase of 180° due to the photodiode's capacitance and the gain bandwidth product of the op amp as it crosses the unity gain point which will cause the circuit to be unstable. To fix this problem, a capacitor should be added in parallel to the feedback resistor, and so

$$B(s) = \frac{R_{FB}C_{FB}s + 1}{R_{FB}(C_{FB} + C_{PD})s + 1}. \quad (B.3)$$

The impedance seen by the current is

$$Z_{FB} = \frac{R_{FB}}{R_{FB}(C_{FB} + C_{PD})s + 1}. \quad (B.4)$$

$H(s)$ can be calculated with the following MATLAB code:

```
function [mag,phase] = photodiodeTransimpedance(cf, rf,cpd,gbw,A,w)
% cf is the capacitor in the feedback loop used to keep the circuit
% stable, and its units are Farads. rf is the feedback resistor
% measured in Ohms. cpd is the photodiodes capacitance (Farads).
    gbw is the
% gain bandwidth of the OP amp, and is measured in Hz. A is the
    Open loop
% gain of the OP amp. w is the angular frequency range that
% the phase and magnitude will be calculated over.
A_s = tf(A,[A/(2*pi*gbw),1]);
B_s = tf([cf*rf,1],[rf*(cf+cpd),1]);
```

```
closedLoopAnalytic = -tf(rf,[rf*(cf+cpd),1])*A_s/(1+A_s*B_s);  
[mag,phase] = bode(closedLoopAnalytic,w);  
mag = squeeze(mag);  
phase = squeeze(phase);  
end
```

Appendix C

FINESSE Simulations

The following simulations were used in chapter 4. Section C.1 shows how the input beam parameter was determined so that single bounce simulations and simulations with arm cavities had the same beam parameters. Section C.2 contains tables of the spacings and properties of the optics for the LLO interferometer.

C.1 Mode Matching the Laser to the Arm Cavity

To find suitable numbers to use for the `gauss` command in the simulation of single bounce and locked arm beam measurements (see section C.3 and figure 4.3), the following simulation was run. It uses the same Y arm and beam splitter as the

main simulation from section C.3, however it does not have an X arm or the SRC. Without the `cav` command, finesse will default to using a 2mm beam with its waist at $z = 0$ with respect to the the node at the laser. The following beam parameters were found with a `bp` detector placed at the lasers node:

- $w_0 = 0.00864064287177005$ m
- $z = 1339.83530090614$ m

To figure out if the sign of z is correct, consider what happens if the beam parameter detector is moved 1m towards the cavity to `bs_a`:

- $w_0 = 0.00864064287177005$
- $z = 1338.83530090614$ m

z increasing suggests moving from `n1` to `bs_a` causes one to be closer to the waist. Intuitively, we know this is the case since the geometry of the cavity means the waist is within the cavity, and moving towards the cavity means you are moving towards the cavity waist. In other words, we know the beam should be getting smaller as we move from the laser to the cavity, which matches the behaviour seen here.

However, setting the beam parameter at the laser with the first set of numbers gives *terrible* mode matching. The simulation below investigates this. A photodiode which is masked from the TM00 mode

```
pd HOM bs_d
mask HOM 0 0 0
```

was placed at `bs_d` , which has the beam reflected from the arm cavity in it. The value of z for the `gauss` was varied and the power incident on `HOM` was measured. To check that enough `HOM` were being used, the simulation was repeated with several `maxtem` commands. The results had a quadratic fitted to them, and the minimum was taken as the optimal value. This is shown in figure C.1. The values:

- $w_0 = 0.00864064287177005$ m
- $z = -1339.83683935$ m

give more than 99.99999% mode matching, which for this work is plenty. This is close to the original values that we obtained, except for a minus sign. This is due to the way `FINESSE` computes the beam parameter.

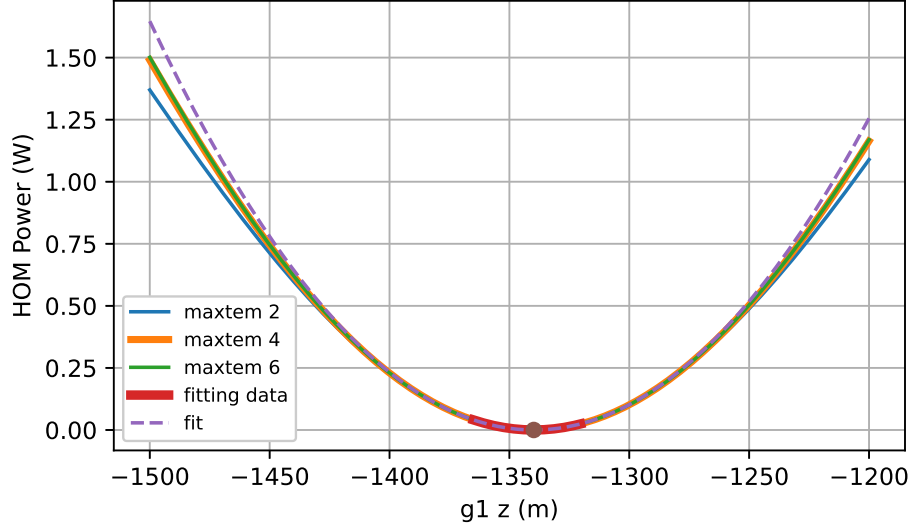


Figure C.1: Optimisation of parameters used in the `gauss` command to mode match the laser to the arms in the simulations described in chapter 4 and section C.3. The blue, green and orange lines show how the mode matching changes as the beam waist location is changed. As the mode matching becomes larger, more HOM are required, as the blue and green/orange lines diverge. A quadratic was fitted to the data highlighted in red, and the minimum value was used as the waist location parameter for the `gauss` command.

C.2 Interferometer Parameters

The simulated interferometer uses ‘design’ arms and the LLO SRC. The simulated arm cavity geometry is based on the design values to avoid contrast defects. The following subsections contain tables specifying the design and measured values for each component of the simulation.

C.2.1 Radii of Curvature of optical components

The optics in the SRC are simulated with the values from table C.2. The arm cavity optics have been simulated using the values from table C.1.

Optic Name	RoC (m)	Focal length (m)	Reference
SRM	−5.6938	−2.8469	[208]
SR2	−6.427	−3.2135	[208]
SR3	36	18	[208]
ITM	1934	967.0	[208]
ETM	2245	1122.5	[208]

Table C.1: The aLIGO design values of the RoC of the optics inside the SRC and the arm.

Optic Name	RoC (m)	Focal length (m)	Reference
SRM-02	-5.673 ± 0.002	-2.8365 ± 0.001	[209]
SR2-04	-6.406 ± 0.006	-3.203 ± 0.003	[210], [209]
SR3-01	35.97 ± 0.012	17.985 ± 0.006	[211], [209]
ITM-08	1938.44	969.22	[209]
ETM-15	2245.4	1122.7	[209]

Table C.2: The measured values of the RoC of the optics at LLO that are inside the SRC and the arm.

C.2.2 Spacings

The nominal values with a diagram can be found in [77] for the H1/L1 spacings. This information is also in [208]. The L1 Zemax parameters are found in [212]. Table C.3 contains the distances derived from the Zemax coordinates. Most spacings do not require sub mm accuracy, however SR3 to SR2 does since the Rayleigh range of this beam is very small (about 4 cm). The design value spacings are shown in table C.4. The values used in an earlier FINESSE model of LLO [213] are shown in table C.5. The thickness of each optic is found in table C.6.

Space Name	Length (m)
EMTY to ITMY	3994.4850
ITMY to CP	0.02
CP to BS HR	4.8471
BS AR to SR3	19.3659
SR3 to SR2	15.4435
SR2 to SRM	15.7562

Table C.3: Measured spacings between optics at LLO [212]

Space Name	Length (m)
EMTY to ITMY	3994.5
ITMY to CP	0.05
CP to BS HR	4.8497
BS AR to SR3	19.368
SR3 to SR2	15.4607
SR2 to SRM	15.726

Table C.4: Design value for spacings [77]

Space Name	Length (m)
EMTY to ITMY	3994.515
ITMY to CP	0.02
CP to BS HR	4.847
BS AR to SR3	19.3661
SR3 to SR2	15.4435
SR2 to SRM	15.7566

Table C.5: Values used in an earlier FINESSE model of LLO [213]

Optic	Thickness (m)
ITM	0.200
CP	0.100
BS	0.0685
Angled BS	0.1202

Table C.6: Thickness of optics between the ITM and the SRM.

C.2.3 Nominal Simulation

The nominal simulation values is shown in table C.7.

Parameter	Value	Description
SRM RoC	−5.673m	Measured RoC of SRM-02 installed at LLO
SR2 focal length	−3.203 m	Measured focal length of SR2-04 installed at LLO
SR3 focal length	17.985 m	Measured focal length of SR3-01 installed at LLO
ITM RoC	1934m	Design Value
ETM RoC	2245m	Design Value
ITM thermal lens	50 km	
ITM to ETM distance	3994.5m	Design Value
ITM to BS Y	4.8471m	As-built Value for Y arm
ITM to BS X	4.8471m	As-built Value for Y arm
BS to SR3	19.3659m	As-built Value
SR3 to SR2	15.4435m	As-built Value
SR2 to SRM	15.7562m	As-built Value
ITM T	0.0148	Power transmission coefficient
SRM T	0.324	Power transmission coefficient
ITM L	10 ppm	Scatter loss of ITM HR surface
ETM L	10 ppm	Scatter loss of ETM HR surface

Table C.7: Nominal values for the major components of the simulation. Design values were used for the arms to remove the contrast defect that would occur otherwise. The as-built values for the SRC were used to replicate the behaviour of the SRC at LLO more closely. The ITMs that are simulated have a thickness of 0.1 m and a refractive index of 1.45. The thermal lens is implemented as shown in figure 4.3 .

C.3 FINESSE Code

This section contains the FINESSE code used in chapter 4.

C.3.1 Michelson Interferometers with Thin Beamsplitter

This FINESSE code describes a Michelson interferometer where the beam splitter is implemented with a single `bs` command. This simulation will produce an incorrect result as the phase at the beamsplitter is not properly accounted for.

```
l l1 1 0 n1
s sin 1 n1 na

bs thin_bs 0.5 0.5 0 0 na nb nc nd

s Yarm 1 nb nY
m Ymirror 1 0 0 nY dump

s Xarm 1 nc nX
m Xmirror 1 0 0 nX dump

pd AS nd

xaxis Xmirror phi lin 0 180 100
```

C.3.2 Michelson Interferometers with Thick Beamsplitter

This FINESSE code describes a Michelson interferometer where the beam splitter is implemented with 3 `bs` and 2 `space` commands. This is the proper way to implement a beamsplitter in FINESSE.

```
l l1 1 0 n1
s sin 1 n1 na

bs thick_bs_1 0.5 0.5 0 0 na nb nsub1 nsub3
s bs_subX 1p nsub1 nsub2
s bs_subY 1p nsub3 nsub4
bs thick_bs_X 0 1 0 0 nsub2 dump nc dump
bs thick_bs_AS 0 1 0 0 dump nsub4 dumo nd

s Yarm 1 nb nY
m Ymirror 1 0 0 nY dump

s Xarm 1 nc nX
m Xmirror 1 0 0 nX dump

pd AS nd

xaxis Xmirror phi lin 0 180 100
```

C.3.3 Main Simulation

The following FINESSE code was used to simulate mode mismatches in the SRC.

```
l 11 5M 0 n1
gauss beamParaml1 11 n1 0.00864064287177005 -1339.83683935

maxtem 12

s space1 1 n1 bs_a

#####
#bs beam_splitter 0.5 0.5 $phi_BS 0 bs_a bs_b bs_c bs_d

bs BSfront 0.5 0.5 0 0 bs_a bs_b nBSi1 nBSi3
s BSsubstrate1 1p 1.45 nBSi1 nBSi2
s BSsubstrate2 1p 1.45 nBSi3 nBSi4
bs BSback1 0 1 0 0 nBSi2 dump bs_c dump
bs BSback2 0 1 0 0 nBSi4 dump bs_d dump

#####
## Y ARM

s s_bs_to_ITMy 4.8471 bs_b nITMyARa

#ITMy
m1 ITMyAR 1 0 $phi_ITMy nITMyARa nITMyARb
s sITMy_thickness 0.05 1.45 nITMyARb nITMyTLa
lens ITMyTL 50k nITMyTLa nITMyTLb
s sITMy_thickness2 0.05 1.45 nITMyTLb nITMyHRa
```

```
m1 ITMy 0.0148 10u $phi_ITMy nITMyHRa nITMyHRb
attr ITMy Rc -1934
```

```
s space_arm_y 3994.5 nITMyHRb nETMy
```

```
m1 ETMy 0 10u $phi_ETMy nETMy dump
attr ETMy Rc 2245
```

```
cav army ETMy nETMy ITMy nITMyHRb
```

```
#####
```

```
## X ARM
```

```
s s_bs_to_ITMx 4.8471 bs_c nITMxARa
```

```
#ITMx
```

```
m1 ITMxAR 1 0 $phi_ITMx nITMxARa nITMxARb
s sITMx_thickness 0.05 1.45 nITMxARb nITMxTLa
lens ITMxTL 50k nITMxTLa nITMxTLb
s sITMx_thickness2 0.05 1.45 nITMxTLb nITMxHRa
m1 ITMx 0.0148 10u $phi_ITMx nITMxHRa nITMxHRb
attr ITMx Rc -1934
```

```
s space_arm_x 3994.5 nITMxHRb nETMx
```

```
m1 ETMx 0 10u $phi_ETMx nETMx dump
attr ETMx Rc 2245
```

```
cav armx ETMx nETMx ITMx nITMxHRb
```


#####

SRC

s bs_to_SR3 19.3659 bs_d nSR3a

lens SR3 17.985 nSR3a nSR3b

s SR3_to_SR2 15.4435 nSR3b nSR2a

lens SR2 -3.203 nSR2a nSR2b

s SR2_to_SRM 15.7562 nSR2b nSRMa

m1 SRM 0.324 0 0 nSRMa nSRMb

s SRM_substrate 0.07 1.45 nSRMb nSRMc

m SRM_AR 0 1 0 nSRMc nSRMd

attr SRM Rc -5.673

#cav src ITMx nITMxHRa SRM nSRMa

#####

#N.B these tunings don't represent how an IF0 is kept on the dark
fringe

const phi_ITMx 0

const phi_ITMy 0

const phi_ETMx 0

const phi_ETMy 0

const phi_BS 0

#####

beam SRM_trans nSRMd
#mask SRM_trans 0 0 0

bp SRM_trans_w0 x w0 nSRMd
bp SRM_trans_w x w nSRMd
bp SRM_trans_r x r nSRMd

bp laser_w x w0 n1
bp laser_z x z n1

beam ITMy_refl nITMyHRa
beam ITMy_refl nITMyHRa

bp ITMy_refl_w x w nITMyHRa
bp ITMx_refl_w x w nITMxHRa

pd p_in bs_a*
pd p_ref bs_a

pd p_out nSRMd

pd p_out_hOM nSRMd
mask p_out_hOM 0 0 0

ad ad_HOM_2_y 2 0 0 nSRMd
ad ad_HOM_2_x 0 2 0 nSRMd

```
gouy src_gouy y sITMx_thickness sITMx_thickness2 s_bs_to_ITMx  
bs_to_SR3 SR3_to_SR2 SR2_to_SRM
```

```
trace 64
```

```
phase 2
```

```
xaxis SRM_trans x lin -10 10 200
```

Appendix D

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures D.2 and D.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [173] (figure D.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained

since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure D.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

D.0.1 Method

This section delineates the method described in section D and can be visualised using figure D.2.

Define the target mode at the optic 3. In figure D.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure D.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure D.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure D.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure D.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure D.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure D.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure D.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found. The number of modes that

are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptres were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

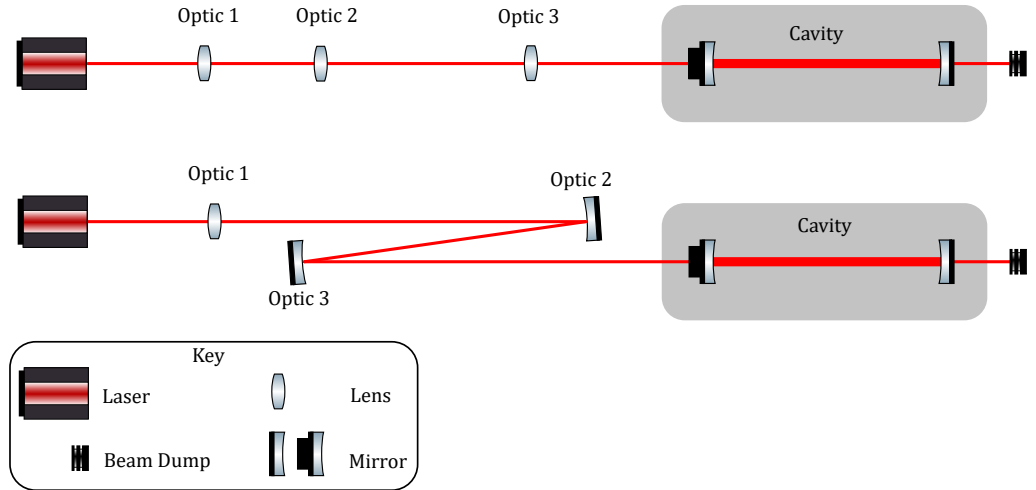


Figure D.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure D.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

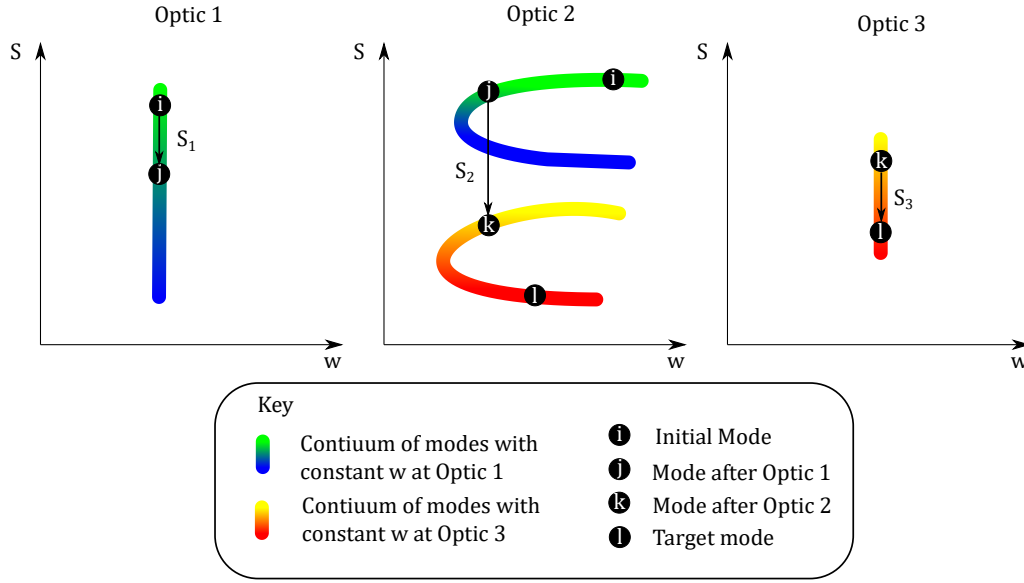


Figure D.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure D.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix E

Mode Matching Between 2 Astigmatic Beams

E.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \tag{E.1}$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{E.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{E.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{E.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{E.5})$$

The numerator of equation E.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{E.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{E.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation E.7 can be written as:

$$\int_{-\infty}^{-\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{-\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{E.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{E.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{E.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{E.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{E.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{E.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}} \quad (\text{E.14})$$

allows equation E.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi (2q_{x1}^* q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi (2q_{x1} q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{E.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{E.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{E.17})$$

This means that the numerator of equation E.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{E.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation E.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{E.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp\left(\frac{ikx^2(q_x - q_x^*)}{2q_x q_x^*}\right) \times y \text{ term}. \quad (\text{E.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{E.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{E.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{E.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{E.24})$$

E.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{E.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{E.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{E.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{E.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of

having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{E.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

Appendix F

Light and Gaussian Modes

This chapter will show the maths...

- check normalisation constants
- check the formula for the electric fields are actually right!
- change formulas to be in terms of a mode function and something that is to do with its propogational along z , and amplitude.

F.1 Plane Waves

Light is electromagnetic radiation. From Maxwell's equations it can be shown that the electric field, $\mathbf{E}$, component of electromagnetic radiation satisfies the wave equation

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{E}. \quad (\text{F.1})$$

A monochromatic, coherent, plane wave solution for $\mathbf{E}$ satisfies equation (F.1):

$$\mathbf{E} = \hat{\mathbf{n}} E \exp(i(\omega t - \mathbf{k} \cdot \mathbf{r} + \varphi)), \quad (\text{F.2})$$

where $\hat{\mathbf{u}}$ indicates the polarisation, E is the complex field amplitude, ω is the angular frequency, $\mathbf{k}$ is the wave vector and φ is the absolute phase. The physical amplitude of the electric field oscillation is obtained from the $\Re(E)$. For optical systems is useful to define an optical axis. When the light is travelling along the optical axis, the $\mathbf{k} \cdot \mathbf{r}$ becomes kz . k is known as the wave number and is related to the wavelength of the light, λ by

$$k = \frac{2\pi}{\lambda}. \quad (\text{F.3})$$

For plane waves there is no dependence of $\mathbf{E}$ on the x and y coordinates. The absolute phase φ becomes relevant in calculations involving interfering beams. By inserting equation (F.2) into equation (F.1), we obtain the dispersion relationship

$$\omega = ck. \quad (\text{F.4})$$

The plane wave solution can be used to analyse many optical phenomena, such as amplitude modulation, phase modulation and interference.

F.2 Gaussian Modes

Light in an optical set-up is normally generated by a laser. Lasers produce almost monochromatic, coherent, light so in this sense they are quite similar to a plane wave, however lasers also have geometric properties. A laser produces light with an approximately circular beam profile which is brightest at its centre. The spread of light power in a laser beam follows a Gaussian distribution, and the width of the beam profile is characterised by the radius at which $\sim 86\%$ of the light power is contained.

As the beam propagates along its optical axis, the beam size changes. A Gaussian beam has a point where it is smallest, this is known as the waist of the beam. A Gaussian beam can be completely characterised by the size of its waist and location along the waist along the optical axis. Whenever a Gaussian mode interacts with a simple optical component such as a lens or a mirror, the current Gaussian mode is transformed into a new Gaussian mode.

The following sections describe Gaussian modes and how they interact with real optical layouts with imperfections. The lowest order Gaussian mode will be derived, and the properties of this mode will be explained. The concept of the

complex Gaussian beam parameter will be introduced as a way of encapsulating all the information needed to determine the properties of a Gaussian mode. Astigmatic Gaussian beams will be defined in terms of two complex Gaussian beam parameters. The first order Hermite-Gaussian (HG) mode will be shown to model the effect of misalignment, and then it will be described how all laser beams can be made up of a set of HG modes. Finally, the way that a Gaussian mode traverses an optical set-up will be expressed in terms of the ABCD matrix formalism.

F.3 The Paraxial Wave Equation

Short intro on what this is To derive the paraxial wave equation, we apply the separation of variables to equation (F.1):

$$\mathbf{E}(\mathbf{r}, t) = E(\mathbf{r})T(t). \quad (\text{F.5})$$

Next we reform equation F.1 in terms of E and T

$$\frac{\nabla^2 E}{E} = \frac{1}{c^2 T} \frac{\partial^2 T}{\partial T^2}. \quad (\text{F.6})$$

By noticing that either side of the equation (F.6) depends on $\mathbf{r}$ or t , the left hand side of this equation must equal a constant, and the Helmholtz equation can be obtained:

$$\frac{\nabla^2 E}{E} = -k^2. \quad (\text{F.7})$$

Rewriting $E(\mathbf{r})$ in terms of two functions, **some how remove units from $u(\mathbf{r})$**)

$$E(\mathbf{r}) = u(\mathbf{r}) \exp(-ikz), \quad (\text{F.8})$$

and substituting this into equation (F.7) yields

$$\left(\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} + \frac{\partial^2 u}{\partial z^2} \right) \exp(ikz) = 0. \quad (\text{F.9})$$

The paraxial approximation means the $\frac{\partial^2 u}{\partial z^2}$ is neglected as it is much smaller than the other terms. Applying this approximation produces the paraxial Helmholtz equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} = 0. \quad (\text{F.10})$$

The physical interpretation of $u(\mathbf{r})$ is a function that describes the spatial distribution of the complex amplitude of the light field. **reword this so it actually makes sense.**

F.4 The fundamental Gaussian mode

Rewrite E in terms of u .

The simplest geometry of a laser beam can be described by the fundamental Gaussian mode. This is a mode which has a circular intensity profile that follows a Gaussian distribution, and the surface of constant phase is curved. Substituting the

simplest solution to (F.10) into equation (F.8) produces the fundamental Gaussian mode **find out what the amplitude constant is**

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{1}{w(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2R_C(z)} - \frac{x^2 + y^2}{w^2(z)}\right) \exp(-ikz). \quad (\text{F.11})$$

This is characterised by a function $w(z)$, which gives the size of the beam at position z the optical axis, by $\phi(z)$, which gives the Gouy Phase of the beam at z along the optical axis, and by $R_c(z)$, which gives radius of curvature of the beam at z . The minimum value of $w(z)$ is known as the beam waist, or w_0 .

It is useful to express $w(z), \phi(z)$ and $R_c(z)$ in terms of the Rayleigh range z_R . The Rayleigh range characterises how fast the beam will spread out. A beam with a short Rayleigh length will expand quicker than a beam with a longer Rayleigh length. For example, in order for the beam to not be significantly clipped by the aperture of the test masses, the Rayleigh length of the Gaussian beam in the arm cavities at LIGO has to be on the order of kilometres. The size of the waist determines the Rayleigh length, a smaller beam waist corresponds to a shorter Rayleigh length. If the waist of a Gaussian beam at $z = 0$, the position along the z axis by which the radius has increased by a factor of $\sqrt{2}$ is known as the Rayleigh range z_R , and is given by

$$z_R = \frac{\pi w_0^2}{\lambda}. \quad (\text{F.12})$$

The radius of the beam as it propagates along the optical axis can be written as

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R} \right)^2}. \quad (\text{F.13})$$

The Rayleigh range can be used to approximately split beam into two regions: the near field for $z < z_R$ and the far field for $z > z_R$. In the far field equation (F.13) becomes linear and the beam size can be approximated as

$$w(z) \approx w_0 \frac{z}{z_R}. \quad (\text{F.14})$$

As a Gaussian beam travels propagates along the optical axis, it acquires an additional phase shift compared to a plane wave. This is known as the Gouy phase shift, $\phi(z)$, and it given by

$$\phi(z) = \tan^{-1} \frac{z}{z_R}. \quad (\text{F.15})$$

At either side of it waist, the fundamental Gaussian mode will be shifted $\pm\pi/2$ relative to a plane wave. An alternative way of putting this is the fundamental Gaussian mode will gain a π change as it goes through from $-\infty$ to ∞ compared to a plane wave. Most of this phase change is gained within $\pm z_R$ of its waist. The Gouy phase results in the the phase velocity at the waist being greater for a Gaussian beam than it is for plane waves.

The radius of curvature, $R_c(z)$, in terms of the Rayleigh range is

$$R = z + \frac{z_R^2}{z}. \quad (\text{F.16})$$

For $z \ll z_R$ the radius of curvature can be approximated as infinite. As the beam approaches its Rayleigh length the radius of curvature decreases until it reaches its minimum at $z = z_R$. After this the radius of curvature begin to flatten out again, and for $z \gg z_R$ the radius of curvature can be approximated as z . The defocus, $S(z)$, of a beam is a more practical way of describing a beam as it is easier to compute the effects of a beam interacting with a focussing element of an optical layout in terms of the defocus than it is with the radius of curvature. The defocus is defined as the inverse of its radius of curvature

$$S(z) = \frac{1}{R(z)} \quad (\text{F.17})$$

The fundamental Gaussian mode can be expressed in terms of a complex beam parameter, $q(z)$. This parameter is useful for calculating how a gaussian beam is transformed by an optical system. From this parameter, all the physical quantities related to a Gaussian beam can be computed. The complex beam parameter is defined as

$$q(z) = z + iz_R. \quad (\text{F.18})$$

The inverse of equation F.18 can be written in terms of the curvature and radius of the beam:

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}. \quad (\text{F.19})$$

Equation (F.11) can be re written in terms of this the complex beam parameter,

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{q_0}{w_0 q(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2q(z)}\right) \exp(-ikz). \quad (\text{F.20})$$

The radius of the beam, gouy phase, and the defocus at a point z along the optical axis can be calculated from $q(z)$ using

$$w(z) = \sqrt{\frac{\lambda}{-\pi \Im \left(\frac{1}{q(z)} \right)}}, \quad (\text{F.21})$$

$$\phi(z) = \tan^{-1} \left(\frac{\Re(q(z))}{\Im(q(z))} \right), \quad (\text{F.22})$$

$$S(z) = \Re \left(\frac{1}{q(z)} \right). \quad (\text{F.23})$$

Mode matching integral

F.4.1 Astigmatism

If a beam has an elliptical rather than circular profile, then the beam is astigmatic. For simple astigmatism, a beam can be described with two complex Gaussian beam parameters, one for the yz plane and one for the xz plane. These are referred to as the tangential (xz) and the saggital (xy) planes. A gaussian beam with simple astigmatism will have a different waist size and location in the tangential and sagittal planes. This type of astigmatism arises due to gaussian beams passing through a cylindrical lens or having a non-zero angle of incidence to a mirror.

An simple astigmatic gaussian beam can be split up into two orthogonal fields

(rewrite in terms of $u(\mathbf{r})$)

$$\begin{aligned}
E(\mathbf{r}) &= E(x, z)E(y, z) \\
&= E_0 \sqrt{\frac{2}{\pi}} \sqrt{\frac{q_{x0}}{w_{x0}q_x(z)}} \sqrt{\frac{q_{y0}}{w_{y0}q_y(z)}} \exp(i\phi_{Av}(z)) \\
&\quad \times \exp\left(-ik\frac{x^2}{2q_x(z)} - ik\frac{y^2}{2q_y(z)}\right) \exp(-ikz). \tag{F.24}
\end{aligned}$$

where $\exp(i\phi_{Av}(z))$ is the average of the Gouy Phase in the saggital and tangential planes. The equations for waist size, equation (F.21), Gouy phase, equation (F.22) and defocus, equation (F.23), remain the same. A beam with simple astigmatism can be though of as two independent beams, and their interaction with the optical layout can be calculated independently.

F.5 Gaussian Beams Traversing optical Layouts

As a Gaussian beam propagates through an optical layout, the beam parameter that describes changes. The effect of simple optical components can be modelled with ABCD matrices, and an entire layout can be modelled by multiplying a series of ABCD matrices together. Components that introduce astigmatism to the beam can be modelled as having separate ABCD matrices for the tangential and saggital planes. This formalism can also be used to compute cavity eigenmodes.

When a beam interacts with an optical component it is transformed from a mode

described by q_1 to a new mode q_2 according to

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \quad (\text{F.25})$$

where $n_{1,2}$ are the refractive indices of the medium $q_{1,2}$ are in. The coefficients A, B, C, D are defined by a coefficient matrix $\mathbf{M}$,

$$\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (\text{F.26})$$

To compute the coefficient matrix for a series of elements numbered 0 to n , the matrices for each individual element should be multiplied together

$$\mathbf{M} = \mathbf{M}_n \mathbf{M}_{n-1} \dots \mathbf{M}_0. \quad (\text{F.27})$$

The ABCD matrices for various components is summarised in table F.1.

F.6 Higher Order Modes and optical imperfections

The fundamental gaussian mode describes the electric field of a beam in an ideal optical system. Imperfections in the optical layout, such as misalignment and mode-mismatch, results in a different mode shape. If two modes are misaligned slightly, one will see a mode profile that looks like two spots instead of one. A small mode-mismatch results in a ring shaped mode profile. These imperfections in the layout

Optical Component	ABCD Matrix
space	$\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}$
Mirror or lens (normal incidence)	$\begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix}$
Mirror (AOI = α , Sagittal Plane)	$\begin{pmatrix} 1 & 0 \\ -\frac{S}{\cos \alpha} & 1 \end{pmatrix}$
Mirror (AOI = α , Tangential Plane)	$\begin{pmatrix} 1 & 0 \\ -S \cos \alpha & 1 \end{pmatrix}$

Table F.1: Table of the ABCD matrices for simple optical components.

and the resulting beam shapes can be modelled as the optical layout scattering light into higher-order optical modes.

There are many solutions to equation F.10. One complete, orthonormal basis set of functions that can be used to construct solutions to F.10 are the Hermite-Gaussian (HG) modes. These modes have xy symmetry, and appear as 'square' shaped. These are useful for modelling misalignments, as this type of imperfection can be naturally broken into xy coordinates. Mode-mismatches are best described by the $r\theta$ coordinates, and so its natural to express these in terms of another orthonormal basis set known as the Laguerre-Gauss (LG) modes. However, if the beam is astigmatic then the mode no longer has the symmetry required in order to be modelled by LG modes.

Equation (F.10) can be solved using HG modes, which have the mathematical form of

$$u_{nm}(x, y, z) = u_n(x, z)u_m(y, z). \quad (\text{F.28})$$

The functions $u_{n,m}$ have the form

$$u_n(x, z) = \left(\frac{2}{\pi}\right)^{1/4} \left(\frac{\exp(i(2n+1)\phi(z))}{2^n n! w(z)}\right)^{1/2} H_n\left(\frac{\sqrt{2}x}{w(z)}\right) \exp\left(-i\frac{kx^2}{2R_c(z)} - i\frac{x^2}{w^2(z)}\right). \quad (\text{F.29})$$

where $H_n(x)$ a Hermite polynomial of the order n . By writing $u_{nm}(x, y, z)$ out explicitly,

$$u_{nm}(x, y, z) = \left(2^{n+m-1} n! m! \pi\right)^{1/2} \frac{1}{w(z)} \exp(i(n+m+1)\phi(z)) H_n\left(\frac{\sqrt{2}x}{w(z)}\right) H_m\left(\frac{\sqrt{2}y}{w(z)}\right) \exp\left(-ik\frac{x^2+y^2}{2R_c(z)} - i\frac{x^2+y^2}{w^2(z)}\right), \quad (\text{F.30})$$

we can see that the higher order mode solutions to equation F.10 have the same waist size and radius of curvature to the fundamental mode, but different intensity distributions and additional Gouy phase. If n and m are set to zero, then equation F.11 is recovered. These HG modes interact with optical elements in the same way as the fundamental mode.

The eigen mode of a cavity is determined by its length and the radius of curvature of its end mirrors. The Gaussian mode that is the eigen mode of the cavity will have the same radius of curvature at the end mirrors of the cavity as the mirrors themselves. As higher order modes share the same radius of curvature as the funda-

mental mode, these modes can also be resonant within the cavity. What separates these modes out in a cavity is the round trip Gouy phase that is acquired. Most cavities are designed such that the round trip Gouy phase is such that only one mode can be resonant at once, and all the other modes destructively interfere.

The Gouy phase separation of higher order modes is also important when designing alignment and mode matching control systems, as different types of errors have different phases. For example, angular and translational misalignment are both first order effects. This means they both transfer light from the HG_{00} mode into the HG_{10} . However, the complex amplitude of these misalignment situations is separated by $\pi/2$ rad of phase. Having two sensors separated by $\pi/2$ of Gouy phase, one can derive alignment control signals.

In general, a mode can be written as a superposition of HG modes

$$E(x, y, z) = \sum_{n=0}^{n=\infty} \sum_{m=0}^{m=\infty} A_m u(x) A_n u(y), \quad (\text{F.31})$$

where A is the mode amplitude. This form is useful when calculating the coupling between modes.

The following example of the amount of power lost due to two arms of an interferometer being misaligned. The field at the AS port of the interferometer can be written as

$$E_{\text{AS}} = \frac{A_0}{\sqrt{2}} \left(\frac{\Delta A_0}{A_{00}} u_0 + \sum_{m=1}^{\infty} \frac{a_m u_m}{A_{00}} \right), \quad (\text{F.32})$$

where A_0 is the incident field amplitude and A_{00} is the amplitude of the field in the

arms. The change in the fundamental power due to an optical imperfection in one of the arms is ΔA_0 . E_{AS} is normalised to the incoming power at the beam splitter. As the modes are orthogonal by calculating the power in the higher order modes, the power lost from the fundamental mode can be calculated. The amplitude of a mode can be written as

$$A_m = \frac{1}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} E(X)u_m(X)e^{-X^2/2}dX \quad (F.33)$$

where Γ is the Gamma function, X is the normalised distance $\sqrt{2}x/w$ and $E(X)$ is an arbitrary field distribution. For a misaligned beam with a purely angular misalignment θ , a linear phase gradient $\varphi(x) = 2\pi\theta x/\lambda$ must be applied to the regular field distribution. The field distribution at the waist can be written as

$$E(X) = \exp\left(iX\frac{\theta}{\theta_c} - \frac{X^2}{2}\right) \quad (F.34)$$

where the critical angle is $\theta_c = \lambda\sqrt{2}/\pi w$. Inserting this $E(X)$ into equation (F.33) gives

$$A_m = \frac{e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(-X+i\frac{\theta}{\theta_c})^2}u_m(X)dX, \quad (F.35)$$

which can be solved analytically by using the relationship

$$\int_{-\infty}^{\infty} e^{-(z^*t)^2}u_n(t)dt = 2^n\sqrt{\pi}z^n \quad (F.36)$$

to give

$$A_m = \frac{(i\frac{\theta}{\theta_c})^m e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)}. \quad (F.37)$$

The fundamental order is reduced by

$$\frac{A_0}{A_{00}} = \exp \left((-\theta/\theta_c)^2 \right) \quad (\text{F.38})$$

F.7 Concluding remarks

Bibliography

- [1] LIGO Virgo Collaboration. Observation of gravitational waves from a binary black hole merger. *Phys. Rev. Lett.*, 116:061102, Feb 2016.
- [2] The LIGO Scientific Collaboration. Advanced LIGO. *Classical and Quantum Gravity*, 32(7):074001, mar 2015.
- [3] LIGO Virgo Collaboration. Gw170814: A three-detector observation of gravitational waves from a binary black hole coalescence. *Phys. Rev. Lett.*, 119:141101, Oct 2017.
- [4] The LIGO Virgo Collaboration. GW190412: Observation of a binary-black-hole coalescence with asymmetric masses. *Physical Review D*, 102(4), aug 2020.
- [5] The LIGO Virgo Collaboration. GW190521: A binary black hole merger with a total mass of $150\text{ m}_{\odot}$. *Physical Review Letters*, 125(10), sep 2020.
- [6] LIGO Virgo Collaboration. Gw170817: Observation of gravitational waves from a binary neutron star inspiral. *Phys. Rev. Lett.*, 119:161101, Oct 2017.

- [7] The LIGO Virgo Collaboration. GW190425: Observation of a compact binary coalescence with total mass $\sim 3.4\,m_{\odot}$. *The Astrophysical Journal*, 892(1):L3, mar 2020.
- [8] The LVK collaboration. GW190814: Gravitational waves from the coalescence of a 23 solar mass black hole with a 2.6 solar mass compact object. *The Astrophysical Journal*, 896(2):L44, jun 2020.
- [9] The LIGO Virgo Collaboration. Gwtc-2: Compact binary coalescences observed by ligo and virgo during the first half of the third observing run.
- [10] The LIGO Virgo Collaboration. The basic physics of the binary black hole merger GW150914. *Annalen der Physik*, 529(1-2):1600209, oct 2016.
- [11] The LIGO Scientific Collaboration and the Virgo Collaboration. Tests of general relativity with binary black holes from the second ligo-virgo gravitational-wave transient catalog.
- [12] Charles W. Misner. *Gravitation*. Princeton University Press, oct 2017.
- [13] Bernard Schutz. *A First Course in General Relativity; 2nd edition*. Cambridge University Press, 2009.
- [14] LVK collaboration. Narrow-band search for gravitational waves from known pulsars using the second LIGO observing run. *Physical Review D*, 99(12), jun 2019.
- [15] The LIGO Virgo Collaboration. Searches for continuous gravitational waves from 15 supernova remnants and fomalhaut b with advanced LIGO. *The Astrophysical Journal*, 875(2):122, apr 2019.

- [16] The LIGO Virgo Collaboration. Search for transient gravitational-wave signals associated with magnetar bursts during advanced LIGO’s second observing run. *The Astrophysical Journal*, 874(2):163, apr 2019.
- [17] B. S. Sathyaprakash and Bernard F. Schutz. Physics, astrophysics and cosmology with gravitational waves. *Living Reviews in Relativity*, 12(1), mar 2009.
- [18] Ernazar Abdikamalov, Giulia Pagliaroli, and David Radice. Gravitational waves from core-collapse supernovae.
- [19] Teng Zhang, Jiří Smetana, Yikang Chen, Joe Bentley, William E. East, Denis Martynov, Haixing Miao, and Huan Yang. Towards observing the neutron star collapse with gravitational wave detectors.
- [20] The LIGO Virgo Collaboration. Search for the isotropic stochastic background using data from advanced LIGO’s second observing run. *Physical Review D*, 100(6), sep 2019.
- [21] Juan García-Bellido. Massive primordial black holes as dark matter and their detection with gravitational waves. *Journal of Physics: Conference Series*, 840:012032, may 2017.
- [22] Simeon Bird, Ilias Cholis, Julian B. Muñoz, Yacine Ali-Haïmoud, Marc Kamionkowski, Ely D. Kovetz, Alvise Raccanelli, and Adam G. Riess. Did LIGO detect dark matter? *Physical Review Letters*, 116(20), may 2016.
- [23] T. Regimbau, M. Evans, N. Christensen, E. Katsavounidis, B. Sathyaprakash, and S. Vitale. Digging deeper: Observing primordial gravitational waves be-

- low the binary-black-hole-produced stochastic background. *Physical Review Letters*, 118(15), apr 2017.
- [24] Yanou Cui, Marek Lewicki, David E. Morrissey, and James D. Wells. Probing the pre-BBN universe with gravitational waves from cosmic strings. *Journal of High Energy Physics*, 2019(1), jan 2019.
- [25] The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration. Prospects for observing and localizing gravitational-wave transients with advanced ligo, advanced virgo and kagra, version 10. 2020.
- [26] F Acernese et al. Advanced virgo: a second-generation interferometric gravitational wave detector. *Classical and Quantum Gravity*, 32(2):024001, dec 2014.
- [27] Yoichi Aso, Yuta Michimura, Kentaro Somiya, Masaki Ando, Osamu Miyakawa, Takanori Sekiguchi, Daisuke Tatsumi, and Hiroaki Yamamoto. Interferometer design of the KAGRA gravitational wave detector. *Physical Review D*, 88(4), aug 2013.
- [28] S M Aston, M A Barton, A S Bell, N Beveridge, B Bland, A J Brummitt, G Cagnoli, C A Cantley, L Carbone, A V Cumming, L Cunningham, R M Cutler, R J S Greenhalgh, G D Hammond, K Haughian, T M Hayler, A Heptonstall, J Heefner, D Hoyland, J Hough, R Jones, J S Kissel, R Kumar, N A Lockerbie, D Lodhia, I W Martin, P G Murray, J O’Dell, M V Plissi, S Reid, J Romie, N A Robertson, S Rowan, B Shapiro, C C Speake, K A Strain, K V Tokmakov, C Torrie, A A van Veggel, A Vecchio, and I Wilmot. Update

on quadruple suspension design for advanced LIGO. *Classical and Quantum Gravity*, 29(23):235004, oct 2012.

- [29] Gregory M Harry, Matthew R Abernathy, Andres E Becerra-Toledo, Helena Armandula, Eric Black, Kate Dooley, Matt Eichenfield, Chinyere Nwabugwu, Akira Villar, D R M Crooks, Gianpietro Cagnoli, Jim Hough, Colin R How, Ian MacLaren, Peter Murray, Stuart Reid, Sheila Rowan, Peter H Sneddon, Martin M Fejer, Roger Route, Steven D Penn, Patrick Ganau, Jean-Marie Mackowski, Christophe Michel, Laurent Pinard, and Alban Remillieux. Titania-doped tantala/silica coatings for gravitational-wave detection. *Classical and Quantum Gravity*, 24(2):405–415, dec 2006.
- [30] S Hild, H Grote, J Degallaix, S Chelkowski, K Danzmann, A Freise, M Hewitson, J Hough, H Lück, M Prijatelj, K A Strain, J R Smith, and B Willke. DC-readout of a signal-recycled gravitational wave detector. *Classical and Quantum Gravity*, 26(5):055012, feb 2009.
- [31] Tobin T Fricke, Nicolás D Smith-Lefebvre, Richard Abbott, Rana Adhikari, Katherine L Dooley, Matthew Evans, Peter Fritschel, Valery V Frolov, Keita Kawabe, Jeffrey S Kissel, Bram J J Slagmolen, and Sam J Waldman. DC readout experiment in enhanced LIGO. *Classical and Quantum Gravity*, 29(6):065005, feb 2012.
- [32] J. Mizuno. *Comparison of optical configurations for laser-interferometric gravitational-wave detectors*. PhD thesis, AEI-Hannover, MPI for Gravitational Physics, Max Planck Society, 1995.

- [33] Gerhard Heinzel. Advanced optical techniques for laser-interferometric gravitational-wave detectors, 1999.
- [34] B.J. Meers. The frequency response of interferometric gravitational wave detectors. *Physics Letters A*, 142(8-9):465–470, dec 1989.
- [35] Charlotte Bond, Daniel Brown, Andreas Freise, and Kenneth A. Strain. Interferometer techniques for gravitational-wave detection. *Living Reviews in Relativity*, 19(1), dec 2016.
- [36] Vladimir B. Braginsky. Energetic quantum limit in large-scale interferometers. In *AIP Conference Proceedings*. AIP, 2000.
- [37] Carlton M. Caves. Quantum-mechanical noise in an interferometer. *Physical Review D*, 23(8):1693–1708, apr 1981.
- [38] Stefan L. Danilishin and Farid Ya. Khalili. Quantum measurement theory in gravitational-wave detectors. *Living Reviews in Relativity*, 15(1), apr 2012.
- [39]
- [40] Brian J. Meers. Recycling in laser-interferometric gravitational-wave detectors. *Physical Review D*, 38(8):2317–2326, oct 1988.
- [41] K. A. Strain and B. J. Meers. Experimental demonstration of dual recycling for interferometric gravitational-wave detectors. *Physical Review Letters*, 66(11):1391–1394, mar 1991.
- [42] D. Schnier, J. Mizuno, G. Heinzel, H. Lück, A. Rüdiger, R. Schilling, M. Schrempel, W. Winkler, and K. Danzmann. Power recycling in the garch-

- ing 30 m prototype interferometer for gravitational-wave detection. *Physics Letters A*, 225(4-6):210–216, feb 1997.
- [43] R. W. P. Drever, J. L. Hall, F. V. Kowalski, J. Hough, G. M. Ford, A. J. Munley, and H. Ward. Laser phase and frequency stabilization using an optical resonator. *Applied Physics B*, 31(2):97–105, Jun 1983.
- [44] Muzammil A. Arain and Guido Mueller. Design of the advanced LIGO recycling cavities. *Optics Express*, 16(14):10018, jun 2008.
- [45] G. Heinzel, J. Mizuno, R. Schilling, W. Winkler, A. Rüdiger, and K. Danzmann. An experimental demonstration of resonant sideband extraction for laser-interferometric gravitational wave detectors. *Physics Letters A*, 217(6):305–314, jul 1996.
- [46] J. Mizuno, K.A. Strain, P.G. Nelson, J.M. Chen, R. Schilling, A. Rüdiger, W. Winkler, and K. Danzmann. Resonant sideband extraction: a new configuration for interferometric gravitational wave detectors. *Physics Letters A*, 175(5):273–276, apr 1993.
- [47] Dhruva Ganapathy, Lee McCuller, Jameson Graef Rollins, Evan D. Hall, Lisa Barsotti, and Matthew Evans. Tuning advanced ligo to kilohertz signals from neutron-star collisions.
- [48] Peter Fritschel Jeff Lewis Sam Waldman Koji Arai, Sam Barnum. LIGO-T1000276-v5.

- [49] M Prijatelj, J Degallaix, H Grote, J Leong, C Affeldt, S Hild, H Lück, J Slutsky, H Wittel, K Strain, and K Danzmann. The output mode cleaner of GEO 600. *Classical and Quantum Gravity*, 29(5):055009, feb 2012.
- [50] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.
- [51] V. B. Braginsky and F. Ya. Khalili. Quantum nondemolition measurements: the route from toys to tools. *Reviews of Modern Physics*, 68(1):1–11, jan 1996.
- [52] Alessandra Buonanno and Yanbei Chen. Quantum noise in second generation, signal-recycled laser interferometric gravitational-wave detectors. *Physical Review D*, 64(4), jul 2001.
- [53] R Flaminio, J Franc, C Michel, N Morgado, L Pinard, and B Sassolas. A study of coating mechanical and optical losses in view of reducing mirror thermal noise in gravitational wave detectors. *Classical and Quantum Gravity*, 27(8):084030, apr 2010.
- [54] Rainer Weiss. Quarterly progress report: Electromagnetically coupled broadband gravitational antenna. DCC LIGO-P720002-v1, April 1972.
- [55] A. Buikema, C. Cahillane, G.L. Mansell, C.D. Blair, R. Abbott, C. Adams, R.X. Adhikari, A. Ananyeva, S. Appert, K. Arai, J.S. Areeda, Y. Asali, S.M. Aston, C. Austin, A.M. Baer, M. Ball, S.W. Ballmer, S. Banagiri, D. Barker, L. Barsotti, J. Bartlett, B.K. Berger, J. Betzwieser, D. Bhattacharjee, G. Billingsley, S. Biscans, R. M. Blair, N. Bode, P. Booker, R. Bork,

A. Bramley, A.F. Brooks, D.D. Brown, K.C. Cannon, X. Chen, A.A. Ciobanu, F. Clara, S.J. Cooper, K.R. Corley, S.T. Countryman, P.B. Co-
 vas, D.C. Coyne, L.E.H. Datrier, D. Davis, C. Di Fronzo, K.L. Dooley, J.C.
 Driggers, P. Dupej, S.E. Dwyer, A. Effler, T. Etzel, M. Evans, T.M. Evans,
 J. Feicht, A. Fernandez-Galiana, P. Fritschel, V.V. Frolov, P. Fulda, M. Fyffe,
 J. A. Giaime, K.D. Giardina, P. Godwin, E. Goetz, S. Gras, C. Gray, R. Gray,
 A.C. Green, E.K. Gustafson, R. Gustafson, J. Hanks, J. Hanson, T. Hard-
 wick, R.K. Hasskew, M.C. Heintze, A.F. Helmling-Cornell, N.A. Holland, J.
 D. Jones, S. Kandhasamy, S. Karki, M. Kasprzack, K. Kawabe, N. Kijbun-
 choo, P.J. King, J.S. Kissel, Rahul Kumar, M. Landry, B.B. Lane, B. Lantz,
 M. Laxen, Y.K. Lecoecuche, J. Leviton, J. Liu, M. Lormand, A.P. Lundgren,
 R. Macas, M. MacInnis, D.M. Macleod, S. Márka, Z. Márka, D.V. Martynov,
 K. Mason, T.J. Massinger, F. Matichard, N. Mavalvala, R. McCarthy, D.E.
 McClelland, S. McCormick, L. McCuller, J. McIver, T. McRae, G. Mendell,
 K. Merfeld, E.L. Merilh, F. Meylahn, T. Mistry, R. Mittleman, G. Moreno,
 C.M. Mow-Lowry, S. Mozzon, A. Mullavey, T.J.N. Nelson, P. Nguyen, L.
 K. Nuttall, J. Oberling, Richard J. Oram, B. O'Reilly, C. Osthelder, D.J.
 Ottaway, H. Overmier, J.R. Palamos, W. Parker, E. Payne, A. Pele, R. Pen-
 horwood, C.J. Perez, M. Pirello, H. Radkins, K.E. Ramirez, J.W. Richardson,
 K. Riles, N.A. Robertson, J.G. Rollins, C.L. Romel, J.H. Romie, M.P. Ross,
 K. Ryan, T. Sadecki, E.J. Sanchez, L.E. Sanchez, T.R. Saravanan, R.L.
 Savage, D. Schaetzel, R. Schnabel, R.M.S. Schofield, E. Schwartz, D. Sellers,
 T. Shaffer, D. Sigg, B.J.J. Slagmolen, J.R. Smith, S. Soni, B. Sorazu, A.P.
 Spencer, K.A. Strain, L. Sun, M.J. Szczepańczyk, M. Thomas, P. Thomas,
 K.A. Thorne, K. Toland, C.I. Torrie, G. Traylor, M. Tse, A.L. Urban, G. Va-

- jente, G. Valdes, D. C. Vander-Hyde, P. J. Veitch, K. Venkateswara, G. Venugopalan, A. D. Viets, T. Vo, C. Vorvick, M. Wade, R. L. Ward, J. Warner, B. Weaver, R. Weiss, C. Whittle, B. Willke, C. C. Wipf, L. Xiao, H. Yamamoto, Hang Yu, Haocun Yu, L. Zhang, M. E. Zucker, and J. Zweizig. Sensitivity and performance of the advanced LIGO detectors in the third observing run. *Physical Review D*, 102(6), sep 2020.
- [56] Carlton M. Caves and Bonny L. Schumaker. New formalism for two-photon quantum optics. i. quadrature phases and squeezed states. *Physical Review A*, 31(5):3068–3092, may 1985.
- [57] Bonny L. Schumaker and Carlton M. Caves. New formalism for two-photon quantum optics. II. mathematical foundation and compact notation. *Physical Review A*, 31(5):3093–3111, may 1985.
- [58] M. Tse, Haocun Yu, N. Kijbunchoo, A. Fernandez-Galiana, P. Dupej, L. Barsotti, C. D. Blair, D. D. Brown, S. E. Dwyer, A. Effler, M. Evans, P. Fritschel, V. V. Frolov, A. C. Green, G. L. Mansell, F. Matichard, N. Mavalvala, D. E. McClelland, L. McCuller, T. McRae, J. Miller, A. Mullavey, E. Oelker, I. Y. Phinney, D. Sigg, B. J. J. Slagmolen, T. Vo, R. L. Ward, C. Whittle, R. Abbott, C. Adams, R. X. Adhikari, A. Ananyeva, S. Appert, K. Arai, J. S. Areeda, Y. Asali, S. M. Aston, C. Austin, A. M. Baer, M. Ball, S. W. Ballmer, S. Bagnagiri, D. Barker, J. Bartlett, B. K. Berger, J. Betzwieser, D. Bhattacharjee, G. Billingsley, S. Biscans, R. M. Blair, N. Bode, P. Booker, R. Bork, A. Bramley, A. F. Brooks, A. Buikema, C. Cahillane, K. C. Cannon, X. Chen, A. A. Ciobanu, F. Clara, S. J. Cooper, K. R. Corley, S. T. Countryman, P. B. Covas, D. C. Coyne, L. E. H. Datrier, D. Davis, C. Di Fronzo, J. C. Driggers,

T. Etzel, T. M. Evans, J. Feicht, P. Fulda, M. Fyffe, J. A. Giaime, K. D. Giardina, P. Godwin, E. Goetz, S. Gras, C. Gray, R. Gray, Anchal Gupta, E. K. Gustafson, R. Gustafson, J. Hanks, J. Hanson, T. Hardwick, R. K. Hasskew, M. C. Heintze, A. F. Helmling-Cornell, N. A. Holland, J. D. Jones, S. Kandhasamy, S. Karki, M. Kasprzack, K. Kawabe, P. J. King, J. S. Kissel, Rahul Kumar, M. Landry, B. B. Lane, B. Lantz, M. Laxen, Y. K. Lecoecue, J. Leviton, J. Liu, M. Lormand, A. P. Lundgren, R. Macas, M. MacInnis, D. M. Macleod, S. Márka, Z. Márka, D. V. Martynov, K. Mason, T. J. Massinger, R. McCarthy, S. McCormick, J. McIver, G. Mendell, K. Merfeld, E. L. Merilh, F. Meylahn, T. Mistry, R. Mittleman, G. Moreno, C. M. Mow-Lowry, S. Mozzon, T. J. N. Nelson, P. Nguyen, L. K. Nuttall, J. Oberling, R. J. Oram, B. O'Reilly, C. Osthelder, D. J. Ottaway, H. Overmier, J. R. Palamos, W. Parker, E. Payne, A. Pele, C. J. Perez, M. Pirello, H. Radkins, K. E. Ramirez, J. W. Richardson, K. Riles, N. A. Robertson, J. G. Rollins, C. L. Romel, J. H. Romie, M. P. Ross, K. Ryan, T. Sadecki, E. J. Sanchez, L. E. Sanchez, T. R. Saravanan, R. L. Savage, D. Schaetzl, R. Schnabel, R. M. S. Schofield, E. Schwartz, D. Sellers, T. J. Shaffer, J. R. Smith, S. Soni, B. Sorazu, A. P. Spencer, K. A. Strain, L. Sun, M. J. Szczepańczyk, M. Thomas, P. Thomas, K. A. Thorne, K. Toland, C. I. Torrie, G. Traylor, A. L. Urban, G. Vajente, G. Valdes, D. C. VanderHyde, P. J. Veitch, K. Venkateswara, G. Venugopalan, A. D. Viets, C. Vorvick, M. Wade, J. Warner, B. Weaver, R. Weiss, B. Willke, C. C. Wipf, L. Xiao, H. Yamamoto, M. J. Yap, Hang Yu, L. Zhang, M. E. Zucker, and J. Zweizig. Quantum-enhanced advanced LIGO detectors in the era of gravitational-wave astronomy. *Physical Review Letters*, 123(23), dec 2019.

- [59] James Lough, Emil Schreiber, Fabio Bergamin, Hartmut Grote, Moritz Mehmet, Henning Vahlbruch, Christoph Affeldt, Marc Brinkmann, Aparna Bisht, Volker Kringel, Harald Lück, Nikhil Mukund, Séverin Nadji, Borja Sorazu, Kenneth Strain, Michael Weinert, and Karsten Danzmann. First demonstration of 6 db quantum noise reduction in a kilometer scale gravitational wave observatory.
- [60] D. F. Walls. Squeezed states of light. *Nature*, 306(5939):141–146, nov 1983.
- [61] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [62] Kentaro Komori, Dhruva Ganapathy, Chris Whittle, Lee McCuller, Lisa Barsotti, Nergis Mavalvala, and Matthew Evans. Demonstration of an amplitude filter cavity at gravitational-wave frequencies. *Physical Review D*, 102(10), nov 2020.
- [63] Joe Bentley, Philip Jones, Denis Martynov, Andreas Freise, and Haixing Miao. Converting the signal-recycling cavity into an unstable optomechanical filter to enhance the detection bandwidth of gravitational-wave detectors. *Physical Review D*, 99(10), may 2019.
- [64] Haixing Miao, Yiqiu Ma, Chunnong Zhao, and Yanbei Chen. Enhancing the bandwidth of gravitational-wave detectors with unstable optomechanical filters. *Physical Review Letters*, 115(21), nov 2015.

- [65] Xiang Li, Maxim Goryachev, Yiqiu Ma, Michael E. Tobar, Chunrong Zhao, Rana X Adhikari, , and Yanbei Chen. Broadband sensitivity improvement via coherent quantum feedback with pt symmetry. *PRL*, To be published.
- [66] Vladimir B. Braginsky, Yuri I. Vorontsov, and Kip S. Thorne. Quantum non-demolition measurements. *Science*, 209(4456):547–557, 1980.
- [67] Patricia Purdue and Yanbei Chen. Practical speed meter designs for quantum nondemolition gravitational-wave interferometers. *Physical Review D*, 66(12), dec 2002.
- [68] T. Zhang, E. Knyazev, S. Steinlechner, F. Y. Khalili, B. W. Barr, A. S. Bell, P. Dupej, J. Briggs, C. Gräf, J. Callaghan, J. S. Hennig, E. A. Houston, S. H. Huttner, S. S. Leavey, D. Pascucci, B. Sorazu, A. Spencer, J. Wright, K. A. Strain, S. Hild, and S. L. Danilishin. Quantum noise cancellation in asymmetric speed metres with balanced homodyne readout. *New journal of physics*, 20(10):103040, 2018.
- [69] S H Huttner, S L Danilishin, S Hild, and K A Strain. Comparison of different sloshing speedmeters. *Classical and Quantum Gravity*, 37(8):085022, mar 2020.
- [70] L. McCuller, C. Whittle, D. Ganapathy, K. Komori, M. Tse, A. Fernandez-Galiana, L. Barsotti, P. Fritschel, M. MacInnis, F. Matichard, K. Mason, N. Mavalvala, R. Mittleman, Haocun Yu, M.E. Zucker, and M. Evans. Frequency-dependent squeezing for advanced LIGO. *Physical Review Letters*, 124(17), apr 2020.

- [71] G. Gasparian. personal communication, September 2018. GPD optoelectronics Corporation.
- [72] T. Hornby. personal communication, September 2018. Laser Components (UK) Ltd.
- [73] R. Miron. personal communication, September 2018. Thorlabs Ltd.
- [74] K. Kalinina. personal communication, September 2018. LED Microsensor.
- [75] Georgia L. Mansell, Terry G. McRae, Paul A. Altin, Min Jet Yap, Robert L. Ward, Bram J. J. Slagmolen, Daniel A. Shaddock, and David E. McClelland. Observation of squeezed light in the $2\mu\text{m}$ region. *Physical Review Letters*, 120(20), may 2018.
- [76] M. J. Yap, D. W. Gould, T. G. McRae, P. A. Altin, N. Kijbunchoo, G. L. Mansell, R. L. Ward, D. A. Shaddock, B. J. J. Slagmolen, and D. E. McClelland. Squeezed vacuum phase control at $2\mu\text{m}$. *Optics Letters*, 44(21):5386, nov 2019.
- [77] P. Fritschel and E. Gustafson. Advanced LIGO Optical Layout. LIGO Document D0902838-v5 <https://dcc.ligo.org/LIGO-D0902838>.
- [78] L. Barsotti, Gras. S, M. Evans, and P. Fritschel. Updated Advanced LIGO sensitivity design curve . LIGO DCC, April 2018. LIGO Document T1800044-v5 <https://dcc.ligo.org/LIGO-T1800044/public>.
- [79] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The A+ design curve. LIGO DCC, April 2018. LIGO Document T1800042-v5 <https://dcc.ligo.org/LIGO-T1800042/public>.

- [80] M Punturo, M Abernathy, F Acernese, B Allen, N Andersson, K Arun, F Barone, B Barr, M Barsuglia, M Beker, N Beveridge, S Birindelli, S Bose, L Bosi, S Braccini, C Bradaschia, T Bulik, E Calloni, G Cella, E Chassande Mottin, S Chelkowski, A Chincarini, J Clark, E Coccia, C Colacino, J Colas, A Cumming, L Cunningham, E Cuoco, S Danilishin, K Danzmann, G De Luca, R De Salvo, T Dent, R De Rosa, L Di Fiore, A Di Virgilio, M Doets, V Fafone, P Falferi, R Flaminio, J Franc, F Frasconi, A Freise, P Fulda, J Gair, G Gemme, A Gennai, A Giazotto, K Glampedakis, M Granata, H Grote, G Guidi, G Hammond, M Hannam, J Harms, D Heinert, M Hendry, I Heng, E Hennes, S Hild, J Hough, S Husa, S Huttner, G Jones, F Khalili, K Kokeyama, K Kokkotas, B Krishnan, M Lorenzini, H Lück, E Majorana, I Mandel, V Mandic, I Martin, C Michel, Y Minenkov, N Morgado, S Mosca, B Mours, H Müller-Ebhardt, P Murray, R Nawrodt, J Nelson, R Oshaughnessy, C D Ott, C Palomba, A Paoli, G Parguez, A Pasqualetti, R Passaquieti, D Passuello, L Pinard, R Poggiani, P Popolizio, M Prato, P Puppo, D Rabeling, P Rapagnani, J Read, T Regimbau, H Rehbein, S Reid, L Rezzolla, F Ricci, F Richard, A Rocchi, S Rowan, A Rüdiger, B Sassolas, B Sathyaprakash, R Schnabel, C Schwarz, P Seidel, A Sintes, K Somiya, F Speirits, K Strain, S Strigin, P Sutton, S Tarabrin, A Thüring, J van den Brand, C van Leewen, M van Veggel, C van den Broeck, A Vecchio, J Veitch, F Vetrano, A Vicere, S Vyatchanin, B Willke, G Woan, P Wolfango, and K Yamamoto. The einstein telescope: a third-generation gravitational wave observatory. *Classical and Quantum Gravity*, 27(19):194002, sep 2010.

- [81] S Hild, M Abernathy, F Acernese, P Amaro-Seoane, N Andersson, K Arun,

F Barone, B Barr, M Barsuglia, M Beker, N Beveridge, S Birindelli, S Bose, L Bosi, S Braccini, C Bradaschia, T Bulik, E Calloni, G Cella, E Chassande Mottin, S Chelkowski, A Chincarini, J Clark, E Coccia, C Colacino, J Colas, A Cumming, L Cunningham, E Cuoco, S Danilishin, K Danzmann, R De Salvo, T Dent, R De Rosa, L Di Fiore, A Di Virgilio, M Doets, V Fafone, P Falferi, R Flaminio, J Franc, F Frasconi, A Freise, D Friedrich, P Fulda, J Gair, G Gemme, E Genin, A Gennai, A Giazotto, K Glampedakis, C Gräf, M Granata, H Grote, G Guidi, A Gurkovsky, G Hammond, M Hannam, J Harms, D Heinert, M Hendry, I Heng, E Hennes, J Hough, S Husa, S Huttner, G Jones, F Khalili, K Kokeyama, K Kokkotas, B Krishnan, T G F Li, M Lorenzini, H Lück, E Majorana, I Mandel, V Mandic, M Mantovani, I Martin, C Michel, Y Minenkov, N Morgado, S Mosca, B Mours, H Müller-Ebhardt, P Murray, R Nawrodt, J Nelson, R Oshaughnessy, C D Ott, C Palomba, A Paoli, G Parguez, A Pasqualetti, R Passaquieti, D Passuello, L Pinard, W Plastino, R Poggiani, P Popolizio, M Prato, M Punturo, P Puppò, D Rabeling, P Rapagnani, J Read, T Regimbau, H Rehbein, S Reid, F Ricci, F Richard, A Rocchi, S Rowan, A Rüdiger, L Santamaría, B Sassolas, B Sathyaprakash, R Schnabel, C Schwarz, P Seidel, A Sintes, K Somiya, F Speirits, K Strain, S Strigin, P Sutton, S Tarabrin, A Thüring, J van den Brand, M van Veggel, C van den Broeck, A Vecchio, J Veitch, F Vetrano, A Vicere, S Vyatchanin, B Willke, G Woan, and K Yamamoto. Sensitivity studies for third-generation gravitational wave observatories. *Classical and Quantum Gravity*, 28(9):094013, apr 2011.

[82] R X Adhikari, K Arai, A F Brooks, C Wipf, O Aguiar, P Altin, B Barr, L Bar-

sotti, R Bassiri, A Bell, G Billingsley, R Birney, D Blair, E Bonilla, J Briggs, D D Brown, R Byer, H Cao, M Constancio, S Cooper, T Corbitt, D Coyne, A Cumming, E Daw, R deRosa, G Eddolls, J Eichholz, M Evans, M Fejer, E C Ferreira, A Freise, V V Frolov, S Gras, A Green, H Grote, E Gustafson, E D Hall, G Hammond, J Harms, G Harry, K Haughian, D Heinert, M Heintze, F Hellman, J Hennig, M Hennig, S Hild, J Hough, W Johnson, B Kamai, D Kapasi, K Komori, D Koptsov, M Korobko, W Z Korth, K Kuns, B Lantz, S Leavey, F Magana-Sandoval, G Mansell, A Markosyan, A Markowitz, I Martin, R Martin, D Martynov, D E McClelland, G McGhee, T McRae, J Mills, V Mitrofanov, M Molina-Ruiz, C Mow-Lowry, J Munch, P Murray, S Ng, M A Okada, D J Ottaway, L Prokhorov, V Quetschke, S Reid, D Reitze, J Richardson, R Robie, I Romero-Shaw, R Route, S Rowan, R Schnabel, M Schneewind, F Seifert, D Shaddock, B Shapiro, D Shoemaker, A S Silva, B Slagmolen, J Smith, N Smith, J Steinlechner, K Strain, D Taira, S Tait, D Tanner, Z Tornasi, C Torrie, M Van Veggel, J Vanheijningen, P Veitch, A Wade, G Wallace, R Ward, R Weiss, P Wessels, B Willke, H Yamamoto, M J Yap, and C Zhao. A cryogenic silicon interferometer for gravitational-wave detection. *Classical and Quantum Gravity*, 37(16):165003, jul 2020.

- [83] The LIGO Virgo Collaboration. Exploring the sensitivity of next generation gravitational wave detectors. *Classical and Quantum Gravity*, 34(4):044001, jan 2017.
- [84] David Reitze, Rana X Adhikari, Stefan Ballmer, Barry Barish, Lisa Barsotti, GariLynn Billingsley, Duncan A. Brown, Yanbei Chen, Dennis Coyne, Robert Eisenstein, Matthew Evans, Peter Fritschel, Evan D. Hall, Albert Laz-

- zarini, Geoffrey Lovelace, Jocelyn Read, B. S. Sathyaprakash, David Shoemaker, Joshua Smith, Calum Torrie, Salvatore Vitale, Rainer Weiss, Christopher Wipf, and Michael Zucker. Cosmic explorer: The u.s. contribution to gravitational-wave astronomy beyond ligo.
- [85] Evan D. Hall, Kevin Kuns, Joshua R. Smith, Yuntao Bai, Christopher Wipf, Sebastien Biscans, Rana X Adhikari, Koji Arai, Stefan Ballmer, Lisa Barsotti, Yanbei Chen, Matthew Evans, Peter Fritschel, Jan Harms, Brittany Kamai, Jameson Graef Rollins, David Shoemaker, Bram Slagmolen, Rainer Weiss, and Hiro Yamamoto. Gravitational-wave physics with cosmic explorer: limits to low-frequency sensitivity. 2020.
- [86] B. Lantz, S. L. Danilishin, S. Hild, Gustafson E., D. Coyne, V. Quetschke, G. D. Hammond, R. Adhikari, M. Evans, and R. Bassiri. Instrument science white paper 2018. LIGO DCC.
- [87] Anja Schroeter, Ronny Nawrodt, Roman Schnabel, Stuart Reid, Iain Martin, Sheila Rowan, Christian Schwarz, Torsten Koettig, Ralf Neubert, Matthias Thürk, Wolfgang Vodel, Andreas Tünnermann, Karsten Danzmann, and Paul Seidel. On the mechanical quality factors of cryogenic test masses from fused silica and crystalline quartz.
- [88] Martin A. Green and Mark J. Keevers. Optical properties of intrinsic silicon at 300 k. *Progress in Photovoltaics: Research and Applications*, 3(3):189–192, 1995.
- [89] I Martin, H Armandula, C Comtet, M M Fejer, A Gretarsson, G Harry, J Hough, J-M M Mackowski, I MacLaren, C Michel, J-L Montorio, N Morgado,

- R Nawrodt, S Penn, S Reid, A Remillieux, R Route, S Rowan, C Schwarz, P Seidel, W Vodel, and A Zimmer. Measurements of a low-temperature mechanical dissipation peak in a single layer of ta 2 o 5 doped with TiO 2. *Classical and Quantum Gravity*, 25(5):055005, feb 2008.
- [90] I W Martin, R Bassiri, R Nawrodt, M M Fejer, A Gretarsson, E Gustafson, G Harry, J Hough, I MacLaren, S Penn, S Reid, R Route, S Rowan, C Schwarz, P Seidel, J Scott, and A L Woodcraft. Effect of heat treatment on mechanical dissipation in ta 2 o 5 coatings. *Classical and Quantum Gravity*, 27(22):225020, oct 2010.
- [91] I W Martin, R Nawrodt, K Craig, C Schwarz, R Bassiri, G Harry, J Hough, S Penn, S Reid, R Robie, and S Rowan. Low temperature mechanical dissipation of an ion-beam sputtered silica film. *Classical and Quantum Gravity*, 31(3):035019, jan 2014.
- [92] Kieran Craig, Jessica Steinlechner, Peter G. Murray, Angus S. Bell, Ross Birney, Karen Haughian, Jim Hough, Ian MacLaren, Steve Penn, Stuart Reid, Raymond Robie, Sheila Rowan, and Iain W. Martin. Mirror coating solution for the cryogenic einstein telescope. *Physical Review Letters*, 122(23), jun 2019.
- [93] Nina Bode, Joseph Briggs, Xu Chen, Maik Frede, Peter Fritschel, Michael Fyffe, Eric Gustafson, Matthew Heintze, Peter King, Jian Liu, Jason Oberling, Richard L. Savage, Andrew Spencer, and Benno Willke. Advanced LIGO laser systems for o3 and future observation runs. *Galaxies*, 8(4):84, dec 2020.

- [94] B. Willke, P. King, R. Savage, and P. Fritschel. Pre-stabilized laser design requirements. DCC LIGO-T050036-v4, July 2011.
- [95] E. A. Shcherbakov, V. V. Fomin, A. A. Abramov, A. A. Ferin, D. V. Mochalov, and V. P. Gapontsev. Industrial grade 100 kW power CW fiber laser. In *Advanced Solid-State Lasers Congress*. OSA, 2013.
- [96] Junsu Lee, Kwang Hyun Lee, Hwanseong Jeong, Minkyu Park, Ji Hoon Seung, and Jung Hwan Lee. 205 kW all-fiber high-beam-quality fiber amplifier with stimulated brillouin scattering suppression incorporating a narrow-linewidth fiber-bragg-grating-stabilized laser diode seed source. *Applied Optics*, 58(23):6251, aug 2019.
- [97] Yusheng Huang, Ping Yan, Zehui Wang, Jiading Tian, Dan Li, Qirong Xiao, and Mali Gong. 219 kW narrow linewidth FBG-based MOPA configuration fiber laser. *Optics Express*, 27(3):3136, jan 2019.
- [98] Pengfei Ma, Rumao Tao, Rongtao Su, Xiaolin Wang, Pu Zhou, and Zejin Liu. 189 kW all-fiberized and polarization-maintained amplifiers with narrow linewidth and near-diffraction-limited beam quality. *Optics Express*, 24(4):4187, feb 2016.
- [99] Craig Robin, Iyad Dajani, and Benjamin Pulford. Modal instability-suppressing, single-frequency photonic crystal fiber amplifier with 811 w output power. *Optics Letters*, 39(3):666, jan 2014.
- [100] Clément Dixneuf, Germain Guiraud, Yves-Vincent Bardin, Quentin Rosa, Mathieu Goeppner, Adèle Hilico, Christophe Pierre, Johan Boullet, Nicholas

- Traynor, and Giorgio Santarelli. Ultra-low intensity noise, all fiber 365 w linearly polarized single frequency laser at 1064 nm. *Optics Express*, 28(8):10960, mar 2020.
- [101] L. Winkelmann, O. Puncken, R. Kluzik, C. Veltkamp, P. Kwee, J. Poeld, C. Bogan, B. Willke, M. Frede, J. Neumann, P. Wessels, and D. Kracht. Injection-locked single-frequency laser with an output power of 220 w. *Applied Physics B*, 102(3):529–538, feb 2011.
- [102] Felix Wellmann, Michael Steinke, Fabian Meylahn, Nina Bode, Benno Willke, Ludger Overmeyer, Jörg Neumann, and Dietmar Kracht. High power, single-frequency, monolithic fiber amplifier for the next generation of gravitational wave detectors. *Optics Express*, 27(20):28523, sep 2019.
- [103] Aaron Buikema, Franklin Jose, Steven J. Augst, Peter Fritschel, and Nergis Mavalvala. Narrow-linewidth fiber amplifier for gravitational-wave detectors. *Optics Letters*, 44(15):3833, jul 2019.
- [104] Nina Bode, Fabian Meylahn, and Benno Willke. Sequential high power laser amplifiers for gravitational wave detection. *Optics Express*, 28(20):29469, sep 2020.
- [105] Fabian Thies, Nina Bode, Patrick Oppermann, Maik Frede, Bastian Schulz, and Benno Willke. Nd:YVO₄ high-power master oscillator power amplifier laser system for second-generation gravitational wave detectors. *Optics Letters*, 44(3):719, jan 2019.
- [106] P. Kwee, C. Bogan, K. Danzmann, M. Frede, H. Kim, P. King, J. Pöld, O. Puncken, R. L. Savage, F. Seifert, P. Wessels, L. Winkelmann, and

- B. Willke. Stabilized high-power laser system for the gravitational wave detector advanced LIGO. *Optics Express*, 20(10):10617, apr 2012.
- [107] the LIGO Scientific Collaboration. Instrument science white paper 2019. LIGO internal document LIGO-T1900409–v3.
- [108] Andrew P. Spencer. *Advanced Techniques in Laser Interferometry for Current and Future Gravitational Wave Detectors*. PhD thesis, University of Glasgow, August 2019.
- [109] Thomas J. Kane and Robert L. Byer. Monolithic, unidirectional single-mode nd:YAG ring laser. *Optics Letters*, 10(2):65, feb 1985.
- [110] I Freitag, A Tünnermann, and H Welling. Power scaling of diode-pumped monolithic nd:YAG lasers to output powers of several watts. *Optics Communications*, 115(5-6):511–515, apr 1995.
- [111] Michèle Heurs, Volker M. Quetschke, Benno Willke, Karsten Danzmann, and Ingo Freitag. Simultaneously suppressing frequency and intensity noise in a nd:YAG nonplanar ring oscillator by means of the current-lock technique. *Optics Letters*, 29(18):2148, sep 2004.
- [112] J. Lui, R. Savage, P. King, L. Zhang, and S. et al. Appert. aLIGO all-bolted PMC summary. LIGO Document T1700543-v2 <https://dcc.ligo.org/LIGO-T1700543>.
- [113] Patrick Kwee, Frank Seifert, Benno Willke, and Karsten Danzmann. Laser beam quality and pointing measurement with an optical resonator. *Review of Scientific Instruments*, 78(7):073103, jul 2007.

- [114] P. Kwee. *Laser Characterization and Stabilization for Precision Interferometry*. PhD thesis, Max Planck Institute for Gravitational Physics, 2010.
- [115] C. Bogan. Intensity Stabilization Servo Module (inner loop). LIGO Document D1001985-v2 <https://dcc.ligo.org/LIGO-D1001985/public>.
- [116] P. Kwee. aLIGO PSL ISS inner-loop photodiode. LIGO Document D1001998-v2 <https://dcc.ligo.org/LIGO-D1001998/public>.
- [117] R. S. Abbott and P. J. King. Diode-pumped nd:YAG laser intensity noise suppression using a current shunt. *Review of Scientific Instruments*, 72(2):1346, 2001.
- [118] B W Barr, K A Strain, and C J Killow. Laser amplitude stabilization for advanced interferometric gravitational wave detectors. *Classical and Quantum Gravity*, 22(20):4279–4283, oct 2005.
- [119] Jameson Rollins, David Ottaway, Michael Zucker, Rainer Weiss, and Richard Abbott. Solid-state laser intensity stabilization at the 10^{-8} level. *Optics Letters*, 29(16):1876, aug 2004.
- [120] I Zawischa, M Brendel, K Danzmann, C Fallnich, M Heurs, S Nagano, V Quetschke, H Welling, and B Willke. The GEO 600 laser system. *Classical and Quantum Gravity*, 19(7):1775–1781, mar 2002.
- [121] F Nocera. LIGO laser intensity noise suppression. *Classical and Quantum Gravity*, 21(5):S481–S485, feb 2004.
- [122] P. Horowitz and W. Hill. *The Art of Electronics*. Cambridge University Pr., 2015.

- [123] G. Mueller. Pointing Requirements for Advanced LIGO. LIGO Document T0900142-v2 <https://dcc.ligo.org/LIGO-T0900142>.
- [124] Euan Morrison, Brian J. Meers, David I. Robertson, and Henry Ward. Automatic alignment of optical interferometers. *Applied Optics*, 33(22):5041, aug 1994.
- [125] H. Kim, P. King, C. Krämer, P. Kwee, J. Pöld, R. Savage, P. Weßels, and B. Willke. Psl final design. DCC LIGO T0900649-v4, February 2010.
- [126] N. Uehara and K. Ueda. Accurate measurement of the radius of curvature of a concave mirror and the power dependence in a high-finesse fabry-perot interferometer. *Applied Optics*, 34(25):5611, sep 1995.
- [127] Henrik Tünnermann, Jan Hendrik Pöld, Jörg Neumann, Dietmar Kracht, Benno Willke, and Peter Weßels. Beam quality and noise properties of coherently combined ytterbium doped single frequency fiber amplifiers. *Optics Express*, 19(20):19600, sep 2011.
- [128] Li-Wei Wei, Frédéric Cleva, and Catherine Nary Man. Coherently combined master oscillator fiber power amplifiers for advanced virgo. *Optics Letters*, 41(24):5817, dec 2016.
- [129] V. Frolov. DARM Optical Gain and Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42231>.
- [130] L. McCuller. Frequency Dependence of Squeezing Loss and Angle, Power Measurement by Radiation Pressure. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=52718>.

- [131] L. McCuller. Squeezing with Mode Mismatch Observations in LIGO . LIGO Document G2001503-v1 <https://dcc.ligo.org/LIGO-G2001503>.
- [132] L. McCuller. Frequency Dependent Squeezing effects in the LIGO Interferometers. LIGO Document G2000863-v2 <https://dcc.ligo.org/LIGO-G2000863>.
- [133] V. Frolov. Arm Cavity Power, Power Recycling Gain, IFO Reflectivity and Arm Cavity Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42639>.
- [134] C. Blair. OMC Scans. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37742>.
- [135] C. Blair and V. Frolov. 40W OMC Scans Method and Preliminary Data. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=45830>.
- [136] M. Kasprzack, A. Effler, A. Mullavey, and M. Tse. HAM6 Work - M12 Move. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37291>.
- [137] A. Effler and M. Kasprzack. LLO Interferometer Output Beam Study. LIGO Document T1800089-v2 <https://dcc.ligo.org/LIGO-T1800089>.
- [138] C. Blair, V. Frolov, and A. Effler. SRM CO2 Chop. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38689>.
- [139] C. Blair. Annular CO2 1W at Various IFO Power. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=43065>.
- [140] C. Blair, V. Frolov, and A. Effler. OMC Mode Matching in Single Bounce. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38698>.

- [141] A. Effler, V. Frolov, and C. Blair. OMC Scans of Cooling Interferometer. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=41428>.
- [142] A. Green, A. Effler, P. Dupej, V. Frolov, P. Fulda, and H. Yu. Finesse Squeezing Mode Matching Study for LLO. LIGO Document T1900159-v1 <https://dcc.ligo.org/LIGO-T1900159>.
- [143] A. Effler, P. Dupej, H. Yu, A. Green, and C. Blair. SQZ SRM Bounce 98% Mode Matching. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42187>.
- [144] Daniel David Brown and Andreas Freise. Pykat, July 2017. <http://www.gwoptics.org/pykat>.
- [145] C. Blair. PRC Gouy phase measurement. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=49409>.
- [146] J. Richardson, M. Tse, and C. Austin. Measurement of SRC Gouy Phase. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=39779>.
- [147] M S Stefszky, C M Mow-Lowry, S S Y Chua, D A Shaddock, B C Buchler, H Vahlbruch, A Khalaidovski, R Schnabel, P K Lam, and D E McClelland. Balanced homodyne detection of optical quantum states at audio-band frequencies and below. *Classical and Quantum Gravity*, 29(14):145015, jun 2012.
- [148] E. Oelker. Personal Communication, 2021.
- [149] Peter Fritschel, Matthew Evans, and Valery Frolov. Balanced homodyne readout for quantum limited gravitational wave detectors. *Optics Express*, 22(4):4224, feb 2014.

- [150] L. McCuller. NOV Squeezing data infers LO/Readout angle of 11.5 ± 2 deg, good side. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=53867>.
- [151] Daniel Sigg and. Status of the LIGO detectors. *Classical and Quantum Gravity*, 25(11):114041, may 2008.
- [152] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [153] D.E. McClelland, N. Mavalvala, Y. Chen, and R. Schnabel. Advanced interferometry, quantum optics and optomechanics in gravitational wave detectors. *Laser & Photonics Reviews*, 5(5):677–696, mar 2011.
- [154] The LIGO Scientific Collaboration. A gravitational wave observatory operating beyond the quantum shot-noise limit. *Nature Physics*, 7(12):962–965, sep 2011.
- [155] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [156] T. Zhang. *Advanced techniques for future generation gravitational wave detectors*. PhD thesis, University of Glasgow, 2019.
- [157] Hang Yu Peter Fritschel, Vivishek Sudhir. LIGO DCC Document G1702485-v1.

- [158] P. Fritschel and S. Hild. Requirements for the HAM Relay Triple Suspension (HRTS). LIGO Document T1800413-v1 <https://dcc.ligo.org/LIGO-T1900817>.
- [159] R. Jones K.A. Strain. Ham relay-optics suspension: pre-conceptual design. LIGO-T1900036-v5 <https://dcc.ligo.org/LIGO-T1900036>.
- [160] B. Lantz, R. Mittleman, J. Kissel, J. Warner, and A. Pele. Typical Motion of HAM Suspension Points in O2. LIGO Document T1800066-v2 <https://dcc.ligo.org/T1800066>.
- [161] Evgenia Kochkina. *Stigmatic and Astigmatic Gaussian Beams in Fundamental Mode: Impact of Beam Model Choice on Interferometric Pathlength Signal Estimates*. PhD thesis, Hannover U., 2013.
- [162] J. Briggs, M. Barton, J. Jones, E. Oelker, S. Webster, and K.A. Strain. A+BHD ISC Additional Design Requirements. LIGO Document E2000072-v2 <https://dcc.ligo.org/LIGO-E2000072-v2>.
- [163] A. Brookes, L. Barsotti, G. Billingsley, K.A. Strain, and M. Zucker. A+o5-ham6 design considerations and requirements. LIGO DCC, April 2020. E1900385-v1.
- [164] A. Brookes, P. Fritschel, F. Matichard, L. McCuller, M. Noh, and C. Torrie. Sams design requirements document. LIGO DCC, February 2019. E1800314.
- [165] Ryan Lawrence, Michael Zucker, Peter Fritschel, Phil Marfuta, and David Shoemaker. Adaptive thermal compensation of test masses in advanced LIGO. *Classical and Quantum Gravity*, 19(7):1803–1812, mar 2002.

- [166] C. Blair. personal communication.
- [167] P. Fulda, R. Goetz, A. Green, R. Martin, and D. B. Tanner. A+ Output Faraday Isolator Final Design. LIGO Document T1900788-v2 <https://dcc.ligo.org/LIGO-T1900788>.
- [168] K. A. Strain, J. Briggs, M. Barton, R. Jones, S. Webster, and E. Oelker. Conceptual design: Balanced homodyne detection for a+. LIGO Document E1900377-v3 <https://dcc.ligo.org/LIGO-E1900377>.
- [169] S. Webster, K. A. Strain, R. Jones, and M. Barton. A+ bhd ham6 layout of sensing and control elements. LIGO Document G2001448-v5 <https://dcc.ligo.org/LIGO-G2001448>.
- [170] Peter Fritschel Jeff Lewis Sam Waldman Koji Arai, Sam Barnum. LIGO-T1000276-v5.
- [171] L. Barsotti. aLIGO ISC Anti-symmetric Port: Beam Parameters and Optical Layout. LIGO Document T1200410-v2 <https://dcc.ligo.org/LIGO-T1200410/public>.
- [172] F. Bayer-Helms. Coupling coefficients of an incident wave and the modes of a spherical optical resonator in the case of mismatching and misalignment. *Applied Optics*, 23(9):1369, may 1984.
- [173] Antonio Perreca, Aidan F. Brooks, Jonathan W. Richardson, Daniel Töyrä, and Rory Smith. Analysis and visualization of the output mode-matching requirements for squeezing in advanced LIGO and future gravitational wave detectors. *Physical Review D*, 101(10), may 2020.

- [174] A.E. Siegman. *lasers*. University Science Books, 1986.
- [175] L. Schnupp. (unpublished).
- [176] T. M. Niebauer, R. Schilling, K. Danzmann, A. Rüdiger, and W. Winkler. Nonstationary shot noise and its effect on the sensitivity of interferometers. *Physical Review A*, 43(9):5022–5029, may 1991.
- [177] Teng Zhang, Denis Martynov, Andreas Freise, and Haixing Miao. Quantum squeezing schemes for heterodyne readout. *Physical Review D*, 101(12), jun 2020.
- [178] Marius Grundmann. *The Physics of Semiconductors*. Springer Berlin Heidelberg, 2010.
- [179] Vladimir Protopopov. *Practical Opto-Electronics*. Springer International Publishing, 2014.
- [180] Ramon U. Martinelli, Thomas J. Zamerowski, and Paul A. Longeway. 2.6 μm InGaAs photodiodes. *Applied Physics Letters*, 53(11):989–991, sep 1988.
- [181] M. S. Alam, M. S. Rahman, M. R. Islam, A. G. Bhuiyan, and M. Yamada. Refractive index, absorption coefficient, and photoelastic constant: Key parameters of InGaAs material relevant to InGaAs-based device performance. In *2007 IEEE 19th International Conference on Indium Phosphide & Related Materials*. IEEE, may 2007.
- [182] Xing you Chen, Yi Gu, and Yong gang Zhang. Epitaxy and device properties of InGaAs photodetectors with relatively high lattice mismatch. In *Epitaxy*. InTech, mar 2018.

- [183] M. Gendry, V. Drouot, C. Santinelli, and G. Hollinger. Critical thicknesses of highly strained InGaAs layers grown on InP by molecular beam epitaxy. *Applied Physics Letters*, 60(18):2249–2251, may 1992.
- [184] Abhay M. Joshi, Gregory H. Olsen, S. M. Mason, M. Kazakia, and Vladimir S. Ban. Near-infrared (1-3 micron) InGaAs detectors and arrays - crystal growth leakage current and reliability. In Harold I. Schiff and Ulrich Platt, editors, *Optical Methods in Atmospheric Chemistry*. SPIE, feb 1993.
- [185] Xiaowei Li, Ning Li, S. Demiguel, J.C. Campbell, D. Tulchinsky, and K.J. Williams. A comparison of front- and backside-illuminated high-saturation power partially depleted absorber photodetectors. *IEEE Journal of Quantum Electronics*, 40(9):1321–1325, sep 2004.
- [186] Y. Uchida, H. Kakibayashi, and S. Goto. Electrical and structural properties of dislocations confined in a InGaAs/GaAs heterostructure. *Journal of Applied Physics*, 74(11):6720–6725, dec 1993.
- [187] Pamela Jurczak, Kimberly A. Sablon, Marina Gutiérrez, Huiyun Liu, and Jiang Wu. 2.5- μm InGaAs photodiodes grown on GaAs substrates by interfacial misfit array technique. *Infrared Physics & Technology*, 81:320–324, mar 2017.
- [188] Lars Zimmermann, Joachim John, Stefan Degroote, Gustaaf Borghs, Chris Van Hoof, and Stefan Nemeth. Extended wavelength InGaAs on GaAs using InAlAs buffer for back-side-illuminated short-wave infrared detectors. *Applied Physics Letters*, 82(17):2838–2840, apr 2003.

- [189] F N Hooge, T G M Kleinpenning, and L K J Vandamme. Experimental studies on 1/f noise. *Reports on Progress in Physics*, 44(5):479–532, may 1981.
- [190] M. D'Hondt, I. Moerman, and P. Demeester. Dark current optimisation of 2.5 microm wavelength, 2% mismatched InGaAs photodetectors on InP. In *Conference Proceedings. 1998 International Conference on Indium Phosphide and Related Materials (Cat. No.98CH36129)*. IEEE, 1998.
- [191] M. López, H. Hofer, K. D. Stock, J. C. Bermúdez, A. Schirmacher, F. Schneck, and S. Kück. Spectral reflectance and responsivity of ge- and InGaAs-photodiodes in the near-infrared: measurement and model. *Applied Optics*, 46(29):7337, oct 2007.
- [192] S. Forrest. Performance of $\text{In}_x\text{Ga}_{1-x}\text{As}_y\text{P}_{1-y}$ photodiodes with dark current limited by diffusion, generation recombination, and tunneling. *IEEE Journal of Quantum Electronics*, 17(2):217–226, feb 1981.
- [193] Thorlabs. *InGaAs Photodiode FD10D*, October 2017. Rev. D.
- [194] Hamamatsu. G12183 series, July 2019.
- [195] Laser Components. Extended ingaas photodiodes ig26 series. Accessed Jan 2021 <https://www.lasercomponents.com/uk/product/ingaas-500-2600-nm/>.
- [196] Laser Components. Extended ingaas photodiodes ig24 series. Accessed Jan 2021 <https://www.lasercomponents.com/uk/product/ingaas-500-2600-nm/>.

- [197] Laser Components. Extended ingaas photodiodes ig22 series. Accessed Jan 2021 <https://www.tme.eu/gb/details/ig22x1000s4i/photodiodes/laser-components/>.
- [198] Hamamatsu. G12182 series, March 2019.
- [199] Henning Vahlbruch, Moritz Mehmet, Karsten Danzmann, and Roman Schnabel. Detection of 15 dB squeezed states of light and their application for the absolute calibration of photoelectric quantum efficiency. *Physical Review Letters*, 117(11), sep 2016.
- [200] M. López, J. C. Molina, H. Hofer, A. Sperling, and S. Kück. Measurement of the nonlinearity of ge-and InGaAs-photodiodes at high irradiance levels. *MAPAN*, 25(1):47–52, mar 2010.
- [201] J R Smith, P Ajith, H Grote, M Hewitson, S Hild, H Lück, K A Strain, B Willke, J Hough, and K Danzmann. Linear projection of technical noise for interferometric gravitational-wave detectors. *Classical and Quantum Gravity*, 23(2):527–537, dec 2005.
- [202] Klaus W. Hodapp, Joseph B. Jensen, Everett M. Irwin, Hubert Yamada, Randolph Chung, Kent Fletcher, Louis Robertson, Joseph L. Hora, Douglas A. Simons, Wendy Mays, Robert Nolan, Matthieu Bec, Michael Merrill, and Albert M. Fowler. The gemini near-infrared imager (NIRI). *Publications of the Astronomical Society of the Pacific*, 115(814):1388–1406, dec 2003.
- [203] Alan R. Beattie. Quantum efficiency in InSb. *Journal of Physics and Chemistry of Solids*, 23(8):1049–1056, aug 1962.

- [204] Ibrahim Kimukin, Necmi Biyikli, and Ekmel Ozbay. InSb high-speed photodetectors grown on GaAs substrate. *Journal of Applied Physics*, 94(8):5414, 2003.
- [205] A.J. Moseley, M.D. Scott, A.H. Moore, and R.H. Wallis. High-efficiency, low-leakage MOCVD-grown GaInAs/AlInAs heterojunction photodiodes for detection to $2.4\mu\text{m}$. *Electronics Letters*, 22(22):1206, 1986.
- [206] Gilberto A. Umana-Membreno. Personal communication, December 2020. University of Western Australia.
- [207] Analog Devices. *AD797 Ultralow Distortion, Ultralow Noise Op Amp*. Rev. K.
- [208] A. A. Muzammil, C. Mueller, and G. Mueller. Optical Layout and Parameters for the Advanced LIGO Cavities. LIGO Document T0900043-v11 <https://dcc.ligo.org/LIGO-T0900043/public>.
- [209] LIGO Optics Database. <https://galaxy.ligo.caltech.edu/optics/>.
- [210] R. Martin and G. Billingsley. Measurement Uncertainties in Determining the Radius of Curvature for SR2-04. LIGO Document T1300653-v1 <https://dcc.ligo.org/LIGO-T1300653>.
- [211] G. Billingsley. SR301 Figure Measurement. LIGO Document E1101231-v1 <https://dcc.ligo.org/E1101231-v1>.
- [212] M. Smith and D. Coyne. Advanced LIGO L1 Optical Layout, ZEMAX. LIGO Document D0902216-v8 <https://dcc.ligo.org/LIGO-D0902216>.

- [213] C. Bond, P. Fulda, A. Freise, D. Brown, K. Kokeyama, L. Carbone, and A. Perreca. Finesse Input Files for the L1 Interferometer. LIGO Document T1300901-v5 <https://dcc.ligo.org/LIGO-T1300901>.